Which one of the following aqueous solutions will have the highest pH?

A. 0.1 M NaOBr
B. 0.1 M HCO_2Na
C. 0.1 M C_6H_5ONa
D. 0.1 M CH_3COONa

Solution:

- The Ka value of the acid is directly proportional to the strength of the acid. That is $\boxed{Ka \ \alpha \ \text{Acid strength}}$
- The pH value is inversely proportional to the strength of the acid.

That is, $\boxed{pH \ \alpha \ \frac{1}{\text{Acid strength}}}$

So, having highest pH value implies the weakest acid. And the weakest acid is the one having lowest Ka value:

(A) $NaOBr \longrightarrow Na^+ + OBr^-$
0.1M 0.1 M 0.1 M

Na^+ has no reaction with water.

$OBr^- + H_2O \rightleftharpoons HBr + OH^-$

Initial conc.	0.1	-	-
change	-x	+x	+x
equilibrium	0.1 – x	x	x

$$K_b = \frac{K_w}{K_a} = \frac{x^2}{0.1-x}$$

At 25°C $K_w = 1 \times 10^{-14}$

$$\frac{1\times10^{-14}}{2.4\times10^{-9}} = \frac{x^2}{0.1-x}$$, using approximation method.

$$\frac{x^2}{0.1-x} \cong \frac{x^2}{0.1}$$

$$\frac{1\times10^{-14}}{2.4\times10^{-9}} = \frac{x^2}{0.1}, \ x = 6.45 \times 10^{-4}$$

$[OH^-] = x = 6.45 \times 10^{-4}$
pOH = 3.91
pH = 14 – pOH = 10.8

(B) $HCOONa \longrightarrow HCOO^-(aq) + Na^+(aq)$

0.1 M 0.1 M 0.1 M

$HCOO^- + H_2O \rightleftharpoons HCOOH + OH^-$

initial conc.	0.1	-	-
change	-x	+x	+x
equil.	0.1 – x	x	x

$$Kb = \frac{Kw}{Ka} = \frac{x^2}{0.1-x} = \frac{1\times10^{-14}}{1.8\times10^{-4}} = \frac{x^2}{0.1}$$ . . . using approximation method

Solving for $x = 2.35 \times 10^{-6}$
$[OH^-] = x = 2.35 \times 10^{-6}$
$pOH = -\log(2.35 \times 10^{-6}) = 5.62$
pH = 8.37

(C) $C_6H_5ONa \longrightarrow C_6H_5O^-(aq) + Na^+(aq)$
0.1 M 0.1M 0.1 M

$C_6H_5O^- + H_2O \rightleftharpoons C_6H_5OH + OH^-$

initial conc.	0.1	-	-
change	-x	+x	+x
equil.	0.1 – x	x	-x

$$Kb = \frac{Kw}{Ka} = \frac{x^2}{0.1-x} = \frac{1\times10^{-14}}{1\times10^{-10}} = \frac{x^2}{0.1-x}$$

Using approximation and solving for x.
$x = 3.16 \times 10^{-3}$
$[OH^-] = x = 3.16 \times 10^{-3}$
pOH = 2.5, pH = 11.5

(D) 0.1 M CH_3COONa

$CH_3COONa \longrightarrow CH_3COO^-(aq) + Na^+$
0.1M 0.1M 0.1M

$CH_3COO^- + H_2O \rightleftharpoons CH_3COO^- + OH^-$

Initial conc.	0.1	-	-
change	-x	+x	+x
equil. conc.	0.1 – x	x	x

$$Kb = \frac{Kw}{Ka} = \frac{x^2}{0.1-x} = \frac{1\times10^{-14}}{1.8\times10^{-5}} = \frac{x^2}{0.1-x}$$

Telegram: https://t.me/ExtremeSeries

Using approximation and solving for x

$x = 7.45 \times 10^{-6}$

$[OH^-] = x = 7.45 \times 10^{-6}$

pOH = 5.12, pH = 8.87

Answer: C

49. Which of the following titrations will have an equivalence point at a pH < 7?

A. strong acid with weak base
B. weak acid with weak base
C. strong acid with strong base
D. weak acid with strong base

Explanation

Titration type	Equivalence point region
Strong acid – weak base	pH < 7
Weak acid – weak base	Difficult to predict
Strong acid – strong base	pH = 7
Weak acid – strong base	pH > 7

Answer: A

50. Which of the following is a conjugate acid/base pair?

A. HCl/OCl^- C. NH_4^+ / NH_3

B. H_3O^+/OH^- D. H_2SO_4 / SO_4^{2-}

Explanation

Conjugate acid: is Bronsted acid formed when the Bronsted base has accepted a proton. Example NH_4^+ is the conjugate acid of NH_3.

Conjugate base: The Bronsted base formed when a Bronsted acid has donated a proton. Example NH_3 is he conjugate base of NH_4^+. Therefore, the correct conjugate acid/base pair of the choices are: HCl/Cl^-, H_3O^+/H_2O, NH_4^+/NH_3, H_2SO_4/ HSO_4^-.

Answer: C

51. All of the following can act as Bronsted-Lowery bases except:

A. I^- B. NH_3 C. HCO_3^- D. H_2CO_3

Explanation: Bronsted-Lowery base is a proton acceptor. For a given species to be Bronsted-Lowery base, it should contain negative charge or should be compound that undergo self ionization (example H_2O, NH_3, CH_3COOH).

Answer: D

52. Consider the following oxidation – reduction equation.

$2H_2O\,(l) + Al\,(s) + MnO_4^-\,(aq) \longrightarrow Al(OH)_4^-\,(aq) + MnO_2\,(s)$

What are the reducing and oxidizing agents in this reaction?

A. Al (s) is the reducing agent and H_2O is the oxidizing agent.
B. Al (s) is oxidizing agent and H_2O is the reducing agent.
C. Al (s) is an oxidizing agent and MnO_4^- (aq) is the reducing agent.
D. Al (s) is the reducing agent and MnO_4^- (aq) is the oxidizing agent.

Explanation

- Oxidizing agent is a substance that causes oxidation (and itself is reduced in the redox reaction).
- Reducing agent is a substance that caused reduction (and is itself oxidized) in a redox reaction.
- Reduction is an electron gain.
- Oxidation is an electron loss.
- A substance that under goes reduction is said to be reduced.
- A substance that under goes oxidation is said to be oxidized.

In order to identify the oxidizing and reducing agents in a given redox reaction, assign the oxidation number values for the reactant and product based on the rue of assuming oxidation number.

$$\overset{+1\ -2}{2H_2O}(\ell) + \overset{0}{Al}(s) + \overset{+7}{MnO_4^-} \longrightarrow \overset{+3}{Al(OH)_4^-} + \overset{+4}{MnO_2}$$

Oxidation | Reducing agent | Reduction oxidizing agent

Answer: D

53. Consider the following reaction:

$I_2O_5\,(s) + 5CO\,(g) \longrightarrow I_2(s) + 5CO_2\,(g)$

What is the magnitude of the change in the oxidation number of the elements?

A. I, +5 to O, C, +2 to +4 C. I, +0 to 0, C, +2, to +4
B. I, +5 to 0, C, +4 to + 4 D. I, +4 to 0, C, +2 to +4

Explanation By applying the rule of assigning an oxidation number assign the oxidation number values for the reactants and products.

Telegram: https://t.me/ExtremeSeries

$$\overset{+5}{I_2}O(s) + 5\overset{+2}{C}O \longrightarrow I_2(s) + 5\overset{+4}{C}O_2$$

54. Which of the following is true? **Answer: A**
A. A battery is rechargeable.
B. A battery produces electricity spontaneously.
C. A battery has generally no liquid components.
D. A battery produces the same amount of electricity, regardless of composition.

Explanation: A battery is a galvanic cell or a series of combined galvanic cells that can be used as a direct current source. It produces electricity spontaneously. Based on rechargebility batteries can be classified as primary and secondary batteries.

55. All of the following when added to water will produce an electrolytic solution **EXCEPT**. **Answer: B**
A. N_2 (g) B. NaI (s) C. HCl D. KOH (s)

Explanation
Electrolytic solution: A solution that conducts electricity.
Electrolytes: A substance that dissolves to give ions (cations and anions) in a solution.
Electrolytic substance include Acids, Bases and Salts.

56. Which one of the following substance is a non-conductor of electricity? **Answer: A**
A. Graphite
B. $MgCl_2$ (s)
C. Silver
D. H_2SO_4 (aq)

Explanation
There are two kinds of conductivity:
(1) Metallic conduction: is conduction through metallic elements by free mobile electrons.
(2) Ionic conduction: is conduction of electricity by the mobility of cations and anions in the solution.
- Ionic compounds do not coduct electricity in soldi state. Salts, acids and bases are electrolytes.

Note: Carbon has two crystalline alltropes.
- Graphite: conducts electricity
- Diamond: do not conduct electricity

Answer: B

57. A solution of 25°C contains the metal ions Ni^{2+}, Pt^{2+}, and Pd^{2+}, all at 1.0M concentrations. Consider the following standard reduction potentials:

$Ni^{2+} + 2e^- \longrightarrow Ni$ $E^\circ = -0.23V$
$Pt^{2+} + 2e^- \longrightarrow Pt$ $E^\circ = 1.20V$
$Pd^{2+} + 2e^- \longrightarrow Pd$ $E^\circ = 0.99V$

Which metal (s) will plate out first when the solution is electrolyzed?
A. Ni B. Pd C. Pt D. Ni and Pd

Explanation
The more positive the reduction potential (E°_{red}), the greater tendency for the substance to be reduced.

58. How can silver be plated onto Nickel? **Answer: C**
A. Electricity to a nickel anode in a solution of silver ions.
B. Electricity to a silver anode in a solution of nickel ions.
C. A solution of nickel ions to react with a piece of silver.
D. Electricity to a nickel cathode in a solution of silver ions.

Explanation
One application of electrolysis is Electroplating. During electroplating one should keep the plating material at the anode and the material to be plated should be kept at cathode. Besides the electrolytic solution be a solution of the plating material ion.

Answer: D

59. For the reaction $A + B \longrightarrow C + D$, $\Delta H^\circ = +40$ kJ and $\Delta S^\circ = +50$J/k. Therefore, the reaction under standard condition is:
A. Spontaneous at all temperatures.
B. Non spontaneous at all temperatures.
C. Spontaneous at temperatures greater than 800k.
D. Spontaneous only at temperatures between 10k and 800k.

Explanation
Finding the Gibbs free energy
$\Delta G = \Delta H - T\Delta S$
$\Delta G = 40 \times 10^3$ J $-$ T(50 J/k), which implies the value of ΔG depends on the value of temperature.
1. If $\Delta H > T\Delta S$, $\Delta G > 0$ and the process will be nonspontaneous.

Telegram: https://t.me/ExtremeSeries

2. At a certain temperature when $\Delta H = T\Delta S$, ΔG becomes zero hence the process is at equilibrium.
3. At a temperature when $\Delta H = T\Delta S$, ΔG becomes negative so the process is spontaneous.

When T = 800 k

$\Delta G = \Delta H - T\Delta S = (40 \times 10^3\ J) - 800k\ (50\ J/k) = 0$ (equilibrium)

When T= 10 k

$\Delta G = 39500$ (nonspontaneous)

∴ The process becomes spontaneous at temperature > 800k.

Answer: C

60. For the conversion of C (diamond) ⟶ C (Gra phite), $\Delta H = -3kJ$. What does this mean?
   A. Both are equally stable.
   B. Graphite is more stable than diamond.
   C. Diamond is more stable than graphite:
   D. Graphite has more stable energy than diamond.

Explanation

The conversion of diamond to graphite is spontaneous, this implies graphite is more stable than diamond.

Answer: B

61. What is the change in internal energy, ΔE of a system if it absorbs 300 kJ heat from the surroundings and does 500 kJ of work on the surroundings?
   A. 100kJ B. −200kJ C. 400kJ D. 500kJ

Solution:

q = +300 kJ (the system absorbs heat energy)

w = −500 kJ (the system does work on the surrounding)

$\Delta E = q + w = 300\ kJ - 500\ kJ = -200\ kJ$ **Answer: B**

62. Which of the following element has the highest melting point?
   A. Iodine C. Mercury
   B. Tungsten D. Bromine

Explanation: Metals in general have high melting and boiling points. Exceptionally, tungsten has the highest melting point above 3000°C.

Answer: B

63. Which of the following compounds is unlikely to contain ionic bonds?
   A. CO B. NaF C. LiCl D. $MgBr_2$

Explanation:

Ionic bond is formed by complete transfer of one or more electrons. It is formed between metals and non – metals and between ammonium ion (NH_4^+) and anions.

Answer: A

64. What is the number of moles of atoms and the number of atoms in a 10.0 g sample of copper?
   A. 0.08 mol Cu atoms and 2.16×10^{23} atoms
   B. 0.16 mol Cu atoms and 9.63×10^{22} atoms
   C. 0.16 mol Cu atoms and 9.63×10^{23} atoms
   D. 0.31 mol Cu atoms and 4.16×10^{23} atoms

Solution:

$$n = \frac{m}{M} = \frac{10\ g}{64\ g/mol} = 0.16\ mol\ Cu\ atoms$$

$1\ mol\ Cu = 6.02 \times 10^{23}$ atoms

$0.16\ mol\ Cu = ?$

$$\Rightarrow ? = \frac{0.16\ mol\ Cu \times 6.02 \times 10^{23}\ atoms}{1\ mol\ Cu}$$

$? = 9.63 \times 10^{22}$ atoms

$\Rightarrow 10\ g\ Cu = 0.16\ mol\ Cu = 9.63 \times 10^{22}$ atoms of Cu

Answer: B

65. Which of the following statements correctly describes a chemical reaction at equilibrium?
   A. The concentrations of the products and reactants are equal.
   B. The change in the concentrations of the products and reactants is constant.
   C. The rate of the forward reaction is less than the rate of the reverse reaction.
   D. The rate of the forward reaction is greater than the rate of the reverse reaction.

Explanation:

At equilibrium:

- Rate of forward reaction is equal to rate of reverse reaction.
- The change in the concentrations of the products and reactants is constant.

Answer: B

Telegram: https://t.me/ExtremeSeries

66. Under what conditions will the enthalpy change of a process equals the amount of heat transferred into or out of the system?
A. Under constant volume.
B. Under constant pressure.
C. Under constant pressure and volume.
D. Under constant pressure and temperature.

Explanation:
Enthalpy is defined as the sum of the internal energy and the pressure – volume product of a system.
$H = E + PV$
$\Delta H = \Delta E + P\Delta V$ (at constant pressure)
But $\Delta E = q - P\Delta V$
$\Rightarrow \Delta H = (q - P\Delta V) + P\Delta V$
$\Delta H = q_P$
q_P – heat transferred at constant pressure.
Enthalpy change is heat transferred at constant pressure. However, change in internal energy is heat transferred at constant volume.

Answer: B

67. What is the hydroxide ion concentration for a solution with a pH of 10 at 25°C?
A. 10^{-14}M B. 10^{-10}M C. 10^{-7}M D. 10^{-4}M

Solution:
$pH = 14 - \log [OH^-]$
$pH - 14 = \log [OH^-]$
$[OH^-] = 10^{pH-14} = 10^{10-14} = 10^{-4}$M

Answer: D

68. Which law relates the concentration of a dissolved gas, C_g, to its partial pressure?
A. Henry's law C. Boyle's law
B. Raoul's law D. Ideal gas law

Explanation:
The quantitative relationship between gas solubility (concentration) and pressure was given by Henry's law. This law states that at a given temperature the solubility of a gas in a liquid is directly proportional to the partial pressure of the gas over the solution.
$C_g = kH.P_g$
C_g – molar concentration of the gas in mol/L

K_H – Henry's law constant in $\frac{mol}{atm.L}$
P_g – partial pressure of the gas

Answer: A

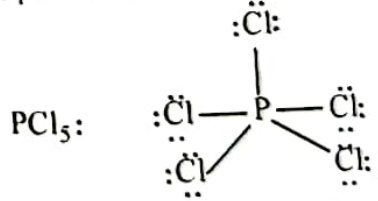
69. How many atoms are present in 22g of CO_2?
A. 3.01×10^{23} C. $2 \times 6.02 \times 10^{23}$
B. 6.02×10^{23} D. $1.5 \times 6.02 \times 10^{23}$

Solution:
$n = \frac{m}{M} = \frac{22g}{44g/mol} = 0.5$ mol
Number of atoms in 22 g of CO_2 = n_{CO2} = (subscript of C + subscript of O) $\times 6.02 \times 10^{23}$ atoms = 0.5 mol $\times$ (1 + 2) $\times 6.02 \times 10^{23}$ atoms = $1.5 \times 6.02 \times 10^{23}$ atoms = 9.03×10^{23} atoms

Answer: D

70. What is the hybridization of phosphorous atom in PCl_5?
A. sp^3d B. sp^3d^2 C. sp^3 D. sp^2

Explanation:
Draw the Lewis structure for PCl_5. Number of hybrid orbitals equal to number of sets of electrons on the central atoms.

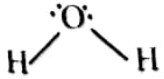


$\Rightarrow$There are five sets of electrons on P. Therefore, the hybridization of P in PCl_5 will be sp^3d.

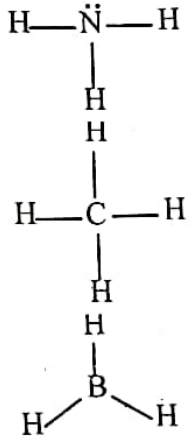
Answer: A

71. Which of the following species has the smallest H – X – H bond angle where X is the central atom?
A. OH_2 B. NH_3 C. CH_4 D. BH_3

Explanation:

Lewis structure	Shape	Bond angle
	Bent (V – shape)	104.5°

Telegram: https://t.me/ExtremeSeries



Trgonal pyramid 107°

Tetrahedral 109°

Trigonal planar 120°

Answer: A

72. The reaction, $2X + Y \longrightarrow Z$, was studied and the following data were obtained.

Expt	[X]	[Y]	Rate (mol. L^{-1}, S^{-1})
1	3.0	3.0	1.8
2	3.0	1.5	0.45
3	1.5	1.5	0.45

What is the proper rate expression?

A. rate = k [X] C. rate = K $[Y]^2$

B. rate = K [Y] D. rate = K [X][Y]

Solution:

Let the rate law be: rate = k $[X]^a[Y]^b$. Find a by using expt 2 and 3 and b by using expt 1 and 2.

$$\frac{rate_2}{rate_3} = \frac{k[x]_2^a[y]_2^b}{k[x]_3^a[y]_3^b}$$

$$\frac{0.45}{0.45} = \frac{(3.0)^a(1.5)^b}{(1.5)^a(1.5)^b}$$

$1 = 2^a$

$2^0 = 2^a$, $a = 0$

$$\frac{rate_2}{rate_3} = \frac{[y]_1^b}{[y]_2^b}$$

$$\frac{1.8}{0.45} = \left(\frac{3.0}{1.5}\right)^b$$

$4 = 2^b$

$2^2 = 2^b$

$b = 2$

$\Rightarrow$ rate = k $[x]^0[y]^2$ = k $[y]^2$

Answer: C

73. What mass of aluminum is produced in one hour by the electrolysis of molten $AlCl_3$ with a current of 10A?

A. 1.5g B. 2.5g C. 3.4g D. 4.3g

Solution:

$Al^{3+} + 3e^- \longrightarrow Al \Rightarrow Z = 3$

$M_{Al} = 27$ g/mol

I = 10A

t = 1 hr = 3600 s

$$\frac{m}{M} = \frac{It}{ZF} = 3600s$$

$$m = \frac{10A \times 3600s \times 27g/mol}{3 \times 96500C/mol} = 3.36\ g$$

Answer: C

74. Suppose reactions A $\longrightarrow$ B and B $\longrightarrow$ A are both elementary of $8 \times 10^2\ s^{-1}$ and $4 \times 10^4\ s^{-1}$, respectively. What is the value of the equilibrium constant for the equilibrium,

$A \rightleftharpoons B$?

A. 2×10^{-2} C. 4×10^{2}

B. 0.5×10^{2} D. 4×10^{-2}

Solution: $A \longrightarrow B$, $r_f = k_f[A]$

$B \longrightarrow A$, $r_r = k_r[B]$

$A \rightleftharpoons B$, At equilibrium, $r_f = r_r \Rightarrow k_f[A] = k_r[B]$

$$\frac{k_f}{k_r} = \frac{[B]}{[A]} = K_C$$

$$K_C = \frac{k_f}{k_r} = \frac{8 \times 10^2 s^{-1}}{4 \times 10^4 s^{-1}} = 2 \times 10^{-2}$$

Answer: A

Telegram: https://t.me/ExtremeSeries

75. Which of the following ion is the most abundant in sea water?
A. Na^+ B. Ca^{2+} C. Cl^- D. HCO_3^-

Explanation:
The most abundant ion in sea water is the sodium ion because the water is salty due to the presence of excess sodium chloride.

Answer: A

76. Which element is added to natural rubber to make it harder and reduce its susceptibility to oxidation and chemical attackers?
A. Sulfur C. Carbon
B. Silicon D. Nitrogen

Explanation:
The properties of natural rubber are improved by heating it with sulphur. This process is called vulcanization.

Answer: A

77. Which of the following is NOT a carbohydrate?
A. Starch C. Glycine
B. Glucose D. Cellulose

Explanation:
Starch – is a poly saccharide
Glucose – is a mono saccharide
Glycine – is an amino acid
Cellulose – is a polysaccharide

Answer: C

78. Why does a pressure cooker reduce cooking time?
A. The heat is more evenly distributed.
B. A stronger flame is used for cooking.
C. The high pressure tenderizes the food.
D. The boiling point of water inside the cooker is increased.

Answer: B

79. The pH at room temperature of a 0.1M solution of formic acid ($HCHO_2$) was measured to be 4.50. What is the hydrogen ion concentration?
A. 3.16×10^{-5}M C. 6.32×10^{-3}M
B. 3.16×10^{-12}M D. 6.32×10^{-4}M

Solution:
$pH = -\log [H^+]$
$[H^+] = 10^{-pH} = 10^{-4.5} = 3.16 \times 10^{-5}$M

Answer: A

80. What is the molarity of a solution containing 10 g o sulfuric acid in 500 mL of solution?
A. 0.02 B. 0.03 C. 0.12 D. 0.2

Solution:

$$M = \frac{m_{H_2SO_4}}{M_{H_2SO_4}}$$

$$M = \frac{10g}{98g/mol \times 0.5L} = 0.2M$$

Answer: D

Telegram: https://t.me/ExtremeSeries

1. Which of the following is NOT a basic SI unit?
   A. Candela C. Mole
   B. Gram D. Second

Explanation:
There are seven basic quantities with seven basic units.

Basic quantity	Basic SI units
Temperature	Kelvin
Mass	Kilogram
Time	Second
Length	Meter
Current	Ampere
Amount of substance	Mole
Luminuous intensity	Candela

Answer: B

2. What is the number of significant figures in 0.0030050?
   A. 4 B. 5 C. 7 D. 8

Explanation: Zeros before non-zero digits are not significant. But between and after non – zero digits, they are significant. As a result, the number of significant figures in 0.0030050 are 5.

Answer: B

3. Which of the following properties of a substance does NOT represent an intensive physical property?
   A. Boiling point C. Density
   B. Color D. Volume

Explanation: Intensive properties are properties which do not depend on the amount of matter present. Example: color, melting point, boiling point, density, hardness, conductivity, odour, specific gravity, freezing point, etc. Extensive physical properties are properties that depend on the amount of substance.
Example: Volume, length thickness, area, weight of substance, diameter etc.

Answer: D

4. What is the kelvin scale (K) corresponding to the temperature readings when the degree Celsius (°C) is identical to the degree Fahrenheit (°F)?
   A. 0°K B. 37K C. 233K D. 273K

Solution: $^{\circ}C = \frac{5}{9}(^{\circ}F - 32)$ $\quad \frac{9}{5}{}^{\circ}C - {}^{\circ}C = -32$

$^{\circ}F = \frac{9}{5}\cdot{}^{\circ}C + 32$ $\quad \frac{4}{5}{}^{\circ}C = 32$

If $^{\circ}C = {}^{\circ}F$ $\quad {}^{\circ}C = -32 \times \frac{5}{4}$

$\Rightarrow {}^{\circ}C = \frac{5}{9}(^{\circ}C - 32)$ $\quad {}^{\circ}C = -40^{\circ}C$

$\frac{9}{5}{}^{\circ}C = {}^{\circ}C - 32$

Therefore, °C = °F at -40
K = °C + 273 = -40 + 273 = 233°K

Answer: C

5. If a piece of aluminum (Al) foil measuring 24 cm by 31 cm has a mass of 10.35 g, (density of Al = 2.70 g/cm^3). What is the thickness of the foil in millimeters?
   A. 5.15×10^{-3} C. 3.833
   B. 5.15×10^{-2} D. 744

Solution:

$\rho = \frac{m}{v}$ $\quad$ ρ – density; m – mass; v – volume

$v = \frac{m}{\rho} = \frac{10.35g}{2.70g/cm^3} = 3.833cm^3$

v = length × width × thickness

$\text{thickness} = \frac{3.833cm^3}{24cm \times 31cm} = 5.15 \times 10^{-3} cm$

1 cm = 10 mm
5.15×10^{-3} cm =?

$? = \frac{5.15 \times 10^{-3} cm \times 10mm}{1\ cm} = 5.15 \times 10^{-2} mm$

⇒ Thickness of the Al foil = 5.15×10^{-2} mm

Answer: B

6. What sizes of particles and velocities can one consider quantum effect?
   A. Particles with very large mass and large velocities.
   B. Particles with large mass and large velocities.
   C. Particles with small mass and large velocities.
   D. Particles with small mass and small velocities.

Telegram: https://t.me/ExtremeSeries

Explanation: The size of particles and velocities of the particles are inversely related.

$v = \frac{h}{m\lambda}$ v – velocity of a particle

λ – wave length
m – mass of the particle
h – plank's constant

Answer: C

The quantum numbers listed below are meant for four different electrons in an atom.

I. $n = 4, \ell = 0, m_\ell = 0, m_s = +½$

II. $n = 3, \ell = 1, m_\ell = 1, m_s = +½$

III. $n = 4, \ell = 2, m_\ell = 0, m_s = +½$

IV. $n = 4, \ell = 1, m_\ell = 0, m_s = -½$

When these sets of quantum numbers are arranged in order to increasing energy, one may set:

A. I < III < II < IV C. II < I < III < IV
B. I < II < III < IV D. IV < III < II < I

Explanation: The atomic sub levels or sub shell are given by the values of n and ℓ.

Sub shell	Value of n	Value of ℓ	Value of m_ℓ	Value of m_s
1s	1	0	0	± ½
2s	2	0	0	± ½
2p	3	1	-1, 0, 1	± ½
3s	3	0	0	± ½
3p	4	1	-1, 0, 1	± ½
4s	3	0	0	± ½
3d	4	2	-2, -1, 0, 1, 2	± ½
4p	5	1	-1, 0, 1	± ½
5s	4	0	0	± ½
4d	5	2	-2, -1, 0, 1, 2	± ½
5p	6	1	-1, 0, 1	± ½
6s	6	0	0	± ½
4f	4	3	-3, -1, -1, 0, 1, 2, 3	± ½
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.

⇒ I. 4s, II. 3p (one of the three p – orbitals)
III. 4d (one of the five d – orbitals)
Iv. 4p (one of the three p – orbitals)
⇒ In increasing order of energy, according to building up principle:
3p < 4s < 4p < 4d i.e., II < I < IV < III **Answer: ___**

(8) What is the wave length associated with an electron of mass, $m = 9.11 \times 10^{-28}$ g, travelling at 40% of the velocity of light?

A. 6.06×10^{-13} m C. 6.06×10^{-12} m
B. 2.42×10^{-13} m D. 2.42×10^{-11} m

Solution

$V_e = \frac{h}{M_e h}$, but, $V_e = 40\%C$

$\lambda = \frac{h}{M_e v_e}$ $V_e = \frac{40}{100} \times 3 \times 10^8 \text{ m/s}$

M_e – mass of an electron $V_e = 1.2 \times 10^8$ m/s
C – speed of light
V_e – velocity of an electron

$$\Rightarrow \lambda = \frac{6.626 \times 10^{-34} \text{ J.s}}{9.11 \times 10^{-31} \text{kg} \times 1.2 \times 10^8 \text{ m/s}}$$

$1J = 1 \text{ kg m}^2/\text{s}^2$

$$\Rightarrow \lambda = \frac{6.626 \times 10^{-34} \text{ kg.m}^2/\text{s}^2}{9.11 \times 10^{-31} \text{kg} \times 1.2 \times 10^8 \text{ m/s}}$$

$\lambda = 6.06 \times 10^{-12}$ m **Answer: C**

(9) An electron has a spin quantum number, $m_s = +½$ and a magnetic quantum number $m_\ell = +1$. In which of the following orbital will it NOT be present?

A. s – orbital C. d – orbital
B. p – orbital D. f – orbital

Explanation:

Each sub level is designated by the azimuual quantum number, ℓ :

ℓ = 0, is an s sub level ℓ = 2, is a d – orbital

ℓ = 1, is a p – orbital ℓ = 3, is an f – orbital

The value of magnetic quantum number, m_ℓ, is from $-\ell$ to $+\ell$.

Telegram: https://t.me/Extremeseries

For $\ell = 0, m_\ell = 0$

For $\ell = 1, m_\ell = -1, 0, 1$

For $\ell = 2, m_\ell = -2, -1, 0, 1, 2$

For $\ell = 3, m_\ell = -3, -2, -1, 0, 1, 2, 3$

$\Rightarrow m_\ell = +1$ is never found in an s – orbital since the value of ℓ is 0.

Answer: A

10. Which of the following represents the general configuration of the transition elements?

A. ns^2np^6 C. $ns\,(n-2)\,f$

B. $ns\,(n-1)\,d$ D. $ns^2np^6\,(n-1)\,d^{10}$

Explanation:

Transition elements are sub – group (IB – VIIIB) or d – block elements that have variable valence. Their general electron configuration is represented by $ns\,(n-1)\,d$. For example:

IB – $ns^1\,(n-1)\,d^{10}$
IIB – $ns^2\,(n-1)\,d^{10}$
IIIB – $ns^2\,(n-1)\,d^1$
IVB – $ns^2\,(n-1)\,d^2$
VB – $ns^2\,(n-1)\,d^3$
VIB – $ns^1\,(n-1)\,d^5$
VIIB – $ns^2\,(n-1)\,d^5$

VIIIB $\begin{cases} ns^2(n-1)d^6 \\ ns^2(n-1)d^7 \\ ns^2(n-1)d^8 \end{cases}$

Answer: B

11. Which of the following quantum number (s) is (are) related to the size and energy of an electron in a hydrogen atom?

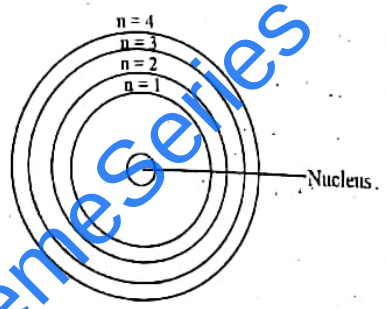
A. n B. n, ℓ C. n, ℓ, m D. n, ℓ, m, s

Explanation:

The size and energy of an electron in a hydrogen atom is affected by the principal quantum number, n. As n increases, the energy and size of an electron increases.

$E_{e^-} = \dfrac{-2.18\times10^{-18}\,J}{n^2}$; E_{e^-} – energy of an electron

The negative sign (–) implies, as n increases the energy also increases.



Answer: A

12. Which of the following statements is TRUE?

A. Ultra violet light has longer wave length than visible light.

B. The energy of radiation decreases as the wave length decreases.

C. The frequency of radiation increases as the wave length decreases.

D. Wave length of an electromagnetic radiation increases as wave length increases.

Explanation:

Gamma rays x-rays UV visible IR →

- Wave length increases
- Energy and frequency decreases

$E = h\nu$ E – energy

$E = \dfrac{hc}{\lambda}$ ν – frequency

c – speed of light

$\Rightarrow$ Energy is directly related with frequency but inversely related with wave length

$c = \lambda\,\nu$ $\nu = \dfrac{c}{\lambda}$

$\Rightarrow$ Frequency is inversely related with wave length.

Answer: C

13. Which of the following ionic compounds is formed from the reaction between magnesium and nitrogen?

A. MgN_2 C. Mg_3N_2

B. Mg_2N_2 D. Mg_2N_3

Telegram: https://t.me/Extremeseries

Explanation:
Magnesium loses two electrons and nitrogen gains three electrons to achieve the nearest noble gas configuration.

$$Mg \longrightarrow \overset{2+}{Mg} + 2e^-$$

$$N + 3e^- \longrightarrow N^{3-}$$

We use a simple criss – cross method to write the formula of ionic compounds by omitting the charge.

$Mg^{2+} \rightleftharpoons N^{3-} = Mg_3N_2$

Answer: C

14. Which of the following molecules represents a non –polar covalent bond?

A. B – Cl C. Cl – Cl
B. C – Cl D. Mg – Cl

Explanation:
Non – polar covalent bond is formed when electrons are shared equally because of similar electronegativity of the bonded atoms.
Example:

H – H O = O F – F I – I etc
N ≡ N Cl – Cl Br – Br

Answer: C

15. Which one of the following groups in the periodic table has paramagnetic atoms?

A. Group zero C. Group IIB
B. Group IIA D. Group IVA

Explanation:
Group IVA elements have unpaired electrons.
Example: ${}_6C: 1s^2 2s^2 2p_x^1 2p_y^1 2p_z^0$

Answer: D

16. Which of the following molecules has a trigonal bi pyramidal structure?

A. SF_4 B. IF_5 C. ICl_4^- D. BrF_5

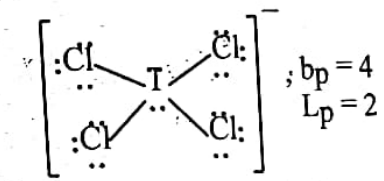
Explanation:
Draw the Lewis structure to determine the structure of the given molecules.

SF_4 : , $L_p = 1$, $b_p = 4$

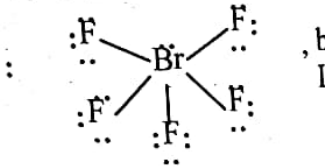
If there are 4 bonding pairs (bp) and 1 lone pair on the central atom, the shape of the molecule is see – saw and trigonal bipyramidal electron arrangement.

IF_5 : , $b_p = 5$, $L_p = 1$

If there are 5 bonding pairs and 1 lone pair on the central atom, the shape (structure) of the molecules is square pyramidal.

ICl_4^- :  ; $b_p = 4$, $L_p = 2$

If there are 4 bonding pairs and 2 lone pairs on the centeral atom, the structure of the molecule is square planar.

BrF_5 :  , $b_p = 5$, $L_p = 1$

There are 5 bonding pairs and one lone on the centeral atom. As a result, the shape of the molecule is square pyramidal.

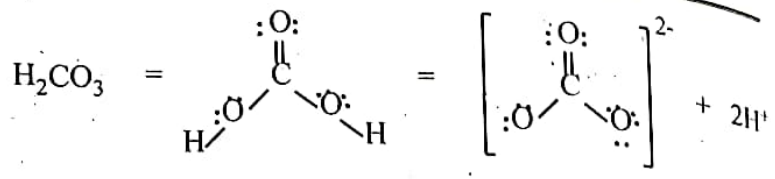
Answer: _____

17. The total number of electrons participating in the bond formation of carbonate ion CO_3^{2-}, in the molecule of carbonic acid are:

A. 16 B. 10 C. 8 D. 5

Explanation:
Draw the Lewis structure of carbonic acid, H_2CO_3:

Telegram: https://t.me/Extremeseries



Four bounding pairs (8 electrons) participate in bond formation of the carbonate ion.

Answer: C

18. Which of the following crystals possess high electrical and thermal conductivities?

A. Ionic crystals C. Molecular crystals
B. Metallic crystals D. Covalent network

Explanation:
Metallic crystals transmit heat and electricity due to free electrons that carry charge.

Answer: B

19. Which of the following hybrid orbitals is favoring the formation of trignal bipyramidal?

A. sp^3d B. sp^3 C. sp^3d^2 D. sp^3d^3

Explanation:

Electron arrangement	Hybridization
Linear	sp
Trigonal planar	sp^2
Tetrahedral	sp^3
Trigonal bipyramidal	sp^3d
Octahedral	sp^3d^2

Answer: A

20. Which of the following molecules/molecular ions is paramagnetic according to the molecular orbital theory?

A. O_2^{2-} B. O_2 C. F_2 D. O_2^{2+}

Explanation: A species with unpaired electron exhibits paramagnetic property. Such species is attracted by an external magnetic field. A species in which all the electrons are paired, exhibits diamagnetism. All species with odd number of electrons are paramagnetic species. For species with even number of electrons, write electron configuration by using molecular orbitals. O_2^{2-}, total electrons $= 2 \times 8 + 2 = 18$

$(\delta 1s)^2(\delta 1s^*)^2(\delta 2s)^2(\delta^* 2s)^2(\delta 2px)^2(\pi 2py)^2(\pi 2py^*)^2(\pi 2p2)^2(\pi^* 2p2)^2$

$\Rightarrow O_2^{2-}$ is diamagnetic species.

O_2, total electrons $= 2 \times 8 = 16$

$\Rightarrow (\delta 1s)^2(\delta^* 2s)^2(\delta 2s)^2(\delta^* 2s)^2(\delta 2px)^2(\pi 2py)^2(\pi^* py)^1(\pi 2p2)^2(\pi 2pz)^1$

O_2 is paramagnetic species because it has two unpaired electrons.

F_2 is similar with O_2^{2-}

O_2^{2+}, total electrons $= 2 \times 8 - 2 = 14$

$\Rightarrow (\delta 1s)^2(\delta 1^* s)^2(\delta 2s)^2(\delta^* 2s)^2(\pi 2py)^2(\pi 2py)^2(\delta 2px)^2(\pi 2p2)^2$

O_2^{2+} is diamagnetic species.

Answer: B

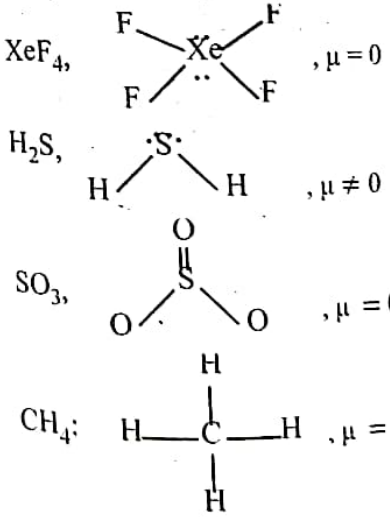
21. Which of the following molecules has a dipole moment?

A. XeF_4 B. H_2S C. SO_3 D. CH_4

Explanation:
Dipole moment, μ, is defined as the magnitude of the separated charges g multiplied by the distance, d, between their centers,

$$\mu = q \times d$$

A molecular dipole is represented by a crossed arrow directed toward the more negative atom.



Answer: B

Telegram: https://t.me/ExtremeSeries

22. What is a valid rate expression for the following reaction?

$2NO + 2H_2 \longrightarrow N_2 + 2H_2O$

A. $\frac{1}{2}\frac{\Delta[H_2]}{\Delta t}$ C. $-\frac{1}{2}\frac{\Delta[NO]}{\Delta t}$

B. $-\frac{1}{2}\frac{\Delta[H_2O]}{\Delta t}$ D. $-\frac{\Delta[N_2]}{\Delta t}$

Explanation:

Rate is the speed at which reactants disappear and products appear. For a general reaction:

$aA + bB \longrightarrow cC + dD,$

$$\text{Rate} = -\frac{1}{a}\frac{\Delta[A]}{\Delta t} = -\frac{1}{b}\frac{\Delta[B]}{\Delta t} = \frac{1}{c}\frac{\Delta[C]}{\Delta t} = \frac{1}{d}\frac{\Delta D}{\Delta t}$$

$$\Rightarrow \text{Rate} = -\frac{1}{2}\frac{[NO]}{\Delta t} = -\frac{1}{2}\frac{\Delta[H_2]}{\Delta t} = \frac{\Delta[N_2]}{\Delta t} = \frac{1}{2}\frac{\Delta[H_2O]}{\Delta t}$$

Answer: C

23. For the reaction;

$2A + B \longrightarrow C$

The following experimental results were obtained.

Experiment	[A]	[B]	Rate [mol $L^{-1}s^{-1}$]
1	0.50	0.50	0.300
2	0.50	0.25	0.075
3	0.25	0.25	0.075

What is the value of the rate constant?

A. 0.6 mol $L^{-1}s^{-1}$ C. 1.2 L mol^{-1} s^{-1}

B. 0.6 mol^{-1} Ls^{-1} D. 2.4 mol L^{-1} s^{-1}

Solution:

Determine the rate law by using initial method. Find the order with respect to A by using experiment 2 and 3. Determine the order with respect to B by using experiment 1 and 2.

Rate = $k[A]^x[B]^y$, k – rate constant

$$\frac{\text{Rate 2}}{\text{Rate 3}} = \frac{k[A]_2^x[B]_2^y}{k[A]_3^x[B]_3^y}$$

$$\frac{0.075}{0.075} = \frac{k(0.50)^x(0.25)^y}{k(0.25)^x(0.25)^y}$$

$$1 = \left(\frac{1}{2}\right)^x$$

$$\left(\frac{1}{2}\right)^0 = \left(\frac{1}{2}\right)^x$$

$x = 0 \Rightarrow$ The order of a reaction with respect to A is zero.

$$\frac{\text{Rate 1}}{\text{Rate 2}} = \frac{k[A]_1^x[B]_1^y}{k[A]_2^x[B]_2^y}$$

$$\frac{0.300}{0.075} = \frac{k(0.50)^x(0.50)^y}{k(0.50)^x(0.25)^y}$$

$4 = 2^y$

$2^2 = 2^y \Rightarrow y = 2$

The reaction is second order with respect to reactant B.

$\Rightarrow \text{Rate} = k[B]^2$

Use either of the experiments to find the value of rate constant, k.

$\text{Rate}_1 = k[B]^2$

$$k = \frac{\text{Rate}_1}{[B]_1^2} = \frac{0.300\text{mol } L^{-1}s^{-1}}{(0.50\text{mol/L})^2} = 1.2\text{Lmol}^{-1}\text{s}^{-1}$$ **Answer: C**

24. Increase in temperature of rate of a given reaction is due to the increase in the:

A. Extent of molecular dissociation.

B. Activation energy of the reaction.

C. Frequency of collisions of the reacting species.

D. Numerical value of the rate constant of the reaction.

Explanation: Temperature is the measure of the average kinetic energy of particles. As temperature increases, the number of collision between reacting species also increases. As a result, the rate of a reaction also increases. **Answer: C**

25. The reaction for the formation of nirosyl chloride:

$2NO\ (g) + Cl_2\ (g) \rightleftharpoons 2NOCl\ (g)$ was studied at 25°C. The value of K_P for this reaction at 25°C is 1.9×10^3 atm^{-1}. What is the value of K_C at 25°C?

A. 1.9×10^{-3} L/mol C. 4.6×10^4 L/mol

B. 3.8×10^{-3} L/mol D. 4.6×10^5 L/mol

Telegram: https://t.me/ExtremeSeries

Solution:

$$K_P = \frac{P_{NOCl}^2}{P_{NO}^2 P_{Cl_2}}$$

$$K_C = \frac{[NOCl]^2}{[NO]^2[Cl_2]}$$

From ideal gas equation, we know that pv = nRT, R – universal gas constant, T – kelvin temperature, P – pressure and v – volume.

$P = \frac{n}{v}RT$, but $\frac{n}{v}$ is concentration

$P = cRT$

$$K_P = \frac{([NOCl]RT)^2}{([NO]RT)^2[Cl_2](RT)^1}$$

$$K_P = \frac{[NOCl]}{[NO)]^2} \cdot (RT)^{2-(2+1)}$$

$K_P = K_C (RT)^{-1}$

$R = 0.082$ atm. L/ mol. K, $T = 25°C = 25 + 273 = 298$ k

$\Rightarrow K_C = 1.9 \times 10^3$ atm^{-1} $\times$ 0.082 atm.L/mol. K $\times$ 298 k

$K_C = 4.6 \times 10^4$ L/mol **Answer: C**

26. Three gases are in equilibrium in a closed chamber sealed with a piston. The following equilibrium is established.
$2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$. What will happen if the piston is pushed into the champer?

A. The mole fraction of N_2 is increases.
B. The mole fraction of N_2 decreases.
C. The mole fraction of N_2 remains same.
D. The mole fraction of N_2 increases and then decreases.

Explanation:

Increasing pressure shifts the equilibrium to the smaller number of molecules and decreasing pressure shifts to the larger number of molecules. In the above chamber, the equilibrium shifts to the left. That is, formation of ammonia (NH_3) is favoured. The amount of N_2 and H_2 decreases.

Answer: B

27. Which of the following statements is TRUE about equilibrium reaction?

A. No more reactants are transformed into products.
B. There are equal amounts of reactants and products.
C. The rate constant for the forward reactions equals that of the reverse reaction.
D. The rate for the forward reaction equals that of the reverse reaction.

Explanation:

Characteristics of chemical equilibrium:

- The rate of the forward reaction is equal to the rate of reverse reaction.
- The equilibrium is dynamic in nature.
- All reactants and products are present at equilibrium.
- The concentration of reactants and products are generally not equal.
- The free energy change of the reaction is zero at equilibrium.

Answer: D

28. What is the process that converts liquid vegetable oils to solid fats?

A. Hydration
B. Hydrogenation
C. Hydrolysis
D. Saponification

Explanation:

Oils can be converted to fats by the addition of hydrogen (hydrogenation) at high pressure in the presence of nickel or palladium as catalyst.

Answer: B

29. What would be the solubility of $HOCH_2(CH_2)_6CH_2OH$ compared to $CH_3(CH_2)_6CH_2OH$?

A. Less soluble in water.
B. More soluble in water.
C. The same solubility in water.
D. More soluble in a non – polar solvent such as dichloro ethane.

Explanation:

Polyhydric alcohol (alcohols with more than one hydroxyl groups) are more soluble than monohydric alcohols in water.

Answer: B

Telegram: https://t.me/Extremeseries

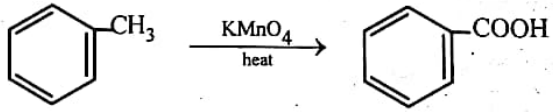
30. Which of the following reactions will produce an alkyl carboxylic acid?
A. Heating methyl ketone with an acid and iodine.
B. Reacting an alkyl halide with hydrogen gas and platinum.
C. Reacting an alcohol with ozone.
D. Oxidation of a primary alcohol with hot permanganate or chromate.

Explanation:
Carboxylic acids are organic compounds that contain at least one carboxyl group in their structure. Carboxylic acids are prepared by oxidation reaction. Alkyl carboxylic acids are manufacture by oxidation of primary alcohols, while aromatic carboxylic acids are prepared by oxidation of alkyl benzene.

$R-CH_2-OH \xrightarrow{KMnO_4} RCOOH$
1° alcohol — Carboxylic acid

Example

$CH_3-CH_2OH \xrightarrow{KMnO_4} CH_3-COOH$
Ethanol — Ethanoic acid



Methyl benzene — Benzoic acid

Answer: D

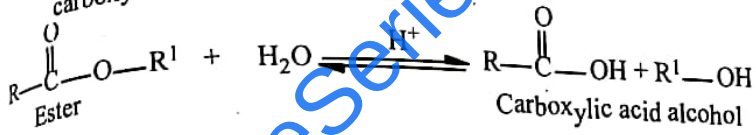
31. Which of the following statements is NOT true?
A. Naturally derived soaps consist of a soluble salt of a long chain fatty acid.
B. Triacyl glycerols are esters of glycerol and long chain carboxylic acids.
C. Long chain caroboxylic acids also known as fatty acids.
D. The major acidic component of vinegar is formic acid.

Explanation:
Formic acid is found in ants and bees. The major acidic components of vinegar is acetic acid.

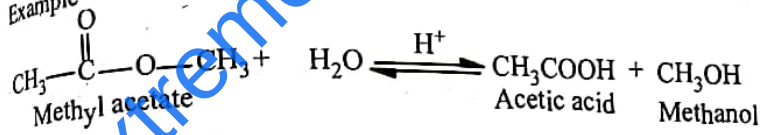
Answer: D

32. What is the product of the hydrolysis of esters in the presence of a mineral acid catalyst?
A. Alcohol
B. Carbon dioxide
C. Ether
D. Ketones

Explanation:
Hydrolysis of esters yields the corresponding alcohols and carboxylic acids.



Example



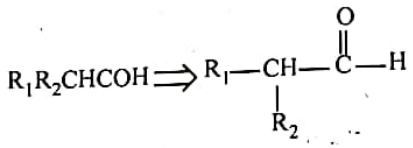
Answer: A

To which organic functional group does the following molecular representation, i.e., R_1R_2CHCOH belong? (R_1 and R_2 represent different alkyl chains).
A. Amides
B. Aldehydes
C. Ethers
D. organic acids

Explanation:
Functional group is part of the compound that determines its property.

Organic compound	General formula	Functional group
Carboxylic acid	R—C(=O)—OH	—COOH
Ester	R—C(=O)—O—R¹	—C(=O)—O—C
Ether	R—O—R¹	—C—O—C—
Aldehyde	R—C(=O)—H	—C(=O)—H
Ketone	R—C(=O)—R¹	—C(=O)—
Alcohol	R—OH	—OH

Telegram: https://t.me/ExtremeSeries



Answer: B

34. What is the name of triacylglycerol (triglycerides) that is solid at room temperature?
A. Lecithin B. Fat C. Wax D. Oil

Explanation:
Fats and oils belong to a class of biomolecules called lipids. They are triesters of glycerol which are collectively known as triglycerides or triacylglycerols. Fat is solid at room temperature while oil is liquid.

Answer: B

35. Which of the following reactions will convert carboxylic acids to primary amines?
A. Decarboxylation with HBr and peroxide, then reaction of the alkyl bromide with ammonia.
B. Reduction of the acid to the alcohol with NaOH formaldehyde, then reaction.
C. A two – step conversion, first to the amide with ammonia and heat and then by reduction with lithium aluminum hydride or hydrogen plus catalyst.
D. Lactase fermentation in the presence of ammonia atmosphere.

Explanation:
Carboxylic acids are weak organic acids that possess all the properties of acids.

Answer: C

36. A solution was prepared by dissolving 3.75 g pure hydrocarbon in 95 g of cylclohexane. The boiling point of pure cyclohexane was observed to be 80.70°C. What is the approximate molecular weight of the hydrocarbon? (k_b for cyclohexane = 2.79 °C/m).
A. 71.0 g/mol C. 147 g/mol
B. 105 g/mol D. 312 g/mol

Solution:

$\Delta T_b = mk_b = T_b - T_b^o$

ΔT_b – boiling point elevation
T_b – boiling point of solution
T_b^o – boiling point of pure solvent

m – molality
k_b – boiling point elevation constant for the solvent
$\Delta T_b = 81.45°C - 80.70°C = 0.75°C$

$$m = \frac{\Delta T}{kb} = \frac{0.75°C}{2.7g°C/m} = 0.2\,m$$

$$m = \frac{n}{\text{kg of solvent}}$$

n = m × kg of solvent
n = 0.27 mol/kg × 0.095 kg = 0.0255 mol

$$n = \frac{m}{M}$$

$$M = \frac{m}{n} = \frac{3.75g}{0.0255mol} = 147g/mol$$

Answer: C

37. A solution was prepared by adding 48 g of methanol into 81 g of water (H_2O). What is the mole fraction of methanol in this solution?
A. 0.25 B. 0.75 C. 1.5 D. 4.5

Solution:

$$n_{CH_3OH} = \frac{m}{M} = \frac{48g}{32g/mol} = 1.5mol$$

$$n_{H_2O} = \frac{m}{M} = \frac{81\,g}{18\,g/mol} = 4.5\,mol$$

$$X_{CH_3OH} = \frac{n_{CH_3OH}}{n_{CH_3OH} + n_{H_2O}}$$

$$X_{CH_3OH} = \frac{1.5\,mol}{(1.5+4.5)mol} = 0.25$$

Answer: A

38. What is the number of chloride ions (Cl^-) present in 1.0×10^{-5} mol of $AlCl_3$?
A. 1.80×10^{19} C. 6.02×10^{23}
B. 6.02×10^{18} D. 6.02×10^{28}

Solution:
Number of Cl^- ions=number of mole of $AlCl_3$ × subscript of Cl×6.02×10^{23}
$= 1.0 \times 10^{-5} \times 3 \times 6.02 \times 10^{23} = 1.80 \times 10^{19}$

Answer: A

Telegram: https://t.me/ExtremeSeries

39. What is the quantity of water in mL, required to prepare 0.5M of HCl from a concentrated solution of 3.5M in 50mL?
A. 50 mL B. 100mL C. 300mL D. 350mL

Solution:
$M_1 = 3.5M$ $M_2 = 0.5M$
$V_1 = 50\ mL$ $V_2 = ?$
$M_1V_1 = M_2V_2$

$$V_2 = \frac{M_1V_1}{M_2} = \frac{3.5M \times 50mL}{0.5M} = 350mL$$

V_{H_2O} added $= V_2 - V_1 = 350\ mL - 50\ mL = 300\ mL$ **Answer: C**

40. Which of the following is the most important type of solute – solvent interaction in a solution of n – butanol in water?
A. Dispersion C. Dipole – dipole
B. Ion – dipole D. Hydrogen bonding

Explanation: Alcohols possess hydrogen bond, that is why they are soluble in water. However, as carbon number increases, solubility of alcohols in water decreases. Ionic compounds dissolve in water due to ion – dipole interaction. Non – polar solutes dissolve in non –polar solvents due to dispersion or London force. Polar compounds are soluble in water due to dipole – dipole interactions. **Answer: D**

41. What is the mass of one molecule of water?
A. 3.0×10^{-23} g C. 0.0003 g
B. 1.8×10^{-22} g D. 18.0 g

Solution: 1 mole = molecular mass $= 6.02 \times 10^{23}$ molecules
1 mole of H_2O = 18 g $H_2O = 6.02 \times 10^{23}$ molecules H_2O
$\Rightarrow$ 18 g $H_2O = 6.02 \times 10^{23}$ molecules of H_2O
? = 1 molecule H_2O

$$? = \frac{18\ g\ H_2O \times 1\ molecule\ H_2O}{6.02 \times 10^{23}\ molecules\ H_2O}$$

$? = 2.99 \times 10^{-23}$ g H_2O
$\Rightarrow$ 1 molecule of water weighs 2.99×10^{-23}g **Answer: A**

42. Which one of the mixture of the following pairs will NOT give a buffer solution?
A. HCN and NaCN C. H_3PO_4 and KH_2PO_4
B. NH_3 and NH_4Cl D. HNO_3 and $NaNO_3$

Explanation:
Buffer solution is a solution that resists pH change upon the addition of small amount of acid or base. Buffer solution is formed from weak acid and its conjugate base or from weak base and its conjugate acid.
HA/A^-, acidic buffer
B/BH^+, basic buffer
A buffer cannot be formed from strong acid and strong base because there is no unionized form in the solution.

Answer: D

43. Which of the following is TRUE for salts formed from strong acids and strong bases?
A. No hydrolysis takes place.
B. Produces ions which are proton acceptors.
C. Produces ions that are proton donors.
D. Depends on the P_{ka} and P_{kb} of the parent acids and bases.

Explanation: Salts formed from strong acids and strong bases do not undergo hydrolysis reaction. Because the cations and anions are very weak acids and bases respectively.
Example

$$NaOH + HCl \longrightarrow NaCl + H_2O$$
Strong base strong acid Natural salt

$NaCl \longrightarrow Na^+ + Cl^-$
$Na^+ + H_2O \longrightarrow$ no reaction
$Cl^- + H_2O \longrightarrow$ no reaction
Adding such types of salts to water does not affect the pH of water. **Answer: A**

44. Given the following equilibira and equilibrium constants:
(I) $HC_2H_3O_2 + H_2O \rightleftharpoons H_3O^+ + C_2H_3O_2^-$; $Ka = 1.8 \times 10^{-5}$
(II) $H_2CO_3 + H_2O \rightleftharpoons H_3O^+ + HCO_3^-$; $Ka = 4.2 \times 10^{-7}$
(III) $NH_4^+ + H_2O \rightleftharpoons H_3O^+ + NH_3$; $Ka = 5.6 \times 10^{-10}$
(IV) $HCOOH + H_2O \rightleftharpoons H_3O^+ + HCOO^-$; $Ka = 1.80 \times 10^{-4}$
What is the strength of the acids in decreasing order?
A. I, IV, II and III C. III, II, I and IV
B. II, III, IV and I D. IV, I, II and III

Telegram: https://t.me/Extremeseries

Explanation: The strength of an acid is directly related with the acid dissociation constant, K_a. The larger the value of Ka, the stronger is the acid. $\Rightarrow HCOOH > HC_2H_3O_2 > H_2CO_3 > NH_4^+$

Answer: D

45. Given the reaction:

$$H_2PO_4^- + H_2O \longrightarrow H_3O^+ + HPO_4^{2-}$$

Which of the following represents a conjugate acid – base pair?

A. $H_2PO_4^-$ and H_2O C. $H_2PO_4^-$ and H_3O^+

B. $H_2PO_4^-$ and HPO_4^{2-} D. H_2O and HPO_4^{2-}

Explanation:

Conjugate acid is formed when a base accepts a proton.

Conjugate acid = base + hydrogen ion

Conjugate base is formed when an acid donates a proton.

Conjugate base = acid – hydrogen ion

The conjugate base of HPO_4^- is $HPO_4^- - H^+ = HPO_4^{2-}$

The conjugate acid of H_2O is $H_2O + H^+ = H_3O^+$.

Answer: B

46. Which of the following statements is TRUE about colligative properties?

A. Both vapour pressure and freezing point increase when a non – volatile solute is added to a solvent.

B. Both freezing point and boiling point increase when a non-volatile solute is added to a solvent.

C. Both freezing point and boiling point increase when a non – volatile solute is added to a solvent.

D. Colligative properties depend only upon number of solute properties in a solution and not upon their identity.

Explanation:

- Colligative properties depend on the concentration of the solute particles that are present in a given amount of solvent and not on their chemical identity. These are: vapor pressure lowering, boiling point elevation, freezing point depression and osmotic pressure.
- When a non – volatile solute is added to a solvent, vapor pressure and freezing point decrease while boiling point and osmotic pressure increase.

Answer: D

47. What is the pH of 0.005M solution of $Ca(OH)_2$?

A. 2.3 B. 10 C. 12 D. 14

Solution:

$$Ca(OH)_2 \longrightarrow 2OH^- + Ca^{2+}$$

1M 2M

0.005M ?

$$\Rightarrow [OH^-] = \frac{2M \times 0.005M}{1M} = 0.01M = 10^{-2}$$

$pH = 14 + \log [OH^-]$

$pH = 14 + \log 10^{-2} = 12$

Answer: C

48. The indicator Bromothymol blue (HBb) is a weak acid with Ka = 1.0×10^{-7} ionizes as follows:

$$HBb\ (aq.\ yellow) \rightleftharpoons H^+\ (aq.\ colorless) + Bb^-\ (aq.\ blue)$$

Which way will the equilibrium shift when NaOH is added and what will the color of NaOH solution be containing this indicator?

A. Equilibrium will shift to the right and the color of NaOH solution will turn green.

B. Equilibrium will shift to the right and the color of NaOH solution will turn blue.

C. Equilibrium will shift to the left and the color of NaOH solution will turn yellow.

D. Equilibrium will shift to the left and the color of NaOH solution will turn blue.

Explanation:

Acid – base indicators are weak organic acids or organic bases that indicate whether a solution is acidic or basic.

$$HBb\ (aq,\ yellow) \rightleftharpoons H^+\ (aq,\ colorless) + Bb^-\ (aq.\ blue)$$

If the indicator is in sufficiently acidic medium, the equilibrium shifts to the left and the predominant color is that of HBb, on the other hand, in basic medium, the equilibrium shifts to the right and the color of the solution will be that of Bb^-.

Answer: B

Telegram: https://t.me/EXtremeSeries

49. To 0.2M solution of a weak monoprotic acid, HA, enough quantity of its sodium salt, NaA, was dissolved to give a concentration of 0.2M of the salt. What will be the acid concentration, $[H_3O^+]$ in the final solutions? (Ka of HA = 1.80×10^{-5})

A. 3.60×10^{-6} M C. 1.80×10^{-5} M
B. 1.00×10^{-5} M D. 1.90×10^{-3} M

Solution:

The common ion effect is caused by the addition of a compound having an ion in common with the dissolved substance that shift the equilibrium. This is very important to adjust the concentration of hydronium ion.

$NaA \longrightarrow Na^+ + A^-$
0.2M 0.2 M 0.2M

$\Rightarrow$ Initially the $[A^-]$ is 0.2M.

$HA + H_2O \rightleftharpoons H_3O^+ + A^-$

Initial	0.20	0	0.2
Change	-x	+x	+x
Equilibrium	0.2 – x	x	0.2 + x

$$Ka = \frac{[H_3O^+][A]^-}{[HA]}$$

$$1.8\times10^{-5} = \frac{x(0.2+x)}{0.2-x}, \text{ approximation}$$

$0.2 + x \approx 0.2, 0.2 - x \approx 0.2$

$$1.8 \times 10^{-5} = \frac{x(0.2)}{0.2}$$

$x = [H_3O^+] = 1.8 \times 10^{-5}$ M **Answer: C**

50. A 50 mL solution of H_2SO_4 0.205 M is titrated with NaOH solution of unknown concentration. The end point against phenolphthalein indicator was known after 41 mL of NaOH added. What is the concentration of NaOH solution?

A. 0.10MB. 0.25 C. 0.41M D. 0.50M

Solution:

Write the neutralization reaction and determine the number of mole of NaOH from the given number of moles of H_2SO_4

$n_{H_2SO_4} = M \times V = 0.205\ M \times 50\ mL = 10.25\ m\ mol$

10.25 m mol x

$H_2SO_4 + 2NaOH \longrightarrow Na_2SO_4 + 2H_2O$

1 2

$$\frac{0.25mol}{1} = \frac{x}{2}, x = n_{NaOH} = 20.5\ mol \Rightarrow M_{NaOH} = \frac{n_{NaOH}}{V_{NaOH}}$$

$$M_{NaOH} = \frac{20.5\ mmol}{41.0\ mL} = 0.50M$$ **Answer: D**

51. What is the pH of an aqueous solution prepared to contain 1.3×10^{-3} M sodium nitrite ($NaNO_2$) if the acid dissociation equilibrium constant, Ka, for nitrous acid (HNO_2) is 5.1×10^{-4}? (Kw = 1.0×10^{-14})

A. 3.1 B. 5.1 C. 6.0 D. 7.3

Solution:

$NaNO_2 \longrightarrow Na^+ + NO_2^-$

$\Rightarrow [NaNO_2] = [NO_2^-] = 1.3 \times 10^{-3}$ M

$NO_2^- + H_2O \rightleftharpoons HNO_2 + OH^-$

Initial	1.3×10^{-3}	0	0
Change	-x	+x	+x
Equilibrium	1.3×10^{-3}	x	x

$$K_b \text{ for } NO_2^- = \frac{K_w}{Ka \text{ of } HNO_2} = \frac{[HNO_2][OH^-]}{[NO_2^-]}$$

$$\frac{1.0\times10^{-14}}{5.1\times10^{-4}} = \frac{x.x}{1.3\times10^{-3}-x}$$

$$1.96 \times 10^{-11} = \frac{x^2}{1.3\times10^{-3}-x}, 1.3 \times 10^{-3} - x \approx 1.3 \times 10^{-3}$$

$2.548 \times 10^{-14} = x^2$

$x = [OH^-] = 1.6 \times 10^{-7}$ M

$pH = 14 + \log [OH^-]$

$pH = 14 + \log^{1.6\times10^{-7}} = 7.3$ **Answer: D**

Telegram: https://t.me/Extremeseries

52. How many mL conc. HNO_3 and how many mL of water are required to prepare 500 mL of 0.1 M HNO_3 from a conc. 13M HNO_3?

A. 1 mL HNO_3 and 496.15 mL of H_2O
B. 3 mL HNO_3 and 500 mL H_2O
C. 3.85 mL HNO_3 and 500 mL H_2O
D. 3.85 mL HNO_3 and 496.15 mL H_2O

Solution:

V_1 =? V_2 = 500 mL
M_1 = 13 M M_2 = 0.1M

From law of dilution, we know that

$M_1V_1 = M_2V_2$

$$V_1 = \frac{M_2V_2}{M_1} = \frac{0.1M \times 500mL}{13M} = 3.85mL$$

V_{H_2O} added $= V_2 - V_1$

V_{H_2O} added = 500 mL – 3.85 mL = 496.15 mL **Answer: D**

53. Which of the following reactions is expected to have negative value of entropy change (Δs)?

A. $C_6H_6(s) + 6O_2(g) \longrightarrow 6CO_2(g) + 6H_2O(l)$
B. $CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$
C. $N_2O_4(g) + Cl_2(g) \longrightarrow 2NOCl(g) + O_2(g)$
D. $2SO_2(g) + O_2(g) \longrightarrow 2SO_3(g)$

Explanation:

Entropy (s) is the thermodynamic property of a system that is related to its degree of randomness of disorder. The entropy of gas >> liquid > solid. If the number of moles of gaseous product is greater than that of gaseous reactant, the entropy change is expected to be positive on the other hand, if the number of moles of gaseous reactant is greater than the number of moles of gaseous product, the entropy change will be expected to be negative. **Answer: D**

54. Gaseous petrol in a combustion system has done 375 kJ of work during its expansion in the piston. Simultaneous, it absorbed 586 kJ of heat from the engine. What is the internal energy change during the process?

A. +211 kJ C. -211 Kj
B. +961 kJ D. -961 kJ

Solution:

Work done by the system has a negative sign while work done by the surroundings has a positive sign. Heat absorbed by a system is positive while heat released by the system is negative.

$\Delta E = W + q$

ΔE = -375 kJ + 586 kJ = +211 kJ **Answer: A**

55. Consider the following gaseous reaction at 25°C:

$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(g)$

ΔH = -802 kJ

Which energy change would occur if 3.2 moles of CH_4 is completely combusted?

A. 2.57×10^2 kJ will be released
B. 2.57×10^2 kJ will be absorbed
C. 6.43×10^3 kJ will be released
D. 6.43×10^3 kJ will be absorbed

Solution:

Combustion reaction is exothermic reaction. The energy released when 1 mole of CH_4 is combusted is 802 kJ. Therefore, when 3.2 moles of CH_4 is combusted the energy released 3.2 × 802 kJ = 2566.4 kJ = 2.57×10^2 kJ. **Answer: A**

56. What is the quantity of heat required to raise the temperature of 80 g of ethanol from 25°C to 75°C?
(Specific heat of ethanol = $2.46 Jg^{-1}k^{-1}$)

A. 2.46 kJ C. 9.84 kJ
B. 4.0 kJ D. 18.68 kJ

Solution:

$\Delta T = T_f - T_i$ = 75°C - 25°C = 50°C = 50 K
m = 80 g
c = 2.46 $Jg^{-1}k^{-1}$
$q = mc\Delta T$
q = 80 g × 2.46 J $g^{-1}k^{-1}$ × 50 k = 9840 J
q = 9.84 kJ **Answer: C**

57. How many kilo joules of heat are absorbed when 20 g of NaCl (s) is decomposed into Na (s) and Cl_2 (g) at constant pressure according to the following reaction?

$2Na(s) + Cl_2(g) \longrightarrow 2NaCl(s)$, ΔH -821.8 kJ

A. -281.0 C. +140.5
B. -140.5 D. +281.0

Telegram: https://t.me/ExtremeSeries

Solution:
When we reverse the given reaction, the sign of ΔH will be changed.

$\Rightarrow 2NaCl\,(s) \longrightarrow 2Na\,(s) + Cl_2\,(g),\ \Delta H = +821.8\ kJ$

$2 \times 58.5\ g = 821.8\ kJ$
NaCl

20 g of =?
NaCl

$$? = \frac{20g \times 821.8kJ}{117g} = 140.5kJ$$

⇒ The amount of heat absorbed to decompose 20 g of NaCl in to Na and Cl_2 is 140.5 kJ. **Answer: C**

58. Enthalpy is defined as the heat content of the system at constant:
A. Heat B. Moles C. pressure D. Volume

Explanation:
Enthalpy is defined as the sum of the internal energy and the pressure – volume product of a system.

$H = E = PV$

$\Delta H = \Delta E = P\Delta V$ (at constant pressure)

but $\Delta E = q - P\Delta V$

$\Rightarrow \Delta H = q - P\Delta V + P\Delta V;\ \Delta H = q_P$

Enthalpy change is heat transferred at constant pressure.

Answer: C

59. Which of the following metals is NOT obtained by commercial electrolytic process?
A. Ag B. Al C. Cu D. Na

Explanation:
Most silver (Ag) produced now a days is as a by – product of the production of other metals such as zinc, copper and lead. Cyanidation process is also used to manufacture silver.

$$Zn\,(s) + 2Ag(CN)_2^- + 4OH^- \longrightarrow 2Ag + 4CN^- + Zn(OH)_4^{2-}$$

Answer: A

60. What is the equivalent weight of HNO_3, as an oxidizing agent, in the following balanced reaction?

$$3Fe^{2+} + 4H^+ + NO_3^- \longrightarrow 3Fe^{3+} + NO + 2H_2O$$

A. 10.50 B. 15.75 C. 21.00 D. 31.50

Solution:
The oxidation state of N in NO_3^- is +5 while in NO is +2. The total electrons gained will be 3.

$$E_{HNO_3} = \frac{M_{HNO_3}}{Z}$$

E – equivalent mass
Z – electrons gained or lost
M – molar mass

$$E_{HNO_3} = \frac{63\ g/mol}{3\ eq/mol} = 21.00\ g/eq$$

Answer: C

61. In the electroplating of nickel from a solution containing Ni^{2+} ion, what will be the weight of the metal deposited on the cathode by a current of 8A flowing for 500 minutes?
A. 73 g B. 103 g C. 117.4 g D. 145 g

Solution:
From Faraday's first law of electrolysis:

$$m = \frac{MIt}{ZF},\ m - mass$$

M – molar mass
I – current
t – time
z – electrons gained or lost
F – Faraday's constant

$Ni^{2+} + 2e^- \longrightarrow Ni \Rightarrow Z = 2$

$t = 500\ min = 500 \times 60\ sec. = 30{,}000\ sec.$

$$m = \frac{5g \times 8A \times 30{,}000sec}{2 \times 96500A.C} = 73.37\ g\ of\ Ni$$

Answer: A

62. A 1 M solution of $Cu(NO_3)_2$ is placed in a beaker with a strip of Cu metal. A 1M of $SnSO_4$ is placed in a second beaker with strip of Sn metal then connected by a salt bridge and the two metal electrodes are connected by wires to a voltmeter. Which of the following electrode serve as the anode and which electrode gains mass?
Given that $E°\ Cu^{2+}/Cu = 0.34\ V$ and $E°\ Sn^{+2}/Sn = -0.14V$
A. Anode, Sn, Sn electrode gains mass
B. Anode, Sn, Cu electrode gains mass

Telegram: https://t.me/ExtremeSeries

C. Anode, Cu, Sn electrode gains mass
D. Anode, Cu, Cu electrode gains mass

Explanation:
The more reactive metal, with more negative reduction potential serves as anode. The less reactive metal with more positive reduction potential used as cathode.
⇒ Sn is anode and Cu is cathode. Mass of cathode increases since copper is deposited on the surface of cathode.

63. Consider the following unbalanced, redox reacton in acidic **Answer: B**
Solution:

$MnO_4^- + Fe^{2+} \longrightarrow Mn^{2+} + Fe^{3+}$

What is the change in oxidation state for both the substances oxidized and reduced and the coefficients of Fe^{2+} and Mn^{2+}, respectively after balancing?

A. 2 and 7, 2 and 5
B. 3 and 2, and 5 and 1D.
C. 1 and 5, and 5 and 1
1 and 5, and 5 and 2

Solution: $(8H^+ + MnO_4^- + 5e^- \longrightarrow Mn^{2+} + 4H_2O) \times 1$

$Fe^{2+} \longrightarrow Fe^{3+} + 3 \times 5$

$8H^+ + MnO_4^- + 5e^- \longrightarrow Mn^+ + 4H_2O$

$5Fe^{2+} \longrightarrow 5Fe^{3+} + 5e^-$

$8H^+ + 5Fe^{2+} + MnO_4^- \longrightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$

64. What reactions occur at the anode and cathode when an aqueous solution of Na_2SO_4 is electrolyzed? **Answer: C**

I. $S_2O_8^{2-} + 2e^- \longrightarrow 2SO_4^{2-}$ E°red
II. $O_2 + 4H^+ + 4e^- \longrightarrow 2H_2O$ 2.01V
III. $2H_2O + 2e^- \longrightarrow H_2 + 2OH^-$ 1.23 V
IV. $Na^+ + e^- \longrightarrow Na$

A. H_2 at cathode and O_2 at anode
B. Na at cathode and $S_2O_8^{2-}$ at anode
C. H_2 at cathode and $S_2O_8^{2-}$ at anode
D. Na at cathode and O_2 at anode

Explanation:
Anode reaction: $2H_2O \longrightarrow O_2 + 4H^+ + 4e^-$
Cathode reaction: $4H_2O + 4e^- \longrightarrow 2H_2 + 4OH^-$
Cell reaction: $2H_2O \longrightarrow 2H_2 + O_2$

Answer: A

65. For the following hypothetical equation, in aqueous solution, what is the correct representation of the cell notation?

$A(s) + B^{2+}(aq) \longrightarrow A^{2+}(aq) + B(s)$

A. $A(s) \mid A^{2+}(aq) \parallel B^{2+}(aq) \mid B(s)$
B. $B^{2+}(aq) \mid B(s) \parallel A^{2+}(aq) \mid A(s)$
C. $A^{2+}(aq) \mid A(s) \parallel B(s) \mid B^{2+}(aq)$
D. $B(s) \mid B^{2+}(aq) \parallel A^{2+}(aq) \mid A(s)$

Explanation:
Cell notation is short hand representation of a galvanic cell without drawing the diagram. It is represent by the following general formula.
Anode | Anode electrolyte (M) || Cathode electrolyte (M)| cathode
The anode is the above hypothetical reaction is A since it is oxidized and cathode is B since it is reduced.
⇒ $A(s) \mid A^{2+}(aq) \parallel B^{2+}(aq) \mid B(s)$ **Answer: A**

66. Which of the following is the most important ingredients used for production of DAP fertilizers?

A. Ammonia and phosphoric acid.
B. Nitric acid, urea and phospheric acid
C. Phosperic acid, urea and ammonia
D. Sulphuric acid, ammonia and urea

Explanation:
Diammonium mono hydrogen phosphate (DAP) a white crystalline compound that is completely soluble in water. It is produced when ammonia reacts with phosphoric acid.

$3NH_3 + 2H_3PO_4 \longrightarrow NH_4H_2PO_4 + (NH_4)_2HPO_4$

$NH_4H_2PO_4 + NH_3 \longrightarrow (NH_4)HPO_4$
DAP

DAP is used as fertilizer and as a yeast nutrient in wine making and beer brewing. **Answer: A**

67. Which of the following metal alloys does NOT contain tin?

A. Brass
B. Bronze
C. Pewter
D. Plumber's solder

Telegram: https://t.me/ExtremeSeries

Explanation:
Brass is an alloy of copper and zinc that does not contain tin. Bronze is an alloy of copper and tin pewter is an alloy that contains 85 – 90% tin with the remainder consisting of copper, antimony, bismuth. Plumber's solder is an alloy of tin and lead.

Answer: A

68. Which of the following plant nutrient will be produced as a result of nitrogen fixation?
A. Carbohydrates C. Mineral
B. Cellulose D. Protein

Explanation:
Nitrogen is used to make proteins.

Answer: D

69. Which of the following elements is the second most abundant element in the earth's crust?
A. Aluminum C. Oxygen
B. Iron D. Silicon

Explanation:
The first four abundant elements in the earth crust in decreasing order are oxygen, silicon, aluminum and iron respectively.

Answer: D

70. Which of the following metals has the highest electrical and thermal conductivities?
A. Ag B. Co C. Cu D. Ni

Solution:
Silver (Ag) is the best conductor of heat and electricity so far known. However, we do not use silver metal for this purpose because it is expensive.

Answer: A

71. Which of the following gases is manufactured using the Haber process?
A. Ammonia C. Nitrogen
B. Nitric oxide D. Nitrogen dioxide

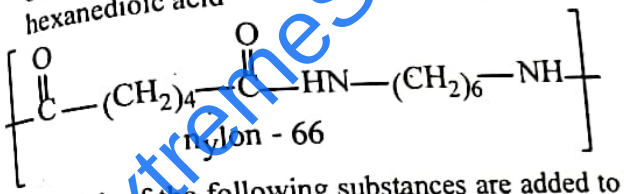
Explanation:
Ammonia is industrially manufactured from hydrogen and nitrogen gases in the presence of iron catalyst at high pressure and moderate temperature by Haber process.

$$3H_2(g) + N_2(g) \underset{}{\overset{Fe}{\rightleftharpoons}} 2NH_3(g)$$

Answer: A

72. What are the raw materials required to synthesize nylon – 66, a specific kind of nylon?
A. Diacids C. Diacids and Diamines
B. Diamines D. Polyethylene

Solution:
Nylons are polyamides. One of the most polyamides is nylon – 66. It is prepared from 1, 6 – amino hexane and 1, 6 – hexanedioic acid



Answer: C

73. Which of the following substances are added to natural rubber to toughen it?
A. Calcium C. Nitrogen
B. Carbon D. Sulfur

Explanation:
The substance used to vulcanize rubber is sulphur: Vulcanized rubber has the following properties:
- It is hard and non – stickey.
- It has higher elasticity.
- It has high tensil strength.
- It is insoluble in solvents.
- It is not oxidized readily.

Answer: D

74. What structural feature is usually needed to present in order for an addition polymer to be produced?
A. A carbon – carbon sigma bond.
B. A carbon – oxygen pi bond.
C. A carbon – oxygen sigma bond.
D. A carbon – carbon pi bond.

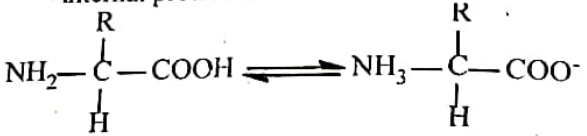
Explanation:
Addition polymerization requires a large number of monomers which can be polymerized by addition reactions. The monomers usually contain carbon – carbon multiple bonds. Pi bonds occur in double and triple bonds. All single bonds are sigma bonds.

Answer: D

Telegram: https://t.me/ExtremeSeries

75. Which of the following is molecules forms a Zwitter ion at higher or lower pH?
A. Cellulose C. Protein
B. Glucose D. Starch

Solution:
Amino acids are monomers of proteins that form zwitter ion by internal protonation.

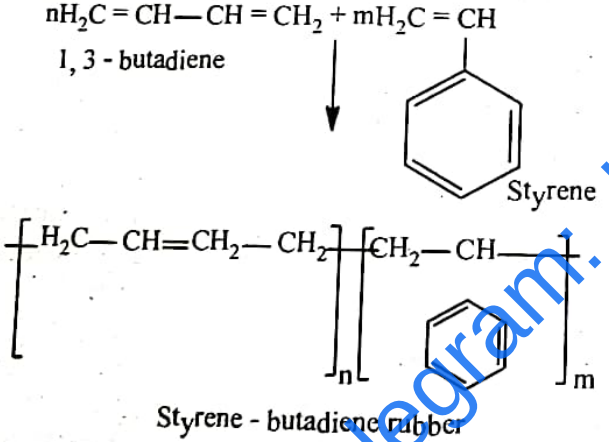


Zwitter ion

Answer: C

76. Which of the following form of synthetic rubbers can be vulcanized to greatly enhance its mechanical strength?
A. Neoprene C. Butyl rubber
B. Isoprene D. Styrene – butadiene rubber

Explanation:
Styrene – butadiene rubber is a copolymer of styrene and butadiene.



Styrene - butadiene rubber

It is vulcanized in a similar manner as natural rubber. This rubber is superior to natural rubber with regard to mechanical strength.

Answer: D

77. What is the range in the number of carbon atoms of the mono saccharides that are found in nature?
A. 3 to 6 C. 4 to 12
B. 4 to 10 D. 5 to 12

Solution:
Mono saccharides are simple sugars that cannot be hydrolyzed in to smaller compounds. They usually contain 3 to 6 carbon atoms.

Answer: A

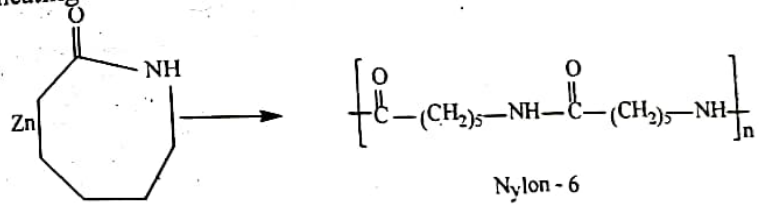
78. What are the two principal poly saccharides forms of starch?
A. Alde hexose and keto pentose
B. Amylose and amylopectin
C. Maltose and cellobiose
D. Sucrose and lactose

Explanation:
Starch is used for energy storage in plants. It is separated into two principal poly saccharides, amylose and amylopectin.

Answer: B

79. Which synthetic polymer is produced from caprolactam?
A. Nylon – 6 C. Teflon
B. Nylon 6, 10 D. Terylene

Solution:
Nylon – 6 is made from caprolactam, which polymenzes on heating in water.



Caprolacton

Nylon - 6

The use and properties of nylon 6 is similar with that of nylon 66.

Answer: A

80. How many types of cubic unit cells are known?
A. 2 B. 3 C. 4 D. 5

Explanation:
There are three types of cubic unit cells. These are:
i. A simple cubic unit cell or a primitive cubic unit cell – has lattice points only at the corners.
ii. A body centered cubic unit cell – has lattice points at the corners plus a lattice point in the center of the cell.
iii. A face centered cubic unit cell – has lattice points at the eight corners and one in the center of each of its six faces.

Answer: B

Telegram: https://t.me/ExtremeSeries

1. Which of the following is correct?
   A. $1L = 1dm^3$ C. $10L = 1dm^3$
   B. $1L = 10dm^3$ D. $1L = 1m^3$
   Solution: $1m^3 = 1000dm^3 = 1000L = 10^6cm^3 = 10^6mL$
   $1cm^3 = 1mL$
   Answer: A

2. Which of the following is NOT a conjugate acid – base pair?
   A. $HNO_3|NO^-_3$ C. $NH_3|NH^-_2$
   B. $H_2SO_4|HSO^-_4$ D. $H_3O^+|OH^-$
   Explanation: The conjugate acid – base pairs differ by a single hydrogen. The acid has one more hydrogen than its conjugate base. ⇒The conjugate base of H_3O^+ is H_2O but not OH^-
   Answer: D

3. Commercially, liquid vegetable oils are converted to solid fats such as margarine by:
   A. Hydrogenation C. saponification
   B. Hydration D. oxidation
   Explanation: Vegetable oils are converted to fats by adding hydrogen to them. This reaction is termed as hydrogenation reaction.
   Answer: A

4. What is the chemical name for Aspirin?
   A. Acetyl salicylic acid C. methyl salicylate
   B. Salicylic acid D. sodium salicylate
   Explanation: Aspirin is formed when salicylic acid reacts with acetic acid. Its chemical name is acetyl salicylic acid.
   Answer: A

5. Which of the following statements (s) is |are| true of an ideal liquid – liquid solution?
   I. It obeys PV = nRT
   II. It obeys Raoult's law
   III. Solute – solute, solvent – solvent and solute solvent interactions are very similar
   IV. Solute – solute, solvent – solvent and solute – solvent interactions are quite different

   A. I, II and III C. II and III
   B. I, II and IV D. II and IV
   Explanation: Ideal solutions have the following properties
   - There is no expansion or contraction on mixing
   - Solute – solute, solvent – solvent and solute – solvent interactions are similar
   - Obey Raoult's law
   - The components have similarity in their structure
   - The vapor pressure above the solution is equal to the sum of the individual particle vapor pressures.

   Answer: C

6. The unit cell in a certain lattice consists of a cube formed by an anion at each corner, an anion in the center and a cation at the center of each face. How many cations and how many anions does the unit cell have?
   A. 5 anions and 6 cations C. 2 anions and 3 cations
   B. 5 anion and 3 canions D. 3 anions and 4 cations
   Answer: A

7. In a reaction, A + B ⟶ product, the rate is doubled when the concentration of B is doubled and the rate increases by a factor of 8 when the concentration of both the reactants A and B are doubled, the rate law for the reaction can be written as:
   A. Rate = k[A] [B] C. Rate = $K[A]^2[B]$
   B. Rate = $K[A] [B]^2$ C. Rate = $K[A]^2 [B]^2$
   Solution: Let the rate law be:rate = $K[A]^x [B]^y$ when the concentration of B is doubled, the rate is doubled. This is true for first order reaction. Thus, the value of Y is 1.
   $rate_1 = K[A]^x_1 [B]_1$
   when $[A]_2 = 1[A]_1$ and $[B]_2 = 2[B]_2$
   $rate_2 = 8\ rate_1$

$$\frac{rate_1}{8rate_1} = \frac{k[A]_1^x[B]_1}{k(2[A]_1)^x 2[B]_1}$$

Telegram: https://t.me/ExtremeSeries

$$\frac{1}{8}=\frac{1}{2^x.2}$$

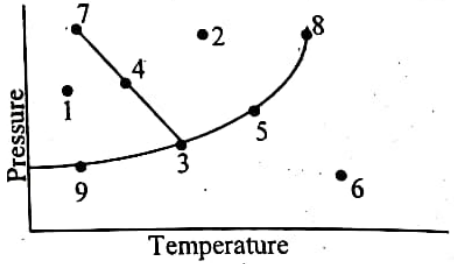
$$\left(\frac{1}{2}\right)^3=\left(\frac{1}{2}\right)^{x+1}$$

$$x+1=3\Rightarrow x=2$$

Therefore, the rate law is: rate = $k[A]^2[B]$.

Answer: C

(8.) Answer the following question using the phase diagram below.



At which point can only the solid and liquid phase coexist?

A. 1 B. 2 C. 3 D. 4

Explanation: Solid and liquid coexist at point 4, solid and vapor coexist at point 9 liquid and vapor coexist at point 5 and the three states coexists at equilibrium at point 3.

Answer: D

(9.) Which electron transition in a hydrogen atom releases the largest energy?

A. $n=2\longrightarrow n=1$ C. $n=6\longrightarrow n=3$

B. $n=4\longrightarrow n=2$ D. $n=7\longrightarrow n=6$

Solution:

A. $\Delta E=2.18\times10^{-18}J\left(\frac{1}{n_i^2}-\frac{1}{n_f^2}\right)$

$$\Delta E=2.18\times10^{-18}J\left(\frac{1}{2^2}-\frac{1}{12}\right)=-1.625\times10^{-18}J$$

B. $\Delta E=2.18\times10^{-18}J\left(\frac{1}{4^2}-\frac{1}{2^2}\right)=-4.09\times10^{-18}Js$

C. $\Delta E=2.18\times10^{-18}J\left(\frac{1}{6^2}-\frac{1}{3^2}\right)=-1.82\times10^{-19}J$

D. $\Delta E=2.18\times10^{-18}J\left(\frac{1}{7^2}-\frac{1}{6^2}\right)=-1.66\times10^{-20}J$

Answer: A

(10.) The simplest formula for a compound containing Mn^{4+} and O^{2-} is

A. MnO B. MnO_2 C. Mn_2O_4 D. Mn_4O_2

Explanation: $Mn^{4}O^{2}=Mn_2O_4$

The simplest formula is MnO_2

Answer: B

(11.) What is the ratio of the energy of a photon of 300nm wavelength radiation to that of 600nm radiation?

A. 1:2 B. 1:1 C. 2:1 D. 3:1

Solution:

$E=h\nu$ but $\nu=\frac{C}{\lambda}$

$$E=\frac{hC}{\lambda}$$

$$E_1=\frac{hC}{\lambda_1}\text{ and }E_2=\frac{hC}{\lambda_2}\Rightarrow\frac{E_1}{E_2}=\frac{hC/\lambda_1}{hC/\lambda_2}$$

$$\frac{E_1}{E_2}=\frac{\lambda_1}{\lambda_2}=\frac{600nm}{300nm}=\frac{2}{1}$$

$E_1:E_2=2:1$

Answer: C

(12.) Which one of the following chemicals is used to disinfect water?

A. Fluorine C. oxygen

B. Nitrogen D. chlorine

Explanation:

Chlorine is added to water to kill bacteria and other microorganisms.

Answer: D

Telegram: https://t.me/ExtremeSeries

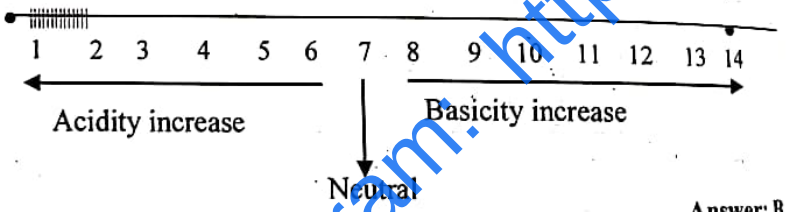
13. During the titration of a known volume of a strong acid with a strong base, there is
A. A steady increase in pH
B. A sharp increase in pH around the end point.
C. A steady decrease in pH
D. A sharp decrease in pH around the end point.

Explanation: Features of the titration of a strong acid with a strong base:
- The pH is low at the beginning of the titration
- The pH change slowly until just before the equivalent point.
- Just before the equivalence point, the pH rises sharply.
- At the equivalence point, the pH is 7.
- Just past the equivalence point, the pH continues its sharp rise.

Answer: B

14. A solution with a pH of 7.5 would be described as:
A. Very basic
B. Slightly basic
C. slightly acidic
D. very acidic

Explanation:



Answer: B

15. The electrolysis of molten NaCl is an industrial process. What does the electrolysis produce?
A. Na and Cl_2
B. H_2 and O_2
C. Na^+ and Cl^-
D. NaOH and Cl_2

Explanation: $NaCl \xrightarrow{\text{Melting}} Na^+ + Cl^-$

Anode reaction: $2Cl^- \longrightarrow Cl_2 + 2e^-$

Cathode reaction: $2Na^+ + 2e^- \longrightarrow 2Na$

Cell reaction: $2Na^+ + 2Cl^- \longrightarrow 2Na + Cl_2$

Answer: A

16. Butane burns in oxygen according to the equation below.
$2C_4H_{10}(g) + 13O_2(g) \longrightarrow 8CO_2(g) + 10H2O(1)$
If 11.6g of butane is burned in 11.6g of oxygen, which is the limiting reactant?
A. Butane
B. Oxygen
C. Neither
D. Both oxygen and butane

Solution: Limiting reactant is a reactant that determines the yield and has smaller total number of moles.

$$\text{Total number of mole } (n_T) = \frac{\textit{given mass}}{\textit{coefficient} \times \textit{Molar mass}}$$

$$n_T(C_4H_{10}) = \frac{11.6g}{2 \times 58g/mol} = 0.10mol$$

$$n_T(O_2) = \frac{11.6g}{13 \times 32g/mol} = 0.03mol$$

Thus, the limiting reactant is oxygen (O_2). **Answer: B**

17. A beaker filled to the 100mL mark with salt (the salt has a mass of 100g) and another beaker to the 100mL mark with water (the water has a mass of 100g) are mixed together in a bigger beaker until the salt is completely dissolved. What will be the mass of the solution?
A. It will be much more than 200g
B. It will be much smaller than 200g
C. It will be exactly 200g
D. It will be slightly more than 200g

Explanation:
mass of solution = mass of solute + mass of solvent mass of solution = 100g + 100g = 200g. But, volume is not always additive.

Answer: C

18. Which one of the following is NOT an intensive property?
A. Mass
B. Temperature
C. Colour
D. Density

Telegram: https://t.me/ExtremeSeries

Explanation:

Extensive properties	Intensive properties
Depend on size of sample of matter.	Independent on size
Example	**Example**
• volume	• density
• mass	• melting point
• area	• conductivity
• thickness	• color
	• temperature

Answer: A

19. Which of the following is natural polymer?
A. Nylon B. PVC C. Cotton D. Dacron

Explanation: Natural polymers include:
- Polysaccharides such as cellulose, starch
- Proteins
- Natural rubber

Cotton is a cellulose **Answer: C**

20. Which of the following metals forms a volatile compound during extraction process?
A. Fe B. Co C. Ni D. Cu

Explanation: Volatile nickel carbonyl, $Ni(CO)_4$, a poisonous liquid of boiling point 43^0C is formed during the production of nickel. **Answer: C**

21. Which one of the following atoms in its ground state has the greatest number of unpaired electrons?
A. $_{13}Al$ B. $_{14}Si$ C. $_{15}P$ D. $_{16}S$

Explanation: $_{13}Al$: $[Ne]\ 3S^2\ 3P^1{}_x\ 3P^0{}_y\ 3P^0{}_z$
Number of unpaired electron = 1

$_{14}Si$: $[Ne]\ 3S^2\ 3P^1{}_x\ 3P^1{}_y\ 3P^0{}_z$
Number of unpaired electrons = 2

$_{15}P$: $[Ne]\ 3S^2\ 3P^1{}_x\ 3P^1{}_y\ 3P^1{}_z$
Number of unpaired electrons = 3

$_{16}S$: $[Ne]\ 3S^2\ 3P^2{}_x\ 3P^1{}_y\ 3P^1{}_z$
Number of unpaired electrons = 2

Answer: C

22. Which statement is true about chemical reactions at equilibrium?
A. The forward and back ward reactions proceed at equal rates
B. The forward and backward reactions have stopped
C. The concentrations of the reactants and products are equal.
D. The forward reaction is exothermic **Answer: A**

23. Which compound contains both covalent and ionic bonds?
A. Sodium carbonate, Na_2CO_3
B. Magnesium bromide, $MgBr_2$
C. Dichloromethane, CH_2Cl_2
D. Ethanoic acid, CH_3COOH

Explanation: Compounds containing metal ion and polyatomic ions have both ionic and covalent bonds. **Answer: A**

24. Which set of quantum numbers (n, ℓ, mℓ, m_s) is not possible?
A. $1, 0, 0, \frac{1}{2}$ C. $1, 0, 0 \frac{-1}{2}$
B. $1, 1, 0, \frac{1}{2}$ D. $2, 1, -1, \frac{1}{2}$

Explanation: n = 1, 2, 3, 4,....
$\ell = 0$ to $n-1$, its maximum value is $n-1$
me = $-\ell$ to $+\ell$
$ms = \pm\frac{1}{2}$ **Answer: B**

25. Which of the following particles contains more electrons than neutrons?
I. $^{1}_{1}H$ II. $^{35}_{17}Cl^-$ III. $^{39}_{19}K^+$
A. I only C. I and II only
B. II only D. II and III only

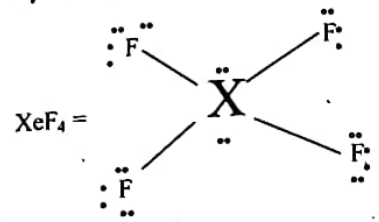
Solution: $^{1}_{1}H$: e = 1 , n = A − p = 1 − 1 = 0, thus e > n
$^{35}_{17}Cl^-$: e = 18 , n = 35 − 17 = 18, thus, e = n
$^{39}_{19}K^+$: e = 18, n = 39 − 19 = 20, thus, e < n **Answer: A**

Telegram: https://t.me/ExtremeSeries

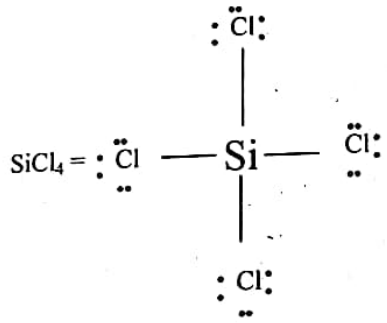
(26) Which molecule or ion does NOT have a tetrahedral shape?

A. XeF_4 B. $SiCl_4$ C. BF^-_4 D. NH^+_4

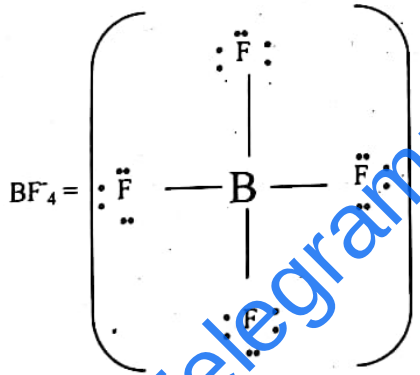
Explanation:

XeF_4 = 

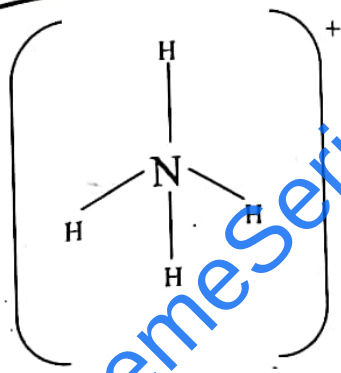
, square planar

$SiCl_4$ = 

, tetrahedral

BF^-_4 = 

, tetrahedral

NH^+_4 = 

, tetrahedral

Answer: A

(27) Why are metals soft and malleable?

A. Because they are very shiny

B. Because they experience electrostatic repulsion

C. Because of the presence of mobile electrons

D. Because the metal cations can slip over each other fairly easily.

Answer: D

(28) In which region of the periodic table would the element with the electronic structure below be location?

$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 4d^6 5s^2$

A. Group 6 C. s – block

B. Noble gases D. d – block

Explanation: write its correct electron configuration based on building up principle $\Rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6 4s^2 3d^{10}\ 4p^6 5s^2 4d^6$

Group = 2 + 6 = VIIIB

Period = 5

Block = d – block

Answer: D

(29) Which factors will influence the rate of the reaction shown below? $NO_2(g) + CO(g) \rightleftarrows NO(g) + CO_2(g)$

I. The number of collisions per second

II. The energy of collisions

III. The geometry with which the molecules collide

A. I only C. I and II only

B. II only D. I, II and III

Answer: A

Telegram: https://t.me/ExtremeSeries

How many π bonds are present in CO_2?

A. One B. Two C. Three D. Four

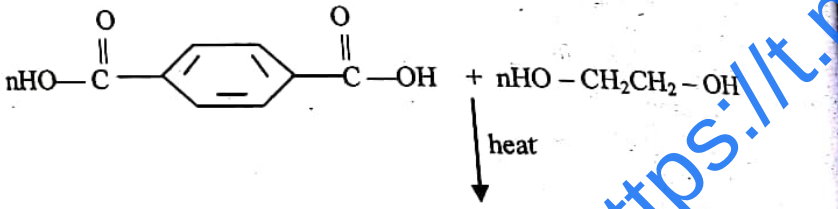
Explanation: Draw the Lewis structure for $CO_2 \Rightarrow O = C = O$
One double contains one π and one σbonds. Since there are two double bonds in CO_2, 2 π and 2σ bonds are present.

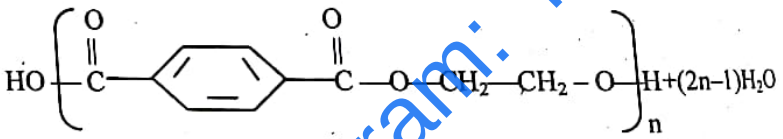
Answer: B

31. Which pair of monomers forms polyesters?

A. Difunctional alcohol, difunctional organic acid
B. Difunctional alcohol, difunctional amino acid
C. Ethylene, ethylene
D. Amino acid, amino acid

Explanation: Polyester (polyethylene tarephthalate or Dacron) is a copolymer formed when 1, 4 – benzene dicarboxylic acid and 1,2 – ethnodiol react





Polyester

Answer: A

32. Which changes will increase the amount of $SO_3(g)$ at equilibrium? $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$, $\Delta H = -197KJ$

I. Increasing the temperature
II. Decreasing volume
III. Adding a catalyst

A. I only C. I and II only
B. II only D. I, II and III

Explanation: The amount of SO_3 is increased by:

- decreasing temperature
- increasing pressure
- increasing amount of SO_2 or O_2 or both
- decreasing volume

Answer: B

33. Which compound is a carboxylic acid?

A. CH_3COOH C. $(CH_3)_2CHOOCH_3$
B. $(CH_3CO)_2O$ D. $(CH_3)_2O$

Explanation: The general formula for carboxylic acid is RCOOH

Answer: A

34. The mechanism of a reaction is shown below.

$HCOOH + I^- \longrightarrow HOI + OH^-$ (slow)
$HOI + I^- \longrightarrow I_2 + OH^-$ (fast)
$2OH^- + 2H_3O^+ \longrightarrow 4H_2O$ (fast)

What is the rate law based on this mechanism?

A. Rate = K[HCOOH] $[I^-]$
B. Rate = K$[HCOOH]^2$ $[I^-]$
C. Rate = K[HCOOH] $[I^-]^2$
D. Rate = K[HCOOH]

Explanation:The slowest step is the rate determing step.
⇒ Rate law, rate = K[HCOOH] $[I^-]$

Answer: A

35. What is the equilibrium constant expression for the following reaction?

$2Hg(g) + O_2(g) \rightleftharpoons 2HgO(s)$

A. $K = \dfrac{1}{[Hg]^2[O_2]}$ C. $K = \dfrac{[HgO]^2}{[Hg]^2[O_2]}$

B. $K = [Hg]^2[O_2]$ D. $K = \dfrac{[2HgO]}{[2Hg][O_2]}$

Telegram: https://t.me/ExtremeSeries

Explanation: Pure solids and liquids (solvent) do not appear in the equilibrium constant expression.

$$\Rightarrow K = \frac{1}{[Hg]^2[O_2]}$$

Answer: A

36. A triacylglycerol that is solid at room temperature is called
A. Lecithin B. Fat C. Wax D. Oil

Answer: B

37. Which compound is an ester?
A. CH_3COOH C. C_2H_5CHO
B. $CH_3O\,C_2H_5$ D. $HCOOCH_3$

Explanation: The general formula of an ester is $RCOOR^1$

Answer: D

38. If a solute dissolves in an endothermic process,
A. H bonds must exist between solvent and solute
B. Strong ion – dipole forces must exist in solution
C. The entropy of the solution must be greater than that of its pure components
D. The solute must be a gas.

Explanation: If a solute dissolves in an endothermic process the entropy of the solution must be greater than that of pure components.

Answer: C

39. A solution is made by dissolving 250g of potassium chromate crystals (K_2CrO_4 , molar mass, 194.2g) in 1kg of water. What will be the freezing point of the solution? (K_f for water is $1.86^0c\ m^{-1}$)
A. -8.87^0c b. -7.18^0c c. -5.73^0c d. -1.86^0c

Solution: $\Delta T_f = Tf^0 - Tf = imk_f$

For electrolytes, i ≈ number of moles of ions.

$$\Rightarrow K_2CrO_4 \longrightarrow 2K^+ + CrO^{2-}_4 \Rightarrow i \approx 3$$

$$T_f = O°C - 3 \times \left(\frac{350g}{194.2\,g/mol \times 1kg}\right) \times 1.86^0c\ m^{-1} = -7.18°C$$

Answer: B

40. How many moles of sodium hydroxide are present in 2.5L of 0.5M aqueous solution?
A. 0.2 B. 0.5 C. 1.25 D. 12.5

Solution: $M = \frac{n}{v}$

$$n = m \times v = 0.5\,\frac{mol}{L} \times 2.5L = 1.25mol$$

Answer: C

41. Which species cannot act as a Lewis acid?
A. NH_3 B. BF_3 C. Fe^{2+} D. $AlCl_3$

Explanation: Lewis acids are electron pair acceptors. Cations, molecules below and above octet are Lewis acids.

Answer: A

42. Which of the following statements is true?
A. A universal indicator is a mixtures of indicators that will give a different color for a different pH.
B. Phenolphthalein is a universal indicator
C. A universal indicator can only be used in either strongly acidic or basic solution
D. The color of a universal indicator is red in a weak acid.

Explanation:
A universal indicator gives different color for different pH.For example, in strong acid red, in medium orange and in weak acid yellow.

Answer: A

43. Three acids, HA, HB, HC have the following ka values ka(HA) = 1×10^{-5}, Ka (HB) = 1×10^{-4} (H C) = 1×10^{-6} what is the correct order of increasing acid strength (weakest first)?
A. HA, HB. HC C. HC, HA, HB
B. HC, HB, HA D. HB, HA, HC

Explanation:
As Ka increases, the strength of acid also increases.
$\Rightarrow HC < HA < HB$

Answer: C

Telegram: https://t.me/ExtremeSeries

44. If the solute – solvent interaction are greater than the solute – solute and solvent – solvent interactions, what will be the total vapor pressure of the solution?

A. Greater than that calculated from Raoult's law
B. Less than that calculated from Raoult's law
C. The same as calculated from Raoult's law
D. Raoult's law cannot be applied for such calculations.

Explanation: If the solute – solvent interactions is greater than the solute – solute and solvent – solvent interactions, the total vapor pressure is less than the sum of the vapor pressures as predicted by Raoult's law.

Answer: B

45. Which of the following procedures will produce a buffered solution?

I. Equal volumes of 0.5M NaOH and 1M HCl solutions are mixed
II. Equal volumes of 0.5MNaOH and 1M CH_3COOH solutions are mixed
III. Equal volumes of 1M $NaCH_3CO_2$ and 1M CH_3COOH solutions are mixed
IV. Equal volumes of 1.0MNaOH and 1M HCl solutions are mixed.

A. I B. III C. I and II D. II and III

Explanation:

Buffered solution is formed from:

i. Weak acid and its conjugate base
ii. Weak base and its conjuoate acid
iii. Excess weak acid and limited amount of strong base
Example: 1M CH_3COOH and 0.5MNaOH
iv. Excess weak base and limited amount of strong acid
Example: 1MNH_3 and 0.5MHCl

Answer: D

46. For which conversion is an oxidizing agent required?

A. $2H^+(aq) \longrightarrow H_2(g)$ C. $SO_3(g) \longrightarrow SO^{2-}{}_4$
B. $2Br^-(aq) \longrightarrow Br_2(aq)$ D. $MnO_2(s) \longrightarrow Mn^{2+}(aq)$

Explanation: Oxidizing agent is a species being reduced but cause oxidation

$2H^+ + 2e^- \longrightarrow H_2$ is reduction and needs reducing agent

$2Br^- \longrightarrow Br_2 + 2e$ is oxidation half – reaction and requires oxidizing agent.

$SO_3 \longrightarrow SO^{2-}{}_4$ is neither oxidation nor reduction. It does not need reducing and oxidizing agent.

$MnO_2 + 2e \longrightarrow Mn^{2+}$ is reduction half reaction and needs reducing agent

Answer: B

47. What volume of 0.500M NaOH is required to neutralize 25.0mL of 1.2M H_2SO_4? (Assume complete ionization of the acid)

A. 60mL B. 90mL C. 100mL D. 120mL

Solution: From the balanced chemical equation find the number of moles of the base (n_b).

$n_a = M_a \times V_a$

$n_a = 1.2m \times 25mL = 30mmol$

30mmol nb

$H_2SO_4 + 2NaOH \longrightarrow Na_2SO_4 + 2H_2O$

1 2

$$\frac{30mmol}{1} = \frac{n_b}{2}$$

$n_b = 60mmol$

$$Mb = \frac{n_b}{V_b}$$

$$Vb = \frac{n_b}{V_b} = \frac{60mmol}{0.5M} = 120mL$$

Answer: D

48. A gas is confined to a cylinder under a constant atmospheric pressure. When the gas undergoes a particular chemical reaction, it releases 135kJ of heat to its surrounding and does 63kJ of P – V work on its surroundings. What are the values of ΔH and ΔE for the process

A. $\Delta H = 125KJ, \Delta E = 63KJ$
B. $\Delta H = 135KJ, \Delta E = -63KJ$
C. $\Delta H = 135KJ, \Delta E = 198KJ$
D. $\Delta H = 135KJ, \Delta E = -198KJ$

Telegram: https://t.me/Extremeseries

Explanation: ΔH is heat transferred at constant pressure.

$\Rightarrow \Delta H = qp = -135kJ - 63kJ = -198kJ$

$\Delta H = q + w = -135kJ - 63kJ = -198kJ$ **Answer: D**

49. The oxidation number of nitrogen in NH_3, HNO_3 and NO_2 are _____ respectively.

A. $-3, -5, +4$ C. $-3, +5, -4$

B. $+3, +5, +4$ D. $-3, +5, +4$

Solution:

$\overset{x}{N}\overset{+1}{H_3}$ $\quad$ $\overset{+1}{H}\overset{x}{N}\overset{-2}{O_3}$ $\quad$ $\overset{x}{N}\overset{-2}{O_2}$

$x + 3(+) = 0$ $\quad$ $1 + x + 3(-2) = 0$ $\quad$ $x + 2(-2) = 0$

$x + 3 = 0$ $\quad$ $1 + x - 6 = 0$ $\quad$ $x - 4 = 0$

$x = -3$ $\quad$ $x = +5$ $\quad$ $x = +4$

Thus, the oxidation states of N in NH_3, HNO_3 and NO_2 are -3, $+5$ and $+4$ respectively. **Answer: D**

50. Which statement about the following reaction is correct?

$2Fe(s) + 3CO_2(g) \longrightarrow Fe_2O_3(s) + 3CO(g)$,

$\Delta H^O = +26.6kJ$

A. 26.6KJ of energy are released for every mole of Fe reacted

B. 26.6KJ of energy are absorbed for every mole of iron reacted.

C. 53.2kJ of energy are released for every mole of Fe reacted

D. 13.3KJ of energy are absorbed for every mole of Fe reacted.

Solution: The reaction is endothermic because of ΔH^0 is positive. 26.6kJ of heat is absorbed for 2 moles of iron reacted. As a result, 13.3kJ of heat is absorbed for every mole of iron reacted.

Answer: D

51. Which of the following metals is extracted by thermal reduction process?

A. Cu B. Fe C. Al D. Mg

Answer: A

52. The two standard electrode potentials involved in the nickel – cadmium rechargeable cell are given below, calculate the ΔG^0 in kJ of the cell.

$NiO_2(s) + 2H_2O(l) + 2e \longrightarrow Ni(OH)_2(s) + 2OH^-$

$E^0 = +0.49V$

$Cd(OH)_2 + 2e \longrightarrow Cd(s) + 2OH^-(aq)$

$E^0 = -0.76V$

A. -184 B. -153 C. -241 D. -206

Solution:

E^0 cell $= E^\circ$ cathode $- E^0$ anode

E^0 cell $= 0.49V - (-0.76V) = 1.25V$

$\Delta G^\circ = -nFE^0$ cell

$\Delta G^\circ = -2 \times 96500 \times 1.25 = -241250J$

$\Delta G^\circ = -241.250kJ$ **Answer: C**

53. Which of the following is a synthetic rubber produced from caprolatam (CPL)?

A. Nylon 6, 10 C. Teflon

B. Teryene D. Nylon 6 **Answer: D**

54. Which one of the following reactions is NOT a redox × reaction?

A. $Ag^+(aq) + Cl^-(aq) \longrightarrow AgCl(s)$

B. $2Na(s) + Cl_2(g) \longrightarrow 2NaCl(s)$

C. $Mg(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + H_2(g)$

D. $Cu^{2+}(aq) + Zn(s) \longrightarrow Cu(s) + Zn^{2-}(aq)$

Explanation: Redox reaction involves transfer of electrons. The presence of free atom or molecule indicates that the reaction is redox **Answer: A**

55. The half – reaction for formation of magnesium metal upon electrolysis of molten $MgCl_2$ is

$Mg^{2+} + 2e^- \longrightarrow Mg$

What is the mass of magnesium formed upon a passage of a current of 60A for a period of 2.00×10^3S?

A. 5g B. 10g C. 15.1g D. 30.2g

Telegram: https://t.me/ExtremeSeries

Solution: $m = \frac{MIt}{2F}$

$$m = \frac{24g/mol \times 60A \times 2 \times 10^3 S}{2 \times 96500 \frac{A.S}{mol}} = 15g$$

Answer: C

56. An aqueous solution is 70% nitric acid (HNO_3) by mass. What is the concentration of HNO_3 expressed in molality?

A. 0.55gm B. 8.62m C. 11.1m D. 37m

Solution:

70% by mass HNO_3 implies 70g HNO_3 in 30g of H_2O

$$\Rightarrow m = \frac{n_{HNO_3}}{kgH_2O} = \frac{70g/63g/mol}{0.030kg}$$

m – 37.04m

Answer: D

57. Which of the following is natural polymer?

A. Polyethylene C. Nylon

B. Polysaccharides D. Terylene

Explanation: Natural polymers include the following

- Proteins
- Polysaccharides such as cellulose, starch and glycogen
- Natural rubber

Answer: B

58. What is the ionization energy of all ion atom if it requires a radiation of 276nm to completely remove its outermost electron in the gaseous state? (plank's constant, h = 6.626 ×10J.S, speed of light, $C=3\times10^8 ms^{-1}$)

A. 7.21×10^{-19}J C. 7.21×10^{19}J

B. 7.21×10^{-19}kJ D. 7.21×10^{19}kJ

Solution: $E = hu$ but $v = \frac{c}{\lambda}$

$$\Rightarrow E = \frac{hc}{\lambda} = \frac{6.626\times10^{-34} J.s \times 3\times10^8 m/s}{276\times10^{-9} m} = 7.2 \times 10^{-19}J$$

Answer: A

59. Which of the following configurations describes the ground state electron configuration of Cl^-?

A. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ C. $1s^2\ 2s^2\ 2p^6\ 3p^1$

B. $1s^2\ 2s^2\ 2p^6\ 3s^1$ D. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p_x^2 3p_4^1$

Explanation: Total number of electrons = 17 + 1 = 18

Electron configuration = $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$

Answer: A

60. When 0.68 is divided by 14.364, the actual answer is 0.0473405. what will be the correct answer:

A. 0.05 C. 0.0473

B. 0.047 D. 0.04734

Explanation: During multiplication and division of a measured quantity, the last answer is reported with the least number of significant figures. Thus, the answer is written with two significant figures.

Answer: B

61. What is the correct molecular electronic configuration for the molecular ion, B^+_2?

A. $\sigma1s^2\sigma^*1s^2\sigma2s^2\sigma^*2s^2\sigma2p^2$

B. $\sigma1s^2\sigma^*1s^2\sigma2s^2\sigma^*2s^2\pi2p_y^2$

C. $\sigma1s^2\sigma^*1s^2\sigma2s^2\delta2s^{*2}\pi2p_x^1$

D. $\sigma1s^2\sigma^*1s^2\sigma2s^2\sigma^*2s^2\pi2p_x^1\pi2^1p_y$

Explanation: Total electrons = 2 × 5 – 1 = 9

$\Rightarrow \sigma1s^2\sigma^*1s^2\sigma2s^2\sigma^*2s^2\pi2p^1$

Answer: C

62. Which of the following molecular ions will exhibit delocalized bonding?

NO^-_2, NH^+_4, N^-_3

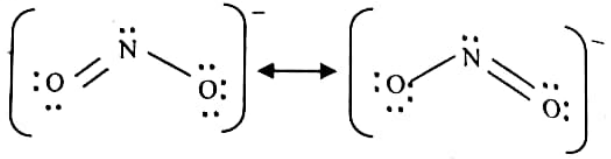
A. NO^-_2 and N^-_3 C. NO^-_2

B. NH^+_4 and N^-_3 D. NO^-_2 and NH^+_4

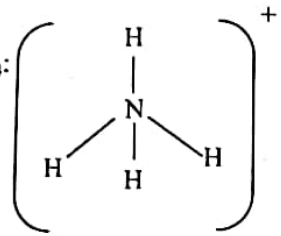
Telegram: https://t.me/ExtremeSeries

Explanation:

NO^-_2 :



NH^+_4:



Answer: A

(63.) Based on molecular orbital theory, the bond orders of H_2, H_2^+, and H_2^-, are _____ respectively.

A. 1,0 and 0 C. 1,0, and ½

B. 1,1/2 and 0 D. 1,1/2 and ½

Solution:

H_2: Total electrons = 2XI = $2\sigma ls^2$

B0= ½ (nb-na)= ½ (2-0) =1

H_2+: total electrons =2X1-1=1,σls^1

Bo=1/2 (nb-na) = ½(1-0)=1/2

H_2-: Total electrons =2X1+1=3,σls^2,σs^1

B0=1/2 (nb-na) =1/2(2-1) =1/2

Answer: D

(64.) How many 3d electrons are present in the ground state of chromium atom?

A. 4 B.5 C.6 D.10

Explanation

24Cr: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$

Number of electrons in 3d is 5

Answer: B

(63) When the following substances are arranged in the order of increasing melting point (lowest melting point first),the correct order is :

A. $CH_3CH_2\ CH_3, CH_3COCH_3, CH_3\ CH_2\ CH_2OH$

B. $CH_3CH_2CH_3, CH_3CH_2OH, CH_3COCH_3$

C. $CH_3COCH_3, CH_3CH_2CH_2OH, CH_3CH_2CH_3$

D. $CH_3CH_2CH_2OH, CH_3CH_2CH_3, CH_3COCH_3$

Explanation: Boiling Point of :

Alkane < Ketone < Alcohol <Carboxylicacid **Answer: A**

(66) The half-life for the first order decomposition of nitromethane, CH_3NH_2at 500K is 650 seconds. If the initial concentration of CH_3NO_2 is 0.5 M, What will its concentration be in (M) after 1300 seconds have elapsed ?

A. 0.125 B. 0.140 C. 0.250 D. 0.425

Solution: $t_{1/2} = \dfrac{0.693}{k}$

$$k = \frac{0.693}{t_{1/2}} = \frac{0.693}{650} s^{-1}$$

$$\log \frac{[CH_3NO_2]_o}{(CH_3NO_2)_t} = \frac{kt}{2.303}$$

$$\frac{[CH_3NO_2]_o}{(CH_3NO_2)_t} = 10^{kt/2.303}$$

$$[CH_3NO_2]_t = \frac{0.50M}{10^{\frac{0.693}{650}\times 1300/2.303}} = \frac{0.50M}{10^{0.6}} = 0.25M$$

Answer: C

(67) In a zero-order reaction for every 10^0c rise of temperature, the rate is double. If the temperature is increased from 10^0C to 100^0C, the rate of the reaction will become.

A. 164 times C. 256 times

B. 128 times D. 512 times

Telegram: https://t.me/ExtremeSeries

Solution: Determine the number of 10^0c

10^0C to 100^0C.

^0C	rate	^0c	rate
1.	10-20=2	6.	61-70=2
2.	21-30=2	7.	71-80=2
3.	31-40=2	8.	81-90=2
4.	41-50=2	9.	81-90=2
5.	51-60=2	10.	91-100=2

Rate = 2^9 =512 times

Answer: D

(68) Which of the following mathematical relationships between K, K_1 and K_2 is correct?

$CO_2(g) + H_2(g) \rightleftharpoons CO(g) + H_2O(g)$ K

$Fe(s) + CO_2(g) \rightleftharpoons FeO(s) + CO(g)$ K_1

$Fe(s) + H_2O(g) \rightleftharpoons FeO(s) + H_2(g)$ K_2

A. $K = K_1 + K_2$ C. $K = K1/K_2$

B. $K = K_1 X K_2$ D. $K = K_2/K_1$

Solution:

To obtain the first reverse the third and add it with the second.

$$Fe(s) + CO_2 \overrightarrow{\leftarrow} FeO(s) + CO(g) k_1$$

$$FeO(s) + H_2(g) \overrightarrow{\leftarrow} Fe(s) + H_2O(g) \frac{1}{k_2}$$

$$CO_2(g) + H_2(g) \overrightarrow{\leftarrow} CO(g) + H_2O(g)$$

$$K = k_1 \times \frac{1}{k_2} = \frac{k_1}{k_2}$$

Answer: C

(69) Which of the following gives the correct order of decreasing acidity of carboxylic acids?

A. $Cl_3CCOOH, Cl_2CHCOOH, FCH_2COOH, CH_3COOH$

B. $FCH_2COOH, CH_3COOH, Cl_2CHCOOH, Cl_3CCOOH$

C. $CH_3COOH, FCH_2COOH, Cl_2CHCOOH, Cl_3CCOOH$

D. $Cl_2CHCOOH, CH_3COOH, FCH_2COOH, Cl_3CCOOH$

https://t.me/Extremesers

Explanation

If halogens replace hydrogen from the carboxylic acids, their strength increased

Example

$CH_3COOH < CH_2C1COOH < CHCl_2COOH < CC1_3COOH$

Answer: C

(70) The value of k_c for the following reaction is 4.0 at a temperature of 373k. $CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$

What mass of ethyl ester ($CH_3COOC_2H_5$) would be present in the equilibrium was established at this temperature

A. 5.2 B. 10.1 C. 12.6 D. 14.1

Solution: $k_c = \frac{[CH_3COOC_2H_5]}{[CH_3COOH][C_2H_5]}$

$$k_c = \frac{n_{CH_3COOC_2H_5}}{n_{CH_3COOH} \cdot n_{C_2H_5OH}}$$

$$4 = \frac{n_{CH_3COOC_2H_5}}{n_{CH_3COOH} \cdot n_{C_2H_5OH}}$$

$n_{CH_3COOC_2H_5} = 0.25mol$

$m_{CH_3COOC_2H_5} = 0.25mol \times 88g/mol = 22g$

Answer: 22g

71. Which of the following is true about of an open system?

A. A system that exchanges both energy and matter with its surroundings.

B. A system that cannot exchange both matter and energy with its surrounds.

C. A system that only exchanges matter with its surroundings

D. A system that only exchanges energy with its surroundings

Explanation:

- Open system exchanges both matter and energy
- Close system exchanges energy only
- Isolated system cannot exchanges both energy and matter.

Answer: A

Telegram

72. Which of the following statements is true about the percent ionization of a weak acid?

A. It decreases with increasing concentration
B. It increases with decreasing concentration
C. It increases with increasing concentration
D. It decreases with decreasing concentration

Explanation:

$$\text{Percent of ionization of acid} = \frac{\text{concentration of } H^+}{\text{Initial concentration of acid}} \times 100$$

Answer: A

73. Which is the value of ΔH for the reaction: $S(s) \longrightarrow S(g)$

$S(s) + O_2(g) \longrightarrow SO_2(g)$ $\quad \Delta H = -395KJ$

$S(g) + O_2(g) \longrightarrow SO_2(g)$ $\quad \Delta H = -618KJ$

A. $-1013KJ$ C. $+223KJ$
B. $-223KJ$ D. $+1013KJ$

Solution:

Reverse the second and add it with the first

$$S(s) + O_2(g) \rightarrow SO_2(g) \quad \Delta H_1 = -395KJ$$
$$SO_2(g) \rightarrow S(g) + O_2(g) \quad \Delta H^1 = +618KJ$$
$$S(s) \rightarrow S(g), \Delta H = \Delta H_1 + \Delta H^1$$

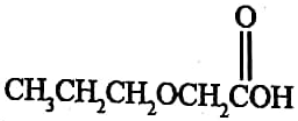
$\Delta H = -395KJ + 618KJ$

$\Delta H = +223KJ$

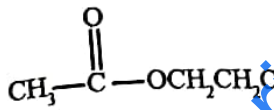
Answer: C

74. Which of these compounds is the ester formed from the reaction of acetic and 1 – Propanol?

A. $CH_3CH_2OOCH_2CH_3$
B. $CH_3CH_2CH_2CH_2COH$
C.

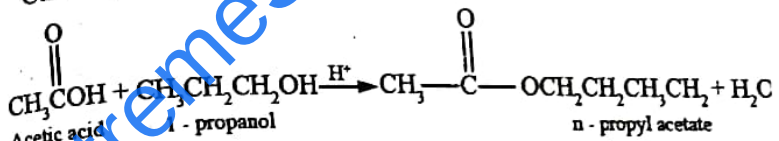


D.



Explanation:

Carboxylic acid + Alcohol $\xrightarrow{H^+}$ $\xrightarrow{\text{Ester}}$ + water



Answer: D

75. A lab instructor is preparing 5.0liters of a 0.10MPb $(NO_3)_2$ (molecular mass 331) solution. What is the mass required?

A. 165.5g of $Pb(NO_3)_2$ and add 5.0kg of H_2O
B. 165.5 g of $Pb(NO_3)_2$ and add H_2O until the solution has a volume of 5.0 liters
C. 33.1g of $Pb(NO_3)_2$ and add H_2O until the solution has a volume of 5.0 liters
D. 33.1g of $Pb(NO_3)_2$ and add 5.0 liters of H_2O

Solution:

$m = M \times M_{wt} \times V$

$$m = 0.10 \frac{mol}{L} \times 331 \frac{g}{mol} \times 5.0L = 165.5g$$

Thus, 165.5g of Pb $(NO_3)_2$ and add water until the solution has a volume of 5.0 liters.

Answer: B

76. Which of the following is NOT a characteristic of theelectrolytic cell containing aqueous solution of NaCl used in the manufacture of sodium hydroxide?

A. The sodium hydroxide solution is produced in the electrolytic cell.
B. The electrolyte must be a dilute solution of NaCl
C. Hydrogen is produced at the cathode
D. The production of chlorine gas occurs at the anode.

Telegram: https://t.me/ExtremeSeries

Explanation:

- The electrolyte must be concentrated NaCl (brine)
- At anode, Cl_2 gas is librated.
- At cathode, H_2 gas is given off
- NaOH is formed as residue

Answer: D

77. The kinetic data below are for the reaction: $A + B \longrightarrow C$
From these data what are the orders of the reaction with respect to A and B?

[A]	[B]	Initial rate ($moldm^{-3}$ sec^{-1})
0.1	0.1	1×10^{-5}
0.1	0.2	4×10^{-5}
0.2	0.1	1×10^{-5}

A. Order of A = 1, order of B = O
B. Order of A = O, order of B = 2
C. Order of A = O, order of B = 4
D. Order of A = 1, order of B = 2

Solution: Late the rate law be: $r = K[A]^x [B]^y$
Find the value of x by using the first and third data and Y using the first and second data.

$$\frac{rate\,1}{rate\,3} = \frac{K[A]_1^x [B]_1^y}{K[A]_3^x [B]_3^y}$$

$$\frac{1\times10^{-5}}{1\times10^{-5}} = \frac{(0.1)^x (0.1)^y}{K(0.2)^x (0.1)^y} \Rightarrow \left(\frac{1}{2}\right)^x = 1 = \left(\frac{1}{2}\right)^0$$

X = O

$$\frac{rate\,1}{rate\,2} = \frac{[B]_1^y}{[B]_2^y} \Rightarrow \left(\frac{1}{2}\right)^y = \left(\frac{1}{2}\right)^2 = \frac{1\times10^{-5}}{4\times10^{-5}} = \left(\frac{0.1}{0.2}\right)^y$$

y = 2

Thus, rate = K $[A]^o[B]^2 = K[B]^2$ **Answer: B**

78. Which of the following metals has the highest electrical and thermal conductivities of any metal?

A. Ag B. Cu C. Ni D. Co

Explanation: Silver is known to be the best conductors of heat and electricity. **Answer: A**

79. Which of the following is a linear polymer?

A. High density polythene (HDPt)
B. Low density polythene (LDPt)
C. Bakelite
D. Vulcanized rubber

Explanation: Polymers can be linear or branched i.e. cross – linked.

- Linear polymers are held together by London dispersion forces and are flexible, low strength and low melting point.
- Branched or cross – linked polymers are hard and non – ductile.Bakelite, vulcanized rubber and high density polythene are cross – linked polymers. **Answer: B**

80. Which of the following is the most important source for the extraction of iron

A. Hematite C. chalcopyrite
B. Bauxite D. Sphalerite

Answer: A

Telegram: https://t.me/ExtremeSeries

1. According to valence bond theory, which orbitals on bromine atoms overlap in the formation of the bond in Br_2?

A. 3s B. 3p C. 4s D. 4P

Explanation: According to VBT, covalent bond is formed when half-filled orbitals overlap. The half-filled orbital is found in 4p orbital. $Br(z=35):[Ar]4s^2 3d^{10} 4p_x^2 4p_y^2 4p_z^1$

Answer: D

2. The type of compound that is most likely to contain a covalent bond is one that is

A. A solid metal
B. Composed of only non-metals
C. Composed of a metal from the far left and a non-metal from far right of the periodic table
D. Held together by the electrostatic forces between oppositely charged ions.

Explanation: Covalent bond is most likely formed between non-metals.

Answer: B

3. The first step of the scientific method involves

A. forming a hypothesis
B. making observations
C. performing an experiment
D. predicting the result of an experiment

Explanation: Steps in scientific method

Observation → hypothesis → experiment → theory → law

Answer: B

4. The process of solute particles being surrounded by solvent particles is known as

A. saturation C. solvation
B. agglomeration D. dehydration

Answer: C

5. Which of the following represents an acceptable possible set of quantum numbers (in the order n, ℓ, m_ℓ, m_s) for an electron an atom?

A. 2, 1, 0, 0 C. 2, 1, -1, $-\frac{1}{2}$
B. 2, 0, 2, $+\frac{1}{2}$ D. 2, 0, 1, $-\frac{1}{2}$

Explanation: n = 1, 2, 3, …

$\ell = 0$ to n − 1

$m_\ell = -\ell$ to $+\ell$

$m_s = \pm \frac{1}{2}$

Answer: C

6. Of the three types of radioactivity characterized by Rutherford, which of the following are particles?

A. γ – rays
B. β - rays
C. α - rays and β - rays
D. α - rays and β - rays and γ– rays

Explanation:

- α - rays and β - rays are particles.
- α - rays have mass and positive charge
- β - rays have negative charge but no mass
- γ– rays are radiation without mass and charge

Answer: C

7. What kind of energy is converted in a galvanic cell?

A. Chemical energy is converted into electrical energy
B. Chemical energy is converted to heat.
C. Chemical energy is obtained from heat
D. Electrical energy is converted into chemical energy.

Explanation:

Electro chemical cell

Electrolytic cell		Galvanic cell	
Electrical energy	→ Chemical energy	Chemical energy	→ Electrical energy

Answer: A

8. The conversion of nitrogen gas to nitrates by bacteria is called

A. nitrification C. execration
B. nitrogen fixation D. denitrification

Answer: B

9. According to the Arrhenius concept, an acid is a substance that?

A. is capable of donating one or more H+ to any solvent
B. causes an increase in concentration of H^+ in aqueous solutions

Telegram: https://t.me/ExtremeSeries

C. can accept a pair of electrons to form a coordinate covalent bond

D. reacts with the solvent to form the cation formed by autoionization of that solvent.

Explanation: Arrhenius definition

- Acid is a substance that releases a hydrogen ion in water
- It causes an increase in the concentration of hydrogen ion in aqueous solutions.

Answer: B

10. A system which can exchange both matter and energy with its surroundings is said to be a/an

A. Isolated system C. ideal system

B. open system D. closed system

Explanation:

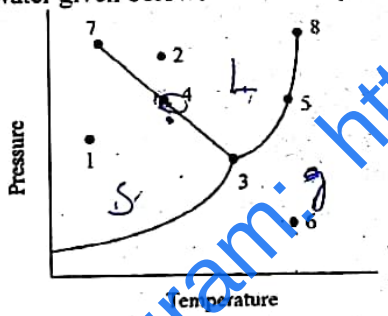
Open system – exchanged matter and energy

Isolated system – exchanges neither mass nor energy

Closed system – exchanges energy but not matter

Answer: B

11. At which point can only the solid and liquid phases coexist in the phase diagram of water given below?



A. 3 B. 4 C. 5 D. 8

Explanation:

→ Solid and liquid phases coexist at equilibrium on fusion curve (point 4)

→ Liquid and vapor phases coexist at equilibrium on vaporization curve (point 5)

→ Solid and vapor phases coexist at equilibrium on sublimation curve.

Answer: B

12. What type of solute-solvent interaction should be the most important in a solution of iodine in carbon tetrachloride?

A. London forces C. Ion-dipole forces

B. Ionic bonding D. Dipole-dipole forces.

Explanation: Iodine is non-polar solute and carbon tetrachloride is non-polar solvent. The solute – solvent interactions responsible for the solubility of non-polar solute in non-polar solvent is London force.

Answer: A

13. The distance between two carbon atoms in a diamond is 154pm. What is the distance between the carbon atoms in millimeters?

A. 7.7×10^{-5} C. 1.54×10^{-7}

B. 7.7×10^{-7} D. 1.54×10^{-9}

Solution:

$$\frac{1\text{pm}}{1\text{mm}}=\frac{10^{-12}\text{m}}{10^{-3}\text{m}}$$

$$\frac{1\text{pm}}{1\text{mm}}=10^{-9}$$

$$1\text{pm}=10^{-9}\text{mm}$$

$$\Rightarrow 154\text{pm}=154\times10^{-9}\text{mm}=1.54\times10^{-7}\text{mm}$$

Answer: C

14. For the acid-base equilibrium,

$$HCO_3^- + H_2O \rightleftharpoons H_2CO_3 + OH^-$$

The Bronsted-Lowry acids are;

A. H_2O and OH^-

B. HCO_3^- and OH^-

C. H_2O and H_2CO_3

D. HCO_3^- and H_2CO_3

Explanation:

- Bronsted-lowry acids are proton (H^+) donors. For the forward reaction, H_2O is proton donor but for the reverse reaction H_2CO_3 is proton donor.

Answer: C

15. How many sigma and pi bonds are present in the following molecule?

$$H_3C-CH=CH-CH_3$$

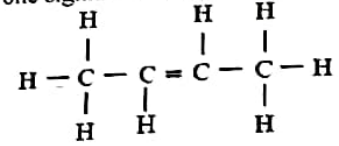
A. 8σ bonds and 1π bonds

B. 8σ bonds and 2π bonds

Telegram: https://t.me/ExtremeSeries

C. 10σ bonds and 2π bonds
D. 11σ bonds and 1π bonds

Solution: Draw detail structural formula to observe all the single bonds. All single bonds are sigma bonds. One double bond contains one sigma and one pi bond.



There are 11σ and 1π bonds. **Answer: D**

16. How many orbitals are there in an atom with $n = 4$?
A. 2 B. 8 C. 16 D. 25

Solution: Number of orbitals in main shell $= n^2 = 4^2 = 16$

Answer: C

17. The most abundant metal on the earth's crust is:
A. Fe B. Al C. Ca D. Na

Explanation:
Al > Fe > Ca > Na

Answer: B

18. Which of the following is correct?
A. $1L = 1dm^3$ C. $10L = 1dm^3$
B. $1L = 10dm^3$ D. $1L = 0.1dm^3$

Solution:

$$1m^3 = 10^3 L = 10^3 dm^3 = 10^6 mL = 10^6 cm^3$$
$$\Rightarrow 1L = 1dm^3,\ 1mL = 1cm^3$$

Answer: A

19. The enthalpy of combustion of solid carbon to form carbondioxide is -383.5KJ/mol carbon, and the enthalpy of combustion of carbon monoxide to form carbondioxide is -283.3KJ/mol CO. what will be the enthalpy change, ΔH for the reaction below?

$$2C(s) + O_2(g) \rightarrow 2CO(g)$$

A. -110.4KJ C. +172.9KJ
B. -220.8KJ D. +1354.0KJ

Solution: $C(s) + O_2(g) \rightarrow CO_2(g), \Delta H_1 = -393.5KJ$

$$CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g), \Delta H_2 = -283.3KJ$$

To obtained the required reaction:

Multiply the first by 2

Reverse the second and multiply by 2

$$\Rightarrow 2C(s) + 2O_2(g) \longrightarrow 2CO_2(g),\ \Delta H_1^1 = 2\Delta H_2$$
$$\Delta H_1^1 = -787KJ$$
$$2CO_2(g) \longrightarrow 2CO(g) + O_2(g)\ \Delta H_2^1 = -2\Delta H_2$$
$$\Delta H_2^1 = -2x(-283.3KJ)$$
$$= 566.6\ KJ$$
$$2C(s) + O_2(g) \longrightarrow 2CO(g),\ \Delta H = \Delta H_1^1 + \Delta H_2^1$$
$$\Delta H = -787kg + 566.6KJ$$
$$\Delta H = -220.4KJ$$

Answer: B

20. The decreasing order of electrochemical characteristics of some metals is given as:
Mg > Al > Zn > Cu > Ag. What will happen if a copper spoon is used to stir a solution of aluminum nitrate ($Al(NO_3)_3$)?
A. There is no reaction
B. The solution becomes blue
C. The spoon will get coated with aluminum
D. An alloy of copper and aluminum is formed.

Explanation:
- There is no reaction because more active metal displaces less active metal but not the reverse.

$Cu + Al(NO_3)_3 \longrightarrow$ no reaction because Cu is less reactive that Al.

$2Al + 3CuSO_4 \longrightarrow Al_2(SO_4)_3 + 3Cu$

Answer: A

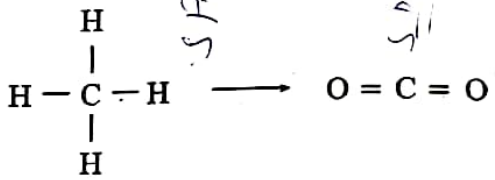
21. What hybridization change does the carbon atom undergo in the combustion of methane?

$$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(g)$$

A. $sp \longrightarrow sp^2$ C. $sp^3 \longrightarrow sp$
B. $sp^2 \longrightarrow sp^3$ D. $sp^2 \longrightarrow sp$

Explanation: $CH_4 \longrightarrow CO_2$

Telegram: https://t.me/ExtremeSeries



- Four sets of electoms hybridization: sp3
- Two sets of electrons
- Hybridization: sp

Answer: C

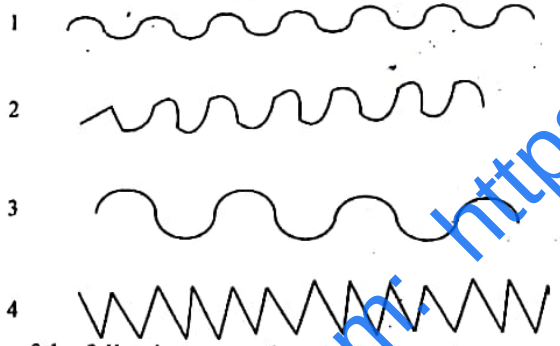
22. Which of the following ionic compounds has the greatest lattice energy?

A. LiF
B. LiCl
C. LiBr
D. LiI

Explanation: Lattice energy is affected by size and magnitude of charge
As size increases, lattice energy decreases.
As magnitude of change increases as, lattice energy also increase.

Answer: A

23. Consider the four electromagnetic waves shown below



Which of the following waves has the highest frequency?

A. 1 B. 2 C. 3 D. 4

Explanation:
As the number of waves increase, frequency also increases.

Answer: D

24. Considering the reaction below in which of the following will the effect of concentration and temperature simultaneous cause an increase in the rate at which products are formed?

$$CaCO_3(s) + 2HCl(aq) \rightarrow CO_2(g) + CaCl_2(aq) + H_2O(l) + heat$$

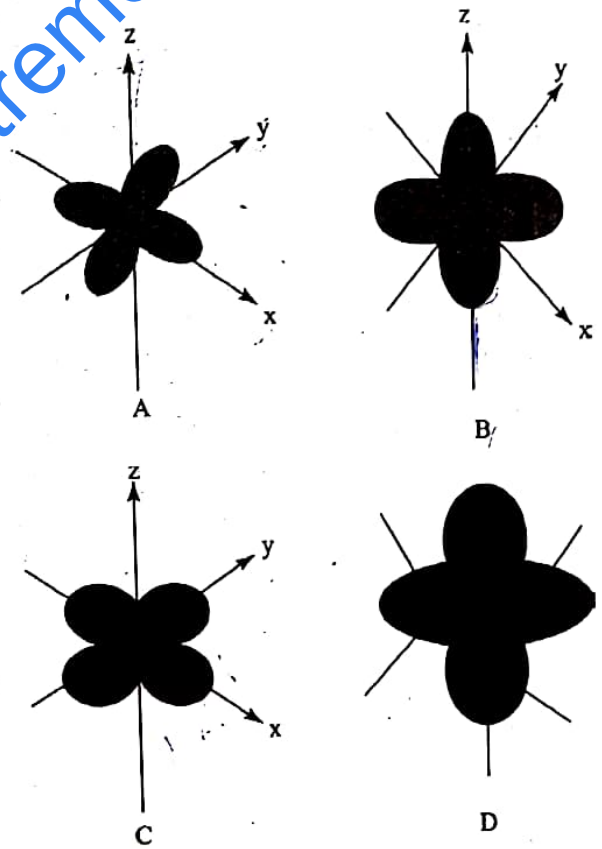
A. Decrease [HCl] and decrease temperature
B. Increase [HCl] and Increase temperature

C. Increase [HCl] and decrease temperature
D. Grind up the $CaCO_3$ and decrease temperature

Explanation:
For exothermic reaction, increasing temperature decrease the rate of a reaction.
Increasing concentration of aqueous solution and gases increase the rate of a reaction.

Answer: C

25. Which of the following diagrams describes the electron density in the dxy orbital?



Solution:

A. dxy
B. dz^2
C. $dx^2 - y^2$
D. dz^2

Answer: A

Telegram: https://t.me/ExtremeSeries

(26) The wave number of an electromagnetic radiation is $1 \times 10^5 cm^{-1}$. The frequency of the –radiation would be

A. $3 \times 10^8 s^{-1}$ C. $3 \times 10^{10} s^{-1}$
B. $3 \times 10^6 s^{-1}$ D. $3 \times 10^{15} s^{-1}$

Solution: Wave number $= \frac{1}{\lambda}$

$$c = \lambda v = v = \frac{c}{\lambda} = C \times \text{wave number}$$

$$v = 3\times10^{10}\frac{cm}{s}\times1\times10^5 cm^{-1} = 3\times10^{15} s^{-1}$$

Answer: D

(27) For the gas phase reaction

$$N_2 + O_2 \rightleftharpoons 2NO,\ \Delta H = +180 KJ/mol$$

The value of K changes with the

A. change in pressure
B. introduction of NO
C. change in concentration of N_2
D. change in temperature

Explanation:

The value of equilibrium constant is affected by:

- temperature
- reversing the reaction
- change in coefficients of reactants and products

Answer: D

(28) In the reaction ($2SO_2 + O_2 \rightleftharpoons 2SO_3$, $K_{eq} = 100$). What will be the concentration of O_2, if the concentration of SO_2 is the same as that of SO_3?

A. $[O_2] = [SO_2]$ C. $[O_2] = 100M$
B. $[O_2] = 0.01M$ D. $[O_2] = 0.1M$

Solution: Given $[SO_2] = [SO_3]$

$$K_{eq} = \frac{[SO_3]^2}{[SO_2]^2[O_2]}$$

$$K_{eq} = \frac{1}{[O_2]}$$

$$[O_2] = \frac{1}{K_{eq}} = \frac{1}{100} = 0.01\ M$$

Answer: B

(29) The decomposition of nitrosyl chloride was studied as:

$$2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$$

The following data were obtained where

$$Rate = \frac{-\Delta[NOCl]}{\Delta t}$$

$[NOCl]_o$ (molecules/cm^3)	Initial Rate (molecules/cm^3.s)
3.0×10^{16}	5.98×10^4
2.0×10^{16}	2.66×10^4
1.0×10^{16}	6.64×10^3
4.0×10^{16}	1.06×10^5

What is the rate law in the above decomposition

A. $r = K[NOCl]^2$ C. $r = K[NOCl]\ [NO]$
B. $r = K[NOCl]$ D. $r = K[NOCl]\ [Cl]$

Solution:

Select any two simple data to find the order of the reaction with respect to NOCl.

$$Rate = K[NOCl]^x$$

For example use data 2 and 4

$$\frac{r_2}{r_4} = \frac{K[NOCl]_2^x}{K[NOCl]_4^x}$$

$$\frac{2.66\times10^4}{10.6\times10^4} = \frac{(2.0\times10^{16})^x}{(4\times10^4)^x}$$

$$\frac{1}{4} = \left(\frac{1}{2}\right)^x \Rightarrow \left(\frac{1}{2}\right)^2 = \left(\frac{1}{2}\right)^x$$

$x = 2$

Thus, rate $= K[NOCl]^2$

Answer: A

(30) Considering the mechanism for a reaction below, which of the following statement is correct?

Step1: $HBr + O_2 \rightarrow HOOBr$
Step 2: $HBr + HOOBr \rightarrow 2HOBr$
Step 3: $2HOOBr + 2HBr \rightarrow 2Br_2 + 2H_2O$

A. Br_2 is a reactant
B. HBr is a product

Telegram: https://t.me/ExtremeSeries

C. HOBr is a catalyst
D. HOOBr is a reaction intermediate

Explanation:
The over all reaction is
$4HBr + O_2 \rightarrow 2Br_2 + 2H_2O$
$\Rightarrow$ HBr is reactant
Br_2 is product
HOBr and HOOBr are intermediates.

Answer: D

31. The acid-base indicator bromocresol green is a weak acid. The yellow acid and blue base forms of the indicator are present in equal concentrations in a solution when the PH is 4.68. What is the PKa of bromocresol green?
A. 4.48 B. 4.68 C. 5.68 D. 6.68

Solution: Let the formula of the indicator be HIn
$HIn \rightleftharpoons H^+ + In^-$
Given, $[HIn] = [In^-]$

$$P^H = PKa + 10g\frac{[In^-]}{[HIn]}$$

$4.68 = PKa + 10^1$
$\Rightarrow PKa = 4.68$

Answer: B

32. If $NaNO_2$ is added to a solution of HNO_2, which of the following statement is true?
A. The pH of the solution will increase
B. The pH of the solution will decrease
C. The pH will remain the same
D. The equilibrium will not be affected

Explanation: $NaNO_2$ is basic salt, because it is obtained from strong base. NaOH and weak acid, HNO_2. When a basic salt is added to a given solution, the pH of the solution will increase. **Answer: A**

33. A lipid is any substance of biochemical origin that is
A. soluble in both water and non-polar solvents
B. insoluble in both water and non-polar solvents
C. soluble in both water but insoluble in non-polar solvents.
D. soluble in non-polar solvents and insoluble in water

Explanation: Fats and soils are collectively known as lipids. They are organic compounds that dissolve in non-polar solvents but not in polar solvents such as water.

Answer: D

34. A solution in an electrolytic cell contains Cu^{2+} $(\epsilon^\circ = 0.34V)$. Ag^+ $(\epsilon^\circ = 0.80V)$ and Zn^{2+} $(\epsilon^\circ = -0.76V)$. If the voltage is initially very low and slowly increased, in which order will the metals be plated out onto the cathode?
A. $Zn^{2+} > Cu^{2+} > Ag^+$
B. $Cu^{2+} > Zn^{2+} > Ag$
C. $Ag^+ > Zn^{2+} > Cu^{2+}$
D. $Ag^+ > Cu^{2+} > Zn^{2+}$

Explanation: The more positive the reduction potential, the easier is to be reduced at cathode.
$\Rightarrow Ag^+ > Cu^{2+} > Zn^{2+}$

Answer: D

35. What is the concentration of sodium chloride in water needed in order to produce an aqueous solution that has an identical Osmotic pressure (isotonic) with blood ($\pi = 7.70$ atm at 25°C)?
A. $0.003molL^{-1}$ C. $0.1575molL^{-1}$
B. $0.006molL^{-1}$ D. $0.315molL^{-1}$

Solution: $\pi = i$ MRT
- The solution is electrolytic: $NaCl \rightarrow Na^+ + Cl^-$
- The value of Van't Hoff factor, i is almost equal to the number of ions produced.

$i = 2$, $T = 25°C = 298K$

$$R = 0.082\ \frac{atm.L}{mol.K}$$

$$M = \frac{\pi}{iRT} = \frac{7.70atm}{2 \times 0.827\ \frac{atm.L}{mol.K} \times 298K}$$

$M = 0.1575mol.L^{-1}$

Answer: C

36. What is the molarity of a 5g hydrogen peroxide (H_2O_2) in 100mL solution that is used for hair bleaching?
A. 0.015M C. 1.5M
B. 0.15M D. 3M

Solution: $M = \frac{m}{M_{wt} \times V}$

Telegram: https://t.me/ExtremeSeries

$$M = \frac{5g}{34\frac{g}{mol} \times 0.1L} = 1.47M$$

Answer: C

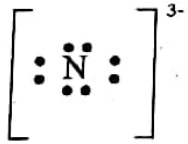
37. The maximum number of electrons in p-orbital with $n = 6$, $m\ell = 0$ is
A. 2 B. 6 C. 16 D. 14

Explanation:
The maximum number of electrons in a p-orbital is 2. The given quantum numbers represent one of three 6p orbital.

Answer: A

38. How many unpaired electrons are there in the Lewis structures of a N^{3-} ion?
A. 0 B. 1 C. 2 D. 3

Explanation: structure for N^{3-} is:



$\Rightarrow$ No unpaired electron

Answer: A

39. Which one of the following metals is extracted by thermal reduction process:
A. Al B. Cu C. Fe D. Mg

Answer: C

40. Which of the following transitions will omit maximum energy in the hydrogen atom?
A. $n = 4 \rightarrow n = 3$ C. $n = 2 \rightarrow n = 1$
B. $n = 4 \rightarrow n = 2$ D. $n = 3 \rightarrow n = 2$

Solution: $\Delta \epsilon = 2.18 \times 10^{-10} J\left(\frac{1}{4^2} - \frac{1}{3^2}\right) = -1.06 \times 10^{-19} J$

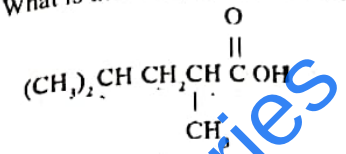
$\Delta \epsilon = 2.18 \times 10^{-18} J\left(\frac{1}{4^2} - \frac{1}{2^2}\right) = -4.10 \times 10^{-19} J$

$\Delta \epsilon = 2.18 \times 10^{-18} J\left(\frac{1}{2^2} - \frac{1}{1^2}\right) = -1.635 \times 10^{-18} J$

$\Delta \epsilon = 2.18 \times 10^{-18} J\left(\frac{1}{3^2} - \frac{1}{2^2}\right) = -1.40 \times 10^{-19} J$

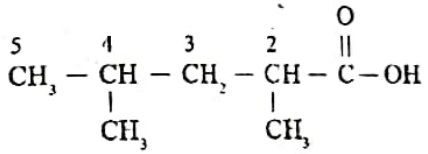
Answer: C

41. What is the IUPAC name for the compound



A. 2, 4 – dimethyl pentanoic acid
B. 1, 1, 3-trimethyl butanoic acid
C. 1-hydroxyl -2, 4 – dimethyl pentanone
D. 2-carboxy isohexone

Explanation:
The compound is carboxylic acid. Start form the carbonyl carbon to give IUPAC name.
The above structural formula is the same as the structure below



2, 4 – dimethyl pentanoic acid

Answer: A

42. The conventional equilibrium constant expression (K_C) for the system.

$2ICl(s) \rightleftharpoons I_2(s) + Cl_2(g)$ is

A. $\frac{[I_2][Cl_2]}{[ICl]^2}$ C. $[Cl_2]$

B. $\frac{[I_2][Cl_2]}{2[ICl]}$ D. $\frac{[I_2]+[Cl_2]}{2[ICl]}$

Explanation: Pure solids and liquids /solvents/ are not included in the equilibrium constant expression.

$\Rightarrow K_C = [Cl_2]$

Answer: C

43. Which of the following compound does Not follow the octet rule?
A. CS_2 B. PBr_3 C. IBr D. BrF_5

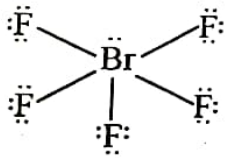
Explanation: If the number of electrons on the centeral atom is not equal to eight, such type of compounds donot obey the octet rule.

Telegram: https://t.me/Extremeseries

CS_2 : S = C = S :, obeys the octet rule

PBr_3 : Br — P — : Br : , obeys the octet rule
(with : Br : below P)

IBr : I — Br : , obeys the octet rule

BrF_5 

, does not obey the octet rule because the number of electrons on Br are 12

Answer: D

44. Chemically, fats and oils are
A. Acids
B. Alcohols
C. Esters
D. Alkenes

Answer: C

45. The reaction A + 3B → 2C + D is first order with respect to reactant A and second order with respect to reactant B. If the concentration of A is doubled and the concentration of B is halved, the rate of the reaction would _____ by a factor of _____
A. Increase, 2
B. decrease, 2
C. Increase, 4
D. decrease, 4

Solution: Given $[A]_2 = 2[A]$, $[B]_2 = \frac{1}{2}[B]_1$

$r_1 = K[A]_1 [B]_1^2$

$r_2 = K[A]_2 [B]_2^2$

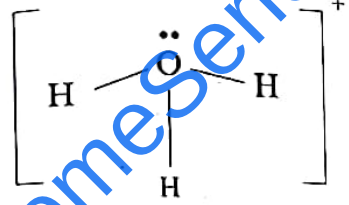
$r_2 = K2[A]_1\left(\frac{1}{2}[B]_1\right)^2 = K2[A]_1 \frac{1}{4}[B]_1^2$

$r_2 = \frac{1}{2} r_1$

Answer: B

46. The molecular geometry of the ion H_3O^+ is
A. linear
B. tetrahedral
C. bent
D. trigonal pyramidal

Explanation:

H_3O^+ 

, trigonal pyramidal

Answer: D

47. If a student wishes to prepare approximately 100mL of an aqueous solution of 6M HCl using 12M HCl. Which procedure is correct?
A. Adding 50mL of 12M HCl to 50mL of water while stirring the mixture steadily.
B. Adding 25mL of 12M HCl to 50mL of water while stirring the mixture steadily.
C. Adding 50mL of water to 50mLof 12M HCl while stirring the mixture steadily.
D. Adding 25mL of water to 50 mL of 12M HCl while stirring the mixture steadily.

Solution:

$M_1 = 12M$ $M_2 = 6M$
$V_1 = ?$ $V_2 = 100mL$
$M_1V_1 = M_2V_2$

$$V_1 = \frac{M_2V_2}{M_1} = \frac{6M \times 100mL}{12M} = 50mL$$

V_{H2O} added $= V_2 - V_1$
$= 100mL - 50mL = \mathbf{50mL}$

Answer: A

48. During dilution of concentrated acid, acid is added to water not water to acid.
What is the final concentration of Cl^- ion when 250mL of 0.20M $CaCl_2$ solution is mixed with 250mL of 0.40M KCl solution? (Assume additive volumes)
A. 1.60M
B. 0.40M
C. 0.20M
D. 0.60M

Telegram: https://t.me/ExtremeSeries

Solution:

$$n_{CaCl_2} = M \times V = 0.20\frac{mol}{L} \times 250mL = 50mmol$$

$$CaCl_2 \longrightarrow Ca^{2+} + 2Cl^-$$

$$\Rightarrow n_{Cl^-} = 2 \times 50m\ mol = 100m\ mol$$

$$n_{KCl} = M \times V = 0.40\frac{mol}{L} \times 250mL = 100m\ mol$$

$$KCl \longrightarrow K^+ + Cl^-$$

$V_T = (250 + 250)\ mL = 500mL$

$n_T = 100m\ mol + 100m\ mol = 200m\ mol$

$$\left[Cl^-\right] = \frac{n_T}{V_T} = \frac{200m\ mol}{500mL} = 0.40mol/L$$

Answer: B

49. Which of the following is used in the reaction called saponification?

A. Strong base C. Hydrogen
B. Strong acid D. Nickel

Explanation:
Long chain organic acids + NaOH/KOH → Soap strong bases NaOH and KOH are used to make hard and soft soaps respectively.

Answer: A

50. The concentration of nitrate ion in a solution that contains 0.900M aluminum nitrate is:

A. 0.90 B. 0.45 C. 0.30 D. 2.70

Solution: The formula of aluminum nitrate is $Al(NO_3)_3$

$$\left[NO_3^-\right] = 0.900M \times 3 = 2.70M$$

Answer: D

51. Electrolysis of dilute aqueous NaCl solution was carried out by passing 10milliampere current. The time required to liberate 0.01mol of H_2 gas at the cathode is

A. $9.65 \times 10^4 s$ C. $28.95 \times 10^4 s$
B. $19.3 \times 10^4 s$ D. $38.6 \times 10^4 s$

Solution: $2H^+ + 2e \longrightarrow H_2$

$2F = 1mol$

$? = 0.01mol$

2 Farad of electricity are required to produce 1mole of H_2

$2F = 2 \times 96500C = 193000C$

To produce 0.01mol of H_2, the quantity of electricity required, Q is

$$Q = \frac{0.01mol \times 193000C}{1mol}$$

$Q = 1930C$

$Q = It$

$I = 10mA = 10 \times 10^{-3}A = 1 \times 10^{-2}A$

$$t = \frac{Q}{I}$$

$$t = \frac{1930Q}{1 \times 10^{-2}A} = 1.93 \times 10^5 s$$

Answer: B

52. How many electrons will appear when the following half-reaction is balanced?

$$S_4O_6^{2-} \longrightarrow S_2^{2-}O_3$$

A. 3 B. 2 C. 4 D. 1

Explanation:
Balance the atoms first by inspection method.

$$S_4O_6^{2-} \longrightarrow 2S_2^{2-}O_3$$

Balance the charge by adding electrons to the deficient side

$$2e^- + S_4O_6^{2-} \longrightarrow 2S_2O_3^{2-}$$

Answer: B

53. PKa values of three acids X, Y and Z are 4.5, 3.5 and 6.5 respectively. Which of the following represents the correct order of acid strength?

A. X > Y > Z C. Y > X > Z
B. Z > X > Y D. Z > Y > X

Explanation: PKa and strength of acid are inversely related.

Answer: C

54. Standard electrode potential for Sn^{4+}/Sn^{2+} couple is +0.15V and that of the Cr^{3+}/Cr couple is -0.74V. These two couples in their standard state are connected to make a spontaneous electrochemical reaction. The cell potential will be

A. +1.83V C. +0.89V
B. +1.19V D. +0.18V

Solution: For spontaneous reaction, the more positive reduction potential used as cathode and the other as anode

$$E^\circ_{cell} = E^\circ_{(cathode)} - E^\circ_{(anode)}$$
$$= 0.15V - (-0.74V) = +0.89V$$

Answer: C

Telegram: https://t.me/ExtremeSeries

55. The enthalpies of formation of gaseous N_2O and NO at 298K are 82 and 90KJ/mol, respectively. The enthalpy change for the reaction

$$N_2O(g) + \frac{1}{2}O_2(g) \longrightarrow 2NO(g)$$

A. −8KJ C. −74KJ
B. 98KJ D. 8KJ

Solution:

$$\Delta H° = \sum n\Delta H^o_f \text{ products} - \sum m\Delta H^o_f \text{ reactants}$$

$$\Delta H° = 2\times\Delta H^o_f NO - 1\times\Delta H^o_f N_2O$$

$$\Delta H° = 2\times 90 - 1\times 82$$

$$\Delta H° = 98KJ$$

Answer: B

56. The following data was collected at the end point of a titration performed to find the molarity of an HCl solution

Volume of acid (HCL) used = 14.4mL
Volume of base (NaOH) used = 22.4mL
Molarity of standard base (NaOH) = 0.20M

On the basis of the above data, what is the molarity of the acid solution?

A. 1.6M C. 0.31M
B. 0.64M D. 0.13M

Solution: $NaOH + HCl \longrightarrow NaCl + H_2O$

$$n_{NaOH} = M_{NaoH} \times V_{NaOH} = 22.4mL \times 0.20\frac{mol}{L}$$

$$n_{NaOH} = 4.48m\ mol$$

$$\overset{4.48m\ mol}{\underset{1}{NaOH}} + \overset{x}{\underset{1}{HCl}} \longrightarrow NaCl + H_2O$$

$$\frac{4.48m\ mol}{1} = \frac{x}{1}$$

$$\Rightarrow n_{HCl} = 4.48m\ mol$$

$$M_{HCl} = \frac{n_{HCl}}{V_{HCl}} = \frac{4.48m\ mol}{14.4mL} = 0.31M$$

Answer: C

57. The hybridization of the centeral atom in the XeF_4 molecule is

A. sp^2 B. sp^3 C. sp^3d D. sp^3d^2

Explanation: Draw the Lewis structure and the number of sets of electrons is equal to the number of hybrid orbitals

There are eight sets of electrons on Xe. Hybridization: sp^3d^2

Answer: D

58. For the reaction

$$N_2 + 3H_2 \longrightarrow 2NH_3$$

The rate of disappearance of H_2 is $0.01molL^{-1}.min^{-1}$. The rate of appearance of NH_3 would be:

A. $0.01\ molL^{-1}\ min^{-1}$
B. $0.02\ molL^{-1}\ min^{-1}$
C. $0.007\ molL^{-1}\ min^{-1}$
D. $0.02\ molL^{-1}\ min^{-1}$

Solution: $r_{NH_3} = \frac{coeff.NH_3}{coeff.H_2} \times r_{H_2}$

$$r_{NH_3} = \frac{2}{3} \times 0.01molL^{-1}min^{-1}$$

$$r_{NH_3} = 0.0067molL^{-1}min^{-1}$$

Answer: C

59. Bakelite is obtained from phenol by reacting with

A. HCHO C. CH_3CHO
B. $(CH_2OH)_2$ D. CH_3COCH_3

Explanation:
Bakelite is formed from phenol and formaldehyde. **Answer: A**

60. In which one of the following numbers are all the zeros significant?

A. 100.090090 C. 0.1000
B. 0.143290 D. 00.0030020

Explanation: Zeros before non-zero digits are not significant, however, after and between non-zero digits they are significant

Answer: A

61. An aqueous solution contains 0.100M NaOH at 25°C. the P^H of the solution is:

A. 0.100 C. 7.00
B. 1.00 D. 13.0

Solution: $[OH^-] = [NaOH] = 0.10M = 10^{-1}M$

$$P^H = 14 + 10g[OH^-]$$

$$P^H = 14 + 10g10^{-1} = 13$$

Answer: D

Telegram: https://t.me/ExtremeSeries

62. The solubility of oxygen gas in water at 25°C and 1.0atm pressure of oxygen is 0.04g/L. the solubility of oxygen in water at 3.0atm and 25°C is _____ g/L

A. 0.014 B. 0.31 C. 0.041 D. 0.123

Solution: $C_{O2} = K_H \cdot P_{O2}$ KH - Henry's constant

$$K_H = \frac{C_{O2}}{P_{O2}} = \frac{0.041 g/L}{1 atm} = 0.041 g/L$$

When $P_{O2} = 3atm$, $C_{O2} = ?$

$$C_{O2} = K_{H(O2)} \cdot P_{O2} = 0.041 \frac{g}{L.atm} \times 3atm = 0.123\ g/L$$

Answer: D

63. The decomposition of carbondisulfide, CS_2, to carbon monosulfide, CS, and sulphur is first order with $K = 2.8 \times 10^{-7} S^{-1}$ at 1000°C. What is the half-life of the reaction below at 1000°C?

A. $5.0 \times 10^7 s$ C. $3.8 \times 10^5 s$

B. $4.7 \times 10^{-6} s$ D. $2.5 \times 10^6 s$

Solution: $t_{1/2} = \frac{0.693}{K}$

$$t_{1/2} = \frac{0.693}{2.8 \times 10^{-7} S^{-1}} = 2.475 \times 10^6 s$$

Answer: D

64. The PKa of a weak monoprotic acid is 4.8. What should be the ratio of [Acid]/[salt] of a buffer, if PH = 5.8 is required.

A. 0.1 B. 1.0 C. 2.0 D. 10

Solution: $PH = PKa + \log \frac{[salt]}{[Acid]}$

$$\log \frac{[salt]}{[Acid]} = PH - PKa = 5.8 - 4.8 = 1$$

$$\frac{[salt]}{[Acid]} = 10$$

$$\frac{[Acid]}{[Salt]} = \frac{1}{10} = 0.1$$

Answer: A

65. The dissolution of water in octane (C_8H_{18}) is prevented by

A. dipole – dipole attraction between octane molecules

B. hydrogen bonding between water molecules

C. London dispersion forces between octane molecules.

D. Repulsion between like-charged water and octane molecules

Explanation: water is polar and octane is non-polar. As a result, the two do not mix due to the like dissolves like principle. The weak London force in octane prevents the mixing of water in it.

Answer: C

66. At 445°C, K_c for the following reaction is 0.020.

$$2HI(g) \rightleftharpoons H_2(g) + I_2(g)$$

A mixture of H_2, I_2 and HI in a vessels at 445°C has the following concentrations: $[HI] = 2.0M$; $[H_2] = 0.50M$ and $[I_2] = 0.10M$. Which of the following statements concerning the reaction quotient, Q_c, is true for the above system?

A. Q_c, is less than K_c; more HI will be produced

B. Q_c, is greater than K_c; more HI will be produced

C. Q_c, is less than K_c; more H_2 and I_2 will be produced

D. Q_c, is greater than K_c; more H_2 and I_2 will be produced

Solution: $Q_c = \frac{[H_2][I_2]}{[HI]^2}$

$$Q_c = \frac{(0.50)(0.10)}{(2.0)^2} = 0.0125$$

$\Rightarrow Q_c < K_c$, formation of products is favored

Answer: C

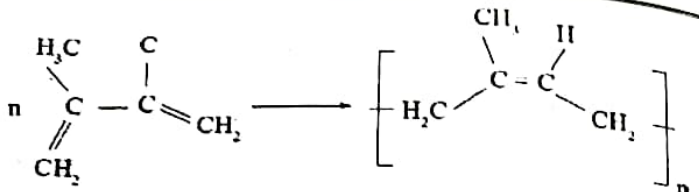
67. Natural rubber is a polymer of

A. butadiene C. neoprene

B. isoprene D. styrene

Explanation: Natural rubber, obtained from the milky sap or latex of a rubber tree, is a polymer of 2-methyl -1, 3-butadiene which has a common name **isoprene.**

Telegram: https://t.me/ExtremeSeries



Isoprene natural rubber

Answer: B

68. Which of the following statements is true?

A. All forms of electromagnetic radiation are visible.
B. Radio waves have shorter wavelengths than visible light.
C. Ultraviolet light has longer wavelengths than visible light
D. The frequency of radiation increases as the wavelengths decreases

Explanation: Natural rubber, obtained from the milky sap or latex of a rubber tree, is a polymer of 2-methyl -1, 3-butadiene which has a common name isoprene.

Frav, x-ray, UV, Visible, IR, Microwaves, Radio waves

- Wave length increase
- Frequency and energy decreases

Answer: D

69. If the enthalpy change for a certain reaction A → B is -2kJ at 300K. What would be the entropy change in the surroundings?

A. -40J/K C. -3.6×10^6J/K
B. 40J/K D. 3.6×10^6J/K

Solution: $\Delta G = \Delta H - T\Delta S$

Dividing by T gives all values pertaining to the system.

$$\frac{\Delta G}{T} = \frac{\Delta H}{T} - \Delta S$$

It can be shown that

$$\boxed{\Delta S_{sur} = -\frac{\Delta H_{sys}}{T}} \text{ (at constant P and T)}$$

$$\frac{\Delta G_{sys}}{T} = \Delta S_{sur} - \Delta S_{sys}$$

$$-\frac{\Delta G_{sys}}{T} = \Delta S_{sur} + \Delta S_{sys}$$

$\Delta S_{unit} = -\frac{\Delta G_{sys}}{T}$ because

$\Delta S_{unit} = \Delta S_{sur} + \Delta S_{sys}$

ΔS_{unit} - change in entropy of the universe

ΔS_{sur} - change in entropy of the surrounding

ΔS_{sys} - change in entropy of the system

ΔG - Gibb's free energy change

T- Kelvin temperature

$$\Delta S_{sur} = -\frac{\Delta H_{sys}}{T} = -\frac{(-2\times10^3 J)}{300K} = 1.7\frac{J}{K}$$

70. During the electrolysis of an aqueous solution of copper sulphate using platinum electrodes, the reaction takes place at the anode is?

A. $Cu^{2+} + 2e^- \longrightarrow Cu$
B. $Cu \longrightarrow Cu^{2+} + 2e^-$
C. $2H_2O \longrightarrow 4H^+ + O_2 + 4e^-$
D. $4H^+ + O_2 + 4e^- \longrightarrow H_2O$

Explanation:

→ Using platinum electrode:

Anode reaction: $2H_2O \longrightarrow 4H^+ + O_2 + 4e^-$

Cathode reaction: $Cu^{2-} + 2e^- \longrightarrow Cu$

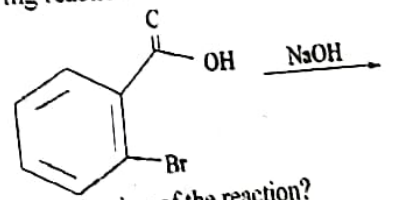
→ Using copper electrode

Anode reaction: $Cu \longrightarrow Cu^{2-} + 2e^-$

Cathode reaction: $Cu^{2+} + 2e^- \longrightarrow Cu$

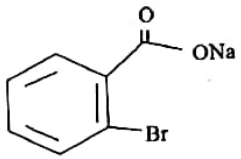
Answer: C

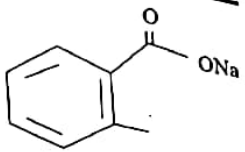
71. Given the following reaction

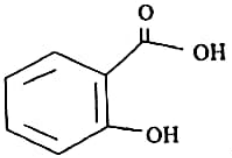


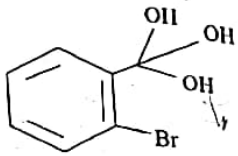
What is the major production of the reaction?

Telegram: https://t.me/ExtremeSeries

A. 

C. 

B. 

D. 

Answer: A

72. Which of the following is an acceptable IUPAC name for the organic compound shown below?

A. Octanoic acid
B. 7-methyl heptanoic acid
C. 8-methyl heptanoic acid
D. 7-methyl octanoic acid

Explanation:

Each angle represents carbon atom

—— methyl

ethyl

- The parent name is octanoic acid
- The IUPAC name of the compound is 7-methyl octanoic acid.

Answer: D

Which of the following molecules has a dipole moment?

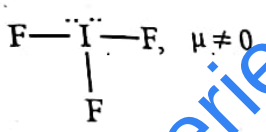
A. XeF_2 C. BF_3

B. IF_3 D. SF_5^+

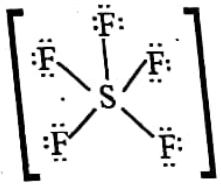
Explanation:

- Polar molecules have dipole moment different different from zero.
- Molecules whose molecular shape is linear, trigonal pyramid, tetrahedral, trigonal bipyramid, octahedral and square planar have zero dipolemoment.

XeF_2, F—Xe—F, $\mu = 0$

IF_3  $\mu \neq 0$

BF_3 F, $\mu = 0$

SF_3^+  , $\mu = 0$

Answer: B

74. Which of the following describes the balanced molecular equation when perchloric acid is mixed with solid iron (III) hydroxide?

A. $HClO_4(aq) + Fe(OH)_3(s) \rightarrow H_2O(l) + FeClO_4(aq)$
B. $3HClO_4(aq) + Fe(OH)_3(s) \rightarrow 3H_2O(l) + Fe(ClO_4)_3(aq)$
C. $3HOClO_2(aq) + Fe(OH)_3(s) \rightarrow 3H_2O(l) + Fe(ClO_3)_3(aq)$
D. $HOClO_2(aq) + Fe(OH)_3(s) \rightarrow H_2O(l) + Fe(ClO_3)(aq)$

Answer: B

75. At 298K the following two gaseous equilibria involving SO_2, SO_3 and O_2 are established.

$$SO_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons SO_3(g), K_1$$

$$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g), K_2$$

Which of the following equilibrium expression is correct?

A. $K_1 = K_2$ C. $K_2 = \frac{1}{K_1^2}$

B. $K_2 = K_1$ D. $K_2 = \frac{1}{K}$

Telegram: https://t.me/ExtremeSeries

Solution: The second reaction is obtained by reversing and multiplying by 2.

$\Rightarrow K_1 = \frac{1}{K_2^{\,2}}$ or $K_2 = \frac{1}{K^1}$

Answer: C

76. A compound is formed by the combination of V and X as follows.

V W X Y Z

What is the empirical formula for the compound?

A. VX C. V_3X

B. V_2X D. VX_2

Explanation: Valence number of V = 1

Valence number of X = 8 – 5 = 3

Formula:

$V^1 \times X^3 = V_3X$

Answer: C

77. ΔH for solid to liquid transitions for compound A is 2.73K cal/mol and for compound B is 3.0K cal /mol. The melting point for compound A is 0°C and the melting point for compound B is 30°C. The entropy changes ΔS_A and ΔS_B at the two transition temperatures are related as:

A. $\Delta S_A = \Delta S_B$ C. $\Delta S_B < \Delta S_A$

B. $\Delta S_A < \Delta S_B$ D. $\Delta S_A = \frac{300\Delta S_A}{273}$

Answer: C

78. Which of the following metals forms a volatile compound that is taken as an advantage for its extraction?

A. Co B. Fe C. Ni D. W

Answer: C

79. A 0.1M solution of HCl is dissolved in water. What species of ions are present at equilibrium and what will be their equilibrium concentrations?

A. $[H_3O^+] = 0.1M$ $[OH^-] = 10^{-13}M$ $[Cl^-] = 0.1M$

B. $[H_3O^+] = 0.1M$ $[OH^-] = 0.1M$ $[Cl^-] = 0.1M$

C. $[H_3O^+] = 0.1M$ $[OH^-] = 10^{-13}M$ $[Cl^-] = 0M$

D. $[H_3O^+] = 10^{-13}M$ $[Cl^-] = 0.1M$ $[OH^-] = 0.1M$

Explanation:

For strong monoprotic acid such as HCl

$[H_3O^+] = [Cl^-] = [HCl] = 0.1M$

$[OH^-] = \frac{K_w}{[H_3O^+]}$

$[OH^-] = \frac{1\times10^{-14}}{0.1} = 10^{-13}M$

Answer: A

80. The following bases and their conjugate acids (as the chlorides) are available in the laboratory:

NH_3, $K_b = 1.8 \times 10^{-5}$

C_6H_5N, $K_b = 1.7 \times 10^{-9}$

$CH_3CH_2NH_2$, $K_b = 6.4 \times 10^{-4}$

CH_3NH_2, $K_b = 4.4 \times 10^{-4}$

Which of these acid-base pairs are best to prepare a buffer solution having a PH of about 9?

A. NH_3 + conjugate acid

B. C_6H_5N + conjugate acid

C. $CH_3CH_2NH_2$ + conjugate acid

D. CH_3NH_2 + conjugate acid

Explanation:

Best buffer can be prepared from weakest base and its conjugate acid.

Answer: B

Telegram: https://t.me/Extreme_series

1. Which of the following is correct when 34495 is rounded to three significant figures?
A. 345 B. 34500 C. 344 D. 3840

Explanation:
When 34495 is rounded to three significance, the result will be 3.45×10^4 or 34500

Answer: B

2. What is the first step of the scientific method?
A. Making observations
B. Forming a hypothesis
C. Performing an experiment
D. Predicting the result of an experiment.

Answer: A

3. Which of the following are **NOT** electromagnetic waves?
A. Infrared waves C. Radio waves
B. Gamma waves D. Sound waves

Explanation:
Gamma rays, x-ray, ultraviolet ray, visible, Infrared, microwaves, Radio waves and Television waves are electromagnetic waves.

Answer: D

4. What is the distance that a radio wave will travel in 0.250s?
A. 1.2×10^7m C. 7.5×10^7m
B. 12×10^7m D. 5.6×10^7m

Solution: $C = \lambda v = \frac{\lambda}{T}$ because $v = \frac{1}{T}$

$\lambda = CT = 3 \times 10^8 \text{m/s} \times 0.25\text{s} = 7.5 \times 10^7\text{m}$

T – period

Answer: C

5. Which one of the following is **NOT** a form of chemical bonding?
A. Covalent bonding C. Ionic bonding
B. Metallic bonding D. Hydrogen bonding

Explanation:
- Ionic, covalent and metallic bonds are the common types of bonds. Hydrogen bond, dipole-dipole force, London force and ion-dipole force are intermolecular forces.

Answer: D

6. Which of the following statements is **NOT** true about covalent bonding?
A. Covalent bonds are least likely to be formed between atoms of the same element.
B. Covalent bonds are least likely to be formed between atoms of different elements on the right side of the periodic table.
C. Covalent bonds are least likely to be formed between an element in group I and an element in Group VII.
D. Covalent bonds are least likely to be formed by head of the group elements with height ionization energies.

Explanation:
- Covalent bond is formed between non-metals, and between less reactive metals and non-metals.
- It is not formed between alkali metals and halogens.

Answer: A

7. If we increase the concentration of a reactant, what happens to the collisions between particles?
A. There are more collisions.
B. There are fewer collisions.
C. There are the same number of collisions, but they have less energy.
D. There are the same number of collisions, but they have more energy.

Explanation: Concentration of reactants and number of collisions are directly related.

Answer: A

8. A drug decomposes by zero-order kinetics with a rate constant of $2\text{mg}.\text{mL}^{-1}$ month^{-1}. If the initial concentration is 100mg mL^{-1}, how long will it take for the drug to decompose by 10%?
A. 2 months C. 5 months
B. 3 months D. 4 months

Solution: $[A]_t - [A]_0 = -kt$

$[A]_0 - [A]_t = kt$ but $\frac{[A]_t}{[A]_0} = 90\% = \frac{90}{100} = \frac{9}{10}$

$[A]_t = \frac{9}{10}[A]_0^9$

$[A]_0 - \frac{9}{10}[A]_0 = kt$

Telegram: https://t.me/Extremeseries

$$\frac{1}{10}[A]_o = kt$$

$$t = \frac{[A]_o}{10k} = \frac{100 mg/mL}{10 \times 2 mg\ mL^{-1}.month^{-1}}$$

$t = 5$ months

Answer: C

9. The value of K_{eq} for the following reaction is 0.25.

$SO_2(g) + NO_2(g) \rightleftharpoons SO_3(g) + NO(g)$

What is the value of K_{eq} at the same temperature for the reaction below: $2SO_2(g) + 2NO_2(g) \rightleftharpoons 2SO_3(g) + 2NO(g)$?

A. 0.25 B. 0.062 C. 0.50 D. 16

Solution: If the given reversible reaction is multiplied by 2, the neal equilibrium constant is the original the power of 2.

$\Rightarrow K_{eq} = (0.25)^2 = 0.0625$

Answer: B

10. Which of the following would react to form pentylethanoate?

A. 1-propanol and pentanoic acid
B. Ethanol and pentanoic acid
C. 1-pentanol and ethanoic acid
D. Ethanol and ethanoic acid

Solution:

$$CH_3CH_2CH_2CH_2CH_2OH + CH_3COOH \xrightleftharpoons{H^+}$$

1-pentanol Ethanoic acid

$CH_3CH_2CH_2CH_2CH_2OOC\ CH_3^+H_2O$

pentylethanoate

Answer: C

11. The difference between fats and oils is that:

A. oils are liquid at room temperature.
B. oils have more calories.
C. oils are solid at room temperature.
D. fats are liquid at room temperature.

Explanation:

Fats are solid while oils are liquid at room temperature.

Answer: A

12. What kind of solution forms when gasoline evaporates in air?

A. gas in gas solution C. Liquid in liquid solution
B. gas in liquid solution D. Liquid in gas solution

Explanation: When gasoline evaporates it turns into gas. Air is gas. Thus, the solution is gas-gas solution.

Answer: A

13. What is the solvent in 70% alcohol solution?

A. Water B. Alcohol C. Sugar D. Kerosene

Explanation: 70% alcohol solution means 70% alcohol and 30% water. Water is universal polar solvent.

Answer: A

14. How many moles of H_2SO_4 are needed to prepare 5.0 liters of a 2.0M solution of H_2SO_4?

A. 2.5 B. 5.0 C. 20 D. 10

Solution: $n = M \times u = 2\frac{mol}{L} \times 5L = 10 moles$ **Answer: D**

15. What is the balanced **net ionic equation** for the reaction of $CaCl_2$ and $AgNO_3$?

A. $CaCl_2(aq) + 2\ AgNO_3(aq) \longrightarrow Ca(NO_3)_2(aq) + 2AgCl(s)$
B. $Ca^{2+}(aq) + 2Cl^-(aq) + 2Ag^+\ 2NO_3^-(aq) \longrightarrow Ca^{2+}(aq) + 2NO_3^-(aq) + 2AgCl(s)$
C. $Cl^-(aq) + Ag^+(aq) \longrightarrow 2AgCl(s)$
D. $2Cl^-(aq) + Ag^+(aq) \longrightarrow 2AgCl(s)$

Solution:

→ $CaCl_2(aq) + 2AgNO_3(aq) \longrightarrow 2AgCl(s) + Ca(NO_3)_2(aq)$ molecular equation

→ $Ca^{2+}(aq) + 2Cl^-(aq) + 2Ag^+(aq) + 2NO_3^-(aq) = 2AgCl(s) + Ca^{2+}(aq) + 2NO_3^-(aq)$ over all ionic equation

$= 2AgCl(s) + Ca^{2+}(aq) + 2NO_3^-(aq)$

→ $2Ag^+(aq) + 2Cl^-(aq) \longrightarrow 2AgCl(s)$, net ionic equation

Ca^{2+} and NO_3^- are spectator ions.

Answer: D

Telegram: https://t.me/Extremeseries

16. The Ka of hypochlorous acid (HClO) is 3.0×10^{-8} at 25°C. What is the % ionization of hypochlorous acid in a 0.015M aquous solution of HClO at 25°C.

A. 2.1×10^{-5} C. 1.4×10^{-2}
B. 0.14 D. 3.3×10^{-1}

Solution:

	$HClO \rightleftharpoons$	H^+	$+ OCl^-$
initial	0.015	0	0
change	-x	+x	+x
equilibrium	0.015 − x	x	x

$$Ka = \frac{[H^+][OCl^-]}{[HClO]}$$

$$3\times10^{-8} = \frac{x \cdot x}{0.015 - x} \quad \text{approximation: } 0.015 - x = 0.015$$

$$x^2 = 0.015 \times 3 \times 10^{-8}$$

$$x^2 = \sqrt{4.5\times10^{-10}} = \sqrt{4.5}\times10^{-5}M = [H^+] \Rightarrow [H^+] = 2.12\times10^{-5}M$$

$$\%HClO = \frac{[H^+]}{[HClO]}\times100 = \frac{2.12\times10^{-5}}{0.015}\times100 = 0.141\%$$ **Answer: B**

17. Which one of the following combinations **CANNOT** produce a buffer solution?

A. $HClO_4$ and $NaClO_4$ C. HNO_2 and $NaNO_2$
B. HCN and NaCN D. NH_3 and $(NH_4)_2SO_4$

Explanation: Buffer solution is a combination of weak acid and its conjugate base or weak base and its conjugate acid.

Answer: A

18. How many moles of NH_4Cl must be added to 1.5L of 0.2M solution of NH_3 to form a buffer whose pH is 9.00 ($Kb = 1.8 \times 10^{-5}$)?

A. 0.36 B. 0.65 C. 0.54 D. 0.8

Solution: $pH = 14 - pOH \Rightarrow pOH = 14 - 9 = 5$

$$pOH = pKb + 10g\frac{[NH_4Cl]}{[NH_3]}$$

$$pOH = -10gKb + 10g\frac{[NH_4Cl]}{[NH_3]}$$

$$5 = -10g\ 1.8\times10^{-5} + 10g\frac{[NH_4Cl]}{[NH_3]}$$

$$5 - 4.74 = 10g\frac{[NH_4Cl]}{0.2}$$

$$0.26 = 10g\frac{[NH_4Cl]}{0.2}$$

$$[NH_4Cl] = 10^{0.26} \times 0.2M = 0.36M$$

$$n_{NH_4Cl} = M \times V = 0.36M \times 1.5L = 0.54mol$$ **Answer: C**

19. What mass of magnesium is plated out upon electrolysis from molten $MgCl_2$ using a current of 60 A for a period of 4000 seconds?

A. 30g B. 24g C. 60g D. 72g

Solution: $Mg^2 + 2e^- \longrightarrow Mg$

$$M = \frac{MIt}{ZF} = \frac{24g/mol \times 60A \times 4000s}{2\times96485A.s/mol.} = 29.85g$$ **Answer: A**

20. For a voltaic (or galvanic) cell using Ag, Ag+(1.0M) and Zn, Zn^{2+} (1.0M) half-cells, which of the following staatements is **INCORRECT?**

A. The zinc electrode is the anode.
B. Electrons will flow through the external circuit from the zinc electrode to the silver electrode.
C. The mass of the zinc electrode will decrease as the cell operates.
D. Reduction occurs at the zinc electrode as the cell operates.

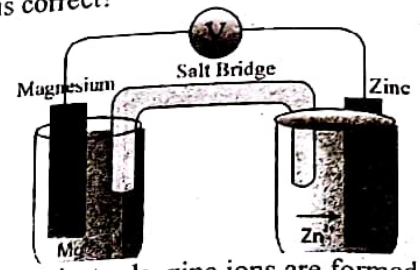
Explanation:

- More reactive metal is anode (Zn)
- Less reactive metal becomes cathode (Ag)
- Electrons flow from anode to cathode in the external circuit.
- reduction occurs at cathode and oxidations occurs at anode.
- mass of anode decreases and that of cathode increases.

Answer: D

Telegram: https://t.me/ExtremeSeries

21. For the galvanic cell shown below, which one of the following statements is correct?



A. At the zinc electrode, zinc ions are formed.
B. The electrode potential is measured by the voltmeter.
C. The following reaction takes place at the magnesium electrode: $Mg^2 + 2e^- \longrightarrow Mg$
D. Electrons flow from the zinc electrode to the magnesium electrode.

Explanation: Mg – anode, Zn – cathode

Anode reaction: $Mg \longrightarrow Mg^{2+} + 2e^-$

Cathode reaction: $Zn^{2+} + 2e^- \longrightarrow Zn$

Answer: B

22. Which is the most common ore used for the extraction of copper?
A. CuO B. $CuSO_4$ C. $CuCO_3$ D. $CuFeS_2$

Solution: The principal ore of copper is chalcopyrite, $CuFeS_2$

Answer: D

23. Which of the following material has maximum ductility?
A. Nickel B. Aluminum C. Mild steel D. Copper

Answer: D

24. Which of the following statements about polyvinyl chloride is **NOT** correct?
A. PVC can be used in making water pipes.
B. PVC is stiff.
C. PVC is softened on heating.
D. The monomer of PVC is CHCl = CHCl.

Explanation: The monor of PVC is vinyl chloride, $CHCl = CH_2$

Answer: D

25. Which of the following is a natural polymer?
A. Kertain C. Cellulose
B. Polythene D. Polymethyl methacrylate

Answer: A, C

26. Which of the following does **NOT** affect the solubility of a gas dissolved in a liquid?
A. Nature of solute and solvent
B. pressure
C. temperature
D. rate at which the gas dissolves

Answer: D

27. When a small amount of crystal solute is added to the supersaturated solution, the solute crystals will:
A. grow bigger. C. remain unchanged.
B. slightly dissolve. D. dissolve completely.

Explanation: Super saturated solution develops a crystal easily.

Answer: A

28. What is the molality of a solution that contains 51.2g of naphthalene, $C_{10}H_8$, in 500mL of carbon tetrachloride? The density of CCl_4 is 1.60g/mL.
A. 0.750m B. 0.500m C. 0.840m D. 1.69m

Explanation:

$m_{solution} = d \times v = 1.60g/mol \times 500mL = 800g$

$m_{solvent} = m_{solution} - m_{solvent} = 800 - 51.2 = 748.8g = 0.7488kg$

$$n = \frac{m}{M} = \frac{51.2g}{128g/mol} = 0.4mol$$

$$m = \frac{n}{Kg\ solvent} = \frac{0.4mol}{0.7488kg} = 0.534m$$

Answer: B

29. What is the ionization constant for a weak acid, HA, that is 1.60% ionized in 0.0950M solution?
A. 2.69×10^{-3} C. 2.47×10^{-5}
B. 3.77×10^{-2} D. 9.91×10^{-6}

Explanation: $\%HA = \frac{[H^+]}{[HA]_o} \times 100 \Rightarrow [H^+] = \frac{1.60 \times 0.095M}{100}$

$[H^+] = 1.52 \times 10^{-3}M$

At equilibrium, $[H^+] = [A^-] = 1.52 \times 10^{-3}M$

Telegram: https://t.me/ExtremeSeries

$$Ka = \frac{[H^+][A^-]}{[HA]-[H^+]} = \frac{(1.5\times10^{-3})(1.52\times10^{-3})}{0.095-0.00152}$$

$$Ka = \frac{2.3104\times10^{-6}}{0.09348} = 2.47\times10^{-5}$$

Answer: C

30. Which of the following is correct?
A. $1Pa = 10Nm^{-2}$
B. $1N = 10kg\ m\ s^{-2}$
C. $0.00072 = 7.2 \times 10^{-3}$
D. $1L = 1dm^3$

Answer: D

31. Which of the following represents a tentative explanation of a certain scientific law?
A. Hypothesis
B. Observation
C. Experimentation
D. Theory

Answer: D

32. In order to advance to the level of a theory, a hypothesis should
A. be obviously accepted by most people.
B. be repeatedly confirmed by experimentation.
C. be a fully functional experiment
D. report the past experience.

Answer: B

33. Which of the following types of rays combine to form atoms of helium?
A. gamma (γ) rays
B. beta (β) rays
C. alpha (α) rays
D. x-rays

Explanation: Alpha particle is known to be a doublyionized helium atom.

Answer: C

34. What is the relationship between frequency (ν), wavelength (λ) and the speed of light (c)?

A. $\frac{\lambda}{\nu} = c$ B. $\nu c = h\lambda$ C. $\frac{hc}{\lambda} = \nu$ D. $c = \nu\lambda$

Solution: $c = \nu\lambda$

Answer: D

35. What is the magnitude of quantum energy and the frequency for an object whose wavelength is 0.6×10^{-6}m?
A. 3.31×10^{-19}J, $5 \times 10^{14}s^{-1}$
B. 3.98×10^{-40}J, $2 \times 10^{15}s^{-1}$
C. 1.99×10^{-25}J, $3.98 \times 10^{-40}s^{-1}$
D. 9.94×10^{-12}J, $1.99 \times 10^{-25}s^{-1}$

Solution: $E = \frac{hc}{\lambda}$ because $\nu = \frac{c}{\lambda}$

$$\nu = \frac{3\times10^8 m/s}{6\times10^{-7}m} = 5.0\times10^{14}s^{-1}$$

$E = 6.626\times10^{-34}J.s \times 5.0\times10^{14}s^{-1} = 3.313\times10^{-19}J$ **Answer: A**

36. What new concept did Bohr adapt and use to formulate his modelof the atom?
A. Electromagnetic theory developed by Maxwell.
B. The quantum concept developed by Planck.
C. Photoelectric theory developed by Thompson.
D. Neutron theory developed by Chadwick.

Explanation: Bohr made the revolutionary assumption that a certain properties of the electron in a hydrogen atom based on the work of plank and Einstein.

Answer: B

37. What values of l are permitted for an electron with n = 4?
A. 1, 2, 3
B. 1, 2, 3, 4
C. 0, 1, 2, 3, 4
D. 0, 1, 2, 3

Solution: The values of l are from 0 to n – 1
⇒ When n = 4, ℓ = 0, 1, 2, 3

Answer: D

38. Which of the following electrons, identified only by their n and ℓ quantum numbers have the highest energy?
n = 3, ℓ = 0
n = 4, ℓ = 1
n = 3, ℓ = 2
n = 4, ℓ = 2
A. n = 3, l = 2
B. n = 4, l = 1
C. n = 4, l = 2
D. n = 3; l = 0

Explanation: n = 3, ℓ = 0 ⇒ 3s
n = 4, ℓ = 1 ⇒ 4p
n = 3, ℓ = 2 ⇒ 3d
n = 4, ℓ = 2 ⇒ 4d
The energy of these orbitals in decreasing order is:
4d > 4p > 3d > 3s.

Answer: C

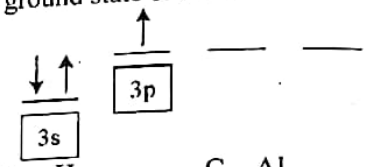
Telegram: https://t.me/ExtremeSeries

39. What is the maximum number of unpaired electrons in a d subshell?
A. 2 B. 5 C. 3 D. 4
Explanation: d – sub shell has five orbitals. Thus, it has a maximum of five unpaired electrons.

1	1	1	1	1

Answer: B

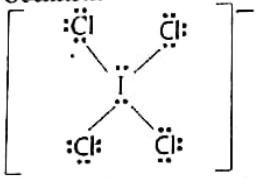
40. The following energy level diagram represents the outermost shell of what ground state element?



A. B B. He C. Al D. Be
Solution: $_5B : 1s^2 2s^2 2p^1$, $_4Be : 1s^2 2s^2$
$_2He : 1s^2$, $_{13}Al : 1s^2 2s^2 2p^6 3s^2 3p^1$

Answer: C

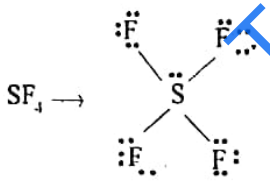
41. How many bonding pairs and lone pairs,, repsectively does the ion ICl_4^- have?
A. 3, 2 B. 4, 2 C. 5, 1 D. 4, 1
Solution:



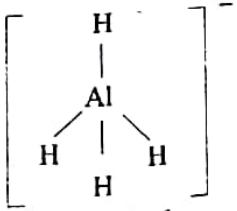
bonding pairs = 4
lone pairs on the central atom = 2
total lone pairs = 14

Answer: B

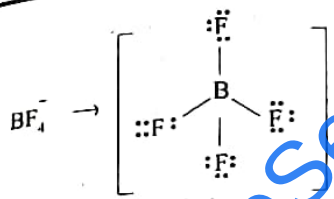
42. Which of the following molecules does **NOT** have a tetrahedral central atom?
A. SF_4 B. AlH_4^- C. BF_4^- D. $SiCl_4$
Solution:



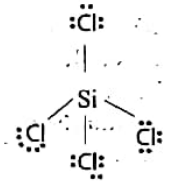
see-saw



tetrahedral



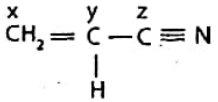
Tetrahedral



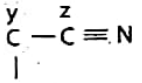
Tetraphedral

Answer: A

43. Acrylonitirile has the follwoig Lewis structure with designation of x, y and z for each carbon atom:



What will be the value of the bond angle and geometry of

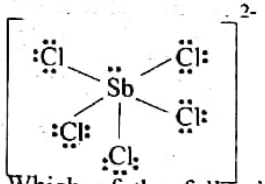


?

A. 109°, tetrahedral C. 180°, linear
B. 120°, trigonal pyramidal D. 90°, T-shaped

Answer: C

44. Antimony (Sb) is a group V element. What will be the molecular gometry and number of lone pair electrons, repsectively that exist in the ion $[SbCl_5]^{2-}$?
A. Seesaw, 1 C. Seesaw, 2
B. Square planar, 2 D. Linear, 3
Explanation:



Molecular geometry = squar pyramid
Lone pair = 1 on the central atom.

Answer:

45. Which of the following molecules does **NOT** have a trigonal bipyramidal electron-pair geometry?
A. SF_4 B. ClF_3 C. XeF_2 D. BrF_5
Explanation:
⇒ SF_4, ClF_3 and XeF_2 have five sets of electrons on their central atom but BrF_5 has six sets of electrons on the central atom. BrF_5 has octahedral electron arrangement. **Answer: D**

Telegram: https://t.me/ExtremeSeries

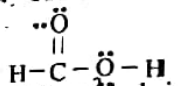
46. How many atomic orbitals are required for an sp^3 hybridization?
A. 2 B. 6 C. 4 D. 8

Solution: One s and three p-atomic orbitals are required for an sp^3 hybridization.

Answer: C

47. Formic acid, which is released by ants, has a molecular formula of HCOOH. What are the possible hybridizations that exist in the molecule?
A. sp^2 and sp^3 C. sp, sp^2 and sp^3
B. sp and sp^3 D. sp and sp^2

Solution: HCOOH has two central atoms, C and O

$\Rightarrow$ 

- C is sp^2 hybridized and 0 – is sp^3 hybridized.

Answer: A

48. What would happen to the O_2 molecule upon ionization to O_2^+?
A. The bond length will increase and the bond energy will increase.
B. The bond length will increase and the bond energy will decrease.
C. The bond length will decrease and the bond energy will increase.
D. The bond length will decrease and the bond energy will decrease.

Explanation:

The bond order of O_2 is 2 but O_2^+ is 2.5. This implies O_2 is less stable than O_2^+. When O_2 is change to O_2^+ bond energy increases but bond length decreases.

Answer: C

49. For a first-order reaction, a plot of _____ versus _____ is linear
A. $\frac{1}{[A]_t}, t$ C. $[A]_t, t$
B. $\text{Ln}\frac{1}{[A]_t}, t$ D. $\text{Ln}[A]_t, t$

Answer: D

50. The rate law of the overall reaction
$A + B \longrightarrow C$
is : rate = $k[A]^2$
Which of the following will NOT increase the rate of the reaction?
A. Increasing the concentration of reactant A.
B. Increasing the concentration of reactant B.
C. Increasing the temperature of the reaction.
D. Adding a catalyst for the reaction.

Explanation:

$A + B \longrightarrow C$, rate = $k[A]^2$ means the reaction is zero order with respect to reactant B. The rate of zero-order reaction is independent on concentration.

Answer: B

51. Which of the following statement (s) is (are) applicable to a balanced chemical equation of an elementary reaction?
i. Order is same as molecularity.
ii. Order is less than the molecularity.
iii. Order is greater than the molecularity.
iv. Molecularity can never be zero.
A. i B. i, ii C. i, iv D. i, iii

Explanation:

- For an elementary reaction order is the same as coefficient of a octant.
- Molecularity cannot to zero.

Answer: C

52. At high pressure, the following reaction is zero order.
$2NH_3(g) \xrightarrow{1130k.pt} N_2(g) + 3H_2(g)$
Which of the following options is (are) correct for this reaction?
i. Rate of reaction = rate constant.
ii. Rate of reaction depends on the concentration of ammonia.
iii. Rate of decomposition of ammonia remains constant until ammonia decomposes completely.
iv. further increase in pressure will change the rate of reaction.
A. i B. i, iii, iv C. i, ii D. i, ii, iv

Answer: B

Telegram: https://t.me/ExtremeSeries

53. Which of the following expressions is correct for the rate of reaction given below?

$$5Br^-(aq) + BrO_3^-(aq) + 6H^+(aq) \longrightarrow 3Br(aq) + 3H_2O(l)$$

A. $\frac{\Delta[Br]}{\Delta t} = 5\frac{\Delta[H]}{\Delta t}$ C. $\frac{\Delta[Br]}{\Delta t} = 6/5\frac{\Delta[H]}{\Delta t}$

B. $\frac{\Delta[Br]}{\Delta t} = 5/6\frac{\Delta[H]}{\Delta t}$ D. $\frac{\Delta[Br]}{\Delta t} = 6\frac{\Delta[H]}{\Delta t}$

Explanation:

$$-\frac{\Delta[Br^-]}{5\Delta t} = -\frac{\Delta[BrO_3^-]}{\Delta t} = -\frac{\Delta[H^+]}{6\Delta t} = \frac{\Delta[Br_2]}{3\Delta t} = \frac{\Delta[H_2O]}{3\Delta t}$$

$$\Rightarrow \frac{\Delta[Br^-]}{\Delta t} = -\frac{5}{6}\frac{\Delta[H^+]}{\Delta t}$$

Answer: B

54. Rate law for the reaction $A + 2B \longrightarrow C$ is found to be

$$Rate = k[A][B]$$

If the concentration of reactant 'B' is doubled, keeping the concentration of 'A' constant, what will be the value of the rate constant?

A. the same C. quadrupled

B. halved D. doubled

Explanation:

Rate constant is affected by temperature only. It is not affected by concentration.

Answer: A

55. The following equilibrium constants were determined at 300°C.

$$2N_2O(g) \rightleftharpoons 2N_2(g) + O_2(g), \quad K_c = 4.0\times10^{18}$$

$$N_2O + O_2(g) \rightleftharpoons 2NO(g), \quad K_c = 4.0\times10^{-31}$$

What will be the equilibrium constant at 300°C for the gaseous reaction of

$$2N_2O(g) + O_2(g) \rightleftharpoons 4NO(g)?$$

A. 3.2×10^{-12} C. 5.0×10^{50}

B. 2×10^{-13} D. 1.6×10^{-49}

Solution: Multiply the second by 2 and add the two reactions to get the required.

$$\Rightarrow 2N_2O(g) \rightleftharpoons 2N_2(g) + O_2(g) \qquad K_1 = 4.0\times10^{18}$$

$$2N_2(g) + 2O_2(g) \rightleftharpoons 4NO(g) \qquad K_2 = \left(4.0\times10^{-31}\right)^2$$

$$2N_2O(g) + O_2(g) \rightleftharpoons 4NO(g) \qquad K_C = K_1 \times K_2$$

$$K_C = 4.0\times10^{18}\left(4.0\times10^{-31}\right)^2$$

$$K_C = 6.4\times10^{-43}$$

56. When 0.50 mol of N_2O_4 is placed 4,0 liter reaction vessel and heated to 400K, 80% of the N_2O_4 decomposes to NO_2 gas as follows: $N_2O_4(g) \rightleftharpoons 2NO_2(g)$

What will be the value of K_p, in units of pressure, at 400K for this reaction?

A. 2.62 B. 13.12 C. 50.48 D. 16.20

Solution: $0.5mol \times \frac{80}{100} = 0.4mol$ of N_2O_4 decomposed

nN_2O_4 left $= 0.5mol - 0.4mol = 0.1mol$

$nNO_2 = 2\times0.4mol = 0.8mol$

$$[N_2O_4] = \frac{0.1mol}{4l} = 0.025M$$

$$[NO_2] = \frac{0.8mol}{4l} = 0.2M$$

$$K_C = \frac{[NO_2]^2}{[N_2O_4]} = \frac{(0.2)^2}{0.025} = 1.6mol/L$$

$$K_p = K_C(R_T)^{\Delta n}, \Delta n = 2-1 = 1$$

$$= 1.6\frac{mol}{L}\times\left(0.82\frac{atm.L}{mol.K}\times400k\right)^1 = 52.48atm$$

Answer: C

Telegram: https://t.me/ExtremeSeries

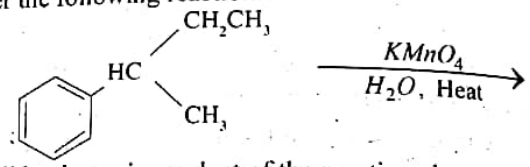
57. Which of the following is **NOT** true about carbonyl compounds?
A. Carbonyl compounds contain 3σ-bond and 1π-bond.
B. The carbon oxygen bond is both longer and weaker.
C. The bond angle in carbonyl is about 120°.
D. Carbonyl compounds may be hydrolyzed.

Explanation: Aldehyde and ketone are carbonyl compounds.

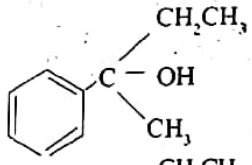
The formula of carbonyl carbon is $-\overset{\overset{O}{\|}}{C}-$. The carbonyl carbon is joined with 3σ and 1π. The carbon oxygen bond is short and strong.

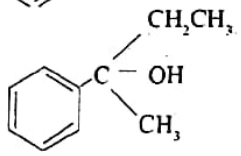
Answer: B

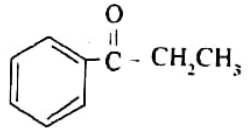
58. Consider the following reaction:

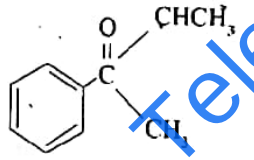


What will be the main product of the reaction above?

A.


B.


C.


D.


Answer: C

59. Which of the following statements concerning the carbonyl group in aldehydes and ketones is **NOT** true?
A. The bond is polar, with a slight negative charge on the oxygen atom.
B. The bond angles about the central carbon atom are 120°.
C. In condensed form, the carbonyl group can be written as CHO.
D. The bond is polar. Therefore, carbonyl groups readily form hydrogen bonds with each other.

Answer: D

60. Which of the following statements concerning fats and oils is **INCORRECT**?
A. They are also called triacylglycerols.
B. They are also called triglycerides.
C. They are fatty acid salts.
D. They are glycerol trimesters.

Explanation:
- Fats and oils are triglycerides, triacylglycerols or trimesters of glycerol.

Answer: C

61. Which of the following types of compounds are expected products from the saponifcation of a fat?
A. Glycerol and fatty acid salts.
B. Glycerol and fatty acids.
C. Fatty acid salts and fatty acids.
D. Glycerol, fatty acid salts and fatty acids.

Explanation: Saponfification of a fat results soap (fatty acid salt) and glycerol.

Answer: A

62. Which one of the following statements best describes the standard enthalpy of formation of any element?
A. The value of $\Delta H°_f$ (element) depends on temperature.
B. The value of $\Delta H°_f$ (element) is zero for any element in the standard state.
C. The value of $\Delta H°_f$ (element) is zero only for elements in the solid state.
D. The value of $\Delta H°_f$ (element) is zero only at absolute zero temperature.

Answer: B

Telegram: https://t.me/ExtremeSeries

63. Equal masses of He and Ne are placed in a sealed container. What is the partial pressure of Ne, if the total pressure is 6 atm?
A. 2 B. 3 C. 1 D. 5

Solution: Let the mass of He and Ne be m.

$$n_{He} = \frac{m}{4}$$

$$n_{Ne} = \frac{m}{20}$$

$$x_{He} = \frac{n_{He}}{n_{He} + n_{Ne}} = \frac{\frac{m}{4}}{\frac{m}{4} + \frac{m}{20}} = \frac{\frac{m}{4}}{\frac{6m}{20}} = \frac{20}{6m} \times \frac{m}{4} = \frac{5}{6}$$

$$x_{Ne} = 1 - \frac{5}{6} = \frac{1}{6}$$

$$p_{Ne} = x_{Ne} \cdot p_T = \frac{1}{6} \times 6\text{atm} = 1\text{atm}$$

Answer: C

64. What is the molarity of a solution made by dissolving 10g of glucose ($C_6H_{12}O_6$) in sufficient water to form 200mL solution?
A. 0.18 B. 0.251 C. 0.362 D. 0.278

Solution: $n = \frac{m}{M} = \frac{10g}{180g/mol} = \frac{1}{18}\text{mol}$

$$M = \frac{n}{v} = \frac{\frac{1}{18}\text{mol}}{0.20L} = 0.278M$$

Answer: D

65. In which direction will the following equilibrium shift, if a solution of CH_3CO_2Na is added?

$$CH_3COOH(aq) \rightleftharpoons CH_3CO_2^-(aq) + H^+(aq)$$

A. The equilibrium shifts to the right (more products)
B. No change
C. The equilibrium shifts to the left (more reactant)
D. Cannot be predicted

Explanation: CH_3COONa is a source of $CH_3CO_2^-$. As a result, concentration of $CH_3CO_2^-$ increase and the equilibrium will shift to the left.

Answer: C

66. A 1.0M solution of a weak acid is found to dissociate by only 1.3%. Which of the following acid is it likely to be?
A. HF $K_a = 7.2 \times 10^{-4}$
B. HFO_2 $K_a = 6.9 \times 10^{-4}$
C. CH_3COOH $K_a = 1.8 \times 10^{-5}$
D. HCOOH $K_a = 1.9 \times 10^{-4}$

Solution: Let the weak acid he HA

$$\%HA = \frac{[H^+]}{[HA]_o} \times 100\%$$

$$[H^+] = \frac{\%HF \cdot [HA]_o}{100} = \frac{1.37 \times 1M}{100} = 0.0137M$$

$$K_a = \frac{[H^+][A^-]}{[HA]_o - [H^+]}, \quad [H^+] = [A^-]$$

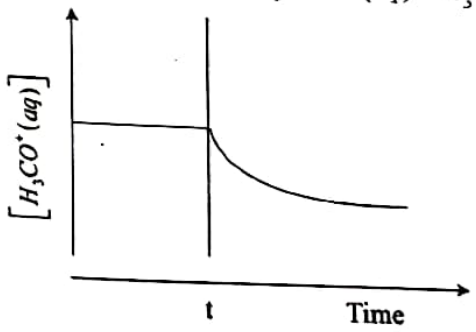
$$K_a = \frac{(0.0137)^2}{1 - 0.0137} = 1.9 \times 10^{-4}$$

Thus, it is the Ka value of HCOOH.

Answer: D

67. Consider the following graph, which relates to the equilibrium system:

$$CH_3COOH(aq) + H_2O \rightleftharpoons CH_3COO^-(aq) + H_3O^+(aq) \Delta H < 0$$



Telegram: https://t.me/ExtremeSeries

Which of the following actions caused the change in the concentration of $[H_3O^+(aq)]$ at time t?

A. Addition of $CH_3COO^-(aq)$
B. Addition of HCl
C. Decreasing of temperature
D. Increasing the volume of the container

Answer: A

68. Sodium acetate spontaneously crystallizes out of a supersaturated solution o n standing. Which of the following is true for the thermodynamic quantities of this system for such a process?

A. $\Delta S < 0, \Delta H < 0$
B. $\Delta G < 0, \Delta H > 0$
C. $\Delta S > 0, \Delta G < 0$
D. $\Delta S > 0, \Delta H > 0$

Explanation: When sodium acetate crystallizes out of a supersaturated solution, entropy decreases.

Answer: A

69. In which of the following systems will the position of equilibrium shift to the left upon an increase in pressure, but to the right upon an increase in temperature?

A. $CO_2(g)+H_2(g)\rightleftharpoons CO(g)+H_2O(g) \quad \Delta H>0$
B. $C_2H_2(g)+H_2O(g)\rightleftharpoons C_2H_5OH(g) \quad \Delta H<0$
C. $C_2H_6(g)\rightleftharpoons C_2H_4(g)+H_2(g) \quad \Delta H>0$
D. $2SO_2(g)+O_2(g)\rightleftharpoons 2SO_3(g) \quad \Delta H<0$

Answer: C

70. The hydrogen used in the Haber process is made by the following reaction:

$$CH_4(g)+H_2O(g)\rightleftharpoons CO(g)+3H_2(g) \quad \Delta H^\circ=+206kJ$$

Which of the following sets of conditions will favor the formation of H_2?

A. Low pressure and high temperature.
B. Low pressure and low temperature.
C. High pressure and low temperature.
D. High pressure and high temperature.

Explanation:
The equilibrium shifts to the right by lowering pressure and increasing temperature.

Answer: A

71. What is the molar solubility of $Fe(OH)_3$ in a solution that is buffered at pH = 3.50 at 25°C? ($K_{sp}(Fe(OH)_3 = 4\times10^{-38}$).

A. 1×10^{-5} B. 1.1×10^{-6} C. 2.0×10^{-6} D. 1.26×10^{-6}

Solution:

$POH = 14 - pH = 14 - 3.5 = 10.5, [OH^-] = 10^{-POH} = 10^{-10.5}$

$\Rightarrow [OH^-] = 3.16\times10^{-11}M$

$Fe(OH)_3 \longrightarrow Fe^{3+} + 3OH^-$

$K_{sp} = [Fe^{3+}][OH^-]^3 \Rightarrow [Fe^{3+}] = \dfrac{K_{SP}}{[OH^-]^3}$

$$[Fe^{3+}] = \frac{4\times10^{-38}}{(3.16\times10^{-11})^3} = 1.26\times10^{-6}\ mol/L$$

Answer: D

72. What is the purpose of a salt bridge in an electrochemical cell?

A. To provide a source of ions to react at the anode and cathode.
B. To maintain electrical neutrality in the half-cell through migration of ions.
C. To provide means of electrons to travel from the cathode to the anode.
D. to provide means of electrons to travel from the anode to the cathode.

Explanation:

- Salt bridge maintains electrical neutrality between the two solutions.
- It prevent mixing of the true half-cells solution.

Answer: B

73. What is galvanized iron?

A. Iron that is coated with tin.
B. Iron that is coated with zinc.
C. Iron that is coated with chromium.
D. Iron that is coated with aluminum.

Explanation:
Iron that is coated with zinc is called galvanized iron.

Answer: B

Telegram: https://t.me/ExtremeSeries

74. The standard cell potential (E°) for the reaction below is 1.10V. What is the cell potential for this reaction hen $[Cu^{2+}] = 1\times10^{-5}M$ and $[Zn^{2+}] = 1M$?

$Zn(s) + Cu^{2+}(aq) \longrightarrow Zn^{2+}(aq) + Cu(s)$

A. 1.10 B. 0.95 C. 1.20 D. 1.35

Solution:

$$E_{cell} = E°_{cell} - \frac{0.0592V}{n}\log Q$$

$$E_{cell} = 1.10v - \frac{0.0592V}{2}\log\frac{[zn^{2+}]}{[Cu^{2+}]}$$

$$E_{cell} = 1.10v - 0.0296\log\frac{1M}{10^{-5}M}$$

$$E_{cell} = 0.952V$$

Answer: B

75. Which of the following statements is true?

A. The more positive the value of $E_{red}°$, the greater the driving force for reduction.

B. The more exothermic the value of $E_{red}°$, the greater the driving force for reduction.

C. The more endothermic the value of $E_{red}°$, the greater the driving force for reduction.

D. The more negative the value of $E_{red}°$, the greater the driving force for reduction.

Explanation: The more positive the standard reduction potential, the easier to be reduced.

Answer: A

76. Which of the following types of elements are good oxidizing agents?

A. Alkali metals C. Lanthanides

B. Halogens D. Transition elements

Solution: Halogens (group (VIIA) non-metals are powerful oxidizing agents. **Answer: B**

77. Which of the following statements concerning petroleum is **INCORRECT?**

A. It is a renewable energy source.

B. It is a fossil fuel.

C. It is a mixture consisting mainly of hydrocarbons.

D. It was formed from marine organisms, which died millions of years ago. **Answer: A**

78. A 1.0×10^{-4}M solution has pH of 10.00. The solute is a

A. weak acid C. strong base

B. weak base D. strong acid

Solution:

$pH = 14 + \log[OH^-]$

$pH = 14 + \log 1.0\times10^{-4}$

$pH = 14 - 4 = 10$. The solute must be strong base because $[OH^-]$ = [base] for strong base. **Answer: C**

79. The four most abundant metals in the earth's crust in decreasing order of abundance are:

A. Oxygen, Silicon, Aluminum, and Iron.

B. Aluminum, Iron, Calcium, and Magnesium.

C. Iron, Aluminum, Silicon, and Oxygen.

D. Silicon, Aluminum, Magnesium, and Sodium.

Explanation: The four abundant metals in decreasing order of abundance are aluminum, iron, calcium and magnesium. However, from elements oxygen is the first. **Answer: B**

80. A polysaccharide is a polymer made up of which kind of monomers?

A. Amino acids

B. Nucleotides

C. Simple sugars

D. Alternating sugar and phosphate groups

Explanation: Monosaccharides (simple sugars) such as glucose, galactose and fructose are monomers of polysaccharides.

Answer: C

Telegram: https://t.me/ExtremeSeries

1. Which one of the following measurements has the greatest number of significant figures?
A. 0.00046 B. 0.00460 C. 460 D. 4.6×10^6

Solution:
- 0.0046 and 4.6×10^6 have two significant figures.
- 0.00460 has three significant figures.
- 460 is very difficult to determine its significant figure because no one is sure about the last zero.

Answer: B

2. How many significant figures has the sum of 2.453, 4.6 and 8.14?
A. 2 B. 3 C. 4 D. 5

Solution: 2.453 + 4.6 + 8.14 = 15.193 but the answer is 15.2 because during addition and subtraction of significant figures, the answer should be reported with the fewer decimal place.

Answer: B

3. Which of the following is the correct relationship between picometer (pm) and nanometer (nm)?
A. 1 pm = 10 nm C. 1 nm = 10 pm
B. 1 pm = 100 nm D. 1 nm = 10000 pm

Solution:

$$\frac{1pm}{1nm}=\frac{10^{-12}m}{10^{-9}m}$$

$$\frac{1pm}{1nm}=10^{-3}$$

$1pm = 10^{-3}nm \Rightarrow 10^{3}pm$

Answer: D

4. Which of the following are the correct steps of the Scientific method?
A. Observation – Hypothesis – Experimentation – Formulation of Laws – Theories
B. Observation – Formulation of Laws – Experimentation – Hypothesis – Theories
C. Observation – Experimentation – Hypothesis – Formulation of Laws – Theories
D. Observation – Experimentation – Formulation of Laws – Hypothesis – Theories

Solution:
- The correct steps of the scientific method are

  - Making observation
  - Hypothesis
  - Theories
  - Scientific laws

5. Which of the following is **NOT** a laboratory safety rule?
A. You should tie back loose hair.
B. You should never add acid to water.
C. Do not suck solution in the pipette by mouth.
D. When lighting a Bunsen Burner, you should light the match stick before turning on the gas.

Solution:
To dilute concentrated acid, we have to add acid to water but not water to acid.

Answer: B

6. Which of the following particle was discovered by the cathode rays experiment?
A. Electron C. Nucleus
B. Neutron D. Proton

Answer: A

7. What is the frequency of a radiation (in Hz) from an atom if its wavelength is 1.0×10^{-9}nm?
A. 3.3×10^{-27} C. 3.0×10^{8}
B. 3.0×10^{17} D. 3.0×10^{26}

Solution:

$$u=\frac{c}{\lambda}=\frac{3\times10^{8}m/s}{1\times10^{-18}m}=3\times10^{26}Hz$$

Answer: D

8. What is the frequency associated with one photon of a microwave radiation having a frequency of $2.45 \times 10^{9}s^{-1}$?
A. $1.62 \times 10^{24}J$ C. $2.45 \times 10^{-9}J$
B. $2.45 \times 10^{9}J$ D. $1.62 \times 10^{-24}J$

Solution:
$E = h\nu = 6.626 \times 10^{-34}\ J.s \times 2.45 \times 10^{9}s^{-1} = 1.62 \times 10^{-24}\ J$

Answer: D

9. Which of the following transitions will have minimum wavelength?
A. $n_4 \rightarrow n_1$ C. $n_4 \rightarrow n_2$
B. $n_2 \rightarrow n_1$ D. $n_3 \rightarrow n_1$

Telegram: https://t.me/Extreme_series

Solution:
The smaller the transition gap, the higher is the energy. As energy increases, wave length decreases.
$n_2 \rightarrow n_1$ is the smallest gap and nearest to the nucleus. This transition corresponds to minimum wavelength.

Answer: B

10. The Heisenberg uncertainty principle states that
    A. Two atoms of the same element must have the same number of protons.
    B. No two electrons in the same atom can have the same set of four quantum numbers.
    C. It is impossible to determine both position and momentum of an electron simultaneously.
    D. Electrons of atoms in ground state enter energetically equivalent set of orbitals singly before they pair up in any orbitals of the set.

Answer: C

11. How many different types of orbitals are there in the third energy level?
    A. 2 B. 3 C. 6 D. 7
    Solution:
    Number of orbitals = n^2
    Types of orbitals = n = 3 (s, p and f)

Answer: B

12. The normal human eye responds to visible light of wavelength ranging from about 390 to 710 nm. What is the frequency range of the human eye?
    A. $7.7 \times 10^{14} s^{-1}$ to $4.2 \times 10^{14} s^{-1}$
    B. $4.2 \times 10^{14} s^{-1}$ to $7.7 \times 10^{14} s^{-1}$
    C. $1.3 \times 10^{-15} s^{-1}$ to $2.4 \times 10^{-15} s^{-1}$
    D. $2.4 \times 10^{-15} s^{-1}$ to $1.3 \times 10^{-15} s^{-1}$
    Solution:

$$v_{390nm} = \frac{c}{\lambda} = \frac{3\times10^8 m/s}{3.90\times10^{-7} m} = 7.7\times10^{14} Hz$$

$$v_{710nm} = \frac{c}{\lambda} = \frac{3\times10^8 m/s}{7.10\times10^{-7} m} = 4.2\times10^{14} Hz$$

Answer: A

13. Which of the following is the correct order for electrons filling in orbitals?
    A. 3s, 3p, 4s, 3d
    B. 2p, 3s, 3p, 3d
    C. 3p, 4s, 3d, 5s
    D. 4s, 3d, 4p, 4d

Answer: A

14. What is the ground state electronic configuration of Cu^{2+}?
    A. $[Ar]4s^1 3d^8$
    B. $[Ar]4s^2 3d^7$
    C. $[Ar]4s^1 3d^{10}$
    D. $[Ar]3d^9$
    Solution:
    Electron = 29 – 2 = 27, the two electrons are lost from 4s and 3d.
    ⇒ Electron configuration: $[Ar]3d^9$

Answer: D

15. Which of the following is the most reasonable valence – bond structure of the nitrocyl chloride (NOCl)?
    A. :Ö–N̈–C̈l:
    B. Ö=N–C̈l:
    C. Ö=N–C̈l:
    D. Ö–N=C̈l
    Solution:
    Total number of valence electrons = 5 + 6 + 7 = 18 pairs = 9, central atom = N.
    ⇒ The structure of NOCl

Ö=N–C̈l:

Answer: C

16. What is the chemical formula of the compound formed by aluminum (III) and sulfate ion, SO_4?
    A. $AlSO_4$
    B. Al_2SO_4
    C. $Al_2(SO_4)_3$
    D. $Al(SO_4)_3$
    Solution:

$$Al^{3} \times SO_4^{2} \Rightarrow Al_2(SO_4)_3$$

Answer: C

17. Which of the following is the correct Lewis structure for oxygen atom?
    A. Three lone – pairs of valence electrons.
    B. One lone – pair of valence electrons.
    C. Two lone – pairs of valence electrons and one bonding site.
    D. Two lone – pairs of valence electrons and two bonding sites.

Telegram: https://t.me/ExtremeSeries

Solution:

$\ddot{O}=\ddot{O}$

On one oxygen atom, 2 bp and 2Lp are present. **Answer: D**

18. How many electron are shared between two atoms in a double bond?
A. 2 B. 3 C. 4 D. 6

Solution:
In a double bond formation two pairs of electrons or four electrons are shared. **Answer: C**

19. The forces that hold different atoms or ions together are called?
A. Electric forces C. Chemical bonds
B. Physical bonds D. Atomic forces

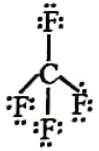
Intra molecular forces (chemical bonds) are forces that hold together atoms or ions. **Answer: C**

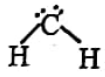
20. Which of the following covalent bonds is the most polar?
A. H – F B. H – C C. H – H D. H – N

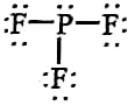
Solution:
The higher the electronegativity difference between the bonded atoms, the more polar is the bond. **Answer: A**

21. Which of the following molecule is **INCORRECTLY** matched with the electronic geometry of the central atoms?
A. CF_4: tetrahedral C. PF_3: pyramidal
B. H_2O: tetrahedral D. $BeBr_2$: linear

Solution:

CF_4:  , tetrahedral, sp^3 hybridized

H_2O:  , bent or angular, sp^3 hybrdized

 , trigonal pyramidal, sp^3hydridized

$BeBr_2$: Br ___ Be ___ Br, linear **Answer: B**

22. What kind of hybrid orbitals are utilized by the carbon atom in CF_4 molecules?
A. Sp B. sp^2 C. sp^3 D. sp^3d

Answer: C

23. What is the bond order of a superoxide anion, with the chemical formula O_2^-?
A. 3.0 B. 2.5 C. 2.0 D. 1.5

Solution:
Electron $= 2 \times 8 + 1 = 17$
$(\sigma 1s)^2(\sigma 1s^*)^2(\sigma 2s)^2(\sigma 2s^*)^2(\sigma 2px)^2(\pi 2py)^2(\pi 2pz)^2(\pi 2py^*)^2(\pi 2pz^*)$

$$B_o = \frac{1}{2}(nb - na) = \frac{1}{2}(10 - 7) = 1.5$$ **Answer: D**

24. What is (are) the F-S-F bond angle (s) SF_6?
A. 109.5° C. 90° and 120°
B. 120° D. 90° and 180°

Solution:
- In SF_6 all bonds are 90° except the one made by the bonds between the central atom and the two atoms diametrically opposite to each other. That angle is 180°.

Answer: D

25. Which of the following refer to a reaction that is **NOT** dependent on the concentration of a reactant?
A. Second order reaction C. Zero order reaction
B. First order reaction D. Third order reaction

Solution:
For zero – order reaction, with rate law, rate = k, the rate is independent on the concentration of reactants. **Answer: C**

26. Consider the following reactions:

$$2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$$

At a certain temperature, the rate of decomposition of $N_2O_5(g)$ is 2.5×10^6 mol/s. What is the rate of formation of NO_2?
A. 1.0×10^{-5} mol/s C. 1.3×10^{-6} mol/s
B. 2.5×10^{-6} mol/s D. 5.0×10^{-6} mol/s

Solution:

$$\text{Rate of } NO_2 = \frac{\text{coefficient of } NO_2}{\text{coefficient of } N_2O_5} \times \text{ rate of } N_2O_5$$

$$= \frac{4}{2} \times 2.5 \times 10^6 \text{ mol/s} = 5.0 \times 10^6 \text{ mol/s}$$

Answer: ___

Telegram: https://t.me/ExtremeSeries

27. The rate of a reaction increases with increasing the concentration of reactant molecules, because
A. The average kinetic energy of the molecules increases.
B. The frequency of molecular collisions increases.
C. The order of the reaction increases.
D. The activation energy increases.

Solution:
As concentration of reactants increases, the number of collision between reacting species also increases. This increases the rate of a reaction. **Answer: B**

28. Given the following data for the reaction:

$$NH_4^+(aq) + NO_2^-(aq) \longrightarrow N_2(g) + 2H_2O(l)$$

Experiment	$[NH_4^+]$	$[NO_2^-]$	Rate
1	0.010M	0.020M	0.020 M/s
2	0.015M	0.020M	0.030M/s
3	0.010M	0.010M	0.005M/s

What will be the rate law for the given reaction?

A. Rate = $k[NH_4^+][NO_2^-]$

B. Rate = $k[NH_4^+]^2[NO_2^-]$

C. Rate = $k[NH_4^+]^2[NO_2^-]^2$

D. Rate = $k[NH_4^+][NO_2^-]^2$

Solution:

Let the rate law be: Rate = $k[NH_4^+]^x[NO_2^-]^y$

- Find the values of x using experiment 1 and 2.

$$\frac{\text{Rate 1}}{\text{Rate 2}} = \frac{k[NH_4^+]_1^x[NO_2^-]_1^y}{k[NH_4^+]_2^x[NO_2^-]_2^y}$$

$$\frac{0.020\,m/s}{0.030\,m/s} = \frac{(0.010)^x(0.020)^y}{(0.015)^x(0.020)^y}$$

$$\frac{2}{3} = \left(\frac{2}{3}\right)^x \Rightarrow x = 1$$

- Use experiments 1 and 3 to find the value of y.

$$\frac{\text{Rate 1}}{\text{Rate 3}} = \frac{[NO_2^-]_1^y}{[NO_2^-]_3^y}$$

$$\frac{0.020}{0.005} = \frac{(0.020)^y}{(0.010)^y} \Rightarrow 4 = 2^y$$

$\therefore$ Rate = $k[NH_4^+][NO_2^-]^2 \Rightarrow y = 2$ **Answer: D**

29. The initial rate of a reaction A + B → products was measured for several different starting concentrations of A and B, with the following results.

Exp	[A]	[B]	Initial Rate of Formation of C M/s
1	0.04	0.04	9.6×10^{-6}
2	0.08	0.04	1.92×10^{-5}
3	0.08	0.02	9.6×10^{-6}

What is the initial rate of reaction when $[A]_o$ = 0.12M and $[B]_o$ = 0.015M?

A. 0.006 Ms^{-1}
B. 0.006$M^{-2}s^{-1}$
C. 1.08×10^{-5} Ms^{-1}
D. 1.08×10^{-4} Ms^{-1}

Solution:

Determine the rate law first. Rate = $[A]^x[B]^y$

- Using experiment 1 and 2, when the concentration of A doubles, the initial rate also doubles. This works for first order reaction only. Thus, the value of x is one.
- Using experiment 2 and 3, when concentration of B halves, the rate also halves. As a result, the order in B is also one.

⇒Rate = k[A][B], then find the value of k

$$k = \frac{\text{Rate}_1}{[A]_1[B]_1} = \frac{9.6\times10^{-6}\,M/s}{(0.04\,M)(0.04)\,M}$$

$k = 6 \times 10^{-3}M^{-1}.S^{-1}$

Rate = k[A][B]

$= 6 \times 10^{-3}M^{-1}.S^{-1} \times 0.12M \times 0.015M = 1.08 \times 10^{-5}MS^{-1}$

Answer: C

Telegram: https://t.me/ExtremeSeries

30. A possible mechanism for the reaction $A + 2B \rightarrow AB_2$ is

Step 1: $A + B \rightarrow AB$ slow

Step 2: $AB + B \rightarrow AB_2$ fast

Overall: $A + 2B \rightarrow AB_2$

According to the mechanism, what will be the rate law?

A. Rate = $k[A][B]$
B. Rate = $k[B]^2$
C. Rate = $k[A][B]^2$
D. Rate = $k[A]$

Solution:

- The slowest step is the rate determining step and the order of the reaction is the same as the coefficient of the reactants because it is an elementary step.

$\Rightarrow$ Rate = $k[A][B]$ **Answer: A**

31. Which one of the following is correct when a reaction reaches equilibrium?

A. The reaction rate for the forward direction is decreased to zero.
B. The rate of the forward and reverse reaction becomes equal.
C. The rate of the reverse reaction is reached a maximum.
D. The amounts of reactants and products becomes equal.

Solution:

At equilibrium, rate of forward reaction is equal to rate of backward reaction. **Answer: B**

32. What is the equilibrium expression in (M)?

$2C(s) + O_2(g) \leftrightarrows 2CO(g)$

A. $K = \dfrac{2[CI]}{2[C]+[O_2]}$
C. $K = \dfrac{2[CO]}{[O_2]}$
B. $K = \dfrac{[CO]^2}{[C]^2[O_2]}$
D. $K = \dfrac{[CO]^2}{[O_2]}$

Solution:

Solids and liquids (solvents) are not included in the equilibrium constant expression.

$\Rightarrow K_C = \dfrac{[CO]^2}{[O_2]}$ **Answer: D**

33. For the reaction: $2C(s) + O_2(g) \rightleftharpoons 2CO(g)$ $\Delta H < 0$ if the reaction is at equilibrium with excess C(s) remaining, what change will increase the quantity of CO(g) for the reaction at equilibrium?

1) Adding C(s)
2) Decreasing the temperature
3) Increasing the pressure

A. Only 1
B. Only 2
C. 1 and 2
D. 2 and 3

Solution:

$2C(s) + O_2(g) \rightleftharpoons 2CO(g)$, $\Delta H < 0$, formation of CO(g) is favored

- By lowering temperature and pressure.
- By increasing concentration of O_2.
- By removing CO and increasing volume.

Answer: B

34. For one component system, when would the degree of freedom becomes one?

A. When only one phase is resent.
B. When two phases are present equilibrium.
C. When three phases are present at equilibrium.
D. At all times.

Solution:

$F = C - P + 2$

$P = C - F + 2$, $C = 1$, $F = 1$... given

$P = 1 - 1 + 2$

$P = 2$, thus, for one component system, degree of freedom becomes on when two phases are present at equilibrium.

Answer: B

35. Consider the following reaction at equilibrium.

$2CO_2(g) \rightleftharpoons 2CO(g) + O_2(g)$ $\Delta H° = -514$ kJ

What does Le Chatelier's principle predict when the temperature for the reaction is increased?

A. The value of the equilibrium constant will increase.
B. The value of the equilibrium constant will decrease.
C. The partial pressure of $O_2(g)$ at equilibrium will increase.
D. The partial pressure of $CO_2(g)$ at equilibrium wll increases.

Telegram: https://t.me/ExtremeSeries

Solution:

$$2CO_2(g) \rightleftharpoons 2CO(g) + O_2(g),\ \Delta H = -514\ kJ$$

Increasing temperature shifts the equilibrium to the left. That is, formation of $CO_2(g)$ is favored.

Answer: D

36. Consider the reaction

$$AB_3(g) \rightleftharpoons A(g) + 3B(g)$$

What is the equilibrium constant expression if the initial concentration of AB_3 is 0.1 M and the equilibrium concentration of A is expressed by x? (Assume the initial concentrations of A and B are both zero.)

A. $\frac{x-3x}{0.1-x}$ C. $\frac{x-x^3}{(0.1-x)^3}$

B. $\frac{x.3}{(0.1-3x)^3}$ D. $\frac{x.(3x)^3}{0.1-x}$

Solution:

$$AB_3(g) \rightleftharpoons A(g) + 3B(g)$$

Initial conc.	0.1	0	0
Change conc.	-x	+x	+3x
Equilibrium conc.	0.1 – x	x	3x

$$K_C = \frac{[A][B]^3}{[AB_3]}$$

$$K_C = \frac{(x)(3x)^3}{0.1-x}$$

Answer: D

37. What products will be formed when a carboxylic acid reacts with an alcohol?

A. Ketone and Water C. Acid chloride and water

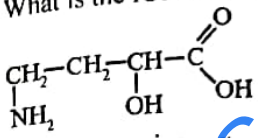
B. Amide and water D. Ester and water

Solution: The reaction between carboxylic acid and alcohol yields ester and water.

$$\underset{\text{Carboxylic acid}}{RCOOH} + \underset{\text{alcohol}}{R'OH} \overset{H^+}{\rightleftharpoons} \underset{\text{Ester}}{RCOOR'} + \underset{\text{Water}}{H_2O}$$

Answer: D

38. What is the IUPAC name for the following molecules?



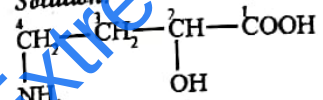
A. γ – amino – α – hydroxybutyric acid

B. 4 – amino – 2 – hydroxybutanoic acid

C. 1 – amino – 3 – hydroxybutanoic acid

D. α – amino – γ – hydroxybutyric acid

Solution



4 – amino – 2 – hydroxyl butanoic acid – IUPAC name

γ – amino – α – hydorxybutyric acid – common name

Answer: B

39. The solubility of compounds containing the carboxylic acid group can be increased by reaction with

A. Sulfuric acid C. Water

B. Sodium hydroxide D. Benzoic acid

Answer: A

40. Which of the following methods is used to convert oils into fats?

A. Esterification reaction using strong bases like NaOH or KOH

B. Saponification reaction using strong bases like NaOH or KOH

C. Hydrolysis reaction in the presence of water and a catalyst.

D. Hydrogenation reaction at high temperature and in the presence of a catalyst.

Solution:

Oils are converted into fats by hydrogenation reaction at high temperature and in the presence of a catalyst.

Answer: D

41. Which of the following general class of organic molecules do triglycerides belong to?

A. Ketones C. Esters

B. Carbonxylic acid D. Ethers

Solution: Triglycerids are fats and oils and they belong to esters.

Answer: C

Telegram: https://t.me/ExtremeSeries

42. The chemical makeup of oil is esters of glycerol with three
A. Identical saturated fatty acids.
B. Predominantly saturated fatty acids.
C. Predominantly unsaturated fatty acids.
D. Identical unsaturated fatty acids.

Solution:
- Fats are esters of glycerol and mostly saturated fatty acids.
- Oils are liquid esters primarily derived from unsaturated fatty acids and glycerol.

Answer: C

43. Which of the following types of solution are possible?
1) Solid dissolved in a liquid
2) Gas dissolved in a liquid
3) Gas dissolved in a gas
4) Solid dissolved in a solid

A. 1 and 2 C. 1, 2 and 4
B. 1, 2, 3 and 4 D. 1

There are six main types of solutions based on the states of solute and solvents.
- Gas – gas
- Gas – liquid
- Liquid – liquid
- Liquid – solid
- Solid – liquid
- Solid – solid

Answer: B

44. Which of the following pairs of substances are miscible?
A. Water and gasoline C. Water and salt
B. Water and oxygen D. Water and ethanol

Solution:
Polar dissolves polar and non – polar dissolves non – polar. For solid solute we say soluble but for gas and liquid solutes we use the term miscible, water and ethanol are miscible.

Answer: D

45. Which of the following will increase the rate at which a solid dissolved in a liquid?
A. Supersaturating the solution.
B. Grinding the solid into smaller pieces.
C. Lowering the temperature of the solvent.
D. Placing the solution in opaque container.

Solution:
The rate of dissolution of solid solute can be by increased its surface area. This can be done by grinding or powdering the solid.

Answer: B

46. The solubility of NaCl in water at 20°C is 36g in 100g of water. A solution that contains 18 g of NaCl in 50g of water at 20°C is
A. Saturated C. Supersaturated
B. Unsaturated D. Dilute

Solution:
The solubility of NaCl in water at 20°C is 36g in 100g of water. This is saturated solution. But at the same temperature, 36g/100g is the same as 18g/50g.

Answer: A

47. What will be the normality of a 2.3M sulphuric acid solution?
A. 0.23N B. 0.46N C. 2.3N D. 4.6N

Solution:

$$H_2SO_4 \longrightarrow 2H^+ + SO_4^{2-}$$

$\therefore Z = 2$ eq/mol

$N = M.Z = 2.3$ mol/L $\times$ 2 eq/mol $= 4.6N$

Answer: D

48. A certain 500g toothpaste sample has 0.2g of fluoride. What is the concentration of the fluoride in ppm?
A. 200 B. 250 C. 400 D. 1,000

Solution:

$$\text{ppm } F^- = \frac{\text{mass of } F^-}{\text{mass of tooth paste}} \times 10^6$$

$$= \frac{0.2g}{500g} \times 10^6 = 400\text{ppm}$$

Answer: C

49. How many grams of methanol would have to be added to water to prepare 150mL of solution which is 2.0M methanol?
A. 2.4 B. 9.6 C. 9.6×10^3 D. 4.3×10^2

Solution:

$$M = \frac{m}{M_{wt}.V}$$

$M = M.M_{wt}.V = 2$ mol/L $\times$ 32 g/mol $\times$ 0.150L

$M = 9.6$g of methanol (CH_3OH)

Answer: B

50. If 0.454 kg of nitroglycerin explodes according to the following reaction:

$$C_2H_5(NO_2)_3(l) \rightarrow CO_2(g) + N_2(g) + H_2O(g) + O_2(g)$$

What is the total volume of gases in dm^3 produced at STP?
A. 44.8 B. 163 C. 325 D. 650

Telegram https://t.me/ExtremeSeries

42. The chemical makeup of oil is esters of glycerol with three
A. Identical saturated fatty acids.
B. Predominantly saturated fatty acids.
C. Predominantly unsaturated fatty acids.
D. Identical unsaturated fatty acids.

Solution:

- Fats are esters of glycerol and mostly saturated fatty acids.
- Oils are liquid esters primarily derived from unsaturated fatty acids and glycerol.

Answer: C

43. Which of the following types of solution are possible?
1) Solid dissolved in a liquid
2) Gas dissolved in a liquid
3) Gas dissolved in a gas
4) Solid dissolved in a solid

A. 1 and 2 C. 1, 2 and 4
B. 1, 2, 3 and 4 D. 1

There are six main types of solutions based on the states of solute and solvents.

- Gas – gas
- Gas – liquid
- Liquid – liquid
- Liquid – solid
- Solid – liquid
- Solid – solid

Answer: B

44. Which of the following pairs of substances are miscible?
A. Water and gasoline C. Water and salt
B. Water and oxygen D. Water and ethanol

Solution:

Polar dissolves polar and non – polar dissolves non – polar. For solid solute we say soluble but for gas and liquid solutes we use the term miscible, water and ethanol are miscible.

Answer: D

45. Which of the following will increase the rate at which a solid dissolved in a liquid?
A. Supersaturating the solution.
B. Grinding the solid into smaller pieces.
C. Lowering the temperature of the solvent.
D. Placing the solution in opaque container.

Solution:

The rate of dissolution of solid solute can be by increased its surface area. This can be done by grinding or powdering the solid.

Answer: B

46. The solubility of NaCl in water at 20°C is 36g in 100g of water. A solution that contains 18 g of NaCl in 50g of water at 20°C is
A. Saturated C. Supersaturated
B. Unsaturated D. Dilute

Solution:

The solubility of NaCl in water at 20°C is 36g in 100g of water. This is saturated solution. But at the same temperature, 36g/100g is the same as 18g/50g.

Answer: A

47. What will be the normality of a 2.3M sulphuric acid solution?
A. 0.23N B. 0.46N C. 2.3N D. 4.6N

Solution:

$$H_2SO_4 \longrightarrow 2H^+ + SO_4^{2-}$$

$\Rightarrow Z = 2$ eq/mol

$N = M.Z = 2.3$ mol/L $\times$ 2 eq/mol $= 4.6N$

Answer: D

48. A certain 500g toothpaste sample has 0.2g of fluoride. What is the concentration of the fluoride in ppm?
A. 200 B. 250 C. 400 D. 1,000

Solution:

$$\text{ppm } F^- = \frac{\text{mass of } F^-}{\text{mass of tooth paste}} \times 10^6$$

$$= \frac{0.2\text{g}}{500\text{g}} \times 10^6 = 400\text{ppm}$$

Answer: C

49. How many grams of methanol would have to be added to water to prepare 150mL of solution which is 2.0M methanol?
A. 2.4 B. 9.6 C. 9.6×10^3 D. 4.3×10^2

Solution:

$$M = \frac{m}{M_{wt}.V}$$

$M = M.M_{wt}.V = 2$ mol/L $\times$ 32 g/mol $\times$ 0.150L

$M = 9.6$g of methanol (CH_3OH)

Answer: B

50. If 0.454 kg of nitroglycerin explodes according to the following reaction:

$$C_3H_5(NO_2)_3(l) \rightarrow CO_2(g) + N_2(g) + H_2O(g) + O_2(g)$$

What is the total volume of gases in dm^3 produced at STP?
A. 44.8 B. 163 C. 325 D. 650

Telegram: https://t.me/ExtremeSeries

Solution:
By mass – volume relationship find the answer. At STP, the volume of one mole of any gas is 22.4L (22.4dm^3). However the reaction is incorrect. The reaction is $2C_3H_5(NO_2)_3(l) + 5O_2(g) \rightarrow 6CO_2(g) + 3N_2(g) + 5H_2O(g)$

51. When the molecular equation:
$CaCl_2(aq) + Na_2CO_3(aq) \rightarrow CaCO_3(s) + 2NaCl(aq)$
is written in terms of ionic equation, which one of the following pairs will be the spectator ions?
A. Na^+ and Cl^- C. CO_3^{2-} and Cl^-
B. Ca^{2+} and Cl^- D. Ca^{2+} and CO_3^{2-}

Solution:
$Ca^{2+}(g) + 2Cl^-(aq) + 2Na^+(aq) + CO_3(aq) \rightarrow CaCO_3(s) + 2Na^+(aq) + 2Cl^-(aq)$
Thus, Na^+ and Cl^- are spectator ions. **Answer: A**

52. What will be the ratio of the boiling point elevation of a 0.1M solution of sucrose, sodium sulphate and potassium bromide?
A. 1:3:1 B. 1:1:2 C. 1:3:2 D. 1:2:2

Solution:
- For no n- electrolytes, the value of Vans Hoff factor i is one. But for electrolytes, it is equal to the number of ions produced.
  - ⇒ For sucrose solution, i = 1
  - ⇒ For sodium sulphate (Na_2SO_4) solution, i ≈ 3
    $Na_2SO_4 \rightarrow 2Na^+ + SO_4^+$
- For potassium bromide solution i ≈ 2
  $KBr \rightarrow K^+ + Br^-$
  $\Delta T_b = imk_b$, therefore the ratio becomes 1: 3: 2

Answer: C

53. Which of the following is the conjugate acid of the HS^- ion?
A. H_2S^+ B. S^{2-} C. S^- D. H_2S

Solution:
Conjugate acid = Base + H^+ = HS^- + H^+ = H_2S

Answer: D

54. Which one of the following is a Lewis acid?
A. NH_3 B. CCl_4 C. $AlCl_3$ D. LiCl

Answer: C

55. Among the following acids, which has the lowest pK_a value?
A. CH_3CH_2COOH C. $(CH_3)_2CHCOOH$
B. HCOOH D. CH_3COOH

Solution:
As carbon number increases, strength of carboxylic acids decreases. The stronger the acid, the lower is the Pka value.

Answer: B

56. Nitrous acid, HNO_2 is 0.37% ionized in 3.0M solution. What is the K_a for this acid?
A. $\frac{3.0(0.0037)^2}{99.63}$ C. $\frac{3.0(0.0037)^2}{0.9963}$
B. $\frac{(0.0037)^2}{0.9963}$ D. $\frac{0.9963}{(0.0037)^2}$

Solution:
$[H^+] = 0.37\% \cdot [HNO_2]_0$

$[H^+] = \frac{0.37}{100} \times 3.0M = 0.0111M$

For monoprotic weak acid: $Ka = \frac{[H^-]^2}{[Acid]_o - [H^-]}$

$Ka = \frac{(0.0111M)^2}{3M - 0.011M} = 4.12 \times 10^{-5}$ **Answer: C**

57. The suppression of ionization of a weak acid or a weak base by adding one of its own ions is known as:
A. Buffer effect C. Common ion effect
B. Ionization effect D. Association effect

Answer: C

58. To a 0.0mL of a buffer solution that is 0.25 M in HF an d0.10M in NaF, if 5.0mL of 0.050 M NaOH is added, what si the final pH of this buffer solution? Given the pKa HF = 3.177
A. 2.77 B. 2.80 C. 2.74 D. 3.17

Solution: $HF + NaOH \rightarrow NaF + H_2O$
$n_{HF} = M_{HF} V_{buffer} = 0.25$ mol/L × 5mL = 0.25mmol
$n_{NaOH} = M_{NaOH} \cdot V_{NaOH} = 0.05$ mol/L × 5mL = 0.25mmol
$n_{NaF} = M_{NaF} \cdot V_{buffer} = 0.10$ mol/L × 50mL = 5mmol
after the reaction

Telegram: https://t.me/ExtremeSeries

n_{HF} left = 12.5 – 0.25 = 12.25 mmol
n_{NaF} = 0.25mmol + 5mmol = 5.25mmol
$n_{NaOH} = 0$

$$[HF] = \frac{n_{HF}}{V_{NaOH} + V_{buffer}} = \frac{5.25mmol}{55mL} = 0.095M$$

$$pH = Pka + \log \frac{[NaF]}{[HF]}$$

$$pH = 3.17 + \log \frac{0.095}{0.223} = 2.80$$

Answer: B

59. Which statement about indicators is always correct?
A. The mid – point of the pH range of an indicator is 7.0.
B. The pH range is greater with higher pK_a values.
C. The color red indicates an acidic solution.
D. The pK_a values of the indicator is within its pH range.

Answer: B

60. A 0.217 g sample o HgO reacts with excess iodide ions according to the reaction: $HgO + 4I^- + H_2O \rightarrow HgI_4^{2-} + 2OH^-$

How many mL of 0.1M HCl is required to reach equilibrium point during the titration
A. 1.0 B. 10 C. 20 D. 50

Solution:
By mass – mole relationship find the number of mole of OH^-

$$\underset{1\times217g}{\overset{0.217g}{HgO}} + 4I^- + H_2O \longrightarrow HgI_4^{2-} + \underset{2mol}{\overset{x}{2OH^-}}$$

$$\frac{0.217g}{217g} = \frac{x}{2mol}$$

$$x = n_{OH^-} = 0.002mol$$

$$2HCl + 2OH^- \longrightarrow 2H_2O + 2Cl^-$$

For 1 mole HCl, 1 mole OH^- is required. Thus, for 0.002 mol OH^-, 0.002 mole HCl is required $\Rightarrow n_{HCl} = 0.002mol$

$$V_{HCl} = \frac{n_{HCl}}{M_{HCl}} = \frac{0.002mol}{0.1mol/L} = 0.02L = 20mL$$

Answer: C

61. What is a thermodynamic system that exchanges both energy and matter with the surroundings?
A. Adiabatic C. Isolated
B. Closed D. Open

Answer: D

62. At what condition (s) is the equation, $\Delta H = \Delta U + p\Delta V$, applicable?
A. For constant pressure and temperature processes.
B. Only for constant pressure processes.
C. Only for constant temperature processes.
D. Only for constant volume processes.

Solution:
$H = U + PV$
$\Delta H = \Delta U + \Delta(PV)$
$\Delta H = \Delta U + \Delta PV + P\Delta V$, at constant pressure
$\Delta H = \Delta u + P\Delta V$

Answer: B

63. Based on the following information:
$H_2(g) + ½\ O_2(g) \rightarrow H_2O(l)$, $\Delta H° = -286kJ$
$2Na(s) + ½\ O_2(g) \rightarrow Na_2O(s)$, $\Delta H° = -414kJ$
$Na(s) + ½\ O_2(g) + ½\ H_2(g)$, $\Delta H° = -425\ kJ$
What will be the standard enthalpy change for the reaction n
$Na_2O(s) + H_2O(l) \rightarrow 2NaOH(s)$?
A. -978kJ B. -722kJ C. -150kJ D -275kJ

Solution:
Reverse the first and second reactions and multiply the third by 2.

$$H_2O(l) \longrightarrow H_2(g) + \frac{1}{2}O_2(g), \Delta H_1^\circ = 286kJ$$

$$Na_2O(g) \longrightarrow 2Na(s) + \frac{1}{2}O_2(g), \Delta H_2^\circ = 414kJ$$

$$2Na(s) + O_2(g) + H_2(g) \longrightarrow 2NaOH(s), \Delta H_3^\circ$$
$$= 2 \times (-425) = -850kJ$$

$$Na_2O(s) + H_2O(l) \longrightarrow 2NaOH(s),\ \Delta H° = \Delta H_1^\circ + \Delta H_2^\circ + \Delta H_3^\circ$$

$$\Delta H° = 286 + 414 - 850 = -150kJ$$

Answer: C

Telegram: https://t.me/ExtremeSeries

64. The K_p for the reaction: $2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$ is 0.26 at 1,000°C and 40.8 at 1,300°C. Which of the following combinations of ΔH and ΔS are most plausible for the reaction at these temperatures?

	ΔH	ΔS
A.	>0	>0
B.	>0	>0
C.	<0	<0
D.	<0	

Solution:

$$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$$

The reaction is endothermic because it is decomposition reaction ($\Delta H > 0$). As temperature increases entropy also increases or there are more gaseous products than reactant ($\Delta s > 0$).

Answer: A

65. In the following reaction:

$3Cl^-(aq) + 3CrO_4^{2-}(aq) + 23H^+(aq) \rightarrow 3HClO_2(aq) + 4Cr^{3+}(aq) + 10H_2O(l)$

What is the behavior of Cl^- (aq)?

A. An acid
B. A base
C. An oxidizing agent
D. A reducing agent

Solution:

$$3Cl^-(aq) + 4CrO_4^{2-}(aq) + 23H^+(aq) \longrightarrow 3HClO_2(aq) + 4Cr^{3+}(aq) + 10H_2O(l)$$

The oxidation state of Cl is increased from $-1(Cl^-)$ to $+3(HClO_2)$. Thus, Cl^- is reducing agent.

Answer: D

66. Devices in which oxidation – reduction reaction is induced by an electric current are called

A. Galvanic cells
B. Electrolytic cells
C. Voltaic cells
D. Electromotive cells

Solution:

In electrolytic cell, electric current causes non – spontaneous redox – reaction to take place but in Galvanic (voltaic cell) spontaneous redox reaction generates electricity.

Answer: B

67. What mass of magnesium (Mg) is produced in the electrolysis of molten $MgCl_2$, if a current of 7.50×10^2 A is passed through the molten salt for 1.00 hour?

A. 3.4×10^2 g
B. 9.92×10^2 g
C. 4.7×10^2 g
D. 4.96×10^2 g

Solution:

At cathode: $Mg^{2+} + 2e^- \rightarrow Mg$

$Z = 2$

$I = 7.50 \times 10^2$A, $t = 1$ hour $= 3600$s

$$m = \frac{MIt}{zF} = \frac{24.31 g/mol \times 7.50 \times 10^2 A \times 3600 s}{2 \times 96500 A.s/mol}$$

$m = 340$g

Answer: A

68. In the reaction:

$Pb(s) + PbO_2(s) + H_2SO_4(aq) \rightarrow 2PbSO_4(s) + 2H_2O(l)$

Which substance behaves as the oxidizing agent?

A. Pb B. $PbSO_4$ C. PbO_2 D. H_2SO_4

Solution:

$Pb^0(s) + P^{+4}bO_2(l) + H_2SO_4(aq) \rightarrow 2Pb^{+2}SO_4(l) + 2H_2O(l)$

Pb is oxidized and it is reducing agent.

PbO_2 is reduced and it is oxidizing agnet.

Answer: C

69. What product (s) is/are formed during the electrolysis of a concentrated aqueous solution of sodium chloride?

B. $Cl_2(g)$
C. $Cl_2(g)$ and $H_2(g)$
C. NaOH(aq) and $Cl_2(g)$
D. $Cl_2(g)$, $H_2(g)$ and NaOH(aq)

Solution:

In electrolysis of concentrated NaCl solution

Anode reaction: $2Cl^- \longrightarrow Cl_2 + 2e^-$

Cathode reaction: $2H^+ + 2e^- \longrightarrow H_2$

Cell reaction: $2NaCl(aq) + 2H_2O(l) \xrightarrow{\text{electrolysis}} 2NaOH(g) + H_2(g) + Cl_2(g)$

Therefore, the products formed are NaOH, H_2 and Cl_2.

Answer: D

70. For the electrochemical cell represented by:

$Zn/AnSO_4(aq)//CuSO_4(aq)/CU(s)$

the standard reduction potentials for Cu^{2+}/Cu and Zn^{2+}/Zn are +0.34 and -0.75V, respectively. What will be the cell reaction that occurs spontaneously and its cell voltage?

Telegram: https://t.me/ExtremeSeries

		E°(V)
A.	$Cu + Zn^{2+} \rightleftharpoons Cu^{2+} + Zn$	1.10
B.	$Zn + Cu^{2+} \rightleftharpoons Zn^{2+} + Cu$	0.42
C.	$Cu^{2+} + Zn^{2+} \rightleftharpoons Zn + Cu$	0.42
D.	$Cu^{2+} + Zn \rightleftharpoons Cu + Zn^{2+}$	1.10

Solution:

$E°\ cell = E°ox + E°\ red$

Anode is $Zn/ZnSO_4$ and cathode is $Cu/CuSO_4$

Cell reaction: $Zn(s) + Cu^{2+}(aq) \rightleftharpoons Zn^{2+}(aq) + Cu(s)$

$E°cell = 0.76V + 0.34V = 1.10V$

Answer: D

71. Consider the reaction:

$$Cu^{2+}(aq) + Fe(s) \rightleftharpoons Cu(s) + Fe^{2+}(aq),\ E° = 0.78V$$

What is the value of E when $[Cu^{2+}]$ is equal to 0.040 M and $[Fe^{2+}]$ is equal to 0.40M?

A. 0.72V B. 0.75V C. 0.81V D. 0.84V

$$E_{cell} = E°cell - \frac{0.0592V}{n}\log Q$$

$$= 0.78V - \frac{0.0592V}{2}\log\frac{0.40}{0.040}$$

$$= 0.75V$$

Answer: B

72. Which one of the following is needed for photosynthesis?

A. Carbon dioxide C. Oxygen
B. Nitrogen D. Phosphorus

Solution:

Carbon dioxide and water are raw materials for photosynthesis in the presence of sunlight. **Answer: A**

73. In what way is the phosphorous cycle differs from the carbon and nitrogen cycle?

A. Phosphorous is more abundant than carbon and nitrogen.
B. Phosphorous is less important to biological system than carbon or nitrogen.
C. Phosphorous, once used by an organism does not cycle back to the environment.
D. The reservoirs for phosphorus exists in mineral form in rocks than in the atmosphere.

Answer: D

74. Which one of the following metal is used for galvanization of iron sheets?

A. Nickel C. Chromium
B. Zinc D. Aluminum

Solution:

Galvanization is coating metal surface with zinc. **Answer: A**

75. Which of the following reactions in used in the production of industrially important chemical?

A. $H_2S_2O_7(l) + H_2O(l) \rightarrow 2H_2SO_4(aq)$
B. $3NO_2(g) + H_2O(l) \rightarrow 2HNO_3(aq) + NO_2(g)$
C. $(NH_4)_2HPO_4(s) + NH_3(g) \rightarrow NH_4H_2PO_4(s)$
D. $2Cr(s) + Al_2O_3(s) \rightarrow Cr_2O_3(s) + 2Al(s)$

Solution:

The most important chemical is sulphuric acid. **Answer: A**

76. The second step in the production of sulphuric acid is given by the reaction:

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$$

Which condition would increase the yield of this reaction?

A. High temperature and low pressure.
B. Low temperature and high pressure.
C. High temperature and high pressure.
D. Low temperature and low pressure.

Solution:

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g), \Delta H = -196kJ$$

Formation of $SO_3(g)$ is favored at low temperature and high pressure. **Answer: B**

77. What is the monomer of polyvinyl chloride (PVC)?

A. Chloromethane C. Chloroform
B. Ethylene dichloride D. Ethyl chloride

Solution:

The monomer of polyvinyl chloride is vinyl chloride or chloreethene or ethylene chloride.

$$nCH_2=CHCl \xrightarrow{\text{covalent}} \left[CH_2-\underset{Cl}{CH}\right]_n$$

Vinyl Chloride — PVC

Telegram: https://t.me/Extremeseries

78. What is the other name for polymethyl methacrylate (PMMA)?
A. Perspex
B. Teflon
C. Bakelite
D. Nylon – 6

Solution:
- Polymethyl methacrylate (PMMA) is also known Perspex, Lucite or Plexi glass.

Answer: A

79. Which of the following is the most abundant polysaccharide?
A. Amylose
B. Cellulose
C. Starch
D. Sucrose

Answer: B

80. What type of polymerization occurs when amino acids react to form peptides and proteins?
A. Addition polymerization
B. Condensation polymerization
C. Substitution polymerization
D. Hydration polymerization

Solution:
Amino acids are monomers of proteins with two functional groups. Such type of monomer form polymers by condensation polymerization.

Answer: B

Telegram: https://t.me/ExtremeSeries

1. The reaction between alcohols and acyl chloride produces _____.
A. Ethers
B. Carboxylic acids
C. Aromatic salts
D. Esters

Solution:

$$R-\overset{O}{\overset{\|}{C}}-Cl + ROH \rightarrow R-\overset{O}{\overset{\|}{C}}-OR' + HCl$$

Aceylchloride alcohol Ester Hydrochloric acid

Answer: D

2. Which of the following aqueous solutions will have the LOWEST freezing point?
A. Pure H_2O
B. aq. 0.50 m KF
C. aq. 0.24 m FeI_3
D. aq. 0.60 m glucose

Solution: $\Delta T_f = imk_f = T_f^\circ - T_f$

T_f° – freezing point of sovlent

T_f – freezing point of solution

As the value of i and m increase, T_f decreases, i

i. For $K_F = 2 \Rightarrow i.m = 2 \times 0.5 = 1$

ii. For $FeI_3 = 4 \Rightarrow i.m = 4 \times 0.24 = 0.96$

iii. For glucose $= 1 \Rightarrow i.m = 1 \times 0.6 = 0.60$

Thus, aqueous solution of 0.50m kF has the lowest freezing point.

Answer: B

3. Fats and oils are:
A. esters B. Alcohols C. Acids D. Alkenes

Answer: A

4. Who used the Cathode Ray Tube to discover the electron and determine its charge to mass ratio?
A. Robert. A. Millikan
B. Ernes Rutherford
C. James Chadwick
D. J. J. Thomson

Answer: D

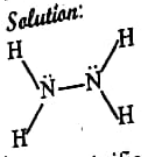
5. Compounds that contain the carboxyl (carbonyl and hydroxyl) group are said to be:
A. Esters
B. Ketones
C. Organic acids
D. Aldehydes

Solution: Compounds containing carboxyl (carbonyl and hydroxyl) group are called carboxylic (organic) acids.

Answer: C

6. When a student draws a plausible Lewis structure for hydrazine molecule (N_2H_4), how many lone pair of electrons are available?
A. 2 B. 1 C. 3 D. 4

Solution:



bp = 5
Lp = 2

Answer: A

7. During nitrification, bacteria convert _____ into _____.
A. N_2, N_2O into nitrous oxide
B. NO_3^-, NH_4^+ or nitrates into ammonium.
C. N_2, nitrous oxide into nitrogen
D. N_2 into nitrates

Answer: D

8. Which of the following metals exists in a free state?
A. Pt B. Mg C. Zn D. Na

Solution: Pt, Au, Pd, Os, Ag and Cu exist in free states.

Answer: A

9. The main ore of lead is called:
A. Cinnabar B. Zinc blend C. Galena D. Chromite

Answer: C

10. Which of the following is **NOT** both a Bronsted – Lowery acid and a Bronsted – Lowery base?
A. HS^- B. HSO_4^- C. OH^- D. HCO_3^-

Solution: Hydroxide ion (OH^-) is a Bronsted – Lowery base but not acid.

Answer C

11. Chlorine has an oxidation number of +5 in:
A. NaClO B. $NaClO_3$ C. $NaClO_2$ D. $NaClO_4$

Solution: The oxidation state of chlorine in NaClO, $NaClO_3$, $NaClO_2$ and $NaClO_4$ is +1, +5, +3 and +7 respectively.

Answer: B

12. A _____ ΔH corresponds to an _____ process.
A. Positive, endothermic C. Negative, endothermic
B. Positive, exothermic D. Zero, exothermic

Solution:
$\Delta H > 0$, endothermic process
$\Delta H < 0$, exothermic process

Answer: A

13. The monomer of neoprene is:
A. Butadiene C. Isoprene
B. 2 – methyl – 1, 3 – butadiene D. Chloroprene

Solution:
- Neoprene is the first synthetic rubber.
- Its monomer is chloroprene (2 – chloro – 1.3 - butadiene),

Answer: D

14. What is the sum of 3.71×10^8 and 4.62×10^7 to the correct significant figure?
A. 4.17×10^8 B. 4.99×10^7 C. 4.172×10^8 D. 4.991×10^7

Solution: $3.71 \times 10^8 + 4.62 \times 10^7 = 37.1 \times 10^7 + 4.62 \times 10^7$
$= (37.1 + 4.62) \times 10^7 = 41.72 \times 10^7 = 4.17 \times 10^8$

Answer: A

15. What is a measure of the closeness of a measurement to its true value?
A. Precision C. Accuracy
B. Reproducibility D. Usefulness

Answer: C

16. When a sodium chromate (Na_2CrO_4) solution is acidified, it is converted to:
A. Cr(s) B. $Cr_2O_7^{2-}$ C. CrO_3 D. Cr_2O_3

Solution: When a sodium chromate (Na_2CrO_4) solution is acidified it is converted to sodium di chromate ($Na_2Cr_2O_7$).
$2Na_2CrO_4 + 2HCl \rightarrow Na_2Cr_2O_7 + 2NaCl + H_2O$

Answer: B

17. Nylons are:
A. Polyamides B. PVC C. Amides D. Polyesters

Solution: Nylons are polyamides. They are condensation polymers.

Answer: A

18. Which of the following is a natural polymer?
A. Perspex B. PVC C. Rubber D. Teflon

Answer: C

19. The maximum kinetic energy of a photoelectron emitted from a metal is 1.03×10^{-19} J when light that has a 656 nm wavelength shines on the surface. What is the threshold frequency for this metal?
A. $4.57 \times 10^{-14}\ s^{-1}$ C. $3.02 \times 10^{-14}\ s^{-1}$
B. $4.57 \times 10^{14}\ s^{-1}$ D. $3.02 \times 10^{14}\ s^{-1}$

Telegram: https://t.me/ExtremeSeries

Solution: $\phi = E_{photon} - k_E = \frac{\lambda C}{\lambda} - kE$

$$= \frac{6.626\times10^{-34}\times3\times10^{8}}{6.56\times10^{-7}} - 1.03\times10^{-19}$$

$$= 3.03\times10^{-19} - 1.03\times10^{-19} = 2\times10^{-19}\,J$$

$\phi = hv_o$

$v_o = \frac{\phi}{\lambda}$

$v_o = \frac{2\times10^{-19}\,J}{6.626\times10^{-34}\,J.s}$

$v_o = 3.02\times10^{14}\,s^{-1}$

Answer: D

20. What is the maximum number of electrons in an atom that can have the principal quantum number n = 4?
A. 32 B. 8 C. 18 D. 34
Solution: Number of electrons $= 2n^2 = 2\times4^2 = 32$

Answer:

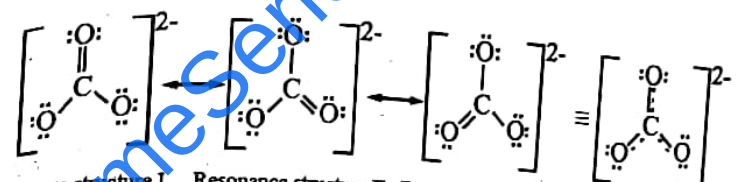
21. Which quantum number is used to determine subshells?
A. Principal quantum number
B. Magnetic quantum number
C. Azimuthal quantum number
D. Spin quantum number
Solution: Angular – momentum or azimuthal quantum number determines sub shells.
$l = 0$, s, $l = 1$, p, $l = 2$, d, $l = 3$, f, $l = 4$, g etc **Answer: C**

22. What skill is a scientist using when he/she listens to the sounds that animals make?
A. Drawing conclusions C. Making a hypothesis
B. Making observations D. Interpreting data
Answer: B

23. Which of the following is fundamentally different from the others?
A. Radio waves C. Light waves
B. Sound waves D. Microwaves
Answer: B

24. Which of the following equations expresses de Broglie's hypothesis?
A. $v = c/\lambda$ B. $\Delta E = hc/\lambda$ C. $\Delta E = c/\lambda$ D. $\lambda = h/mv$
Answer: D

25. The number of resonance structures for CO_3^{2-} are:
A. 3 B. 2 C. 6 D. 9
Solution:



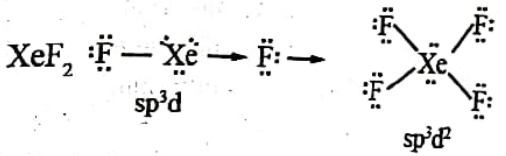
Resonance structure I Resonance structure II Resonance structure III Resonance hybrid

Answer: A

26. In the following equation, what type of hybridization change, if any, occurs at the Xe atom?

$$XeF_2(s) + F_2(g) \rightarrow XeF_4(s)$$

A. sp^3d to sp^3 C. sp^3d to sp^3d^2
B. dsp^2 to sp^3 D. sp^3 to sp^3d
Solution:



Answer: C

27. What is (are) the bond angle (s) in SF_6?
A. 180° B. 109.5° C. 90° and 109.5° D. 90°
Answer: D

28. Which of the following statements about oxygen and fluorine is **NOT** correct?
A. O and F have the same number of core electrons.
B. O has a smaller atomic radius than F.
C. O has a smaller electron affinity than F.
D. O^{2-} has a larger ionic radius than F^-.
Solution:
- $_8O$: $1s^2 2s^2 2p^4$, number of core (inner) electrons = 2
- $_9F$: $1s^2 2s^2 2p^5$, number of core electrons = 2
- Oxygen has larger atomic radius than fluorine.
- O^{2-} has larger ionic radius than F^-

Answer: B

29. What will be the charges on the ions formed when silicon reacts with nitrogen?
A. Si^{2+}, N^{2-} B. Si^{4+}, N^{-3} C. Si^{3+}, N^{3+} D. Si^{4+}, N^{2}

Telegram: https://t.me/ExtremeSeries

Solution: $3Si + 2N_2 \rightarrow Si_3N_4$, thus silicon forms Si^{4+} while nitrogen forms N^{3-}.

Answer: B

30. Which of the following compounds does **NOT** contain an ionic bond?

A. K_2S B. NaOH C. HCl D. LiH

Solution: An ionic bond is formed between metals and non – metals or between ammonium ion and non – metals. It is not formed between two non – metals.

Answer: C

31. The rate equation for the decomposition of nitramide, $H_2HNO_2 \rightarrow N_2O + H_2O$, Rate = $k[H_2NNO_2][H^+]^{-1}$. Which of the following mechanisms is consistent with this rate equation?

A. $H_2NNO_2 \rightarrow N_2O + H_2O$ slow

B. $H_2NNO_2 + H^+ \rightleftharpoons H_3NNO_2^+$ fast equilibrium
$H_3NNO_2^+ \rightarrow N_2O + H_3O^+$ slow
$H_3O^+ \rightleftharpoons H^+ + H_2O$ fast equilibrium

C. $H_2NNO_2 \rightarrow OH^- + NH_4^+$ slow
$NH_4^+ \rightleftharpoons NH_3 + H^+$ fast equilibrium
$H_2O \rightleftharpoons H^+ + OH^-$ fast equilibrium

D. $H_2NNO_2 \rightleftharpoons H^+ + HNNO_2^-$ fast equilibrium
$HNNO_2^- \rightarrow N_2O + OH^-$ slow
$H^+ + OH^- \rightarrow H_2O$ fast

Solution: For a mechanism to be consistent for the reaction:

- The rate law must agree with the experimentally determined rate law.
- The overall reaction must be the same as the given reaction.

A. Cannot be consistent because the rate law does not agree.

B. Similar reason as A.

C. The rate law and the overall reaction are not similar with the given reaction.

D. Rate = $k[HNNO_2^-]$, using the slow step, but $HNNO_2^-$ is an intermediate. Replace it using the fast equilibrium

$r_f = k_f[H_2NNO_2], r_b = k_b[H^+][HNNO_2^-]$ at equilibrium $r_f = r_b$

$k_f[H_2NNO_2] = k_b[H^+][HNNO_2^-]$

$$\Rightarrow [HNNO_2^-] = \frac{k_f[H_2NNO_2]}{k_b[H^+]}$$

$$\Rightarrow \text{Rate} = k\frac{k_f}{k_b}\frac{[H_2NNO_2]}{[H^+]} = k[H_2NNO_2][H^+]^{-1}$$

Answer: D

32. Given the equilibrium constant values:

$N_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons N_2O(g)$, $K_1 = 3.4\times10^{-18}$

$N_2O_4(g) \rightleftharpoons 2NO_2(g)$ $K_2 = 4.6\times10^{-3}$

$\frac{1}{2}N_2(g) + O_2(g) \rightleftharpoons NO_2(g)$ $K_3 = 4.1\times10^{-9}$

What is the value of K_C for the following reaction?

$2N_2O(g) + 3O_2(g) \rightleftharpoons 2N_2O_4(g)$

A. 2.4×10^{-6} B. 1.2×10^{6} C. 1.2×10^{-6} D. 4.8×10^{6}

Solution: Reverse the first and multiply by 2.
Reverse the second and multiply by 2.
Multiply the third by 4.

$$2N_2O(g) \rightleftharpoons 2N_2(g) + O_2(g), k_1' = \left(\frac{1}{3.4\times10^{-18}}\right)^2 = 8.65\times10^{34}$$

$$4NO_2(g) \rightleftharpoons 2N_2O_4(g), k_2' = \left(\frac{1}{4.6\times10^{-3}}\right)^2 = 4.73\times10^{4}$$

$$2N_2(g) + 4O_2(g) \rightleftharpoons 4NO_2(g), k_3' = (4.1\times10^{-9})^4 = 2.83\times10^{-34}$$

$$2N_2O(g) + 3O_2(g) \rightleftharpoons 2N_2O_4(g),\ k_C = k_1'.k_2'.k_3'$$

$$K_C = 1.16\times10^{6}$$

Answer: B

33. The equilibrium constant for the ionization of HCN is 4.9×10^{-10}.

$HCN \rightleftharpoons H^+ + CN^-$, $K = 4.9\times10^{-10}$

Which of the following statements is true regarding this equilibrium?

I. The reaction is product favored.
II. The reaction is reactant favored.
III. Equilibrium lies far to the right.
IV. Equilibrium lies far to the left.

A. II and III C. II and IV
B. I and III D. I and IV

Telegram: https://t.me/ExtremeSeries

Solution:

- K is less than one. This implies, formation of reactant is favored and the equilibrium lies far to the left.

Answer: C

34. Which of the following molecular orbital diagrams is correct for the carbide ion (C_2^{2-})?

A. $\sigma_{1s}^2, \sigma_{1s}^2\sigma^{*2}_{1s}, \sigma_{2s}^2, \sigma^{*2}_{2s}\pi^4_{2p}$

B. $\sigma_{1s}^2\ \sigma^{*2}_{1s}\ \sigma_{2s}^2\ \sigma^{*2}_{2s}\ \pi^4_{2p}\ \sigma_{2p}^2\ \pi^2_{2p}$

C. $\sigma_{1s}^2\ \sigma^{*2}_{1s}\ \sigma_{2s}^2\ \sigma^{*2}_{2s}\ \pi^4_{2p}\ \sigma_{2p}^2\ \pi^{*4}_{2p}$

D. $\sigma_{1s}^2\ \sigma^{*2}_{1s}\ \sigma_{2s}^2\ \sigma^{*2}_{2s}\ \pi^4_{2p}\ \sigma_{2p}^2$

Solution:

C_2^2, electrons $= 2 \times 6 + 2 = 14$

$(\sigma_{12})^2(\sigma^*_{1s})^2(\sigma_{2s})^2(\sigma^*_{2s})^2(\pi_2 py)^2 = (\pi_2 p_z)^2(\sigma_2 p_x)^2$

$\Rightarrow \sigma_{1s}^2\sigma_{1s}^2\sigma_{2s}^2\sigma^{*2}_{2s}\pi^4_{2p}\sigma^2_{2p}$

Answer: D

35. If a reaction is zero order in A, tripling the concentration of A will cause the reaction rate to:

A. Increase by a factor of 27. C. Remain constant

B. Increase by a factor of 3. D. Increase by a factor of 9.

Solution: The rate of zero – order reaction is independent on concentration.

Answer: C

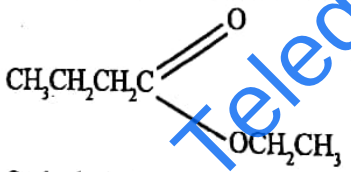
36. Which one of the following factors does **NOT** affect the rate of a chemical reaction?

A. Humidity C. Temperature

B. Concentration D. Nature of reactants

Answer: A

37. An ester has the structural formula



On hydrolysis, the ester would produce:

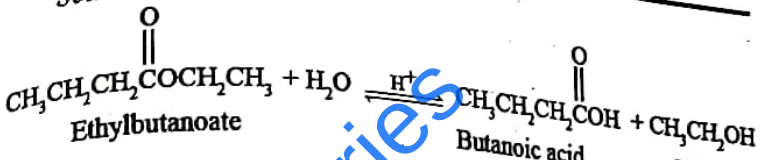
A. Propanoci acid and propan – 1 – ol

B. Ethanoic acid and butan – 1 – ol

C. Butanoic acid and ethanol

D. Propanoci acid and ethanol

Solution:



Ethylbutanoate — Butanoic acid — Ethanol

Answer: C

38. You are given a bottle of solid X and three aqueous solutions of Y, the first saturated, the second unsaturated and the third supersaturated. Which of the following is correct, if you add a small amount of the solid solute to each solution?

A. The solution in which the added solid solute dissolves is the saturated solution.

B. The supersaturated solution is unstable and addition of additional solute causes the excess solute to crystallize.

C. The solution in which the added solid solute remains un-dissolved is the unsaturated solution.

D. In all the three solution; saturated, unsaturated and supersaturated the added solid solute will dissolve.

Solution:

- Unsaturated solution dissolves the solid solute added.
- In saturated solution, the added solute remains un dissolved.
- In supersaturated solution, the size of the added solute will grow bigger.

Answer: B

39. The phrase "like dissolves like" refers to the fact that:

A. Polar solvents dissolve polar solutes and nonpolar solvents dissolve non polar solutes:

B. Polar solvents dissolve nonpolar solutes and vice versa.

C. Solvents can only dissolve solutes of similar molar mass.

D. Gases can only dissolve other gases.

Answer: A

40. For a certain gas phase reaction.

$2A(g) \rightleftharpoons B(g) + C(g)\ \Delta H = +45\ KJ/mol, K = 4.5 \times 10^{-2}$

Which of the following would be true if the temperature was increased from 25°C to 200°C?

I. The value of K would be smaller.

II. The concentration of A(g) would be increased.

III. The concentration of B(g) would increase.

A. III B. II C. I D. I and III

Solution:

Telegram: https://t.me/ExtremeSeries

- The forward reaction is endothermic thus increasing temperature favors formation of B and C.
- The value of equilibrium constant increases with increasing temperature.

Answer: A

41. Which of the following compounds would be the most soluble in H_2O?
A. Ethane C. Octanoic acid
B. Pentane D. Ethanoic acid

Solution: Carboxylic acid with smaller mass are miscible with water due to hydrogen bonding but not hydrocarbons.

Answer: D

42. Consider the following equilibrium:
$2CO_{(g)} + O_{2(g)} \leftrightarrows 2CO_{2(g)}$ $K_{eq} = 4.0 \times 10^{-10}$
What is the value of K_{eq} for $2CO_{2(g)} \leftrightarrows 2CO_{(g)} + O_{2(g)}$?
A. 4.0×10^{-10} B. 2.5×10^{9} C. 5.0×10^{4} D. 2.0×10^{-5}

Solution: If the reaction is reversed, the new K_{eq} will be the reciprocal of the original.

$$\Rightarrow k_{eq} = \frac{1}{k_{eq}} = \frac{1}{4\times10^{-10}} = 2.5\times10^{9}$$

Answer: B

43. What species of ions are present in a 0.1M solution of HCl and what will be their equilibrium concentrations?
A. $[H_3O]^+ = 0.1M$; $[OH]^- = 0.1$ M, $[Cl]^- = 0.1M$
B. $[H_3O]^+ = 0.1M$; $[OH]^- = 0.01M$, $[Cl]^- = 0.1M$
C. $[H_3O]^+ = 0.1M$; $[OH]^- = 10^{-13}M$, $[Cl]^- = 0.1M$
D. $[H_3O]^- = 10^{-13}$; $[OH]^- = 0.1M$, $[Cl]^- = 0.1M$

Solution: For strong monoprotic acid, HA, $[HA] = [H^+] = [A^{-1}]$.
$\Rightarrow$ In 0.1MHCl solution, $[H_3O^+] = [Cl^-] = 0.1M$

$$[OH^-] = \frac{K_w}{[H^-]} = \frac{1\times10^{-14}M^2}{0.1M} = 1\times10^{-13}M$$

Answer: C

44. A solution in which propionic acid is 0.94% ionized has a pH of 2.85. What is the value of the acid ionization constant (Ka) for propionic acid?
A. 4.2×10^{-4} B. 6.8×10^{-4} C. 1.3×10^{-5} D. 8.7×10^{-5}

Solution: $[H^+] = 10^{-pH} = 10^{-2.85} = 1.41 \times 10^{-3}M$

$$Ka = \frac{(1.41\times10^{-3})^2}{0.15-0.00141} = 1.3\times10^{-5}$$

Answer: C

45. A 0.2M solution of a weak acid HA is 1% ionized at 25°C. K_a for the acid is equal to:
A. $\frac{0.19}{0.01\times0.01}$ C. $\frac{0.01\times0.01}{0.19}$
B. $\frac{0.02\times0.02}{0.18}$ D. $\frac{0.002\times0.002}{0.198}$

Solution: $[H^+] = 1\% \times [HA]_o = \frac{1\times0.2M}{100} = \frac{0.2M}{100}$

$$K_a = \frac{[H^+]^2}{[HA]_o - [H^+]} = \frac{\left(\frac{0.2}{100}\right)^2}{0.2-\frac{0.2}{100}} = \frac{\frac{0.2\times0.2}{100\times100}}{\frac{20-0.2}{100}} = \frac{0.02\times0.02}{19.8}$$

$$Ka = \frac{0.002\times0.002}{0.198} = 2.02\times10^{-5}$$

Answer: D

46. 2.3 g of ethanol (CH_3CH_2OH) is added to 500g of water. What is the molality of the resulting solution?
A. 0.01m B. 1.0m C. 0.1m D. 10.0m

Solution: $m = \frac{m_{solute}}{M_{solute}.kg\ of\ solvent}$

$$m = \frac{2.3g}{46g/mol\times0.5kg} = 0.1m$$

Answer: C

47. A 0.5L and 0.1M HNO_3 solution is to be prepared by dilution process from a 13M nitric acid. How many mL conc. HNO_3 and how many mL of water are required to prepare the dilute solution?
A. 3.85 mL HNO_3 and 496.15 mL water
B. 2 mL HNO_3 and 498 mL water
C. 10mL HNO_3 and 490 mL water
D. 20 mL HNO_3 and 490 mL water

Solution: $V_1 = ?$ $V_2 = 0.5L$
$M_1 = 13M$ $M_2 = 0.1M$

$$V_1 = \frac{M_2V_2}{M_1} = \frac{0.1M\times0.5L}{13M} = 3.85\times10^{-3}L = 3.85mL$$

$$V_{H_2O\ added} = V_2 - V_1 = 500mL - 3.85mL = 496.15mL$$

Answer: A

Telegram: https://t.me/ExtremeSeries

48. Commercial concentrated sulfuric acid (density = 1.831 g cm^{-3}) is 94.0% H_2SO_4, by mass. What is the normality of H_2SO_4 solution? (Molar mass of H_2SO_4 = 98 g mol^{-1})
A. 16.8M B. 28.2 C. 40.4M D. 35.0N

Solution: $N = Z.M = \frac{Z\% \times d \times 10}{M_{H_2SO_4}} = \frac{2 \times 94 \times 1.831 \times 10}{98} = 35.1N$

$H_2SO_4 \rightarrow 2H^+ + SO_4^{2-}, Z = 2$ **Answer: D**

49. During the electrolysis of a concentrated aqueous solution of NaCl, what substance is formed at the cathode?
A. Hydrogen C. Chlorine
B. Oxygen D. Sodium
Answer: A

50. In a voltaic cell, electrons flow from the _____ to the _____.
A. Salt bridge, anode C. Anode, slat bridge
B. Salt bridge, cathode D. Anode, cathode
Solution: In both galvanic (voltaic) and electrolytic cell, electrons flow from anode to cathode in the external circuit.
Answer: D

51. During the electrolysis of $CuSO_4$(aq) using carbon electrodes, which of the following is the correct half reaction for the anode electrode?
A. $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$
B. $4OH^-(aq) \rightarrow 2H_2O(l) + O_2(g) + 4e^-$
C. $2H_2O(l) \rightarrow O_2(g) + 4H^+(aq) + 4e^-$
D. $SO_4^{2-}(aq) + O_2(g) + 4H^+(aq) \rightarrow SO_4^-(aq) + 2H_2O(l)$ **Answer: B**

52. Which of the following statements is true?
A. Addition of $(CH_3NH_3)Cl$ to a solution of CH_3NH_2 will decrease the pH.
B. Addition of $NaNO_2$ to a solution of HNO_2 will decrease the pH.
C. Addition to HCl to a solution of $NaC_2H_3O_2$ will increase the pH.
D. Addition of KBr to a solution of HBr will increase the pH.
Solution:
- Addition of acidic salt or acid to a certain solution will decrease the pH but increase the pOH.
- Addition of basic salt or base to a given solution will increase its pH but decrease its pOH.

- Neutral salts will not affect the pH or pOH of a solution.

Answer: A

53. What is the final temperature when 150.0 mL of water at 90.0 °C is added to 100.0mL of water at 30.0°C?
A. 33.0°C B. 45.0°C C. 66.0°C D. 60.0°C
Solution: Heat lost by hot water = heat gained by cold water
⇒ The initial temperature of hot water is the same as, the final temperature of cold water. Let it be T.
$m_1 = d \times v_1$ and $m_2 = d. v_2$
⇒ $m_1 C \Delta T_1 = m_2 C \Delta T_2$, $\Delta T_1 = 90 - T$, $\Delta T_2 = T - 30$
$dV_1 C\Delta T_1 = dv_2 C\Delta T_2$, $V_1(90 - T) = V_2(T - 30)$
$150 (90 - T) = 100 (T - 30)$
Solving for T, T = 66°C **Answer: C**

54. What is the pH of a mixture of 15.0 mL of 0.126M NaOH and 21.0M H_2SO_4?
A. 1.70 B. 13.60 C. 11.81 D. 2.15
No answer because the volume of NaOH is not given.

55. The relationship between picometer (pm) and nanometer (nm) is:
A. 1 pm = 10 nm C. 1 pm = 100nm
B. 1 nm = 1000 pm D. 1 nm = 10 pm

Solution: $\frac{1pm}{1nm} = \frac{10^{-12}m}{10^{-9}nm}$

$\frac{1pm}{1nm} = 10^{-3} \Rightarrow 1pm = 10^{-3}nm$ or $1nm = 10^3 pm$ **Answer: B**

56. What would be the wavelength of a radio wave having a frequency of 3 MHz?
A. 300nm B. 300m C. 100nm D. 100m

Solution: $\lambda = \frac{C}{V} = \frac{3 \times 10^8 m/s}{3 \times 10^6 s^{-1}} = 100m$ **Answer: D**

57. Which of the following correctly lists electromagnetic waves in order from shortest to longest wavelength?
A. Microwaves, ultraviolet, visible light, gamma rays
B. Radio waves, infrared, gamma rays, ultraviolet
C. Gamma rays, ultraviolet, infrared, microwaves
D. Gamma rays, infrared, ultraviolet, microwaves
Solution:
- Based on wavelength: Gamma rays < x – rays < ultraviolet < visible < infrared < microwaves < radio wave **Answer: C**

Telegram: https://t.me/Extremeseries

58. Which of the following is **NOT** the decomposition product of HNO_3?
A. N_2O_4 B. NO_2 C. O_2 D. H_2O

Solution: $4HNO_3 \xrightarrow{\Delta} 4NO_2 + 2H_2O + O_2$ **Answer: A**

59. To determine the volume of an irregularly shaped glass vessel, the vessel is weighed empty (121.3 g) and when filled with CCl_4 (283.2 g). What is the volume capacity of the vessel, given that the density of CCl_4 is 1.59 g/cm^3?
A. 76.29cm^3 B. 257.42cm^3 C. 178.11 cm^3 D. 101.82 cm^3

Solution: $m_{CCl_4} = m_{total} - m_{glass} = 283.2g - 121.3g = 161.9g$

$$V = \frac{m}{d} = \frac{161.9g}{1.59g/cm^3} = 101.82cm^3$$ **Answer: D**

60. The process of vulcanization of rubber makes it _____.
A. Less elastic C. Hard
B. Soft D. More soluble in sovlents

Answer: C

61. For the reaction $C_6H_{14}(g) \rightarrow C_6H_6(g) + H_2(g)$, $\Delta P(H_2)/\Delta t$ was found to be 2.5×10^{-2} atm/s, where $\Delta P(H_2)$ is change in the pressure of hydrogen. Determine $\Delta P(C_6H_{14})/\Delta t$ (in units of atm/s) for this reaction at the same time.
A. -6.2×10^{-3} B. 1.6×10^{-3} C. 2.5×10^{-2} D. 6.2×10^{-3}

Solution: $\Delta P(C_6H_{14})/\Delta t = \frac{\text{coefficient of } C_6H_{14}}{\text{coefficient of } H_2} \times \Delta P(H_2)/\Delta t$

$$= \frac{1}{4} \times 2.5 \times 10^{-2} \text{ atm/s} = 6.25 \times 10^{-3} \text{ atm/s}$$ **Answer: D**

62. The decomposition of a compound at 400°C is first order with a half-life of 1570 seconds. What fraction of an initial amount of the compound remains after 4710 seconds?
A. 1/12 B. 1/6 C. 1/8 D. 1/3

Solution: $t_{1/2} = \frac{0.693}{k}$ $\quad k = \frac{0.693}{1570s}$

$$k = \frac{0.693}{t_{1/2}}$$

$$\log\frac{[A]_0}{[A]_t} = \frac{kt}{2.303}$$

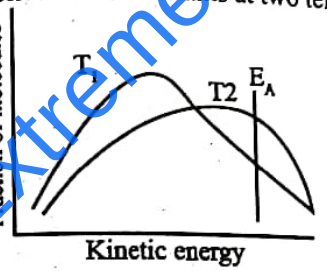
$$\frac{[A]_0}{[A]_t} = 10^{\frac{kt}{2.303}} = 10^{\frac{\frac{0.693}{1570} \times 4710}{2.303}} = 8$$

$$[A]_0 = 8[A]_t$$

$$[A]_t = \frac{1}{8}[A]_0$$

Answer: C

63. The diagram below shows the range of energies of collision of a collection of reactants at two temperatures. T_1 and T_2.



Which of the following is true regarding T1 and T2?
A. T1 = T2, fraction of molecules at both temperatures are equal.
B. T1 < T2, fraction of molecules at T1 is smaller.
C. T2 < T1, fraction of molecules at T2 is smaller.
D. T1 < T2, fraction of molecules at T1 is larger.

Answer: A

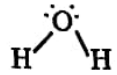
64. From CO_2, H_2O, $BeCl_2$ and N_2O which have the same molecular geometry?
A. CO_2, $BeCl_2$, and N_2O C. CO_2, H_2O, and N_2O
B. CO_2 and $BeCl_2$ only D. H_2O and N_2O only

Solution:

CO_2: $\ddot{O}=C=\ddot{O}$, linear

$BeCl_2$: $:\ddot{Cl}-Be-\ddot{Cl}:$, linear

N_2O: $:\ddot{N}=N=\ddot{O}$, linear

H_2O:  **Answer: A**

65. How many electrons are present in the σ_{2p} molecular orbital of N_2^+?
A. 1 B. 4 C. 3 D. 2

Telegram: https://t.me/ExtremeSeries

Solution: N_2^+ : electrons $= 2\times7-1=13$

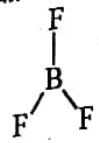
$\sigma_{1s}^2\ \sigma_{1s}^{*2}\ \sigma_{2s}^2\ \sigma_{2s}^{*2}\ \pi_{2py}^2\ \pi_{2pz}^2\ \sigma_{2px}^1$

Answer: A

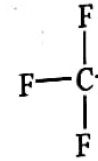
66. Given the following AF_n species, BF_3, BeF_2, CF_4, NF_3, OF what is the correct of F – A – F bond angles?

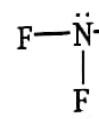
A. $OF_2 < BeF_2 < NF_3 < BF_3 < CF_4$
B. $CF_4 < BF_3 < NF_3 < BeF_2 < OF_2$
C. $OF_2 < NF_3 < CF_4 < BF_3 < BeF_2$
D. $BeF_2 < OF_2 < NF_3 < BF_3 < CF_4$

Solution:

BF_3:  , Bond angle = 120°

BeF_2: F—Be—F , bond angle = 180°

CF_4:  , bond angle = 109.5°

NF_3:  , bond angle = 107°

OF_3: , bond angle = 104.5°

Answer: C

67. Which of the following is true regarding the solution formation process?

A. Intermolecular forces between the solute particles must weaken in which the enthalpy change is exothermic ($\Delta H<0$).
B. Intermolecular forces between the solvent molecules must weaken in which the enthalpy change in exothermic ($\Delta H<0$).
C. Covalent bonds within the solute and solvent molecules must be broken.

D. New Columbic attractions between the solute and solvent form in which the enthalpy change is exothermic ($\Delta H<0$).

Answer: D

68. What volume of 1.40M H_2SO_4 solution is needed to react exactly with 10.0g of aluminum according to the following reaction?

$2Al(s) + 3H_2SO_4(aq) \rightarrow Al_2(SO_4)_3(aq) + 3H_2(g)$

A. 2.643mL B. 26.43mL C. 2643mL D. 264.3mL

Solution: The balanced chemical equation given is wrong correct it. By mass – mass relationship, find the mass of H_2SO_4.

$$\underset{2\times27}{\overset{10g}{2Al(s)}} + \underset{3\times98}{\overset{x}{3H_2SO_4}} \longrightarrow Al_2(SO_4)_3(aq) + 3H_2(g)$$

$$\frac{10g}{2\times27} = \frac{x}{3\times98}$$

$$x = m_{H_2SO_4} = 54.4g$$

$$n_{H_2SO_4} = \frac{m}{M} = \frac{54.4g}{98g/mol} = 0.56mol$$

$$\therefore V_{H_2SO_4} = \frac{n_{H_2SO_4}}{M_{H_2SO_4}} = \frac{0.56\ mol}{1.40\ mol/L} = 0.397L = 397mL$$

Answer: _____

69. A solution of NH_4Cl made by dissolving 3.16g NH_4Cl in 30.14g H_2O has a density of 1.0272g cm^{-3}. What is the mole fraction of NH_4Cl?

A. 0.0341 B. 0.9659 C. 0.6500 D. 0.2100

Solution: $n_{NH_4Cl} = \frac{m}{M} = \frac{3.16g}{53.5g/mol} = 0.059mol$

$$n_{H_2O} = \frac{30.14}{18} = 1.67$$

$$X_{NH_4Cl} = \frac{n_{NH_4Cl}}{n_{NH_4Cl} + n_{H_2O}} = \frac{0.059}{0.059 + 1.67} = 0.0341$$

Answer: A

70. Consider the following equilibrium:

$Cl_2(g) + 2NO(g) \leftrightarrows 2NOCl(g)$ $K_{eq} = 5.0$, at equilibrium $[Cl_2] = 1M$ and $[NO] = 2M$ what is $[NOCl]$?

A. 4.5M B. 0.89 C. 0.80M D. 10M

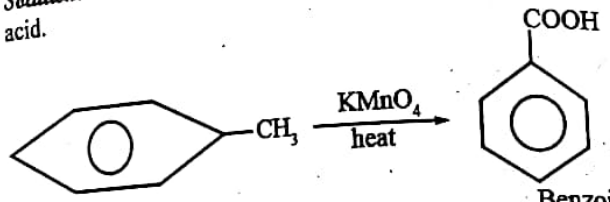
Solution: $k_{eq} = \frac{[NOCl]^2}{[Cl_2][NO]^2}$

Telegram: https://t.me/Extremeseries

$[NOCl] = \sqrt{k_{eq}[Cl_2][NO]^2} = \sqrt{5 \times 1 \times (2)^2} = 4.47M$ **Answer: A**

71. Which acid is produced when toluene is subjected to $KMnO_4$ oxidation?
A. Toluic acid C. Phenyl acetic acid
B. Benzoic acid D. Phthalic acid

Solution: Alkyl benzene reacts with $KMnO_4$ to yield benzoic acid.



Toluene (Methyl benzene) Benzoic acid

Answer: B

72. Which of the following statements is **NOT** true in relation to the triple point on a single component phase diagram?
A. The point at which the solid, liquid and gaseous phases for a substance co-exist.
B. The system must be enclosed so that no vapour can escape.
C. The triple point exists for a substance occurs at a specific temperature and pressure.
D. The triple point exists at a single temperature and is independent of pressure.

Solution: At the triple point:
- Solid, liquid and gaseous phases coexist at equilibrium.
- The system must be enclosed so that no vapor can escape.
- A substance occurs at a specific temperature and pressure.
- The degree of freedom is zero (invariant system).

Answer: D

73. Calculate ΔH for the following reaction using the bond energies given below:
$H-H(g) + I-I(g) \rightarrow 2H-I(g)$
Bond energies: H – H = 436 kJ/mol, I – I = 151 kJ/mol, H – I = 297 kJ/mol
A. +290kJ B. -7kJ C. +7kJ D. -290kJ

Solution: $\Delta H = \sum$ bond broken $- \sum$ bond formed

$= (1 \times H-H + 1 \times I-I) - (2 \times H-I)$
$= (436 + 151) - (2 \times 297) = -7kJ$

74. Two electrolytic cells were placed in series, one of $AgNO_3$ and the other of $CuSO_4$. If 1.273g of Ag is deposited, how much Cu was deposited at the same time? **Answer: B**
A. 0.0118 B. 0.954 C. 0.748 D. 0.374

Solution: $\frac{m_{Cu}}{E_{Cu}} = \frac{m_{Ag}}{E_{Ag}}$

$m_{Cu} = \frac{m_{Ag}}{E_{Ag}} E_{Cu} = \frac{1.273g}{108} \times 32 = 0.377g$ **Answer: D**

75. What is the standard cell potential for the oxidation of ammonia, given below? $4NH_3 + 3O_2 \rightarrow 2N_2 + 6H_2O$
A. 3.51V B. 1.17V C. 7.02V D. 14.04V

Solution: $4N^{3-} \rightarrow 2N_2 + 12e^-$
⇒The number of electrons lost is 12(n = 12)
$\Delta G = -nE^\circ_{cell}F$

$E^\circ_{cell} = \frac{\Delta G}{-nF} = \frac{-356 \times 10^3 J}{-12 \times 96500 J/V} = 1.17V$ **Answer: B**

76. What is the freezing point of a 0.1 m solution of NaCl (in °C)? (K_f = 1.86°C/m and the freezing point of pure water is 0°C).
A. -1.86 B. -0.186 C. -0.372 D. -0.093

Solution: $T_f = T_f^\circ - imk_f$, i for NaCl = 2
$= 0°C - 2 \times 0.1m \times 1.86°C/m = -0.372°C$

Answer: C

77. During a titration, what volume of 0.500M KOH is necessary to completely neutralize 10.0mL of 2.00M CH_3COOH?
A. 10.0mL B. 20mL C. 40.0mL D. 25.0mL

Solution: $n_{CH_3COOH} = M_{CH_3COOH} \cdot V_{CH_3COOH} = \frac{2mol}{2} \times 10mL = 20mmol$

By mole – mole relationship, find the number of mole of kOH.

$\underset{1}{\overset{20mmol}{CH_3COOH}} + \underset{1}{\overset{x}{KOH}} \longrightarrow CH_3COOK + H_2O \Rightarrow n_{KOH} = 20mmol$

$V_{KOH} = \frac{n_{KOH}}{M_{KOH}} = \frac{20mmol}{0.5mol/L} = 40mL$ **Answer: C**

Telegram: https://t.me/ExtremeSeries

78. Which of the following is an organic acid?
A. CH_3CO_2H B. CH_3CH_2OH C. $CH_2=CH_2$ D. CH_3CH_3
Solution: An organic acid contains carboxyl group (–COOH or $-CO_2H$) as a functional groups. **Answer: A**

79. Which of the following statements is true?
A. If the entropy of the system increases during a reversible process, the entropy change of the surroundings will decrease by the same amount.
B. If the entropy of the system increases during a reversible process, the entropy change of the surroundings will also increase by the same amount.
C. If the entropy of the system increases during a reversible process, the entropy change of the surroundings will remain the same.
D. If the entropy of the system decreases during a reversible process, the entropy change of the surroundings will also decrease by the same amount.

Solution: For a reversible process, $\Delta S_{univ} = \Delta S_{syst} + \Delta S_{surr} = 0$
$\Rightarrow \Delta S_{system} = -\Delta S_{surrounding}$ **Answer: A**

80. How long has a current of 3 ampere to be applied through a solution of silver nitrate to a coat a metal surface of 80 cm^2 with 0.05 cm thick layer? Density of silver = 10.5 g cm^3
A. 476s B. 1028s C. 1252s D. 683s
Solution: $m_{Ag} = d_{Ag}.\ V_{Ag} = 10.5\ g/cm^3 \times (80cm^2 \times 0.005cm)$
$m_{Ag} = 4.2g$, $Ag^+ + e^- \rightarrow Ag$
$z = 1$

$$m = \frac{MIt}{ZF}$$

$$t = \frac{mZF}{MI} = \frac{4.2g \times 1 \times 96500A\,s/mol}{108g/mol \times 3A}$$

$t = 1251$ sec **Answer: C**

SECTION 3 – PHYSICS

1. Two non – zero vectors $\vec{A}$ and $\vec{B}$ are related by $\vec{A} = c\vec{B}$, where c is a scalar. If the two vectors have opposite directions, then one of the following is true about c?
A. c is a positive number C. $c = 1$
B. c is a negative number D. $c = 0$
Solution:
When a vector is multiplied by a positive scalar only its magnitude is changed. When a vector is multiplied by a negative scalar both the magnitude and direction changed. When $\vec{A}$ and $c\vec{B}$ are in opposite direction then the scalar c is a negative number.
Answer: B

2. On which of the following does the induced emf in a coil NOT depend when a magnet is inserted into it?
A. Resistance of the coil.
B. Number of turns in the coil.
C. Speed of approach of the magnet.
D. Magnetic moment of the magnet.
Solution:
The induced electromotive force (emf) in a coil during electromagnetic induction depends on:
- The number turns of the coil.
- The speed of the approaching magnet.
- The magnetic field strength.
- The orientation of the coil to the field.

Answer: D

3. A wave has a speed of 50 m/s and a wavelength of 25,000 m. What is the frequency of the wave?
A. 2.0×10^{-3} Hz C. 5.0×10^{2} Hz
B. 6.0×10^{-1} Hz D. 1.8×10^{6} Hz
Solution: $V = 50$ m/s, $\lambda = 25000$ m
$f = ?$
Using the wave equation:

$$V = \lambda f \Rightarrow f = \frac{V}{\lambda} = \frac{50\ m/s}{25000\ m}$$

$f = 2 \times 10^{-3}$ Hz **Answer: A**

Telegram: https://t.me/ExtremeSeries

4. An oscillating system is used to create a succession of waves on a water surface. The rate of wave generation is one every 0.33 seconds. What is the period of these waves?
   A. 0.3 seconds
   B. 1.0 seconds
   C. 2.0 seconds
   D. 3.0 seconds

Solution:
If it generates one wave in every 0.33 second this implies in one second three wave can be produced. So f = 3 Hz

$$T = \frac{1}{f} = \frac{1}{3Hz} = 0.3\,sec$$

Answer: A

5. A vector that represents the position of an object in relation to another object is called
   A. Unit vectors
   B. Position vector
   C. Coplanar vectors
   D. Collinear vectors

Solution:
The vector that represents the position of an object in relation to another object is called position vector. **Answer: B**

6. Consider a series RC circuit for which R = 1.0MΩ, C = 5.0μF and ε = 30.0 V. What is the time constant of the circuit and the maximum charge on the capacitor after the switch is closed respectively?
   A. 5 seconds, 150μC
   B. 7 seconds, 155μC
   C. 10 seconds, 160μC
   D. 9 seconds, 135μC

Solution:
$R = 1Mn = 1 \times 10^6\Omega$
$C = 5\mu F = 5 \times 10^{-6}\ F$
$\varepsilon = 30V$
The time constant 'τ' is given by $\tau = RC\ \tau = 10^6\Omega \times 5 \times 10^{-6} = 5$ second and the maximum charge is $Q = \varepsilon \times C = 30 \times 5 \times 10^{-6}$ $F = 150 \times 10^{-6}C = 150\mu C$
Answer: A

7. What is the direction to which a fish must push the water with its fins in order to propel eastward?
   A. Eastward
   B. Upward
   C. Westward
   D. Downward

Solution:

When the fish needs to propel eastward. It must push the water west ward. This is the consequence of "For every action force there exist an equal and opposite reaction." **Answer: C**

8. Which of the following is correct about the meaning of Photon?
   A. Quantum of matter.
   B. Positively charged particle.
   C. Negatively charged particle.
   D. Quantum of light.

Solution:
A photon is defined as a quantum of energy photons are natural.
Answer: C

9. Suppose two molecules of different mass are kept at the same temperature. Which one of the following statements is NOT true?
   A. Both remain at rest.
   B. Both move with the same speed.
   C. The lighter molecule moves faster.
   D. The heavier molecule moves faster.

Solution:
The molecules have different mass. Let m_1 is the mass of the first molecule and m_2 is the mass of the second molecule

$$m_1 \neq m_2$$

They are at the same temperature at the same temperature, they have the same kinetic energy.

$$E_{k_1} = E_{k_2}$$

Since they have different masses. The higher molecule must move faster than the heavier molecules. Probably both maybe at rest so B and D are necessary false. A = can be true or false and C is true.
Answer: B and D

10. One person drops a ball from the top of a building while another person at the bottom observes its motion. The two people **DISAGREE** on which of the following?
   A. The value of total energy of the ball.
   B. The change in potential energy of the ball.
   C. The value of the gravitational energy of the ball Earth system.
   D. The kinetic energy of the ball at some point in its motion if one of them is walking. S

Solution:

Telegram: https://t.me/ExtremeSeries

To observe the motion of the ball by each of them both must be at the same reference. i.e, both person should be at rest. If one of the person is moving, the speed of the ball, therefore the kinetic energy are different as observe by both person.

Answer: D

11. An air – filled parallel-plate capacitor has a capacitance of 1 pF. The plate separation is then doubled and a wax dielectric is inserted, completely filling the space between the plates. As a result, the capacitance become 2 pF. What is the dielectric constant of the wax?

A. 8.0 B. 4.0 C. 2.0 D. 0.5

Solution:

Let the capacitance initially was $C_1 = 1pF$ the plate separation is d_1 after doubling the separation $d_2 = 2d$, the capacitance c_2 becomes $c_2 = 2pF$, $k = ?$

Initially the capacitance is $c_1 = \frac{\varepsilon_o A}{d_1}$.

Finally, $c_2 = \frac{\varepsilon_o A}{d_2}, d_2 = 2d$

$$c_2 = \frac{\varepsilon_o kA}{2d_1}$$

$$2PF = \frac{\varepsilon_o kA}{2d_1}$$

$$2PF = \frac{1}{2}\left(\frac{A}{d_1}\right) \text{ but } \frac{EA}{d_1} = c_1 = 1\mu F$$

$$2PF = \frac{k}{2}\left(\frac{\varepsilon_o A}{d_1}\right)$$

$$2PF = \frac{k}{2}c_1$$

$$2PF = \frac{k}{2}(1PF)$$

$$k = \frac{2PF \times 2}{1PF} = 4$$

$k = 4$ **Answer: B**

12. A step – up transformer is used to

A. Increase the power C. Increase the voltage

B. Decrease the power D. Decrease the voltage

Solution:

A step up transformer used to increase the voltage supplied. For step up transformer, the voltage in the secondary coil is larger than the voltage in the secondary coil.

$V_s > V_P$ and also $N_s > N_P$

For step down transformer $V_s < V_P$ and $N_s < N_P$ **Answer: C**

13. A 200 – V power line is protected by a 15 – A fuse. What is the maximum number of "220V, 500W" light bulbs that can be operated at full brightness from the line?

A. 0.2 B. 0.4 C. 0.6 D. 0.8

Solution:

The current tolerated by the fuse is 15A. The current supplied by the line is $P = VI$.

$$I = \frac{P}{V} = \frac{500W}{220V}, I = 2.27\ A$$

So $n \times 15A = 2.27\ A$

$$n = \frac{2.27A}{15A} = 0.153 \approx 0.2$$ **Answer: A**

14. In a reversible thermodynamic process, the system

A. Is always close to equilibrium states.

B. Might never be close to any equilibrium state.

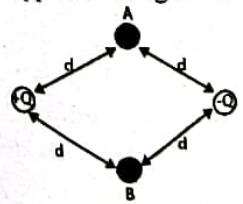
C. Is close to equilibrium states only at the beginning and end

D. Is close to equilibrium states throughout, except at the beginning and end

Solution:

In a reversible thermodynamic process, the system is always close the equilibrium state. **Answer: A**

15. Points A and B are each a distance d from each of two equal and opposite charges as shown below.



Telegram: https://t.me/ExtremeSeries

What will be the work required to move a negative charge q from A to B?

A. Zero

B. $\frac{qQ}{4\pi\varepsilon_o d^2}$

C. $\frac{qQ}{4\pi\varepsilon_o d}$

D. $\frac{qQ}{4\pi\varepsilon_o (4d^2)}$

Solution:

The potential at point A is

$$V_A = \frac{k(+Q)}{d} + \frac{(k)(-Q)}{d} = \frac{kQ}{d} - \frac{kQ}{d}$$

$V_A = 0$

$$V_B = \frac{k(+Q)}{d} + \frac{k(-Q)}{d} = \frac{kQ}{d} - \frac{kQ}{d}$$

$V_B = 0$

The potential difference between points A and B is $V_{AB} = V_A - V_B = 0$. Then the work done is $W = Q\Delta V = Q(0) = 0$,

W = zero

Answer: A

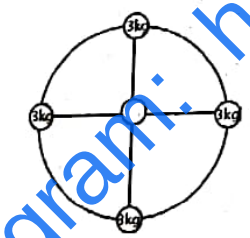
16. The radius of the wheel shown below is 0.75m. What is its moment of inertia?

A. 2.25 kgm^2

B. 3.38 kgm^2

C. 6.75 kgm^2

D. 9.00 kgm^2



Solution:

r = 0.75m, I =?

The total moment of inertia is the sum of the moment of inertia due to each mass.

$I = m_1r_1^2 + m_2r_2^2 + m_3r_3^2 + m_4r_4^2$ but $m_1 = m_2 = m_3 = m_4 = 3$ kg

$r_1 = r_2 = r_3 = r_4 = 0.75$m

$I = mr^2 + mr^2 + mr^2 + mr^2 = 4mr^2 = 4\ (3\ kg)\ (0.75m)^2 = 6.75\ kg$

Answer: C

17. An ordinary refrigerator is turned on and its door is left open in a thermally insulated kitchen. What is the temperature of the room?

A. Increase according to the first law of thermodynamics.

B. Decreases according to the first law of thermodynamics.

C. Remains constant according to the first law of thermodynamics.

D. Remain constant according to the second law of thermodynamics.

Solution:

As the room thermally insulted, there is NO transfer of heat energy between the room and the refrigerator. This is due to the first law of thermodynamics.

Answer: C

18. If $\vec{A} = 8\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{B} = 5\hat{i} - 3\hat{j} + \hat{k}$, then, what is the value of $\vec{B} \times \vec{A}$?

A. $3\hat{i} + 4\hat{j} - 3\hat{k}$

B. $-3\hat{i} - 4\hat{j} + 3\hat{k}$

C. $\vec{A} = -5\hat{i} - 18\hat{j} - 29\hat{k}$

D. $\vec{A} = 5\hat{i} + 18\hat{j} + 29\hat{k}$

Solution:

$\vec{A} = 8\hat{i} + \hat{j} - 2k$

$\vec{B} = 5\hat{i} - 3\hat{j} + k$

B × A =?

$B \times A = (5\hat{i} - 3\hat{j} + k) \times (8i + j - 2k) = 5\ (i \times j) + (-10)(i \times k) + -24\ (j \times i) + 6(j \times k) + 8\ (k \times i) + 1\ (k \times j) = 5k - 10(-j) - 24\ (-k) + 6\ (2) + 8j + 1\ (-i)$

$\vec{B} \times \vec{A} = 6\hat{i} - \hat{i} + 10\hat{j} + 8\hat{j} + 5k + 24k$

$\vec{B} \times \vec{A} = 5\hat{i} + 18\hat{i} + 29k$

Answer: D

19. The coordinate of a particle in meters is given by $x(t) = 25t - 3t^3$, where the time t is in second. At what value t will the particle become momentarily at rest?

A. 2.78s B. 1.67s C. 0.6s D. 0.36s

Solution:

$x(t) = 25t - 3t^3$

The velocity of the particle at any time 't' is:

$$V = \frac{dx}{dt} = \frac{d(25t - 3t^3)}{dt}$$

$V = 25 - 9t^2$

Telegram: https://t.me/ExtremeSeries

$\Rightarrow V = -9t^2 + 25$

To find the time at which the particle comes to rest i.e., the velocity is zero.

$V = -9t^2 + 25$

$O = -9t^2 + 25 \Rightarrow 9t^2 = 25$

$$t^2 = \frac{25}{9}$$

$$t = \sqrt{\frac{25}{9}} = \frac{5}{3}\text{sec}$$

$t = 1.67$ seconds

Answer: B

20. A woman driving a 2000 kg car along a level road at 30 m/s takes her foot off the gas to see how far her car will roll before it slows to a stop. She discovers that it takes 150m. What is the average force of friction acting on the car?

A. 9000N B. 6000N C. 3000N D. 400N

Solution:

$m = 2000$ kg ; $V_i = 30$ m/s; $V_f = 0$; $S = 150$m

$F_{av} = ?$

The acceleration of the car is

$$a = \frac{V_f^2 - V_i^2}{2s} \quad a = \frac{V_i^2}{2s} = \frac{(30\text{m/s})}{2\times 150\text{m}} = \frac{900\text{m}^2/\text{s}^2}{300\text{m}}$$

$a = 3$ m/s^2

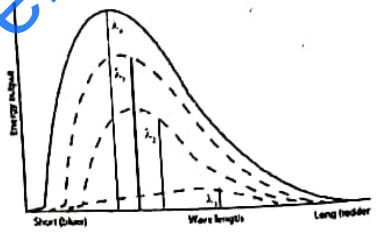
Then From Newton's second law, the force is

$F = ma = 2000$ kg $\times$ 3 m/s^2

$F = 6000$ N

Answer: B

21. The energy spectrum emitted by a black body at different temperature is as shown in the figure below. Which of the following statements is NOT correct about the black body energy spectrum?



A. $T_1 < T_2 < T_3 < T_4$

B. $\lambda_1 > \lambda_2 > \lambda_3 > \lambda_4$

C. Black body radiate energy over the entire electromagnetic spectrum.

D. The wavelength corresponding to the maximum intensity with temperature.

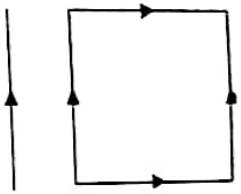
Solution:

As shown in the figure the energy of the spectrum varies directly with the temperature. The energy depends on the intensity of light and frequency and the maximum energy corresponds to the shorter wave length.

Answer: D

22. A rectangular loop carrying a current is situated near a long straight wire such that the wire is parallel to one of the sides of the loop and it is in a plane of the loop. If a steady current is established in the wire as shown in figure, the loop will

A. Remain stationary
B. Move toward the wire
C. Move away from the wire
D. Rotate about an axis parallel to the wire



Solution:

The figure show, the direction of the current in the wire and the side of the loop and the wire are parallel, there is attraction between them the loop attracts the wire and so does the wire i.e., the loop moves towards the wire.

Answer: B

23. Which device uses the principle of electromagnetic induction for its operation?

A. AC generator C. Cathode ray tube
B. Moving coil meter D. Light emitting diode

Solution:

AC generator works with the principle of electromagnetic induction.

Answer: A

24. An electron enters a region of uniform electric field of 5×10^3 N/C. What is the force on the electron?

A. 8×10^{-22} N C. 8×10^{-16} N
B. 5.3×10^{-22} N D. 1.88×10^{22} N

Solution:

$E = 5 \times 10^3$ N/C

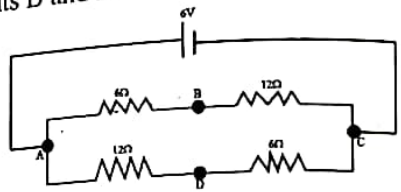
$Q = e = 1.6 \times 10^{-19}$ C

Telegram: https://t.me/ExtremeSeries

$F = EQ = 5 \times 10^3 N/C \times 1.6 \times 10^{-19} C = 8 \times 10^{-16} N$ is the force on the electron.

Answer: C

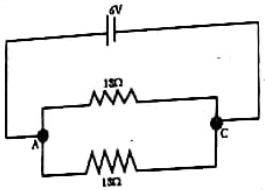
25. In the circuit shown below, what is the potential difference between points B and D?



A. 1V B. 2V C. 3V D. 4V

Solution:

The circuit can be reduced as:



The total resistance is $\frac{1}{R_T} = \frac{1}{18\Omega} + \frac{1}{18\Omega}$

$\Rightarrow R_T = \frac{2}{18\Omega} \Rightarrow R_T = 9\Omega$ are the total current is

$I = \frac{V}{R} = \frac{6V}{9\Omega} = \frac{2}{3}A$ and the current through each of the 18Ω is

$\frac{2}{3}A$ so the current through each resistor in the current is $\frac{1}{3}A$.

The voltage across point B is $V_B = IR = \frac{1}{3}A \times 6\Omega = 2V$ and

voltage across point D is $V_D = IR = \frac{1}{3}A \times 12\Omega = 4V$

The potential difference between B and D is:

$\Delta V = V_D - V_B = 4V - 2V = 2V$

Answer: B

26. A 1.5 kg particle moves in the xy-plane with velocity $v = (4.2\hat{i} - 3.6\hat{j})$ m/s. What is the angular momentum when its position vector is $r = (1.5\hat{i} + 2.2\hat{j})$ m?

A. $-22\hat{k}\ kgm^2/s$ C. $-14.6\hat{k}\ kgm^2/s$

B. $14.6\hat{k}\ kgm^2/s$ D. $22\hat{k}\ kgm^2/s$

Solution:

m = 1.5 kg; $\vec{v} = (4.2\hat{i} - 3.6\hat{j})$ m/s; $\vec{r} = (1.5\hat{i} + 2.2\hat{j})$m

$\vec{L} = ?$

The angular momentum of a particle of mass m is given by:

$\vec{L} = m(\vec{r} \times \vec{v}) = 1.5\ kg\ (1.5i + 2.2j) \times (4.2i - 3.6j)$

$= 1.5\ kg\ (-5.4k + (-9.24k))m^2/s$

$= 1.5\ kg\ (-14.64\hat{k})\ m^2/s = -21.96\ kkgm^2/s \approx -22\ \hat{k}\ kg\ m^2/s$

Answer: A

27. A very large open tank is full of oil whose density is 800 kg/m^3. If the absolute pressure at the bottom of the tank is 1.81×10^5 Pa, then what is the depth of the oil? (Note that atmospheric pressure is 1.01×10^5 Pa).

A. 100.0m B. 75.0m C. 10.0m D. 9.5m

Solution:

$\rho = 800\ kg/m^3$, $P_{tot} = 1.81 \times 10^5$ Pa

$P_{atm} = 1.01 \times 10^5$Pa

h = ?

The total pressure at the bottom of the tank is:

$P_{total} = P_{atm} + \rho gh \Rightarrow \rho gh = P_{tot} - P_{atm}$

$$h = \frac{P_{tot} - P_{atm}}{\rho g} = \frac{1.8\times10^5 Pa - 1.0\times10^5 Pa}{800\ kg/m^3 \times 10\ m/s^2}$$

$$= \frac{8\times10^4 Pa}{8\times10^3 kg/m^2s^2}$$

h = 10m

Answer: C

28. The length of an aluminum rod at a temperature of 20°C is 3m. What will be the ratio of change in length of the rod to its original length if its temperature is raised to 120°C? The coefficient of linear expansion of Aluminum is $2.31 \times 10^{-5} K^{-1}$.

Telegram: https://t.me/ExtremeSeries

A. 23% B. 3.2% C. 3% D. 0.23%

Solution:

Initial (original) length $\ell_o = 3m$

Initial temperature $T_o = 20°C$

Final temperature $T_f = 120°C$

$\alpha = 2.31 \times 10^{-5} K^{-1}$

The change in length of the rod is

$\Delta\ell = \ell_o \times \Delta T = 3m \times 2.31 \times 10^{-5} k^{-1} \times 100°C$

$\Delta\ell = 0.00693$ m

The ratio of $\Delta\ell$ to ℓ_o is: $\frac{\Delta\ell}{\ell_o} = \frac{0.00693}{3} = 0.00231$

In percentage it becomes

$\frac{\Delta\ell}{\ell_o} \times 100\% = 0.00231 \times 100\% = 0.231\%$

Answer: D

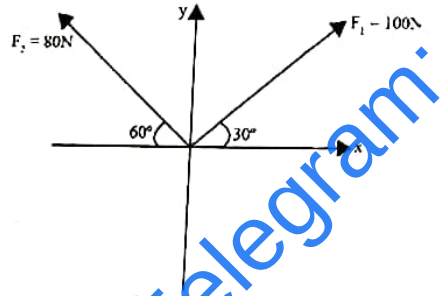
29. A 2 kg object experiences two concurrent forces of 100N at 30° above the positive x-axis 80N at 60° above the negative x-axis. What is the magnitude of acceleration of the object?

A. 68 m/s² B. 64 m/s² C. 48 m/s² D. 46 m/s²

Solution:

m = 2 kg; P_1 = 100N at 30° above positive x

F_2 = 80N at 60° above negative x



The components of the forces are

$F_x = P_{1x} + F_{2x} = F_1 \cos 30 - F_2 \cos 60° = 100\mu\ (0.86) - 80\mu\ (0.5)$

$F_x = 86.6 - 40 = 46.6N$

$F_y = F_{1y} + F_{2y} = F_1 \sin 30 + F_2 \sin 60 = 100\mu\ (0.5) + 80\mu(0.86)$

$F_2 = 50N + 68.8N = 118.8N$

The magnitude of the resultant force is

$|F| = \sqrt{F_x^2 + F_y^2} = \sqrt{(46.6N)^2 + (118.8N)^2}$

$F = 127.5\mu$ then the acceleration is $a = \frac{F}{m} = \frac{127.6\mu}{2m}$

$= 63.8\ m/s^2 \approx 64\ m/s^2$

Answer: B

30. A driver moving at a constant speed of 20 m/s sees an accident up ahead and hits the brakes. If the car decelerates at a constant rate of 05 m/s^2, how far does the car go before it comes to a stop?

A. 120m B. 100m C. 40m D. 20m

Solution:

$V_i = 20$ m/s, $V_f = 0$, $a = -5\ m/s^2$, s =?

The distance moved in the car before coming to rest is given by the equation of uniformly accelerating motion

$s = \frac{V_f^2 - V_i^2}{2a} = \frac{0 - (20m/s)^2}{2x - 5m/s^2} = \frac{-400m^2/s^2}{-10m/s^2}$

s = 40m

Answer: C

31. A series RLC circuit has a resistance of 45.0Ω and an impedance of 75.0Ω. What average power is delivered to this circuit when $\Delta V_{rms} = 210V$?

A. 573W B. 353W C. 255W D. 247W

Solution:

R = 45Ω, Z = 75Ω, $\Delta V_{rms} = 210V$, P_{av} =?

First the root mean square current is

$I_{rms} = \frac{V_{rms}}{Z} = \frac{210V}{75\Omega} = 2.8A$

$I_{rms} = 2.8A$

The average power is given by $P_{av} = (I_{rms})^2 \times R = (2.8A)^2 \times 45\Omega = 352.8W$

$P_{av} \approx 353$ W

Answer: B

32. A container resting on the floor of an elevator contains an incompressible fluid of density ρ. When the elevator has an upward acceleration of magnitude a then what will be the pressure difference between two points in the fluid separated by a vertical distance Δh?

A. ρaΔh B. ρ(g + a)Δh C. ρgΔh D. ρgaΔh

Solution:

Telegram: https://t.me/Extremeseries

Density of fluid = ρ acceleration of elevator = a distance b/n points = Δh when the elevator accelerates upward the net force is $F_{net} = F' - mg$ when F' is the force on the fluid and mg – is its weight $F_{net} = ma$; $F_{net} = F' - mg \Rightarrow ma = F' = ma + mg$; $F' = m(a + g)$ but $m = \rho v = \rho A\Delta h$, $F' = \rho A\Delta h (a + g)$ then the pressure is $P = \frac{F'}{A} = \frac{\rho A\Delta h(a+g)}{A}$; $P = \rho\Delta h (a + g)$.

Answer: B

33. The planet Jupiter is about 318 times as massive as Earth, yet on its surface you would weigh only about 2.4 times your weight on earth. Why is this happened?

A. Your mass is 132.5 times less on Jupiter.
B. You are 132.5 times more weightless there.
C. Jupiter is significantly farther from the sun.
D. Jupiter's radius is about 11.5 times the Earth's radius.

Solution:

$M_{Jupiter} = 318\ Me$

$W_{earth} = 2.4\ W_{Jupiter}$

$\Rightarrow$ The acceleration due to gravity on Jupiter in 2.4 times that of the earth. i.e., $g_{jupiter} = 24\ g_{earth}$

$$g_{earth} = \frac{GM_e}{R_e^2} \ldots\ldots\ldots (i) \qquad g_{jupiter} = \frac{GM_{jupiter}}{R_{jupiter}}$$

$$g_{jup} = \frac{G(318)(Me)}{R_{jup}^2} \ldots (ii)$$

Taking the ratio of (ii) to (i)

$$\frac{g_i}{g_e} = \frac{\frac{G(318)Me}{R_{jup}^2}}{\frac{GM_e}{R_e^2}}$$

$$2.4 = \frac{318GMe}{R_{jup}^2} \times \frac{Re^2}{GMe}$$

$$2.4 = \frac{318Me}{R_{jup}^2} \times \frac{R_e^2}{GMe} \Rightarrow R_{jup}^2 = \frac{318}{2.4} Re^2$$

$$R_{jup} = \sqrt{\frac{318.}{2.4}}\, Re$$

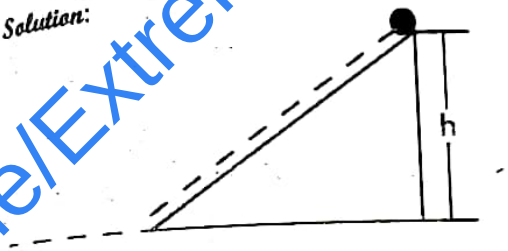
$R_{jup} = 11.5\ Re$

Answer: D

34. A uniform ball of radius r and mass m starts from rest and rolls down a frictionless incline of height h. When the ball reaches the ground, how fast is it going?

A. $\sqrt{\frac{10}{7}gh}$ B. $\sqrt{\frac{9}{10}gh}$ C. $\sqrt{\frac{10}{9}gh}$ D. $\sqrt{\frac{7}{10}gh}$

Solution:



When the sphere rolls down the frictionless inclined plane the total energy is conserved.

$E_{top} = E_{bottom}$

$mgh = E_{krot} + E_{k\ translational}$

$$mgh = \frac{1}{2}Iw^2 + \frac{1}{2}mv^2 = \frac{1}{2}\left(\frac{2}{5}mR^2\right)w^2 + \frac{1}{2}mv^2$$

$$= \frac{1}{5}mR^2w^2 + \frac{1}{2}mv^2$$

$$mgh = \frac{7}{10}mv^2$$

$$V^2 = \frac{10gh}{7}$$

$$V = \sqrt{\frac{10}{7}gh}$$

Answer: A

35. A particle of mass m is fired into a magnetic field of strength B at a speed v. The particle travels in a circular path inside the field

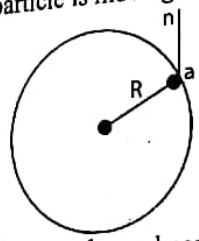
Telegram: https://t.me/ExtremeSeries

with a radius r. Which of the following expressions gives the magnitude of the charge on the particle?

A. $\frac{mr}{v^2B}$ B. $\frac{mv^2}{Br}$ C. $\frac{vB}{mr}$ D. $\frac{mv}{Br}$

Solution:

The particle is moving in a circular path.



The force on charged particle is given by:

$F_m = qvB$

q = charge; v = speed; B = magnetic field also. The force on the particle along a circle is equal to the centripetal force

$$F_C = \frac{mv^2}{R}$$

Equating the centripetal force and the magnetic force $F_m = F_C$

$$qvB = \frac{mv^2}{R}$$

$$\Rightarrow q = \frac{mv^2}{vBR} = \frac{mv}{BR//}$$

$\therefore$ The charge is $q = \frac{mv}{BR}$ **Answer: D**

36. An alternating current is represented by I (t) = (10.0 A) sin (100π t) what are the amplitude and frequency of the alternating current?

A. 100 A, 75 Hz C. 10.0 A, 50 Hz
B. 10.0 A, 100 Hz D. 5.0 A, 25 Hz

Solution:

I (t) = 10.0A sin (500 Hz) the amplitude of the current is 10A to find the frequency the equation is analogues to I (t) = A sin (w t) = A sin (2π ft) $\Rightarrow 2\pi = 100\pi \Rightarrow f = \frac{100}{2} = 50$ HZ

Answer: C

37. If two particles have equal momenta, then which one of the following statements is TRUE about their kinetic energies?

A. Their kinetic energies are always equal.
B. Their kinetic energies are never equal.
C. Their kinetic energies are equal when their masses are the same.
D. Their kinetic energies are equal when the speed of one of the particle is half the speed of the other particle and their masses are the same.

Solution:

Let the particles are 1 and 2. It is given that they have the same momentum.

$$P_1 = P_2$$

If they have the same mass, it is true that they have the same speed. Now, the particles have the same mass and speed therefore they have the same kinetic energy. **Answer: C**

38. What is the capacitance of a capacitor that should be connected with 2 other capacitors having of 4μF and 8μF all in series to produce an effective capacitance of $\frac{24}{11}$μF?

A. 12μF B. 10μF C. 8μF D. 4μF

Solution:

Let $C_1 = 4\mu F$, $C_2 = 8\mu F$, C_3 is unknown

$$C_{total} = \frac{24}{11}\mu F$$

If all the three capacitors are connected in series.

$$\frac{1}{C_{tot}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

$$\frac{1}{\frac{24}{11}\mu F} = \frac{1}{4\mu F} + \frac{1}{8\mu F} + \frac{1}{C_3}$$

Telegram: https://t.me/ExtremeSeries

$$\frac{11}{24}\mu F = \frac{3}{8}\mu F + \frac{1}{C_3}$$

$$\frac{1}{C_3} = \frac{11}{24}\mu F - \frac{3}{8}\mu F = \frac{2}{24}\mu F$$

$$\Rightarrow C_3 = \frac{24}{2}\mu F = 12\mu F$$

Answer: A

39. In Young's double slit experiment both the separation between the slits the distance between the slits and the screen are halved then what is the fringe width?
A. Halved B. unchanged C. Zero D. doubled

Solution:
Let S_1 is the slit separation initially D_1 is the distance between the screen and the slits W_1 is the initial fringe width and S_2, D_2 and W_2 are the quantities after changing. The Yound's double slit experiment provides.

$$W = \frac{\lambda s}{D}, \lambda = \text{wave length}$$

$$\text{So, } W_1 = \frac{\lambda S_1}{D_1}$$

$$W_2 = \frac{\lambda S_2}{D_2} \text{ but } D_2 = \frac{1}{2}D_1$$

$$W_2 = \frac{\lambda S_2}{D_2} = \frac{\lambda \frac{S_2}{1}}{\frac{D_1}{2}} = \frac{\lambda 2 S_1}{2D_1}$$

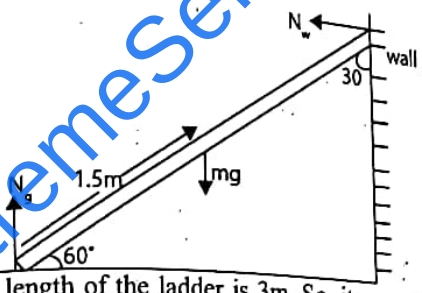
$$W_2 = \frac{2\lambda S_1}{2D_1} = \frac{\lambda S_1}{D_1} = W_1$$

Therefore, the fringes width is not changed. **Answer:** B

40. A ladder of length 3m and mass 20kg leans against a smooth, vertical wall so that the angle between the wall and the ladder is 30°. What are the magnitudes of the normal forces at the contact point on the wall (N_w) and at the contact point on the ground (N_8)?

A. $N_w = 0N, N_8 = 200N$
B. $N_w = 200N, N_8 = 0N$
C. $N_w = 57.7N, N_8 = 200N$
D. $N_w = 200N, N_8 = 57.7N$

Solution:
The ladder can be drawn as below:



The length of the ladder is 3m. So, its center of mass is at 1.5m from the ground. The ladder is in equilibrium, it satisfies the first and second conditions of equilibrium.
First condition:

$$\sum F = 0 \Rightarrow Ng - mg = 0$$

$$N_g = mg = 20\text{ kg} \times 10\text{ m/s}^2$$

$$N_g = 200N$$

Second condition:

$$\sum \tau = 0 \Rightarrow N_w (3m) \sin 60 - mg\,(1.5m)(\cos 60) = 0$$

$$N_w = \frac{mg(1.5)(\cos 60°)}{3m \sin 6} = \frac{200N \times 1.5 \times 0.5}{3 \times 0.86}$$

$$N_w = 57.7N$$

Answer: C

41. Gas at a pressure P is contained in a vessel. If the masses of all the molecules are doubled and their speeds halved, then what will be the resulting pressure?

A. 2P B. $\frac{P}{3}$ C. $\frac{P}{2}$ D. 3P

Solution:
Initial condition pressure = P
Mass = m_1; speed = v_1
Final condition:
Pressure = P_2
Mass = $m_2 = 2m_1$

Telegram: https://t.me/ExtremeSeries

Speed $= v_2 = \frac{v_1}{2}$

$P_2 = ?$

Assuming the volume of the gas is constant.

From kinetic theory of gases

$$PV = \frac{1}{3}mv^2$$

$$P_1V = \frac{1}{3}m_1v_1^2$$

$$\Rightarrow P_1 = \frac{1}{3}\frac{m_1v_1^2}{v} \text{ and } P_2V = \frac{1}{3}\frac{m_2v_2^2}{v} = \frac{1}{3}\frac{(2m_1)}{v}\left(\frac{v_1}{2}\right)^2$$

$$= \frac{1}{3}\left(\frac{2m_1 \times v_1^2}{4v}\right) = \frac{1}{3}\left(\frac{1}{2}\right)\left(\frac{m_1v_1^2}{v}\right)$$

$$P_2 = \frac{1}{2}P \Rightarrow P_2 = \frac{P}{2}$$

Answer: C

42. Refraction of a wave does NOT involve change in its

A. Speed
B. Wavelength
C. Frequency
D. Speed and wavelength

Solution:

During refraction:

- The speed of wave changes
- The wave length changes but the frequency remains constant.

Answer: C

43. A rocket moving in free space has a speed of 3×10^3 m/s relative to the earth. It engines are turned on, and fuel is ejected in a direction opposite the rocket's motion at a speed of 5×10^3 m/s relative to the rocket. What is the speed of rocket relative to the earth once the rocket's mass is reduced to half its mass before ignition?

A. 7×10^3 m/s
B. 5×10^3 m/s
C. 4×10^3 m/s
D. 2×10^3 m/s

Solution:

$V_{re} = 3 \times 10^2$ m/s; $V_{fr} = 5 \times 10^3$ m/s

The mass of the rocket reduced to half of its value using the principle of conservation momentum. $MV_1 = \frac{m}{2}V_2 = \frac{m}{2}V$

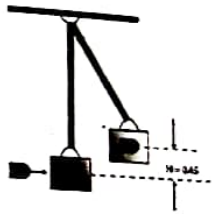
$$m(3 \times 10^3 \text{ m/s}) - \frac{m}{2}(5 \times 10^3 \text{ m/s}) = \frac{m}{2}V$$

$$m(3 \times 10^3 \text{ m/s}) = \left(\frac{m}{2} + \frac{m}{2}\right)(v + 5 \times 10^3 \text{ m/s})$$

$$\Rightarrow v = 2 \times 10^3 \text{ m/s}$$

Answer: D

44. A bullet of mass 10 gm, moving horizontally, strikes and embeddes itself in a box of mass 1 kg suspended from a light string as shown in the figure below. If the composite mass rises to a height, H = 0.45 m, then what is the speed of the bullet before collision?



A. 303 m/s
B. 300 m/s
C. 90.0 m/s
D. 90 m/s

Solution:

m_b = mass of butlet = 10 g = 0.01 kg

m_B = mass of block = 1 kg

H = 0.45 m

U_b = speed of bullet = ?

Using conservation of momentum

$P_{before} = P_{after}$

$$m_bU_b = (m_b + M_b)V \Rightarrow V_b = \frac{(m_b + M_B)V}{m_b} \text{ but}$$

$$V = \sqrt{2gh} = \sqrt{2 \times 10 m/s \times 0.45m}$$

V = 3 m/s

$$\text{Then } V_b = \frac{(m_b + M_B)V}{m_b} = \frac{(0.01kg + 1kg)3m/s}{0.01kg}$$

V_b = 303 m/s

Answer: A

45. Water flows through a fire hose of diameter 6.35 cm at a rate of 0.012 m^3/s. The fire hose ends in a nozzle of inner diameter 2.20 cm. What is the speed with which the water exists the nozzle?

Telegram: https://t.me/Extreme_series

A. 63.2 m/s B. 46.8m/s C. 31.6m/s D. 23.4 m/s

Solution:

Diameter$_1$ = d_1 = 6.35 cm = 0.063 m

Flow rate Q = 0.012 m^3/s

d_2 = 2.20 cm

radius r_2 = 1.10 cm = 0.11 m

The flow rate is constant Q = AV

$$V = \frac{Q}{A} = \frac{0.012 m^3/s}{\pi(0.011m)^2}$$

V = 31.6 m/s

Answer: C

46. Suppose that a voltage surge produces 240V for a moment. By what percentage does the power output of 220-V light bulb increase?

A. 38% B. 36% C. 19% D. 18%

Solution:

Let P_1 is the power when V_1 = 240 and P_2 is theh power when V_2 = 220V

$$P_1 = \frac{V_1^2}{R} \text{ and } P_2 = \frac{V_2}{R}$$

$$P_1 = \frac{(240V)^2}{R}, P_2 = \frac{(220V)^2}{R}$$

$$\frac{P_1}{P_2} = \frac{\frac{(240V)^2}{R}}{\frac{(220V)^2}{R}}$$

$$\frac{P_1}{P_2} = \frac{(240V)^2}{(220V)^2} = \left(\frac{24}{22}\right)^2 = 1.19$$

$$\frac{P_1}{P_2} = 1.19$$

$P_1 = 1.19 P_2$

$P_1 = P_2 + 0.19 P_2$

The increased is 0.19 × 100% = 19%

Answer: C

47. If all the resistors in the circuit shown in the figure have equal resistances, and the current flowing into resistor B is 4A, then what is the current flowing into resistor G?

A. 16A B. 8A C. 2A D. 1A

Solution:

I_B = 4A

Since all the resistors have the same resistance. The current through B is equal to the current through D.

So $I_B = I_D$ = 4A

$I_G = I_B + I_D$ = 4A + 4A

I_G = 8A

Answer: B

48. A wire carries a steady current of 2.4A. A straight section of the wire is 0.75 m long and lies along the x axis within a uniform magnetic field, B = 1.6T along +z axis. If the current is in the −x direction, what is the magnetic force on the section of wire?

A. 2.88 $\hat{j}$ N B. 1.62 $\hat{j}$ N C. -1.62 $\hat{j}$ N D. -2.88 $\hat{j}$ N

Solution:

I = 2.4A; ℓ = 0.75 m along x axis; B = 1.6T along z axis

F =?

$F = I(\ell \times B)$

$F = 2.4A\,(0.75\,\hat{j} \times 1.6k) = 2.4A\,(1.2(-\hat{j})Tm = -2.88\,\hat{j}N$

Answer: D

49. Which one of the following changes to parallel plate capacitor would not increase energy stored in the capacitor at a fixed voltage across the plates?

A. Increasing the area of the plates.

B. Increasing the dielectric constant.

C. Decreasing the charges on the plates.

D. [illegible] the plates

Telegram: https://t.me/ExtremeSeries

Solution:
Given: constant voltage

The energy stored in a capacitor is $E = \frac{1}{2}CV^2$

E increases as 'C' increase and 'C' increases as area A increase according to $C = \frac{\varepsilon_o A}{d}$.

- When a dielectric constant increases, the capacitance and also the energy increases.
- When the charge decrease the energy also decreases.
- Increasing the distance between the plates decrease the capacitance and also the energy.

Answer: D

50. A thin film of oil (n = 1.25) is located on a smooth we floor. When viewed perpendicular to the floor, the film reflects most strongly red light at 640 nm and reflects no blue light at 512 nm. How thick is the oil film?
A. 640 nm B. 612 nm C. 548 nm D. 512 nm

Solution:
$n = 1.25$, $\lambda_{red} = 640$ nm; $\lambda_{blue} = 512$ nm the thickness of the film

$$t = \frac{\lambda_o}{n} = \frac{640nm}{1.25}$$

$t = 512$ nm

Answer: D

1. The distance between two point charges is doubled. What will happen to the force of interaction between the charges?
A. The force will be doubled.
B. The force will be reduced by half.
C. The force will be reduced by a factor of $1/4$.
D. The force will not be affected.

Solution:
The electrostatic force between charged particle is inversely proportional to the square of the distance between them. $F \sim \frac{1}{r^2}$.

Let r_1 = initial distance and r_2 = final distance. Then $r_2 = 2r_1$.

Thus, $F_1 = \frac{kQ_1Q_2}{r_1^2}$ and $F_2 = \frac{kQ_1Q_2}{r_2^2} = \frac{kQ_1Q_2}{(2r_1)^2}$.

$$F_1 = \frac{kQ_1Q_2}{4r_1^2} = \frac{1}{4}F_1$$

Answer: C

2. What does the slope of a velocity – time graph represent?
A. Distance C. Displacement
B. Acceleration D. Speed

Solution:

$$\vec{a} = \frac{\Delta\vec{v}}{\Delta t} = \frac{v_2 - v_1}{t_2 - t_1} \longleftarrow \text{Slope}$$

Answer: B

3. Moving 2.5×10^{-6} coulomb of charge from point A to point B in an electric field requires 6.3×10^{-4} joule of work. The potential difference between points A and B is approximately.
A. 4.0×10^{-3}V C. 1.6×10^{-19}V
B. 2.5×10^{-2}V D. 1.0×10^{-14}V

Solution:
$Q = 2.5 \times 10^{-6}$C, $W = 6.3 \times 10^{-4}$J, $V_{AB} = ?$

$$V_{AB} = \frac{W}{Q} = \frac{6.3\times10^{-4}\,J}{2.5\times10^{-6}\,C} = 2.5\times10^{2}\,V$$

Answer: B

4. A transverse sinusoidal wave is travelling on a string. Which statement is correct a point on the string?
A. The point moves in the same direction as the wave.

Telegram: https://t.me/ExtremeSeries

B. The point moves in simple harmonic motion with a different frequency than that of the wave.
C. The point moves in simple harmonic motion with the same angular frequency as the wave.
D. The point moves I uniform circular motion with a different angular speed than the wave.

Explanation:
The displacement of the particle in a string is perpendicular to the direction of the wave motion. **Answer:**

5. Two non zero vectors **D** and **E** have precisely equal magnitudes. For the magnitude of **D** + **E** to be 3 times larger than the magnitude of D – E, what must be the angle between D and E?
A. 30° B. 37° C. 53° D. 60°

Solution:
We have $|D| = |E|$ and $|D + E| = 3|D - E|$ squaring of magnitude:
$|D + E|^2 = (3|D - E|)^2$
$D^2 + E^2 + 2DE\cos\theta = 9D^2 + 9E^2 - 18DE\cos\theta \ldots (*)$
Since $|D| = |E| \Rightarrow D^2 = E^2$ replacing in (*)
$\Rightarrow D^2 + D^2 + 2D\cos\theta = 9D^2 + 9D^2 + 9D^2 - 18D^2\cos\theta$
$\Rightarrow 2D^2 - 18D^2 = -18D^2\cos\theta - 2D^2\cos\theta$
$-16D^2 = -20D^2\cos\theta \ldots$ simplifying

$$\cos\theta = \frac{16}{20} = \frac{8}{10} = 0.8$$

$\therefore \theta = \cos^{-1}(0.8) = 37°$ **Answer: B**

6. Which element is correct about the pressure under a fluid in a container?
A. It depends on the volume of the fluid.
B. It depends on the density of the fluid.
C. It depends on the shape of the container.
D. It is independent of the depth of the fluid.

Explanation:
The pressure under the fluid in a container depends on the density, depth of the fluid and the acceleration due to gravity. **Answer: B**

7. A capacitor made of parallel plates is charged by connecting it to a battery. The distance between the plates is much less than the surface area of each plate. After fully charged if the plates are disconnected from the battery and then pulled apart a little, what will happen to the total stored energy, the electric field, and the potential difference between the plates?
A. The electric field between the plates increases.
B. The potential difference between the plates decreases.
C. The total energy stored in the capacitor increases.
D. The electric field the potential difference and the stored energy remain the same.

Explanation:
When the capacitor plate is fully charged, the charge on the plates remains constant.

i) Capacitance: $C = \frac{\varepsilon_o A}{d}$

As the distance between the plates increases, the capacitance decrease.

ii) **The voltage (p.d):** $V = \frac{Q}{C}$

As the capacitance decrease the voltage increases. The potential difference increases linearly.

iii) **The electric field:** $E = \frac{v}{d} =$ remains the same. As d – decrease, v – increase.

iv) **Energy stored** in the capacitor: $U_{stored} = \frac{1}{2}CV^2$. The energy stored in the capacitor increase linearly with separation due to the work required to move the plates farther apart. **Answer: C**

8. Which of the following statements is true about the motion of a particle in a circular path?
A. The centripetal acceleration is constant if the particle's speed is constant.
B. The tangential acceleration can be perpendicular to the velocity of the particle.
C. The centripetal acceleration is always in the direction perpendicular to the velocity vector of the particle.
D. The acceleration is always perpendicular to the velocity vector of the particle.

Explanation:

Telegram: https://t.me/ExtremeSeries

When a particle moving in a circular path:

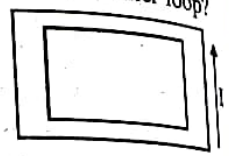
→ Its speed remains constant in magnitude but direction changes continuously.

→ Its centripetal acceleration is perpendicular to the velocity.

→ Its tangential acceleration is parallel to the velocity.

Answer: C

9. Two rectangular loops of wire lie in the plane of this paper as shown in Fig. 1. If the current I in the outer loop is counter clockwise and increases with time, which one of the following statements is correct about the current induced in the inner loop?



A. A current is induced in the inner loop in the clockwise direction.

B. A loop is induced in the inner loop in the counter clockwise direction.

C. The induced current in the inner loop creates a magnetic field inside the area bounded by the inner loop in the same direction of the magnetic field that the outer loop products in the same area.

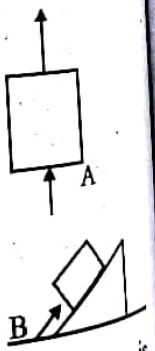
D. It is zero.

Explanation:

As the current I in the outer loop is increasing with time and counter clock wise, the induced current in the inner loop is flows in clock wise direction that opposes the change that caused the current.

Answer: A

10. Two men, A and B, are loading identical boxes onto a truck. A lifts his box straight up from the ground to the bed of the truck, whereas B slides his box up a rough ramp to the truck. Which statement is correct about the change in the potential energy on the box – Earth system?



A. A and B change the potential energy by the same amount.

B. A changes more potential energy than B.

C. B changes more potential energy than A.

D. None of the statements is necessarily true because the force of friction and the angle of the incline is unknown.

Explanation:

As the boxes moves the same height in loading onto the truck the changes in potential energy is the same in each cases. **Answer: A**

11. Two projectiles are fired at an angle of 30 degrees and the other at 60 degrees. The projectile to hit the ground first will be the one fired at (neglect air resistance).

A. 60 degrees

B. 30 degrees

C. Both hit at the same time.

D. Cannot be determined from the given information.

Solution:

The time taken to reach the ground is given by $t = \frac{2v_o \sin\theta}{g}$

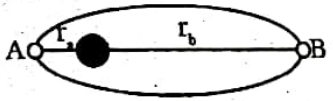
When, $\theta = 30°$, $t_1 = \frac{2v_o \sin 30°}{g} = \frac{v_o}{g}$

When, $\theta = 60°$, $t_2 = \frac{2v_o \sin 60°}{g} = 1.7\frac{v_o}{g}$

$\Rightarrow t_1 < t_2$, so the net that thrown at 30° hits the ground first.

Answer: B

12. A planet of mass m is circling a star in a elliptical orbit as shown below.



If its velocity at point A is 60,000 m/s, and $r_b = 8\ r_a$ what is its velocity at point B?

A. 7500 m/s B. 8,571 m/s C. 60,000 m/s D. 488,000 m/s

Solution:

Using Kepler's law of planetary motion "the principle of conservation of angular momentum"

$L_A = L_B \Rightarrow mr_aV_A = mr_bV_B$

$$V_B = V_A\left(\frac{r_a}{r_b}\right) = (60,000\text{m/s})\left(\frac{r_a}{8r_a}\right) = 7,500$$

Answer: A

13. A small radio receiver uses a batter that delivers a constant current of 25 m/A for 3 hours. Calculate the total charge delivered by the battery.

A. 1080000C C. 540

Telegram: https://t.me/ExtremeSeries

B. 2700000C D. 270C

Solution:

$Q = It = (25 \times 10^{-3}A)(3 \times 3600\ sec) = 270\ C$ **Answer: D**

14. A cup of coffee at a temperature of 80°C is placed on a table in a 20°C room. What happens to be the thermal energy and molecular motion of the coffee when it is left on the table?

A. The thermal energy decreases and the molecules move more slowly.

B. The thermal energy increases and the molecules move more slowly.

C. The thermal energy decreases and the molecules move faster.

D. The thermal energy increases and the molecules move faster.

Explanation:

The coffee is at higher temperature. When placed on a table at lower temperature, the thermal energy of the coffee decreases and its molecular speed decreases. **Answer: A**

15. The vector $2\hat{i}+\hat{j}+\hat{k}$ is perpendicular to $\hat{i}-4\hat{j}+\lambda\hat{k}$ if λ is equal to:

A. 0 B. -1 C. 2 D. 3

Solution:

The scalar product of perpendicular vectors is zero:

$\Rightarrow(2i+j+k).(i-4j+\lambda k)=0$

$\Rightarrow 2-4+\lambda=0 \Rightarrow \lambda=2,$ **Answer: C**

16. A garden hose has an inside diameter 0d 16 cm. The hose can fill a liter bucket in 20 seconds. If the diameter of the nozzle is reduced to 8 cm.

A. Water exits with a speed twice the speed inside the hose.

B. A 10 liter bucked can be filled in 10 seconds.

C. A 20 liter bucket can be filled in 40 seconds.

D. Water exits with a speed half the speed inside the hose.

Solution:

Inside diameter $d_1 = 16$ cm and $d_2 = 8$ cm. Thus $r_1 = 8$ cm, $r_2 = 4$ cm flow rate $= \dfrac{10\ liter}{20\ sec} = 0.5$ liter/sec.

Equation of continuity: $A_1V_1 = A_2V$

$$V = V\left(\frac{A_1}{A_2}\right) = V_1\left(\frac{\pi r_1^2}{\pi r_2^2}\right) = V_1\left(\frac{64}{16}\right) = 4V_1$$

Since the flow rate is 0.5 liter/sec constant everywhere:

- 10 liter bucket can be filled in 20 second.
- 20 liter bucket can be filled in 40 second.

17. A mass suspended from the end of 16 cm. The hose can fill a 10 litter bucket in 20 seconds. If the diameter of the nozzle is reduced to 8 cm. **Answer: C**

A. Water exists with a speed twice the speed inside the hose.

B. A 10 litter bucked can be filled in 10 seconds.

C. A 20 litter bucket can be filled in 40 seconds.

D. Water exists with a speed half the speed inside the hose.

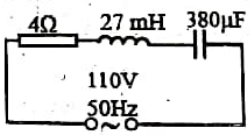
Solution:

$$\text{Frequency} = \frac{\text{number of oscillaiton}}{\text{Time taken}}$$

$$f = \frac{24}{36\ sec} = 0.67Hz$$

$$T = \frac{36\ sec}{24} = \frac{3}{2} = 1.5\ sec$$ **Answer: B**

18. Determine the total impedance for the circuit shown below.



A. 0Ω B. 4Ω C. 27Ω D. 31Ω

Solution:

$R = 4\Omega$, $L = 27mH$, $C = 380\mu F$, $f = 50$ Hz, $V = 110V$

- $X_C = \dfrac{1}{\omega C} = \dfrac{1}{2\pi fC} = \dfrac{1}{2\pi(50Hz)(380\times10^{-6}F)} = 8.4\Omega$
- $X_L = \sqrt{R^2+(X_L-X_C)^2} = \sqrt{16+(8.5-8.4)^2} = 4\Omega$

Answer: B

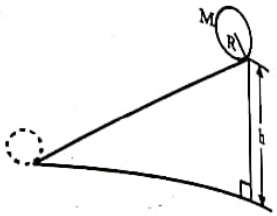
19. A hoop of mass M and radius R is at rest at the top of an incline plane of height h. The hoop rolls down the plane without slipping. When the hoop reaches bottom, its angular momentum about its center of mass is:

Telegram: https://t.me/ExtremeSeries

A. Mgh B. $MR\sqrt{2gh}$ C. $MR\sqrt{gh}/2$ D. $MR\sqrt{gh}$

Solution:

$E_{top} = E_{bottom}$



$$Mgh = \frac{1}{2}MV^2 + \frac{1}{2}I\omega^2$$

But $I = MR^2$, and $\omega = \frac{V}{R}$

$$Mgh = \frac{1}{2}M(\omega^2R^2) + \frac{1}{2}MR^2\omega^2$$

$$gh = \omega^2R^2 \Rightarrow\Rightarrow \omega\frac{\sqrt{gh}}{R}$$

$$L = I\omega = MR^2\frac{\sqrt{gh}}{R} = MR\sqrt{gh}$$

Answer: D

20. What is the resistance of a copper cable if it has a cross - sectional area of 1 cm^2 and a length of 2 km? The resistivity of copper is $2 \times 10^{-8}\Omega m$?

A. 0.0004Ω B. 0.04Ω C. 0.04Ω D. 0.4Ω

Solution:

$R = 1\ cm^2 = 1 \times 10^{-4}m^2$, $L = 2\ km = 2 \times 10^3\ m$ and $\sigma = 2 \times 10^{-8}\Omega m$

$$R = \frac{\rho}{A} = \frac{(2\times10^{-8}\Omega m)(2\times10^{3}\ m)}{1\times10^{-4}\ m^2} = 0.4\Omega$$

Answer: D

21. When a load of 500 kg is hanging from a steel wire of length 3m and cross sectional area 0.20 cm^2, the wire stretches beyond its no – load length. If the Young's modulus for the wire made of steel is $Y = 2 \times 10^{11}Pa$, what is the change in length of the wire beyond its no load length?

A. 3.75 mm B. 2.75mm C.1.75mm D. 0.75mm

Solution: Load = 500 kg, $\ell = 3m$, $A = 0.20\ cm^2 = 0.2 \times 10^{-4}m^2$ and $Y = 2 \times 10^{11}Pa$, $\Delta\ell = ?$

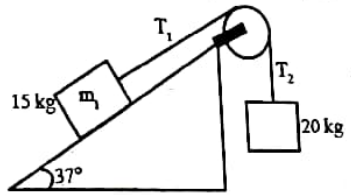
Young's modules, $Y = \frac{\text{stress}}{\text{strain}} = \frac{\frac{F}{A}}{\frac{\Delta f}{\ell}} = \frac{F\ell}{A\Delta\ell}$

$$\Rightarrow \Delta\ell = \frac{F\ell}{Y.A} = \frac{(500kg)(10m/s^2)(3m)}{(2\times10^{11}Pa)(0.2\times10^{-4}m)} = 3.75\times10^{-3}m$$

$\Delta\ell = 3.75 \times 10^{-3}m = 3.75$ mm

Answer: A

22. As shown in figure 5, two blocks are connected by a string of negligible mass passing over a pulley of radius r = 0.2m and moment of inertia I. The block on the frictionless incline is moving with a constant acceleration a = 2.0 m/s^2 up along the incline.



What is the moment of inertia of the pulley?

A. 0.8 kgm^2 B. 1.25 kgm^2 C. 4 kgm^2 D. 40 kgm^2

Solution:

Applying Newton's second law.

For m_2: $T_2 - m_2g = -m_2a$

$\Rightarrow T_2 = m_2g - m_2a = 20\ kg\ (10\ m/s^2 - 2m/s^2) = 160N$

For m_1: $T_1 - m_1\ g\sin 37° = m_1a$

$\Rightarrow T_1 = m_1(g\sin 37° + a) = 15\ kg\ [10(0.6)m/s^2 + 2m/s^2] = 120N$

The net torque on the pulley is:

$\tau_{net} = (T_2 - T_1)r$ and also $\tau = I\alpha$

$$I = \frac{(T_2 - T_2)r}{\alpha} = \frac{(T_2 - T_1)r}{a/r} = \frac{(T_2 - T_1)r^2}{a}$$

$$I = \frac{(160N - 120N)(0.2m)^2}{2m/s^2} = 0.8kgm^2$$

Answer: A

23. A table – tennis ball which is spherical in shape has a diameter of 6 cm and average density to 0.400 g/cm^3. What force is required to hold it completely submerged under water?

A. 0.68N down ward C. 1.13N downward

B. 0.68N upward D. 1.13N upward

Solution:

Telegram: https://t.me/Extremeseries

$F + W = F_b \Rightarrow F = F_b - W$

$F = \rho_f V_{fg} - \rho_o V_{og} = gV_o(\rho_f - \rho_o)$, since $V_f = V_o$

$\Rightarrow F = (10\ m/s^2)\left[\frac{4}{3}\pi \times (0.03m)^3\right][1000kg/m^3 - 400kg/m^3]$

$F = (10)\left(\frac{4}{3}\right)(3.14)(0.03m)^3(600\ kg/m^3) = 0.68N$ downward

Answer: A

24. A viewing screen is separated from a double slit by 4.80m. The distance between the two slits is 0.30 nm. Monochromatic lights is directed toward the double slit and forms an interference pattern on the screen. What is the wave length of the lights?
A. 1250nm B. 1875nm C. 2500nm D. 5625nm

Solution:

D = 4.8nm, d = 0.03mm = 3×10^{-5}m and W = 4.5cm = 4.5×10^{-2}m

Condition for dark – fringes in double slit interference is d sin

$\theta = \left(m + \frac{1}{2}\right)\lambda, m = 0, +1, +2 \ldots$

For small: $\sin\theta = \tan\theta = \frac{W}{D}$ and m = 1

$\frac{Wd}{D} = \left(1 + \frac{1}{2}\right)\lambda \Rightarrow \lambda = \frac{2Wd}{3D}$

$\lambda = \frac{2(4.5 \times 10^{-2}m)(3\times10^{-5}m)}{(3)(4.8m)} = 1.875\times10^{-7}m = 1875\times10^{-8}m$

25. A horizontal pipe of 10 cm in diameter has a smooth reduction to a pipe of 5 cm in diameter. If the pressure of the water in the large pipe is 80 kPa and the pressure in the smaller pipe is 60 kPa, at what rate does the water flow through the smaller pipe?
A. 1.63 m/s B. 6.53 m/sC. 13.6m/s D. 42.67 m/s

Solution:

$d_1 = 10$ cm, $d_2 = 5$ cm, $P_1 = 80$ kPa, $P_2 = 60$ kPa

Equation of continuity

$A_1V_1 = A_2V_2 \Rightarrow V_2 = V_1\left(\frac{A_1}{A_2}\right) = V_2\left(\frac{100}{25}\right) = 4V_1$

Applying Bernoulis principle

$P_1 + \frac{1}{2}\rho V_1^2 = P_2 + \frac{1}{2}\rho V_2^2 \Rightarrow P_1 - P_2 = \frac{1}{2}\rho(V_2^2 - V_1^2)$

$80{,}000\ Pa - 60{,}000\ Pa = \frac{1}{2}(1000kg/m)(14V_1)^2 - V_1^2$

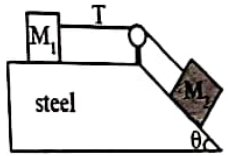
$20{,}000Pa = (7500\ kg/m^3)V_1^2$

$V = \frac{20{,}000}{7500} = 2.7m^2/s^2 \Rightarrow V = \sqrt{2.7m^2/s^2} = 1.63m/s$

But $V_2 = 4V_1 = 4(1.63\ m/s) = 6.53\ m/s$

Answer: B

26. An aluminum block of mass $m_1 = 3$ kg and a copper block of mass $m_2 = 5$ kg are connected by a light string over a frictionless pulley. They sit on a steel surface as shown in Figure 4, where θ = 53°. The coefficient of kinetic friction between the steel surface and each block is 0.3. if the blocks are released from rest. What is the tension T in the string?
A. 29.40N
B. 17.25N
C. 12.25N
D. 8.60N



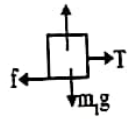
Solution:

Applying Newton's second law for each mass:

i) For M_1, $T - f = m_1a \Rightarrow T = m_1a + f$

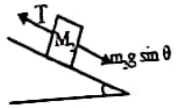
$T = m_1a + \mu m_1g$

$T = 3a + 9 \ldots (1)$



ii) For M_1

$m_2g\sin\theta - T - f = m_2a$, but $f = \mu F_N = \mu mg\cos\theta$



$m_2g\sin53° - T - \mu m_2g\cos53° = m_2a$

$(5\ kg)(10\ m/s^2)(0.8) - T - (0.3)(5\ kg)(10\ m/s^2)(0.6) = 5a$

$40 - T - 9 = 5a \Rightarrow T = 31 - 5a \ldots (2)$

Equating equation (1) and (2)

$\begin{cases} T = 3a + 9N \\ T = -5a + 31N \end{cases} \Rightarrow 3a + 9N = -5a + 31N$

Telegram: https://t.me/EvtremeSeries

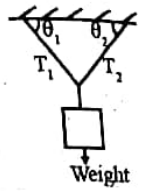
$8a = 22N \Rightarrow a = \frac{22}{8} m/s^2 = 2.75 m/s^2$

To find tension: $T = 9N + 3a = 9N + 3(2.75)N = 17.25N$

Answer: B

27. An object of mass 3 kg hangs from three light inextensible wires as shown in Figure 6. Two of the wires make angles $\theta_1 = 53°$ and $\theta_2 = 37°$ with the horizontal. If the system is equilibrium, what will be the tension T_1 and T_2 in the wires?

A. $T_1 = 18N$ and $T_2 = 24N$
B. $T_1 = T_2 = 18N$
C. $T_1 = T_2 = 24N$
D. $T_1 = 24N$ and $T_2 = 18N$



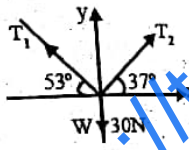
Solution:

$\sum F_x = 0, T_1 \sin 53° + T_2 \sin 37° - 30N = 0$

$\sum F_x = 0, T_1 \cos 53° - T_2 \cos 37° = 0$

From these two equation

$0.8T_1 \div 0.6T_2 = 30$ and



$0.6T_1 = 0.8T_2 \Rightarrow T_2 = \frac{6}{8}T_1 = \frac{3}{4}T_1$

$0.8T_1 + \frac{3}{4}(0.6)T_1 = 30 \Rightarrow T_1 = 24N$ and $T_2 = 18N$

Answer: D

28. An object of mass ms is hung from a string and set into oscillation. The frequency of the oscillation is measured and recorded as f. If the object of mass ms is replaced with an object of mass 2m and set into oscillation, what will be the frequency of the motion?

A. $\frac{f}{\sqrt{2}}$ B. f C. $(\sqrt{2})f$ D. 2f

Solution:

The frequency of spring mass – system:

$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$

If $m \to 2m$; $f' = \frac{1}{2\pi}\sqrt{\frac{k}{2m}} = \frac{1}{\sqrt{2}}\left(\frac{1}{2\pi}\sqrt{\frac{k}{m}}\right) = \frac{f}{\sqrt{2}}$ **Answer: A**

29. A placekicker must kick a football from a point which is at a distance of 36.0 m from the goal. When kicked, the ball leaves the ground with a speed of 20.0 m/s at an angle of 53.0° to the horizontal. If the ball hits the crossbar of the goal at a height h and bounces back what will be the height of the crossbar?

A. 2.45m B. 2.85 C. 3.00m D. 3.15m

Solution:

The equation of motion of a ball from the origin with an initial speed v_o at an angle θ with respect to the horizontal are:

(i) $x = (v_o \cos\theta)t \Rightarrow t = \frac{x}{v_o \cos\theta}$

$t = \frac{36m}{(20m/s)\cos 53°} = \frac{36m}{(20)(0.6)m/s} = 3sec$

(ii) The height of the cross bar of the goal:

$h = (V_o \sin\theta)t - \frac{1}{2}gt$

$h = [(20\ m/s)\sin 53°](3\ sec) - \frac{1}{2}(10\ m/s^2)(9\ sec)$

$h = 48m - 45m = 3m$ **Answer: C**

30. A body moving with constant acceleration covers the distance between two points 60m apart in 5s. It velocity as it passes the second points is 15 m/s. What is the acceleration?

A. $3\ m/s^2$ B. $2.4 m/s^2$ C. $1.8\ m/s^2$ D. $1.2\ m/s^2$

Solution: $V_{av} = \frac{s}{t} = \frac{60m}{5sec} = 12m/s$

Similarly, V_{av} is expressed as

$V_{av} = \frac{V_A + V_B}{2} \Rightarrow 12m/s = \frac{V_A + 15m/s}{2}$

$V_{av} = 24\ m/s - 15\ m/s = 9\ m/s$

Acceleration: $a = \frac{V - V}{t} = \frac{15m/s - 9m/s}{5sec} = 1.2 m\,s^2$

Answer: D

Telegram: https://t.me/ExtremeSeries

31. Which of the following combinations of properties would be most desirable for a heat sink in a machine?
A. High specific heat and conductivity.
B. Low specific heat and high conductivity.
C. High specific heat and high conductivity.
D. Low specific heat and low conductivity. **Answer: A**

32. A 6.00V battery is connected across the primary coil of a transformers having 50 turns. If the secondary coil of the transformer has 100 turns, what voltage appears across the secondary?
A. 24.0V B. 12.0V C. 6.00V D. 3.00V

Solution: Transformer equation: $\frac{N_s}{N_P} = \frac{V_s}{V_P}$

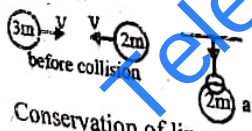
$$\Rightarrow V_s = \frac{N_s V_P}{N_P} = \frac{(100)(6V)}{50} = 12V$$ **Answer: B**

33. The change in entropy is zero for:
A. Reversible isobaric processes.
B. Reversible adiabatic processes.
C. Reversible isothermal processes.
D. Reversible processes during which no work is done.

Solution: In reversible adiabatic process, the change in entropy is zero $\Delta s = \frac{Q}{T} = \frac{0}{T} = 0$ **Answer: B**

34. Two particles with masses 2m and 3m are moving toward each other along the x axis with the same initial speeds v. Particle 2m is travelling to the left, and the particle 3m is traveling to the right. They undergo an electric glancing collision such that particle 2m is moving in the negative y direction after the collision. What are the x component of the final velocity of particle 3m and the kinetic energy of particle 2m, respectively?
A. 0.33V and 0.7 mV^2 C. 0.67V and 0.70mV^2
B. 0.33V and 1.40 mV^2 D. 0.67V and 1.4mV^2

Solution:



Conservation of linear momentum of x – component.

$(P_x)_{before} = (P_x)_{after}$

$$(3m)v - (2m)V = (3m)V_x + \cancel{(2m)V_x} = 0$$

$$mV = (3m)V_x \Rightarrow V_x = \frac{mV}{3m} = \frac{1}{3}V = 0.33V$$ **Answer: B**

35. Two separate coherent sources produce waves whose wavelengths are 0.10m. Which statement is correct about points that are 3.2m away from one source?
A. Constructive interference occurs if the points are 3.0m from the other source.
B. Constructive interference occurs if the points are 3.05m from the other waves.
C. Constructive interference occurs if the points are 2.95m from the other source.
D. Destructive interference occurs if the points are 3.2m from the other source.

Solution:
Since $\lambda = 0.1$ path length difference of construction interference:

$\ell_1 - \ell_2 = m\lambda$, where, $m = 0, 1, 2, 3, \ldots$ $\ell_1 - \ell_2 = \frac{\lambda}{2}, \frac{3}{2}\lambda, \frac{5}{2}\lambda, \ldots$

for $\lambda = 0.1 \Rightarrow \ell_1 - \ell_2 = 0.05, 0.15, 0.25, \ldots$

A. Let $\ell_1 = 3.2m$, $\ell_2 = 3m$, $\ell_1 - \ell_2 = 3.2 - 3 = 0.2$
∴Constructive interference

B. Let $\ell_1 = 3.2m - 2.95m$, $\ell_2 = 3.05$;
$\ell_1 - \ell_2 = 3.2m - 3.05m = 0.15m$
∴Destructive interference

C. $\ell_1 - \ell_2 = 3.2m - 2.95m = 0.25m \leftarrow$ destructive

D. $\ell_1 - \ell_2 = 3.2m - 3.2m = 0 \leftarrow$constructive **Answer: A**

36. A coil formed by wrapping 50 turns of wire in the shape of a square is positioned in a magnetic field so that the normal to the plane of the coil makes an angle of 37° with the direction of the field. When the magnetic field is increased uniformly from 200 mT to 600 mT in 4.00 s an emf of magnitude 160.0mV is induced in the coil. What is the total length of the wire in the coil?
A. 100m B. 200m C. 300m D. 400m

Telegram: https://t.me/ExtremeSeries

$$\varepsilon = NA\frac{\Delta B}{\Delta t}\cos\theta = \frac{NA(B_2 - B_1)\cos 37°}{\Delta t}$$

$$160mV = \frac{(50)(A)(600mT - 200mT)(0.8)}{4\,sec}$$

$$A = \frac{16}{400}m^2 \text{ but } A = L^2 = \frac{16}{400}m^2 \Rightarrow L = \frac{4}{20}$$

Total length = perimeter of square = 4NL

$$\therefore \text{Total length} = (450)\left(\frac{4}{20}\right) = 40m$$

37. The half – time of radium – 224 is 3.5 days. What fraction of the sample remains undecayed after fourteen days?

A. ½ B. ¼ C. 1/8 D. 1/16

Solution:

Half – life; $t_{\frac{1}{2}} = 3.5$ days and $t = 14$ days

Decay constant: $\lambda = \frac{\ln 2}{t_{\frac{1}{2}}} = \frac{.0.693}{3.5\ days}$

Sample after, $t = 14$ days; $N = N_e e^{-\lambda t}$

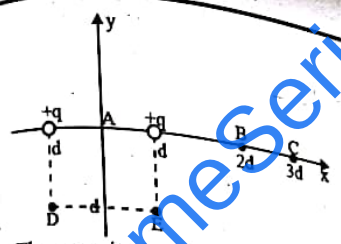
$$\Rightarrow \frac{N}{N_o} = e^{-\lambda t} = e^{\left(\frac{-0.693}{3.5}\right)} = (e^{-0.693})^{\left(\frac{14}{3.5}\right)}$$

$$\frac{N}{N_o} = \left(\frac{1}{2}\right)^4 \text{; recall that: } e^{-\ln 2} = e^{-0.693} = \frac{1}{2}$$

$$\frac{N}{N_o} = \frac{1}{16}$$

Answer: D

38. Two positive charges of magnitude q are such a distance d from the origin A of a coordinate system as shown below.



The magnitude of the electric field is minimum at point.

A. A B. B C. C D. D

Explanation:

An electric field created by a charge always points away from positive charge and toward a negative charge. The magnitude of the electric field that (+q)an d(+q) produce.

i) At point A: $E_A = \frac{k(+q)}{d} - \frac{k(+q)}{d} = 0$

ii) At point B: $E_B = \frac{k(+q)}{d^2} + \frac{k(+q)}{9d^2} = \frac{10kq}{9d^2}$ to right

iii) At point D: $E_D = E_{1D} + E_{2D}$ (non – zero vector sum)

$E_{1D} = \frac{kq}{d^2}$ down ward in the negative y – direction

$E_{2D} = \frac{kq}{\left(\sqrt{2}d\right)^2} = \frac{kq}{2d^2}$ in the x – y direction. **Answer: A**

39. If the terminals of a battery with zero internal resistance are connected across two identical resistors in series, the total power delivered by the battery is 8W. If the same battery is connected across the same resistors in parallel, what is the total power delivered by the battery?

A. 2W B. 4W C. 15W D. 32W

Solution:

In series case: $R_T = R_1 + R_2 = R + R = 2R$

$$\text{Power} = \frac{V^2}{R_T} = \frac{V^2}{2R} = 8W \Rightarrow \frac{V^2}{R} = 16W$$

In parallel case: $\frac{1}{R_{Total}} = \frac{1}{R} + \frac{1}{R} = \frac{2}{R} \Rightarrow R_T = \frac{R}{2}$

Telegram: https://t.me/ExtremeSeries

$$\text{Power} = \frac{V^2}{R_T} = \frac{V^2}{R/2} = \frac{2V^2}{R} \text{ but } \frac{V^2}{R} = 16W$$

$$\therefore P = 2\left(\frac{V^2}{R}\right) = 2(16W) = 32W$$ **Answer: D**

40. Wire B has twice the length and twice the radius of wire A. Both wires are made from the same material. If wire A has a resistance R, what is the resistance of wire B?
A. 4R B. 2R C. R/s D. R/4

Solution:

$\ell_B = 2\ell_A$, $r_B = 2r_B$, $\rho_A = \rho_B$ and $R_A = R$

The resistance of

i) Wire A: $R_A = \frac{\rho_A \ell_A}{A_A} = \frac{\rho \ell_A}{\pi r_A^2} = R$

ii) Wire B: $R_B = \frac{\rho \ell_B}{\pi r_B^2} = \frac{\rho(2\ell_A)}{\pi(2r_A)^2} = \frac{2\rho\ell_A}{4\pi_A^2} = \frac{1}{2}\left(\frac{\rho\ell_A}{\pi_A^2}\right)$

Hence $R_B = \frac{1}{2}R$ **Answer: C**

41. The sound level at a distance of 4.00m from a source is 120 dB. At what distance is the sound level 100 dB?
A. 0.40m B. 4.80m C. 40m D. 48m

Solution:

Describe (dB) is a measurement used when comparing two sound intensities. The intensity level $\beta = (dB)\log\left(\frac{I}{I_o}\right)$.

Hence $\beta_1 - \beta_2 = (10\text{ dB})\log\left(\frac{I}{I_o}\right) - (10\text{ dB})\log\left(\frac{I_2}{I_o}\right)$.

$$120\text{ dB} - 100\text{dB} = (10\text{ dB})\log\left(\frac{I_1/I_o}{I_2/I_o}\right) = (10\text{ dB})\log\left(\frac{I_2}{I_o}\right).$$

$$20\text{ dB} = (10\text{ dB})\log\left(\frac{I_1}{I_2}\right) \Rightarrow \log\frac{I_1}{I_2} = 2,$$

$$\Rightarrow \frac{I_1}{I_2} = 10^2 = 100, \text{ since } I = \frac{P}{4\pi r}$$

Since I is inversely related to the square of radius.

$$\frac{I_1}{I_2} = \frac{r_2^2}{r_1^2}\frac{r_2^2}{(4m)^2} = 100 \Rightarrow r_2^2 = (100)(16m)^2$$ **Answer: C**

42. An ideal gas is at a temperature of 300 K. If we wish to double the root mean square (rms) speed of the molecules of the gas, to what value must we raise its temperature?
A. 450 B. 600k C. 800K D. 1200K

Solution: $V_{rms} = \sqrt{\frac{3kT}{m}}$

The root mean square speed of molecules is directly proportional to the square root temperature. $V_{rms} = \sqrt{\frac{T_1}{T_2}} = \frac{1}{2}$

$$\frac{T_1}{T_2} = \frac{1}{4} \Rightarrow \frac{300k}{T_2} = \frac{1}{4} \Rightarrow T_2 = 1200k$$ **Answer: D**

43. Suppose that the index of refraction of a soap – bubble film is 1.25, what is. The minimum thickness of the soap – bubble film that results in constructive interference in the reflected light if the film is illuminated with light whose wavelength in free space is $\lambda = 600nm$?
A. 120nm B. 150nm C. 300nm D. 450nM

Solution:

In constructive air – film – air interference the film thickness can be calculated by $2nt = \left(m + \frac{1}{2}\right)\lambda, m = 0, 1, 2, \ldots$

For $m = 0$, $t = \frac{\lambda}{4n} = \frac{600nm}{4(1.25)} = 120nm$ **Answer: A**

44. A wire carries a steady current of 2.4 A.A straight section of the wire is 0.75 m long and lies in xy – plane along the x axis within

Telegram: https://t.me/ExtremeSeries

$$\text{Power} = \frac{V^2}{R_T} = \frac{V^2}{R/2} = \frac{2V^2}{R} \text{ but } \frac{V^2}{R} = 16W$$

$$\therefore P = 2\left(\frac{V^2}{R}\right) = 2(16W) = 32W$$ **Answer: D**

40. Wire B has twice the length and twice the radius of wire A. Both wires are made from the same material. If wire A has a resistance R, what is the resistance of wire B?
A. 4R B. 2R C. R/s D. R/4

Solution:

$\ell_B = 2\ell_A$, $r_B = 2r_B$, $\rho_A = \rho_B$ and $R_A = R$

The resistance of

i) Wire A: $R_A = \frac{\rho_A \ell_A}{A_A} = \frac{\rho \ell_A}{\pi r_A^2} = R$

ii) Wire B: $R_B = \frac{\rho \ell_B}{\pi r_B^2} = \frac{\rho(2\ell_A)}{\pi(2r_A)^2} = \frac{2\rho\ell_A}{4\pi_A^2} = \frac{1}{2}\left(\frac{\rho\ell_A}{\pi_A^2}\right)$

Hence $R_B = \frac{1}{2}R$ **Answer: C**

41. The sound level at a distance of 4.00m from a source is 120 dB. At what distance is the sound level 100 dB?
A. 0.40m B. 4.80m C. 40m D. 48m

Solution:

Describe (dB) is a measurement used when comparing two sound intensities. The intensity level $\beta = (\text{dB}) \log\left(\frac{I}{I_o}\right)$.

Hence $\beta_1 - \beta_2 = (10\text{ dB}) \log\left(\frac{I}{I_o}\right) - (10\text{ dB}) \log\left(\frac{I_2}{I_o}\right)$.

$$120\text{ dB} - 100\text{dB} = (10\text{ dB}) \log\left(\frac{I_1/I_o}{I_2/I_o}\right) = (10\text{ dB}) \log\left(\frac{I_2}{I_o}\right).$$

$$20\text{ dB} = (10\text{ dB}) \log\left(\frac{I_1}{I_2}\right) \Rightarrow \log\frac{I_1}{I_2} = 2,$$

$$\Rightarrow \frac{I_1}{I_2} = 10^2 = 100, \text{ since } I = \frac{P}{4\pi r^2}$$

Since I is inversely related to the square of radius.

$$\frac{I_1}{I_2} = \frac{r_2^2}{r_1^2}\frac{r_2^2}{(4m)^2} = 100 \Rightarrow r_2^2 = (100)(16m)^2$$ **Answer: C**

42. An ideal gas is at a temperature of 300 K. If we wish to double the root mean square (rms) speed of the molecules of the gas, to what value must we raise its temperature?
A. 450 B. 600k C. 800K D. 1200K

Solution: $V_{rms} = \sqrt{\frac{3kT}{m}}$

The root mean square speed of molecules is directly proportional to the square root temperature. $V_{rms} = \sqrt{\frac{T_1}{T_2}} = \frac{1}{2}$

$$\frac{T_1}{T_2} = \frac{1}{4} \Rightarrow \frac{300\,k}{T_2} = \frac{1}{4} \Rightarrow T_2 = 1200k$$ **Answer: D**

43. Suppose that the index of refraction of a soap – bubble film is 1.25, what is. The minimum thickness of the soap – bubble film that results in constructive interference in the reflected light if the film is illuminated with light whose wavelength in free space is $\lambda = 600nm$?
A. 120nm B. 150nm C. 300nm D. 450nM

Solution:

In constructive air – film – air interference the film thickness can be calculated by $2nt = \left(m + \frac{1}{2}\right)\lambda, m = 0, 1, 2, \ldots$

For $m = 0$, $t = \frac{\lambda}{4n} = \frac{600nm}{4(1.25)} = 120nm$ **Answer: A**

44. A wire carries a steady current of 2.4 A.A straight section of the wire is 0.75 m long and lies in xy – plane along the x axis within

Telegram: https://t.me/ExtremeSeries

a uniform magnetic field, B = 1.6k T. If the current is in the positive x direction, what is the magnetic force on the section of wire?

A. 2.88N, in the negative y direction.
B. 2.88N, in the positive y direction.
C. 3.84N, in the negative y direction.
D. 3.84N, in the positive y direction.

Solution:

The magnetic force on a current carrying conductor is given by:

$F = I(\ell \times B)$.

$F = (2.4A)[(0.75mi \times 1.6Tk)] = 2.88(-j)N$

$F = 2.88N$ in the negative y – direction

Answer: A

45. Two different capacitors of capacitances $C_1 = 2\mu F$ and $C_2 = 3\mu F$ are connected in series across a 120V supply line. The charged capacitors are disconnected from the line and from each other and reconnected with terminals of like sign together. What will be the final charged C_1 and C_2, respectively?

A. 57.6μC and 86.4μC
B. 115.2μC and 127.8μC
C. 144μC across each
D. 288μC across each

Solution:

Initial: Capacitor in series; $C_T = \dfrac{C_1C_2}{C_1 + C_2} = \dfrac{(2\mu F)(3\mu F)}{2\mu F + 3\mu F} = \dfrac{6}{5}\mu F$

$$Q_T = C_T V = \left(\frac{6}{5}\mu F\right)(120V) = 144\mu F$$

$$V_1 = \frac{Q_T}{C_1} = \frac{144\mu F}{2\mu F} = 72V \text{ and } V_1 = \frac{Q_T}{C_1} = \frac{144\mu F}{2\mu F} = 72V \text{ and}$$

$$V_2 = \frac{Q_T}{C_2} = \frac{144\mu F}{2\mu F} = 48V \text{ is } V = \frac{Q_1 + Q_2}{C_1 + C_2} = \frac{288\mu F}{2\mu F + 3\mu F} = 57.6V$$

$Q_1 = (57.6V)(2\mu F) = 115.2\mu C$ and $Q_2 = (57.6V)(3\mu F) = 172.8\mu C$

46. A glass flask whose volume is exactly 1000 cm^3 at 0°C is totally filled with 1000 cm^3 of mercury at this temperature. When flask and mercury are heated to 100°C, 15.2 cm^3 of mercury is 1.82×10^{-4} per Celsius degree, what should be the coefficient of volume expansion of the glass?

A. 1.52×10^{-4}/°C
C. 3.00×10^{-5}/°C

B. 1.82×10^{-4}/°C
D. 1.00×10^{-5}/°C

Solution:

$V_o = 1000\ cm^3$, $T_o = 0°C$, $V_{o\,mercury} = 1000\ cm^3$, $T_f = 100°C$

$V_{o\ over\ flow} = 15.2\ cm^3$, $\gamma_{mercury} = 1.82 \times 10^{-4}$/°C and

$\Delta T = 100°C - 0°C = 100°C$

The volume of mercury over flow is equal to difference in expansion of mercury and glass.

$V_{o\ over\ flow} = \Delta V_{mercury} - \Delta V_{glass} = V_o\gamma_m\Delta T - V_o\gamma_g\Delta T$

$15.2cm^3 = V_o\Delta T(\gamma_m - \gamma_{glass}) = 1000\ cm^3(100°C)(\gamma_m - \gamma_{glass})$

$$\Rightarrow \gamma_m - \gamma_g = \frac{12.2cm^3}{(1000cm^3)(100°C)} = 1.52 \times 10^{-5}/°C$$

$\gamma_{glass} = \gamma_m - 15.2 \times 10^{-5}/°C = 1.82 \times 10^{-4} - 1.52 \times 10^{-4}$

$\gamma_{glass} = 0.3 \times 10^{-4}/°C = 3 \times 10^{-5}/°C$

Answer: C

47. A heat engine is being designed to have a Carnot efficiency of 65.0% when operating between two energy reservoirs. If the temperature of the cold reservoir is 20.0°C, what must be the temperature of the hot reservoir?

A. 57°C B. 310°C C. 564°C D. 807°C

Solution: Efficiency: $e = 1 - \dfrac{T_{cold}}{T_{not}}$

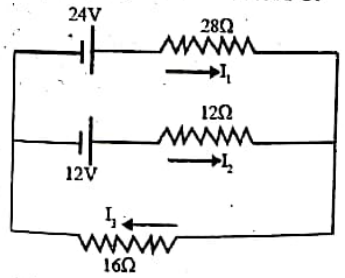
$$0.65 = 1 - \frac{293}{T} \Rightarrow T_{not} = \frac{293}{1 - 0.65} = 873k$$

$T_{hot} = 873 - 273 = 564°C$

Answer: C

48. In Figure 8, the current in the 16Ω resistor is:

A. 0.64A B. 0.15A C. 0.49A D. 1.28A



Solution:

Applying kirchoff's rules: $I_1 + I_2 = I_3$. From kirchoff's loop – rule

The potential drop in upper branch loop

Telegram: https://t.me/ExtremeSeries

$24V - (28\Omega)I_1 + (12\Omega)I_2 - 12V = 0$

$\Rightarrow (28\Omega)I_1 - (12\Omega)I_2 = 12V \ \ldots (1)$

The potential drop in lower branch loop;

$12V - (12\Omega)I_2 - (16\Omega)(I_1 + I_2) = 0$

$(28\Omega)I_2 + (16\Omega)I_1 = 12V \ \ldots (2)$

Solving (1) and (2)

$$7\begin{cases}(28\Omega)I_1 - (12\Omega)I_2 = 12V\\ 3\begin{cases}(16\Omega)I_1 + (28\Omega)I_2 = 12V\end{cases}\end{cases} \Rightarrow \begin{cases}196I_1 - 84I_2 = 84V\\ 48I_1 + 84I_2 = 36V\end{cases}$$

$$244I_1 + 0 = 120V$$

$$I_1 = \frac{120V}{244\Omega} = 0.49A \text{ and } I_2 = 0.15A$$

$I_3 + I_1 + I_2 = 0.49A + 0.15A = 0.64A$

Answer: A

49. A 10.0 m uniform ladder weighing 200N rests against a frictionless wall. The ladder makes 53° angle with the horizontal. If the ladder is just on the verge of slipping when a 600.0N fire fighter is 6.00m along the ladder form the bottom, what is the coefficient of static friction between the ladder and ground?

A. 0.3240 B. 0.4313 C. 0.5750 D. 0.7667

Solution: The forces on the ladder can be shown as below.

$\sum F_x = 0 \Rightarrow F_w - f = 0 \qquad f = F_w \ldots (1)$

$\sum F_y = 0 \Rightarrow F_R - 200N - 600N = 0 \qquad F_B = 800N \ldots (2)$

Applying the second condition of equilibrium considering the ground as an axis:

$F_w \times 10m \sin 53° = 200N \times (5m) \cos 53 + 600 \times 6m \times \cos 53°$

$F_w \times 10 \times 0.8 = 1000 \times 0.6 + 3600 \times 0.6;\ 8F_w = 2760N$

$$F_w = \frac{2760}{8}N = 345N \text{ . The coefficient of friction is } f = \mu F_N$$

$$\Rightarrow \mu = \frac{345N}{800N} = 0.4313 \text{ , since } F_N = F_w$$

Answer: B

50. A block of mass 1.6 kg is attached to a horizontal spring that has a force constant of 2000 N/m. The spring is compressed 2.0 cm and is then released from rest. At what speed will the block pass through the equilibrium position of a constant friction force of 10.0N retards its motion from the moment it is released?

A. 70 m/s B. 50 m/s C. 70 cm/s D. 50 cm/s

$k = 2000$ N/m, $f = 10$N, $x = 2$ cm, $m = 1.6$ kg

As the block starts from rest its velocity is zero at the release point. At the compressed position the total engine is Elastic potential energy.

$$EPe = \frac{1}{2}kx^2 = \frac{1}{2}(2000)(2\times10^{-2}m)^2 = (1000N/m)(4\times10^{-4}m^2) = 0.4J$$

Work done by friction force is $W_f = fs. -fs = 10N \times 0.02m = 0.2J$

The kinetic energy at the equilibrium position is

$$k.E = \frac{1}{2}mV^2 = \frac{1}{2}(1.6kg)V^2 = 0.8V^2$$

Work done by non – conservative force is equal to the lost in mechanical energy: $\Delta M.E = W_{nC}$

$$0.4J - 0.8V^2 = 0.2J \Rightarrow 0.8V^2 = 0.2J \Rightarrow V = \frac{0.2}{0.8} = \frac{1}{4}$$

$\therefore V = 0.5$ m/s $= 50$ cm/s

Answer: D

Telegram: https://t.me/ExtremeSeries

1. What is the angle between vector $\mathbf{A}=\left(a\mathbf{i}+a\sqrt{3}\mathbf{j}\right)$ units $\mathbf{B}=\left(a\sqrt{3}\mathbf{i}+a\mathbf{j}\right)$ units?

A. 90° B. 60° C. 37° D. 30°

Clue:

$$\mathbf{A}\cdot\mathbf{B}=AB\cos\theta$$

$$\Rightarrow\cos\theta=\frac{\mathbf{A}\cdot\mathbf{B}}{AB}=\frac{\left(a\mathbf{i}+a\sqrt{3}\mathbf{j}\right)\cdot\left(a\sqrt{3}\mathbf{i}+a\mathbf{j}\right)}{\left(\sqrt{a^2+\left(a\sqrt{3}\right)^2}\right)\cdot\left(\sqrt{\left(a\sqrt{3}\right)^2+a^2}\right)}$$

$$\cos\theta=\frac{a^2\sqrt{3}+a^2\sqrt{3}}{\left(\sqrt{a^2+3a^2}\right)\left(\sqrt{3a^2+a^2}\right)}$$

$$=\frac{2a^2\sqrt{3}}{4a^2}=\frac{\sqrt{3}}{2}\Rightarrow\theta=\cos^{-1}\frac{\sqrt{3}}{2}=30°$$

Answer: D

2. The magnitudes of two vectors **A** and **B** are 12 units and 5 units, respectively. Which one of the following is **NOT** the possible value for the magnitude of the resultant vector **A + B**?

A. 60 units C. 13 units

B. 17 units D. 7 units

Clue: For any two vectors A and B, the resultant is always, in the range $|A-B|\le R\le|A+B|$, now A = 12 and B = 5

$\Rightarrow|12-5|\le R\le|12+5|\Rightarrow 7\le R\le 17$. Thus, 60 is out of the range

Answer: A

3. What is the vector product $\vec{A}\times\vec{B}$ of two vectors $\vec{A}=7\hat{i}+4\hat{j}-8\hat{k}$ and $\vec{B}=3\hat{i}-2\hat{j}+5\hat{k}$?

A. $36\hat{i}+11\hat{j}-2\hat{k}$ C. $-4\hat{i}+59\hat{j}+26\hat{k}$

B. $4\hat{i}-59\hat{j}-26\hat{k}$ D. $-36\hat{i}-11\hat{j}+2\hat{k}$

Clue:

$$\vec{A}\times\vec{B}=\begin{vmatrix} i & j & k \\ 7 & 4 & -8 \\ 3 & -2 & 5 \end{vmatrix}$$

$$=\left[(4)(5)-(-2)(-8)\right]i+\left[(-8)(3)-(7\times5)\right]j+\left[(7)(-2)-(4)(3)\right]K$$

$$=4i-59j-26k$$

Answer: B

4. Which one of the following experimental errors can be reduced by taking repeated measurements?

A. Parallax errors C. Random errors

B. Systematic errors D. Zeros errors

Answer: C

5. A teacher gave a student a book whose thickness is 4.30 cm, and instructed him to measure the thickness of the book with vernier calipers. The student took four measurements. The average and standard deviation of the measurements were 4.33 cm and 0.04 cm. respectively. Which one of the following is the uncertainty in the accuracy of the student's measurement?

A. 0.03 cm C. 0.04cm

B. 0.035cm D. 0.30cm

Answer: A

6. If a long distance jumper athlete leaves he ground at an angle of 37.0° above the horizontal surface at speed of 10.0m/s how far does he jump in the horizontal direction?

A. 4.8m B. 6m C. 9.6m D. 12m

Clue:

$$R=\frac{2(u\cos\theta)(u\sin\theta)}{g}=\frac{(2)(10)\cos37°-(10)\sin37°}{10}$$

$$=9.6m$$

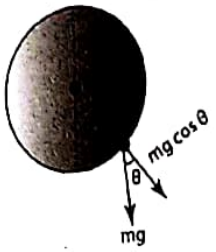
Answer: C

7. A woman is rotating a bucket of water in a vertical circle of radius 0.9m. the mass of bucket and water is 5kg. What is the bucket's minimum speed at the top of the circle if no water is to spill out?

A. 0 B. 1m/s C. 3m/s D. 9m/s

Telegram https://t.me/ExtremeSeries

Clue:



At any time, $T - mg\cos\theta = \frac{mv^2}{r}$, and at the top, $\theta = 180°$ and $T = 0$

$$\Rightarrow mg = \frac{mv^2}{r},$$

$$v = \sqrt{gr} = \sqrt{(0.9)(10)} = 3m/s$$

Answer: C

8. A force **F** of magnitude 20N is applied to a block of mass 2 kg that lies on a rough, horizontal surface as shown in Figure 1. The coefficient of kinetic friction between the block and surface is 0.4. what is the magnitude of the acceleration of the block?

A. $10m/s^2$
B. $5.2\ m/s^2$
C. $4\ m/s^2$
D. $2.8\ m/s^2$

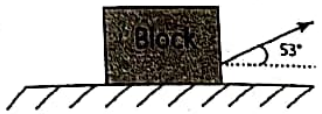


Fig.1

Clue:

$F\cos\theta - f = ma$

$F\cos\theta - \mu F_N = ma$ but $F_N = mg - F\sin\theta$

$F\cos\theta - \mu(mg - F\sin\theta) = ma$

$$\Rightarrow a = \frac{F\cos\theta - \mu(mg - F\sin\theta)}{m}$$

$$a = \frac{(20)\cos 53 - 0.4[(2)(10) - 20\sin 53°)}{2} = 5.2m/s^2$$

Answer: B

9. Which one of the following statements is correct?

A. An object moving toward the east cannot have an acceleration toward the west.

B. If the average velocity of an object is zero in some time interval, the average speed of the object for the time interval is also zero.

C. The velocity-time graph of an object moving with constant acceleration is parallel to the time axis.

D. An object having zero velocity can have acceleration different from zero.

Answer: D

10. A car travels along a straight line at a constant speed of 20m/s for a distance d and then another distance d in the same direction at a constant speed of 36 km/hr. what is the average speed for the entire trip?

A. 23km/h
B. 48km/h
C. 54 km/h
D. 60km/h

Clue: $V_{av} = \frac{S_T}{t_T} = \frac{S_1 + S_2}{t_1 + t_2} = \frac{S_1 + S_2}{\frac{S_1}{V_1} + \frac{S_2}{V_2}}$, but $S_1 = S_2 = d$

$V_1 = 20m/s = (20)(3.6) = 72km/hr$

$$V_{av} = \frac{d + d}{\frac{d}{V_1} + \frac{d}{V_2}} = \frac{2d(V_1V_2)}{d(V_1 + V_2)} = \frac{2V_1V_2}{V_1 + V_2}$$

$$= \frac{(2)(36)(72)}{(36+72)} = 48km/h$$

Answer: B

11. A highway curve is banked (inclined) in such a way that a car travelling at a speed of 13.5m/s can round the curve without skidding, in the absence of friction. If the banking angle is 30° what is the radius of the curve?

A. 10.5m
B. 21.0m
C. 31.7 m
D. 36.7 m

Clue:

$$f = F_c \Rightarrow mg\tan\theta = \frac{mv^2}{r} \Rightarrow r = \frac{v^2}{g\tan\theta} = \frac{(13.5)^2}{(10)\tan 30}$$

$$= 31.6m$$

Answer: C

12. A car mass 1500 kg is travelling at a uniform velocity of 30m/s due east. The driver applies the brakes to slow down the car to a velocity of 10m/s due east. What is the work done in slowing down the car?

Telegram: https://t.me/ExtremeSeries

A. 15kJ
B. 300kJ
C. 600kJ
D. 750 kJ

Clue:

$$W = \Delta K_E = \frac{1}{2} \Rightarrow \left(v^2 - u^2\right) = \frac{1}{2}(1500)\left[10^2 - 30^2\right] = 600k.$$

Answer: C

13. Which one of the following statements is true about a spring-mass system oscillating on a horizontal frictionless surface?
A. The kinetic and potential energies are equal at all times.
B. The kinetic and potential energies are both constant.
C. The maximum potential energy is achieved when the mass passes though its equilibrium position.
D. The maximum kinetic energy and maximum potential energy are equal.

Answer: D

Clue: from conservation of energy,

14. A hard ball moving on a horizontal frictionless surface with a speed of 5m/s in the positive x direction strikes a stationary hard ball of the same mass. After the collision, the first ball moves with a speed of 4m/s along a direction that makes an angle of 37.0° with its initial direction of motion. What is the struck ball's speed and the smallest angle between its direction of motion and the positive x axis after the collision?
A. 3m/s and 53.0°
B. 3m/s and 37.0°
C. 4m/s and 53.0°
D. 4m/s and 37.0°

Clue: from conservation of linear momentum,

$$m_1 = \begin{bmatrix} u_1x \\ u_1y \end{bmatrix} + m_2 = \begin{bmatrix} u_2x \\ u_2y \end{bmatrix} = m_1 \begin{bmatrix} v_1x \\ v_1y \end{bmatrix} + m_2 \begin{bmatrix} v_2x \\ v_2y \end{bmatrix}, m_1 = m_2$$

$$\begin{bmatrix} 5 \\ 0 \end{bmatrix} + 0 = \begin{bmatrix} v_1 \cos 37 \\ v_1 \sin 37 \end{bmatrix} = \begin{bmatrix} v_2x \\ v_2y \end{bmatrix}$$

$$\begin{bmatrix} 5 \\ 0 \end{bmatrix} + 0 = \begin{bmatrix} (4)(0.8) \\ (4)(0.6) \end{bmatrix} = \begin{bmatrix} v_2x \\ v_2y \end{bmatrix} = \begin{bmatrix} 3.2 \\ 2.4 \end{bmatrix} = \begin{bmatrix} v_2x \\ v_2y \end{bmatrix}$$

$$5 = 3.2 + v_2x \Rightarrow v_2x = 1.8\text{m/s and}$$

$$0 = 2.4 + v_2y \Rightarrow v_2y = -2.4\text{ m/s} \Rightarrow v_2 = \sqrt{v_{2x}^2 + v_{2y}^2}$$

$$v_2 = \sqrt{(1.8)^2 + (-2.4)^2} = 3\text{m/s and } \theta\ 53°$$

Answer: A

15. A man standing on a frictionless surface continuously fired 120 bullets horizontally in 1 minute using a machine gun. If each bullet has a mass of 10 g and leaves the muzzle of the gun with a speed of 800m/s, what is the average force exerted by the gun on the man?
A. 8N
B. 16N
C. 960N
D. 16000N

Clue:

$$F\Delta t = nm\Delta v \Rightarrow F = \frac{nm\Delta v}{\Delta t} = \frac{(120)(0.01)(800)}{60} = 16\text{N}$$

Answer: B

16. Two blocks are connected by a string of negligible mass passing over a pulley of radius $r = 0.2$m and moment of inertia I as shown in Figure 3. The b lock on the horizontal frictionless surface is moving with a constant acceleration $a = 20.0\text{m/s}^2$. If the pulley axis is frictionless, the string is inextensible and does not slip on the pulley, what should be the moment of inertia of the pulley?
A. 0.40 k gm^2
B. 0.80 k gm^2
C. 4.00 k gm^2
D. 8.00 k gm^2

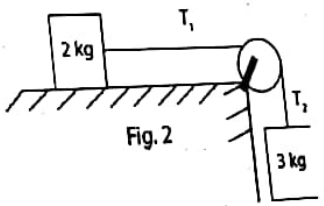


Fig. 2

Clue:

$$T_1 = 4N \text{ and } T_2 = 24N, \Rightarrow \Delta T = F = 20N, \text{ and}$$

$$\tau = Fr = I\alpha, \alpha = \frac{a}{r}$$

$$Fr = I\left(\frac{a}{r}\right) \Rightarrow I = \frac{Fr^2}{a} = \frac{(20)(0.2)^2}{2} = 0.4kgm^2$$

Answer: A

17. An object with mass of $m = 3.0$ kg is attached to the free end of a light string wrapped around a pulley of radius $R = 0.50$ m and moment of inertia $I = 0.75$kg

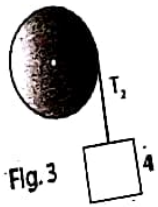


Fig. 3

Telegram: https://t.me/ExtremeSeries

m^2. The pulley is free to rotate in a vertical plane about a fixed frictionless axis passing through its center as shown in Figure 4. The suspended object is released from rest at 3.60 m above the floor. What is the angular momentum of the pulley at the time when the object strikes the floor?

A. 4.5 kg m^2 /s
B. 9.0 kg m^2 /s
C. 12.3 kg m^2 /s
D. 18 kg m^2 /s

Clue: $mgh = \frac{1}{2}mv^2 + \frac{1}{2}Iw^2, w = \frac{v}{R}$

$$\Rightarrow mgh = \frac{1}{2}mv^2 + \frac{1}{2}I = \left(\frac{v}{R}\right)^2 = (3)(36) = 1.5v^2 + 1.5v^2$$

v= 6m/s and $w = v/R$=12 rad/s $\Rightarrow L = Iw = (0.75)(12) = 9$

Answer: B

18. Which one of the following statements is **NOT** correct about a particle that is in equilibrium?

A. It can have a velocity different from zero.
B. Its velocity can be equal to zero.
C. It can be under the action of more than one force.
D. It can have an acceleration different from zero.

Answer: D

19. A block of mass 10 kg is pulled up a rough inclined plane with a force of 100N parallel to the inclined plane. The inclined plane makes an angle of 37° with the horizontal. At the instant when the force is applied the block has a speed of 1.5m/s. If the coefficient of kinetic friction is 0.40, and the block is pulled up 5m, what is the change in the kinetic energy of the block?

A. 20J C. 40J
B. 30J D. 45.5J

Hint:

$$F - mg\sin\theta - \mu mg\cos\theta = ma \Rightarrow a = \frac{F - mg\sin\theta - \mu mg\cos\theta}{m}$$

$$a = \frac{100 - (10)(10)(0.6) - (0.4)(10)(10)(0.8)}{10} = 0.8\text{m/s}^2$$ a

nd

$$v^2 = u^2 + 2as = (1.5)^2 + (2)(0.8)(5) = 10.25$$

$$\Delta K_E = \frac{1}{2}m\left(v^2 - u^2\right) = \frac{1}{2}(10)(10.25 - 2.25) = 40\text{J}$$

Answer: C

20. The particles shown in Figure 2 are connected by rectangular rigid rods of negligible mass. The origin of the xy plane about the z axis with an angular speed of 6.00 rad/s what is the rotational kinetic energy of the system?

A. 2574J
B. 2340J
C. 1287 J
D. 1170J

Hint:

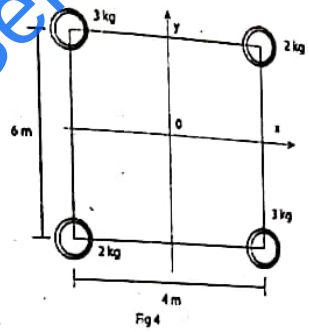


Fig 4

$$K_{Er} = \frac{1}{2}Iw^2 = \frac{1}{2}mr^2w^2 = \frac{1}{2}w^2\left(m_1r_1^2 + m_2r_2^2 + m_3r_3^2 + m_4r_4^2\right)$$

$$K_{Er} = \frac{1}{2}(6)^2\left((3)(13) + 2(13) + 2(13) + (3)(13)\right) = 2340\text{J}$$

Answer: B

21. Two solid spheres made of gold and silver having equal masses are suspended by strings and fully submerged in a tank of water. Given that the density of gold is greater than that of silver, which one of the following statements is true?

A. The buoyant force on each sphere is the same.
B. The volume of water displaced by the gold sphere is larger than that of the silver sphere.
C. The tension in the string supporting each sphere is the same.
D. The buoyant force on the silver sphere is greater than that on the gold sphere.

Hint: Since $\rho_g > \rho_s$ and $m_g = m_s \Rightarrow v_s > v_g$ The buoyant force equals the weight of the displaced fluid

$\Rightarrow F_b = m_d g = \rho_w v_d g$

Answer: D

22. The cross-sectional areas of the small and large pistons of a hydraulic lift

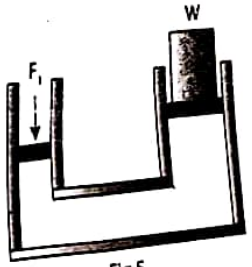


Fig 5

Telegram: https://t.me/ExtremeSeries

shown in Figure 6 are 4.0cm^2 and 0.02 m^2, respectively. What is the magnitude of the force that must be applied to the small piston to lift a weight w – 15,000N place on the large piston.

A. 3.0×10^2 N
B. 1.5×10^4 N
C. 7.5×10^5 N
D. 3×10^6 N

Hint: From pascal's principle,

$$\frac{F_1}{A_1} = \frac{F_2}{A_2} \Rightarrow \frac{F_1}{4\times10^{-2}} = \frac{15000}{2\times10^{-2}} \Rightarrow F_1 = 3\times10^2\,N$$

Answer: B

23. An incompressible fluid flows through a horizontal pipe whose diameter varies along its length. The sum of the kinetic energy per unit volume and pressure of the fluid at different sections of the pipe would _____

A. depend on the length of the pipe.
B. increase as the pipe diameter increases.
C. increase as the pipe diameter decreases.
D. remain the same as the pipe diameter changes.

Hint: From Bernoulli's equation $\frac{p}{\rho} + gh + \frac{v_2}{2} =$ constant

$$\Rightarrow \frac{p_1}{\rho} + gh_1 + \frac{v_1^2}{2} = \frac{p_2}{\rho} + gh_2 + \frac{v_2^2}{2}$$

Answer: D

24. a uniform bar of length 2.0 m and mass 30.0 kg is supported by two mass less ropes which are attached to a fixed horizontal bar, as shown in Figure 5. What are the tensions in each rope when a 700N weight is suspended at $d = 0.50$ m from the left end of the bar?

A. $T_1 = 675$ N and $T_2 = 325$ N
B. $T_1 = 325$ N and $T_2 = 675$ N
C. $T_1 = 500$ N and $T_2 = 500$ N
D. $T_1 = 525$ N and $T_2 = 175$ N

Telegram: https://t.me/ExtremeSeries

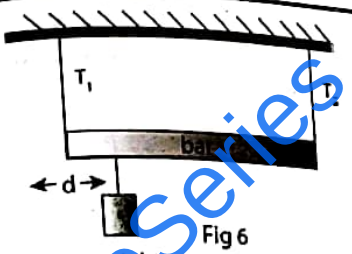


Fig 6

Hint: Taking moment at the left end we have, $\sum\tau = 0$
$\Rightarrow (0.5)(700) + (1)(300) = (T_2)2 = T2 = 325\text{N}\ T_1 = 635\text{N}$

Answer: A

25. Which one of the following laws best describes the behavior of elastic materials?

A. Gay Lussac's law
B. Hook's Law
C. Newton's second law
D. Newton's law of gravitation

Answer: B

26. The first law of Thermodynamics take the form dU = dW. This equation is valid if the process is _____.

A. isobaric
B. isochoric
C. isothermal
D. adiabatic

Hint: $\Delta u = \Delta Q + \Delta w$. When $\Delta u = \Delta w$, adiabatic

Answer: D

27. An ideal gas undergoes an isothermal process starting with a pressure of 2×10^5Pa and a volume of 6cm^3. Which one o the following might be the pressure and the volume of the final state?

A. 4×10^5Pa and 4 cm^3
B. 4×10^5Pa and 3 cm^3
C. 5×10^5Pa and 3 cm^3
D. 10^5Pa and 4 cm^3

Hint: From ideal gas equation $P_1V_1 = P_2V_2$

Answer: B

28. Which one of the following statement is **NOT** correct?

A. The entropy of the universe increases in all natural processes.
B. No heat engine operating in a cycle can absorb energy from reservoir and use it entirely to do work.
C. When a system undergoes a change in state, the change in its internal energy is the difference between the energy transferred to it by heat and the work done on it.

D. The entropy of an isolated system reaches its maximum value when the system is in equilibrium state.

Answer: D

29. The amount of heat required to raise the temperature of an object of mass m = 1kg by 1 Kelvin is known as _____.
A. Specific heat capacity
B. Heat capacity
C. Expansion coefficient
D. Internal energy

Answer: A

30. A given weight of gas has a temperature of 27°C and volume V. keeping the pressure constant, the volume is increased to 3V by varying the temperature. What is the final temperature of the gas?
A. 81°C
B. 546 °C
C. 627 °C
D. 900 °C

Clue: $\frac{P_1V_1}{T_1}=\frac{P_2V_2}{T_2}=\frac{V}{300}=\frac{3V}{T_2}=T_2=900\,k=627°C$

Answer: C

31. A given minimum point (through) of a sinusoidal wave is traveling 480 cm along a rope in 12 s. if the oscillator that generates the wave completes 48 vibrations in 30 s. what is the wave length of the wave?
A. 25 m
B. 15 m
C. 0.25m
D. 0.15m

Hint:

$V=\frac{(4.8)}{(12)}=0.4m/s$ and $f=\frac{48}{30}=1.6\,H_z, \lambda=\frac{V}{f}=0.25\,m$

Answer: C

32. A wave traveling from medium of index of refraction n_1 toward a medium of index of refraction n_2 undergoes partly transmission and partly reflection. If n_2 is greater than n_1 which one of the following statements is not correct?
A. The speed of the wave in the medium with refractive index n_1 is smaller than with n_2.
B. The incident and the reflected waves are in-phase.
C. The angle of reflection is the same as the angle of incidence.
D. The refraction angle is less than the incidence angle.

Answer: A

Hint: from snell's law,

33. In Young's double-slit experiment interference pattern which one of the following activities move the successive bright fringes farther apart.

A. Decreasing the wavelength of the light.
B. Decreasing the distance between the viewing screen and the double slit.
C. Increasing the slit spacing.
D. Decreasing the slit spacing.

Clue: from the relation, $\frac{\lambda}{s}=\frac{w}{D}$, w increases as s decreases.

Answer: D

34. The position of a particle moving with simple harmonic motion is given by an expression $x = A \cos(\omega t + \varnothing)$, ω where A is the amplitude of the oscillation, t is time, ω is angular frequency $\varnothing$ is the phase constant. If $\varnothing = \pi/2$, what will be the sign of the particle's position and velocity at $t = 7T/8$, where T is the period of oscillation?
A. The position is positive, and the velocity is negative.
B. The position and velocity are both positive.
C. The position and velocity are both negative
D. The position is negative, and the velocity is positive.

Clue: $x=A\cos(wt+\phi)=A\left(\frac{2\pi}{T}t+\frac{\pi}{2}\right), t=\frac{7T}{8}$

$x=A\cos\left(\frac{2\pi}{T}\left(\frac{7T}{8}\right)+\frac{\pi}{2}\right)=A\cos\left(\frac{9}{4}\pi\right)$ and $V=A\sin\left(\frac{9\pi}{4}\right)$

Answer: B

35. If a long rope of uniform linear density is hung from a ceiling and waves are sent up the rope from its lower end, which one of the following is true about the speed of the ascending waves?
A. The speed decrease as the wave ascend.
B. The speed increase as the wave ascend.
C. The waves ascend with a uniform speed.
D. The speed of the waves is larger around the middle of the rope than around the beginning and end.

Answer: A

36. A parallel plate capacitor has a capacitance of 10μF. if the distance between the plates is doubled, keeping all other quantities constant, what is the new capacitance of the capacitor?
A. 0.2 μF
B. 5 μF
C. 10 μF
D. 20 μF

Telegram: https://t.me/ExtremeSeries

Clue:

$$C_1 = \epsilon_r \epsilon_0 \frac{A}{d_1}, \text{ if } d_2 = 2d_1, C_2 = \epsilon_r \epsilon_0 \frac{A}{2d_1} = \frac{10}{2} = 5\mu F$$

37. If several resistors are connected in parallel, the equivalent resistance is

Answer: B

A. greater than any of the resistance in the group.
B. is equal to the average resistance of the group
C. less than any of the resistances in the group.
D. equal to the sum of the resistances in the group.

Answer: C

38. If aluminum and copper wires with circular cross sections have equal lengths and equal resistance, then the ratio of the radius of aluminum to that of copper should be the same as that of the _____.

A. Ratio of the resistivity of aluminum to copper
B. Ratio of the resistivity of copper to aluminum
C. Square root of the ratio of the resistivity of aluminum to copper
D. Square root of the ratio of the resistivity of copper to aluminum

Clue: $R = \frac{\rho \ell}{A} \Rightarrow \frac{\rho_A \ell_A}{\pi r_A^2} = \frac{\rho_C \ell_C}{\pi r_C^2} = \frac{r_A}{r_C} = \sqrt{\frac{\rho_A}{\rho_C}}$

Answer: C

39. Which of the following statements is correct?

A. The electric field strength due to two point charges of equal magnitude and the same sign is zero only at one point on the line joining the two charges.
B. If a point charge of mass m is released from rest in a uniform electric field, it will move along a line of force with uniform speed.
C. The variation of the strength of the electric field due to an electric dipole is inversely proportional with the square of the distance from its center of mass.
D. An electric dipole moment $\boldsymbol{P}$ in a uniform external electric field $\boldsymbol{E}$ is in stable equilibrium if $\boldsymbol{P}$ and $\boldsymbol{E}$ are in opposite directions.

Answer: B

40. In a certain region of space, a uniform electric field is directed in the positive x direction. If a particle with negative charge is carried from $x = 20$ cm to $x = 60$ cm, what can we conclude about the electric potential and the electric potential energy of the charge-field system?

A. The electric potential at $x = 20$cm is larger than that at $x = 60$ cm.
B. The electric potential at $x = 20$cm is the same as that at $x = 60$ cm.
C. The electric potential energy decrease.
D. The electric potential energy increase.

Answer: D, $u = EQd$

41. An electron moving at a speed of 1.5×10^7 m/s enters a region of uniform magnetic field of magnitude 2×10^{-3} T, in a direction perpendicular to the field. The path followed by the electron is circular. What is the radius of the circular path?

A. 0.43 cm
B. 4.3 cm
C. 43 cm
D. 430 cm

Hint:

$$qB = \frac{mv}{r} \Rightarrow r\frac{mv}{qB} = \frac{(9.11\times10^{-31})(1.5\times10^{7})}{(1.6\times10^{-19})(2\times10^{-3})} = 4.3\times10^{-2}$$

Answer: B

42. If a wire carrying current in the plane of this paper toward the top of the page experiences a magnetic force toward the right edge of the page what will be the direction of the magnetic field causing this force?

A. In the plane of the page and toward the left edge.
B. In the plane of the page and toward the bottom edge.
C. Perpendicular and directed out of the plane of the page.
D. Perpendicular and directed into the plane of the page.

Hint: using right hand rule.

Answer: B

43. A square flat loop of wire is placed in a uniform magnetic field that is in the y direction. The magnetic flux through the loop is a maximum if the plane of the loop is in the

A. xy plane.
B. xz plane.
C. yz plane
D. *inclined 45° with respect to the xy* plane

Answer: B

Telegram: https://t.me/ExtremeSeries

44. In the circuit shown in Figure 7, what is the potential difference between point b and a?

A. 19.2 V
B. 24.0 V
C. 13.0 V
D. 12. 0 V

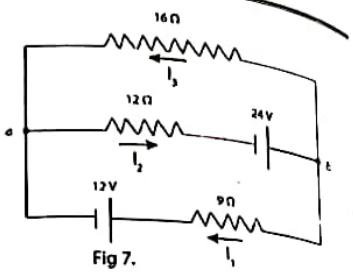


Fig 7.

Hint: from point rule. $I_1 + I_3 = I_2$ and form loop rule $24 = 16I_3 + 12I_2$ and $-24 + 12 = 12I_2 - 9I_1$
$\Rightarrow I_2 = 1A$ and $V_a = I_2R_2 = (1)(12) = 12V$, $V_b = 24_V$,
$\Delta V = V_b - V_a = 12V$

Answer: D

45. Which of the following statements is correct about the effect of a magnetic field?

A. A magnetic field exertes a nonzero force on a stationary charged object.
B. A magnetic field exerts a nonzero force on a conducting wire placed in the field
C. A uniform magnetic field can be created at the point around long current carrying wire
D. A magnetic field can be created in the space surrounding any charged object that is in motion.

Answer: D

46. A series *RLCI* circuit has a total impedance of 75.0 Ω. If the resistance is R = 45Ω. And the average power delivered to the circuit is P = 352.8W, what is the root-mean square value of ac voltage source?

A. 83 V
B. 116 V
C. 210 V
D. 297 V

Clue:

$P = V_{rms} I_{rms} \cos\theta$, $I_{rms} = \frac{V_{rms}}{Z}$ and $\theta = \tan^{-1}\left(\frac{X_L - X_C}{R}\right)$ but

$Z = \sqrt{R^2 + X_L - X_C{}^2} \Rightarrow X_L - X_C = \sqrt{Z^2 - R^2} = \sqrt{75^2 - 45^2}$

$X_L - X_C = 60$. Then,

$$P = (V_{rms}) = \left(\frac{V_{rms}}{Z}\right)\cos\theta = \frac{V_{rms}^2}{Z}\cos\left(\tan^{-1}\left(\frac{60}{45}\right)\right)$$

$$\Rightarrow V_{rms}^2 = \frac{PZ}{\cos 53^\circ} = \frac{(352.8)(75)}{0.6} \Rightarrow V_{rms} = 210V.$$

Answer: C

47. Which of the following phenomena most clearly demonstrates the wave nature of electrons?

A. The photoelectric effect
B. Blackbody radiation
C. The Compton effect
D. Diffraction of electrons by crystals

Answer: D

48. When the radioactive element $^{144}Nd_{60}$ decays to $^{140}Ce_{58}$, which one of the following particles is emitted?

A. A beta particle
B. An alpha particle
C. A proton
D. A gamma ray

Answer: B

49. Which one of the following actions increases the magnetic energy density in the interior of a very long current-carrying solenoid?

A. Increasing the number of turns per unit length of the solenoid
B. Increasing the cross sectional area of the solenoid.
C. Increasing the length of the solenoid keeping the number of turns per unit length fixed.
D. Decreasing the cross sectional area of the solenoid.

Hint:

Since $U_B = \frac{B^2}{2\mu o}$, $B = \mu \times In$, $n = \mu/\ell$ $\in = BLV$ and $I = \frac{\in}{R}$

Answer: A

50. Figure 8 (below) shows a rectangular loop of wire of width L and resistance R. One end of the loop is in a uniform magnetic field of strength B in directed perpendicular into the plane of the

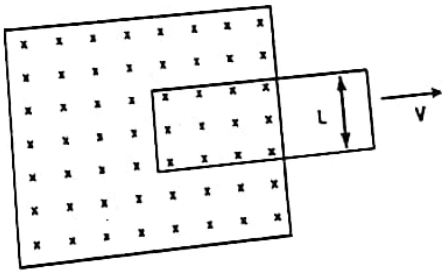


Fig 8.

Telegram: https://t.me/Extremeseries

page. The loop is pulled to the right at a constant speed v. what are the magnitude and direction of the induced current in the loop?

A. $I = BLvR$ and clockwise
B. $I = BLv/R$ and clockwise
C. $I = BLvR$ and counterclockwise
D. $I = BLv/R$ and counterclockwise

Explanation:

The flux is decreasing as the loop leaves the region of the magnetic field. To oppose, a decreasing the induced current flux generate a magnetic field in the same direction as an applied magnetic field into the page. Then the induced current must be clock wise direction.

$\varepsilon = BLV$ and $I = \frac{\varepsilon}{R}$

$\therefore I = \frac{BLV}{R}$ clockwise

Answer: B

Physics

DIRECTIONS: Each of the following questions is followed by four possible alternatives. Read each question and carefully **blacken** the letter of your best choice on the separate answer sheet provided.

You may refer to the information given below when you work on some of the questions.

Constant	Symbol	Value
Acceleration due to gravity	G	$10m/s^2$
Permittivity of vaccum	ε_o	$8.85\times10^{-12}F/m$
Permeability of free space	μ_o	$4\pi\times10^{-7}Tm/A$
Faraday's constant	F	$96500C\ mol^{-1}$
Charge of an electron	ė	1.6×10^{-19}Coulomb
Universal gravitational constant	G	$6.67\times10^{-19}\ \frac{Nm^2}{Kg^2}$
Planck constant	h	$6.63\times10^{-34}Js$

$\sin 30° = \cos 60° = 0.5$

$\sin 37° = \cos 53° = 0.6$

$\sin 53° = \cos 37° = 0.8$

$\sin 60° = \cos 30° = 0.866$

$\frac{1}{\sqrt{2}} = 0.707$

1. The graph below illustrates the position and time for a cat that runs to catch a rat and then returns with it. The cat caught the rat after 2 seconds. What was the cat's average speed as it returned with the rat?

Telegram: https://t.me/ExtremeSeries

A. Its average return speed was 2 m/s.
B. Its average return speed was 4 m/s.
C. Its average return speed was 0.7 m/s.
D. Its average return speed was 0.9m/s.

Explanation: The return average speed is

$$v_{av} = \frac{S_T}{t_T} = \frac{7}{8} = 0.875 \approx 0.9 m/s$$

Answer: D

2. A man is standing on a spring scale in an elevator. when the elevator is at rest, the scale reads 560N. When the elevator starts to move, the scale reads 600N. The elevator has:

A. an upward acceleration
B. a constant downward speed
C. a constant upward speed
D. downward acceleration

Explanation: $T - mg = ma \Rightarrow T = mg + ma$
The scale reading is bigger when the elevator accelerates upward.

Answer: A

3. What is the necessary and sufficient condition for conservation of linear momentum?

A. The presence of more linear than nonlinear forces in the system.
B. Net force on the system is zero.
C. the presence of only non-linear forces
D. Internal forces exceed external forces.

Explanation: The necessary and sufficient condition for the conservation of linear momentum is that, the net force on the system, should be zero.

Answer: B

4. Which statements is **WRONG** about units and dimensions?

A. Two quantities can have the same units but different dimensions.
B. Two quantities can have the same dimensions but different units.
C. It is always possible to add two numbers that have the same units.
D. There are quantities that have no unit and dimension.

Explanation: It is impossible to add speed and velocity even if they have the same unit (m/s) since one is scalar and the other is vector.

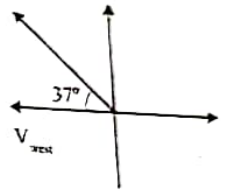
Answer: C

5. An airplane flies at speed of 720.0km/hr at 37° North of west direction. How far does the plane travel to the west in 1 hour?

A. 432 km B. 720 km C. 576 km D. 504 km

Explanation: The displacement of the plane to the west can be computed by

$s = v \Rightarrow s_{(west)} = v_{(west)} \times t$
$s = v_{(cos37)}\, t \Rightarrow s_{(west)} = v_{(west)} \times t$
$s = v_{(cos37)}\,(t) = (720)(cos37)(1)$
$s = 576km$



Answer: C

6. Abrehet (A): Bekele (B) and Chaltu (C) are pulling a box horizontally with forces $\vec{F}_A$; $\vec{F}_B$ and $\vec{F}_C$ respectively as shown below. Which statement is correct if the box is in equilibrium?

A. $\vec{F}_A = \vec{F}_B \times \vec{F}_C$
B. $-\vec{F}_A = \vec{F}_B + \vec{F}_C$
C. $\vec{F}_A = \vec{F}_B \cdot \vec{F}_C$
D. $\vec{F}_A = \vec{F}_B - \vec{F}_C$

Explanation: $\sum F = 0 \Rightarrow \vec{F}_A = \vec{F}_B \times \vec{F}_C = 0$
$-\vec{F}_A = \vec{F}_B + \vec{F}_C$

Answer: B

7. Determine the torque due to force $\vec{F} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ N acting at a point $\vec{r} = 3\hat{i} + 2\hat{j} + 3\hat{k}$ about the origin?

A. $6\hat{i} - 6\hat{j} + 12\hat{k}$ C. $17\hat{i} - 6\hat{j} + 13\hat{k}$

Telegram: https://t.me/ExtremeSeries

B. $-6\hat{j}+6\hat{j}+12\hat{k}$ D. $-17\hat{i}+6\hat{j}+13\hat{k}$

Solution: $\tau=\vec{r}\times\vec{F}=\begin{vmatrix} i & j & k \\ 3 & 2 & 3 \\ 2 & -3 & 4 \end{vmatrix}$

$\tau=17\hat{i}-6\hat{j}+13\hat{k}$

Answer: C

8. An airplane travelled from Addis Ababa to Mekele, 780km, in 45 minutes. What was its average speed?

A. 0.28m/sec C. 17.3 m/min
B. 1040 km/hr D. 585 km/hr

Explanation: $v_{av}=\frac{S_T}{t_T}=\frac{780}{45/60}=1040\text{km/hr}..,\ 45\text{min}=\frac{45}{60}\text{ hr}$

Answer: B

9. An example of a body moving with constant speed but still accelerating is:

A. A body moving on a straight road
B. A body moving on a straight railway track
C. A body moving in a circular path
D. A body falling in a viscous fluid

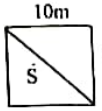
Explanation: A body is said to be accelerating when it changes speed, or direction or both. Thus a body in circular path continuously changes its direction while moving at constant speed.

Answer: C

10. A student moves along the boundary of a square field of side 10m in 40 sec. If he started from one of the corners of the student at the end of 2 minutes and 20 seconds from his initial position?

A. 10m. B. $10\sqrt{2}$m C. 40m D. 30m

Explanation: If they body takes 40 seconds to move the square filed once, he will move the field 3 and half times in 2 minutes and 20 seconds. Thus, his displacement after 140 seconds is



$S=\sqrt{40^2+10^2}=\sqrt{200}$

$S=10\sqrt{2}$m

Answer: B

11. If the impulse of a rotating disc is non-zero, what can you conclude?

A. The change in linear momentum is zero.
B. The torque acting on object is constant.
C. Impulse of the disc equals its moment of inertia.
D. The change in angular momentum is non zero.

Explanation: Impulse, $I=F\Delta t=\Delta P$. or

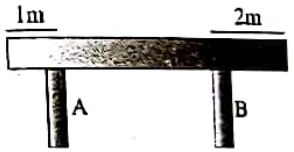
$I=\tau\Delta t=\Delta L$

it is non-zero when $\Delta L\neq 0$.

Answer: D

12. A uniform, 6m long, and 600-N beam, rests on two support, as shown. The force exerted on the beam by the right support B is closest to:

A. 75N
B. 240N
C. 400N
D. 320N

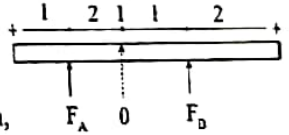


Solution:

from first condition of equilibrium.

$\sum F=0$

$\Rightarrow F_A+F_B=W=600....(*)$



From second condition of equilibrium, taking moment at the center of the rod,

$\Rightarrow 2F_A=F_B$ ...(**)

$\Rightarrow$There fore, from equation (*) and (**)

$$\frac{F_B}{2}+F_B=600\Rightarrow F_B=400\text{N}$$

Answer: C

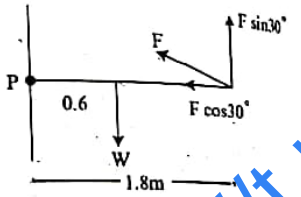
13. A rod PQ is attached at P to a vertical wall, as shown in the figure below. The length of the rod is 1.80m. The weight W of the rod acts at 0.60m from P. The rod is kept horizontal and in equilibrium by a wire attached to Q and to the wall at R. The wire

Telegram: https://t.me/ExtremeSeries

provides a force F on the rod of 50N at 30° to the horizontal. By taking moments about P, determine the weight W of the rod.

A. 130.0N B. 150.0N C. 75.0 D. 50.0N

Solution: Now, using the second condition of equilibrium and taking the pivot point at P, $0.6w = 1.8(50°)(\sin 30°)$.

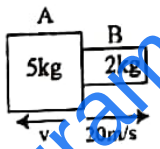


$$w = \frac{(0.8)(50)(0.5)}{0.6}$$

$$w = 75N$$

Answer: C

14. A large mass (A) and a small mass (B) are pushing one another apart, as shown in the figure below. The small mass moves away at a velocity of 20m/s to the right. What is the recoil velocity of the large mass?



A. -20m/s B. 8m/s C. 20m/s D. -8m/s

Explanation: from the conservation of momentum, $\Delta p = 0$

$p_f - p_i = 0$, $p_i = 0$

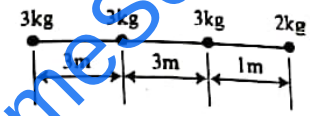
$\Rightarrow p_f = 0$

$m_A v_A + m_B v_B = 0$

$$v_A = \frac{-m_B v_B}{m_A} = \frac{-(2)(20)}{5} = -8m/s$$

Answer: D

15. Find the distance of the centre of mass of the system shown in the diagram from a point A. the rod is massless.



A. 3.73m B. 2.09m C. 2.45m D. 5.86m

Solution: $c \cdot m = \frac{m_1x_1 + m_2x_2 + m_3x_3 + m_4x_4}{m_1 + m_2 + m_3 + m_4}$

measuring the distances of each mass from point A,

$$c \cdot m = \frac{3(0)+3(3)+3(6)+2(7)}{3+3+3+2} = \frac{41}{11} = 3.73m$$

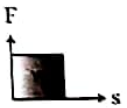
Answer: A

16. The area under a force versus distance graph represents:

A. acceleration
B. the change in energy
C. the change in momentum
D. work done

Explanation: The area under a force versus distance graph represents work done.

since, $w = Fs$



Answer: D

17. The net external non zero force acts on a particle that is moving along a straight line. Which one of the following statement is correct?

A. The velocity, but not the kinetic energy, of the particle is changing.
B. Both the velocity and the kinetic energy of the particle are changing.
C. The kinetic energy, but not the velocity, of the particle is changing.
D. The momentum is conserved.

Telegram: https://t.me/Extremeseries

Explanation: A net external non zero force causes a particle to accelerate and thus to change its velocity and hence its kinetic energy.

Answer: B

18. Which statement is correct about conservative and dissipative forces?
   A. The work done by conservative forces is path dependent.
   B. The work done, along a closed path, by non-conservative forces is nonzero.
   C. Gravitational force is an example of non-conservative force.
   D. Friction is an example of conservative force.

Explanation: The work done, along a closed path, by non-conservative forces such as friction is non zero.

Answer: B

19. The rotational analogue of mass is:
   A. moment of inertia.
   B. torque.
   C. angular momentum.
   D. angular acceleration.

Answer: A

20. A fly wheel is rotating with an angular velocity of 2.0 rad/s and acted by an acceleration of $1/\pi$ rad/s^2. How long will it take to complete three revolutions?
   A. 0.5π s. B. π s. C. 2π s. D. 4π s^2.

Explanation: $\theta = \text{wot} + \frac{1}{2}\alpha t^2$

in 3 revolutions, $\theta = 3(2\pi) = 6\pi$rad

$\Rightarrow 6\pi = 2t + \frac{1}{2}\left(\frac{1}{\pi}\right)(t^2)$, solving this

$t = 2\pi$ sec

Answer: C

21. A device for transforming heat energy to mechanical energy is referred to as:
   A. transformer
   B. refrigerant
   C. generator
   D. engine.

Answer: D

22. the number of joules of heat energy required to raise the temperature of a body by 1 K is:
   A. Heat capacity.
   B. specific heat capacity
   C. expansion coefficient
   D. calorific value

Answer: A

23. A simple pendulum is made using a long, thin metal wire. What happens to the period of the pendulum when the temperature drops?
   A. Increase
   B. Remain the same
   C. Decrease
   D. Period of oscillation has random dependence on temperature.

Explanation: $T = 2\pi\sqrt{\frac{l}{g}}$, as the temperature decreases, the metal wire contracts and decreases in length which inturn causes the period to drop.

Answer: C

24. Pressure at the bottom of a liquid column is independent of:
   A. shape of the column.
   B. the density of the fluid.
   C. the depth of the point.
   D. the acceleration due to gravity.

Explanation: pressure of fluid is given by, $p = \rho gh$

Answer: A

25. The definition of Young's Modulus is given by:
   A. force per unit area.
   B. ratio of strain to stress.
   C. ratio of stress to strain.
   D. force per unit elongation.

Explanation: Youngs modulus $= \frac{\text{stress}}{\text{strain}}$

Answer: C

26. A water pipe has diameter of 5.0cm in the inlet side and 1.6cm in outlet side. Water flows at 0.50m^3 per minute at the inlet point. What is the peed of the water at the outlet point?
   A. 26.67m/s B. 13.33m/s C. 74.1m/s D. 148.1m/s

Telegram: https://t.me/ExtremeSeries

Solution: from continuity equation,
$A_1V_1 = A_2V_2$ = discharge

$$\Rightarrow \frac{\pi\left(1.5\times10^{-2}\right)^2}{4} V_2 = \frac{0.5m^3}{min} = \frac{0.5m^3}{60sec}$$

$V_2 = 47.15m/s$

No answer

27. A metal ball has a diameter that is lightly greater than the diameter of a hole that has been cut into a metal plate. The coefficient of linear expansion for the metal from which the ball is made is greater than that for the metal of the plate. Which of the following procedures can be used to make the ball pass through the hole?

A. Raise the temperatures of the ball and the plate by the same amount.
B. Heat the ball and cool the plate.
C. Lower the temperatures of the ball and the plate by the same amount.
D. There is no way of passing the ball through the hole.

Answer: A

28. A billiard ball of radius 4cm is falling through a liquid of viscosity 0.985 at a velocity of 2.5m/s. The drag force on the ball is

A. 0.591π N
B. 0.118πN
C. 1.182π N
D. 0.059π N

Solution: $F = \pi 6\ y\ vr = \pi 6\ (0.985)(2.5)(0.04) = 0.591\pi$ N

Answer: A

29. Which one of the following statements is correct?

A. Elastic materials retain their shape after deformation.
B. Plastic materials can recover the original size or shape after being distorted.
C. Plastic materials change their shape by pressure.
D. Elastic materials cannot recover the original size or shape after being distorted.

Answer: A

30. Two bodies that are separately in thermal equilibrium with theird body must be in thermal equilibrium with each other. This statement is statement of:

A. the first law of thermodynamics
B. the zeroth law of thermodynamics
C. the second law of thermodynamics
D. the third law of thermodynamics.

Answer: B

31. Which one of the following statements is NOT correct about galvanometer?

A. It is used to measure small amount of current.
B. it has infinite resistance.
C. It can be used to measure voltage if shunted with large load in series.
D. Its working principle is the same as that of the electric motor.

Answer: B

32. Three resistors connected in series have individual voltages labeled ΔV_1, ΔV_2 and ΔV_3 respectively. Which of the following expresses the value of the total voltage Δv_T, taken over the three resistors together?

A. $\Delta V_T = \Delta V_1 + \Delta V_2 + \Delta V_3$
B. $\Delta V_T = \Delta V_1 = \Delta V_2 = \Delta V_3$
C. $V_T = \left(1/\Delta V_1 + 1/\Delta V_2\ 1/\Delta V_3\right)$
D. $V_T = \left(1/\Delta V_1 + 1/\Delta V_2\ 1/\Delta V_3\right)^{-1}$

Explanation: For resistors in series, the total potential drop is equal to the sum of the potential drops across individual resistors.

Answer: A

33. Which instrument should, ideally, have zero resistance?

A. Wheatstone bridge
B. Oscilloscope
C. Voltmeter
D. Ammeter

Answer: D

Telegram: https://t.me/Extremeseries

34. What fundamental frequency can a string vibrate at if it is 30cm long and the velocity of travelling waves along the string is 120m/s?

A. 400 Hz B. 600Hz C. 800Hz D. 200Hz

Solution: $f = \frac{v}{2L} = \frac{120}{2(0.3)} = 200\text{Hz}$

Answer: D

35. The hearing curve, shown below, indicates the variation of the threshold of hearing with frequency. The ear is most sensitive at the point indicated by _____.

A. A B. C C. B D. D

Explanation: The ear is most sensitive to 2000 – 5000Hz.

Answer: C

36. Away length is:

A. a lien that represents the direction of propagation of light.
B. the minimum distance between two points that are in phase.
C. the locus of all points of a travelling wave that are in phase.
D. the number of waves crossing a given plane in a unit time.

Answer: B

37. When viewed in white light, soap bubbles show color because of:

A. interference C. scattering
B. dispersion D. diffraction

Explanation: Colorful patterns in soap bubbles is due to a thin film interference

Answer: A

38. In the sketches below, which best describes the location of the negative charges (electrons) when a negatively charged rod is brought near but not touching is conducting sphere?

A. C B. D C. A D. B

Explanation: Induction causes polarization of charges in a neutral body.

Answer: D

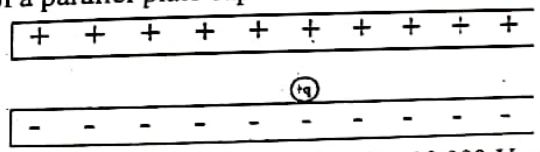
39. What is the potential (V) at a distance of 10m from a charge of +5.0C?

A. -4.45×10^9V C. $+4.45 \times 10^9$V
B. -0.5V D. +0.5V

Solution: $V = \frac{KQ}{r} = \frac{8.9\times10^9(5)}{10} = 4.45\times10^9\,\text{V}$

Answer: C

40. A positive charge of 1.0×10^{-4}C is on the negatively charged plate of a parallel plate capacitor as shown below.



If the potential (V) on the top plate is +10,000 V and that of the bottom plate is -10,000V, how much work (W) is required to move the charge from the bottom to the top plate?

A. 2.0J B. 1.0J C. Zero D. 2.0×10^4J

Solution: $W = Q(\Delta V) = 10^{-4}\left(10,000 - \left(-10,000\right)\right) = 2\text{J}$

Answer: A

41. In photoelectric effect, the energy of photoelectrons is proportional to the:

A. frequency of light. C. speed of light.
B. intensity of light. D. absorbance of the metal

Explanation: $E = hf$

Answer: A

Telegram: https://t.me/ExtremeSeries

42. The activity of a radioactive source falls to one sixteenth of its original value in 32 minutes. What is the half-life, in minutes, of this decay process?

A. 2 min
B. 8 min
C. 4 min
D. 16 min

Explanation: $N = N_o e^{-\lambda t}$, $\frac{1}{16}N_o = N_o e^{-\lambda(32)} \Rightarrow \lambda = \frac{\ln 16}{32}$

for half life, $\frac{1}{2}N_o = N_o e^{-\lambda(t)} \Rightarrow t = \frac{\ln 2}{\lambda} = (32)\left(\frac{\ln 2}{\ln 16}\right) = 8\text{min}$

Answer: B

43. Which statement is **NOT** correct about emissions form radioactive isotopes?

A. Alpha particles are positively charged and less penetrating than beta and gamma.
B. Beta particles are negatively charged and more penetrating than alpha particles.
C. Alpha emissions are least influenced by magnetic field.
D. The most penetrating emissions are gamma particles.

Answer: C

44. The torque experienced by a current carrying loop in an external, magnetic field is;

A. independent of the number of turns of the loop.
B. independent of the current in the loop.
C. dependent on the orientation of the loop.
D. dependent on magnetic field generated according to Lenz's law.

Explanation: $\tau = NIAB\sin\theta$

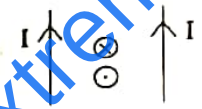
Answer: C

45. Two long wires of the same length and cross-sectional area carry the same amount of current. Which one of the following statements is correct about the magnetic field and the magnetic forces on the wires?

A. The magnetic field is zero at the midpoint between the wires when the currents are in the same direction.

B. The force between the wires is attractive when they carry current in opposite direction.
C. The magnetic fields between the wires reinforce each other when the currents are in the same direction.
D. The magnetic field due to one is proportional to the current through the other wire.

Explanation: The magnetic field produced by wires carrying current in the same directions do have opposite directions. Thus the resultant magnetic field is zero at mid point between the wires is zero.



Since $B_1 = \frac{N_o I}{2\pi r}$ in to and

$B_2 = \frac{N_o I}{2\pi r}$ out of

$\vec{B} = \vec{B}_1 + \vec{B}_2 = 0$

Answer: A

46. Self inductance is quantity that describes:

A. the number of turns in a circuit.
B. the induction of an electric field by an AC current in a lone coil.
C. the induction of a magnetic field by an AC current in a lone coil.
D. the steady state current in a coil

Answer: C

47. Which conversion process forms the basis for the function of an electric generator?

A. alternating current to direct.
B. mechanical energy to electrical.
C. electrical energy to mechanical.
D. low voltage to high or vice versa.

Answer: B

48. What is the change in magnetic flux in a coil of area $5m^2$ as its orientation relative to the perpendicular of a uniform 3.0T magnetic field changes form 45° to 90°?

A. 0 Wb
B. -11 Wb
C. 7.5 Wb
D. 11 Wb

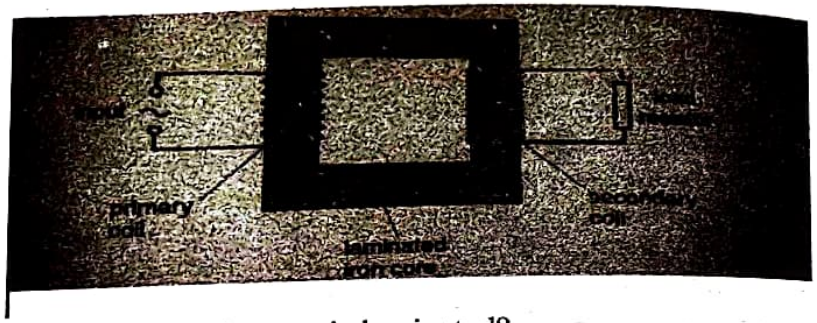
Telegram: https://t.me/ExtremeSeries

Solution: $\phi = \Delta BA\cos\theta = (5)(3)\cos 0° - (5)(3)\cos 45°$

$\phi = 7.5\,Wb$

Answer: C

49. A simple transformer is illustrated in the figure below.



Why is it that the core is laminated?

A. To increase the induced current in the primary coil.
B. To increase the induced current in the secondary coil.
C. To avoid heat exchange between the primary and secondary coils.
D. To reduce the loss in power in the core due to eddy currents.

Answer: D

50. A common unit of energy in atomic physics is:

A. plank's constant
B. electron volt
C. de Broglie wave length
D. uncertainty principle

Answer: B

1. A block of mass 5kg hanging at the end of spring scale is totally submerged in a beaker of water. The beaker is large enough to avoid overflow of water. If the density of the block is $2.5 \times 10^3 Kg/m^3$, then what is the reading of the scale?

A. 3kg B. 2kg C. 1kg D. 0 kg

Solution:

$W_{air} - W_{water} = F_{bounce}$

$W_{water} = W_{air} - F_b = \delta_b V_b g - \delta_\omega V_\omega g$, here $V_\omega = V_B V_b$

$V_b = \frac{m_b}{s_b} = \frac{5}{2500}$

$\Rightarrow W_{water} = 5(10) - 1000\left(\frac{5}{2500}\right)(10) = 50 - 20$

$W_{water} = 30N$ or $W_{water} = 3kg$ **Answer: A**

2. On an unknown scale X, the ice – point temperature is 50°X and the steam – point temperature is 200°X. If the temperature is 60°C, then what is the reading in °X?

A. 90°X B. 140°X C. 15°X D. 333°X

Solution:

$$\frac{T_x - T_x ice}{T_{x,boiling} - T_{x,ice}} = \frac{T_c - T_{c,ice}}{T_{c,boiling} - T_{c,ice}}$$

$$\frac{T_x - 50}{200 - 50} = \frac{T_c - 0}{100 - 0}$$

$\Rightarrow \frac{T_x - 50}{150} = \frac{T_c}{100}$ when $T_c = 60°C$, then

$\frac{T_x - 50}{15} = \frac{60}{60} \Rightarrow T_x = 140°$ **Answer: B**

3. Metal A has a coefficient of linear expansion that is three times the coefficient of linear expansion of metal B. How does their coefficient of areal expansion, γ, compare?

A. $\gamma_A = 2\gamma_B$ C. $\gamma_A = 4\gamma_B$
B. $\gamma_A = 3\gamma_B$ D. $\gamma_A = 6\gamma_B$

Solution:

$\alpha_A = 3\alpha_B$, and we know $\beta = 2\alpha$

Telegram: https://t.me/Extreme series

$$\therefore \frac{\beta_A}{\beta_B} = \frac{2\alpha_A}{2\alpha_B} = \frac{6}{2} \Rightarrow \beta_A = 3\beta_B$$

Answer: B

4. Three metals are in contact and are in thermal equilibrium. The metals have different specific heat capacity. What can be concluded about the temperature of the metals?
   A. Since they have different specific heat capacity, they must have different temperature at thermal equilibrium.
   B. They must have the same temperature at thermal equilibrium.
   C. They must have different internal energy.
   D. They must have different kinetic energy.

Answer: B

5. A certain heat engine draw 500 cal/s from a water bath at 27°C and rejects 400 calories to a reservoir at a lower temperature. What is the efficiency of this engine?
   A. 80% B. 75% C. 25% D. 20%

Solution:

$$\eta = \left(1 - \frac{Q_C}{Q_H}\right)100\% = \left(1 - \frac{400}{500}\right)100\%$$

$$\eta = (0.2)100\% = 20\%$$

Answer: D

6. Which of the following statements is correct?
   A. Irreversible process is a process that can be reserved through finite change in system.
   B. Complete conversion of energy from a hot source into work is possible.
   C. There are processes in which there is an overall decrease in entropy.
   D. Entropy is a measure of the amount of disorder.

Answer: C

7. Which of the following is **NOT** correct about a simple harmonic oscillator?
   A. Total mechanical energy is conserved.
   B. The acceleration is directly proportional to and opposite in direction to displacement.

   C. The period of small oscillation of a pendulum with string length l, a bod mass m attached to its end is $T = 2\pi\sqrt{\frac{l}{g}}$.
   D. The frequency of an oscillating mass attached to a massless spring is independent of the spring constant.

Answer: D

8. Which one of the following indicates with wavelength of a travelling wave?
   A. The distance between an adjacent trough and a peak.
   B. The amplitude of the wave.
   C. The distance between a node and antinodes.
   D. The distance between two adjacent points which are in phase.

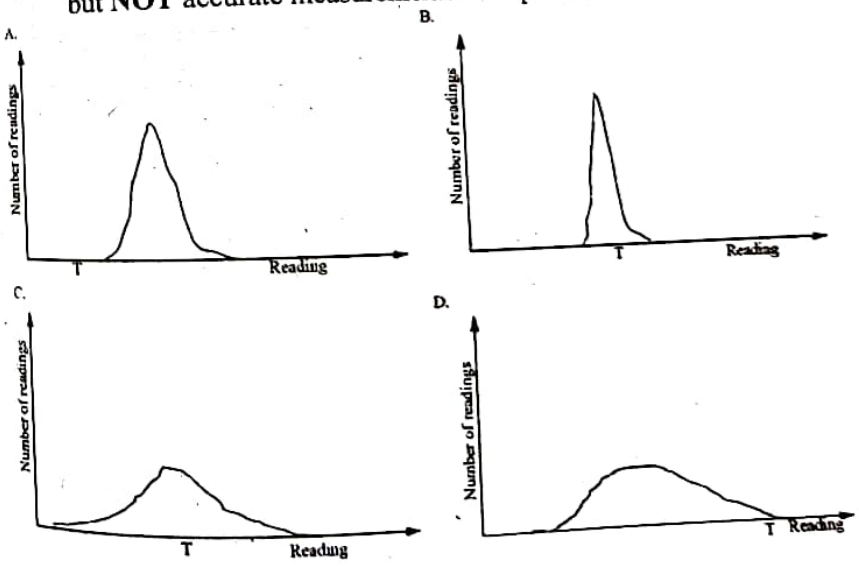
Answer: D

9. If a precision of surveyor measured a distance of 500 meters/between two points in a village/using a meter that has centimeter division on it, then what is the measurement result?
   A. 500m B. 500.0m 500.00m D. 500,000m

Hint: Since centimeter is the smallest division it is 0.01m.

Answer: C

10. From the figures given below, which one of them shows precise but **NOT** accurate measurements? T represents the true value.



Telegram: https://t.me/ExtremeSeries

Hint: Precision is the measure of the closeness of values to each other.

Answer: A

11. If vector $\vec{p}$ is 6m in the x – direction and vector $\vec{q}$ is 8m in the y – direction. Then what is the value of the vector $\vec{p}+\vec{q}$?

A. 10m at 53° from x – direction.
B. $\sqrt{136}$ m at 45° from x – direction.
C. 10m at 37° from x – direction.
D. $\sqrt{136}$ m at 30° from x – direction.

Solution:

$$\vec{R}=\vec{P}+\vec{q}$$

$$|R|=\sqrt{P^2+q^2}=\sqrt{(8)^2+(6)^2}=10m$$

and, $\tan\theta=\frac{8}{6}\Rightarrow\theta=\tan^{-1}\left(\frac{4}{3}\right)=53°$

Answer: A

12. Given vectors $\vec{A}=\sqrt{3}\hat{i}+\hat{j}$ and $\vec{B}=\sqrt{3}\hat{i}-\hat{j}$. What is the smaller angle between the vectors?

A. 60° B. 53° C. 45° D. 30°

Solution: $A.B = AB\cos\theta$

$$\cos\theta=\frac{A.B}{AB}=\frac{(\sqrt{3})(\sqrt{3})+(1)(-1)}{\left(\sqrt{(\sqrt{3})^2+(1)^2}\right)\left(\sqrt{(\sqrt{3})^2+(-1)^2}\right)}=\frac{2}{4}=\frac{1}{2}$$

$$\cos\theta=\frac{1}{2}\Rightarrow\theta=\cos^{-1}\left(\frac{1}{2}\right)=60°$$

Answer: A

13. If $\vec{A}=\hat{x}$ m/s + 3m/s $\hat{z}$, and $\vec{B}$ = 2m/s $\hat{y}$ + 3 m/s $\hat{y}$ + 3m/s $\hat{z}$, then what is the value of $|\vec{A}\times\vec{B}|$?

A. $\sqrt{10}$m/s B. $\sqrt{13}$m/s C. 7m/s D. $\sqrt{130}$m/s

Solution: $A\times B=\begin{vmatrix} i & j & k \\ 1 & 0 & 3 \\ 0 & 2 & 3 \end{vmatrix}$ (i, j, k +ve)

$= (0-6)i + (0-3)j + (2-0)k = -6i - 3j + 2k$

$$|A\times B|=\sqrt{(-6)^2+(-3)^2+(2)^2}=7m/s$$

Answer: C

14. A cricketer throws a ball vertically upwards with initial speed of 20m/s. How long is it in the air before it returns to the cricketer's hands?

A. 2s B. 10s C. 1.5s D. 4s

Solution:

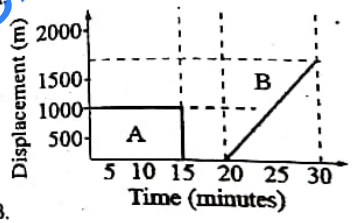
From, $y=ut-\frac{1}{2}gt^2$, $y=0$

$$\Rightarrow t=\frac{2u}{g}=\frac{2(20)}{10}=4\sec$$

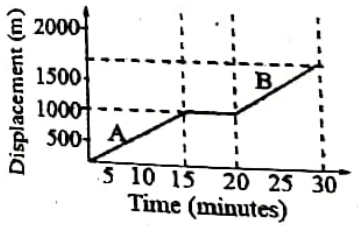
Answer: B

15. Abebe walks to school. She walks 1km in 15 minutes. She meets her friend Makeda – they talk for 5 minutes and then carry on walking to school. They walk 800m in 10 minutes. Which of the following displacement – time graph shows Abeba's journey to school?

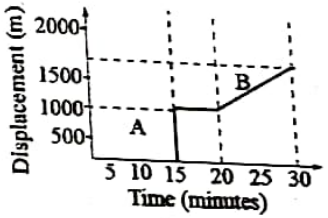
A.



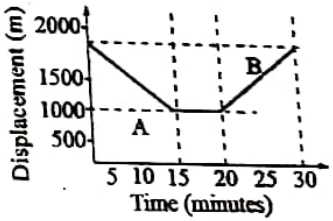
B.



C.



D.



Answer: B

16. A fun making boy swings a blinking ball tied to the end of a string in a vertical circle. If the radius of the circle is 0.5m, then what should be the minimum speed of the ball at the bottom of the circle to make a complete circular motion at the top?

Telegram: https://t.me/ExtremeSeries

A. 5m/s B. $\sqrt{5}$m/s C. $\sqrt{15}$m/s D. $2\sqrt{5}$m/s

Solution: $mg = \frac{mv^2}{r} \Rightarrow v = \sqrt{gr} = \sqrt{(10)(0.5)} = \sqrt{5}$m/s

Answer: B

17. The range of cannon – ball fired horizontally from a laboratory table is equal to 8/3 times the height of the table. What is the direction of velocity vector of the projectile as it strikes the ground?

A. 45° from the horizontal C. 60° from the horizontal
B. 53° from the horizontal D. 37° from the horizontal

Solution: $x = \frac{8}{3}y$

$v_x = \frac{x}{t}$, but $t = \sqrt{\frac{2y}{g}}, x = \frac{8}{3}y$

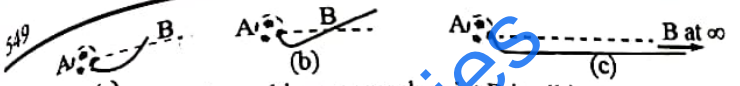
$v_x = \frac{\frac{8}{3}y}{\sqrt{\frac{2y}{g}}} = \frac{8y}{3\sqrt{\frac{2y}{g}}}$, and

$v_y = gt = g\sqrt{\frac{2y}{g}}$

$\tan\theta = \frac{v_y}{v_x} = \frac{g\sqrt{2y/g}}{\frac{8y}{3\sqrt{\frac{2y}{g}}}} = \frac{3}{4} \Rightarrow \theta = \tan^{-1}\left(\frac{3}{4}\right) = 37°$

Answer: D

18. Galileo carried out a 'thought' experiment on motion. The line between A and B represents a surface and the center of the ball is placed at A. What would happen in each of the three situations shown in the figure below, without friction when the ball is released?

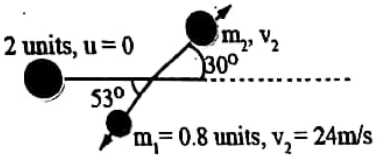


A. The ball would never reach point B in all instances.
B. The ball would reach B in all instances even if point B keeps on receding to infinity in (c).
C. The ball would reach point B in (a) and (b) and not in (c).
D. The ball would reach point B only in (a).

Hint: In the absence of external force, the ball will move forever.

Answer: B

19. A Uranium atom initially at rest has a mass of 2 units and spontaneously disintegrated into two fragments. One of the fragments has a mass of 0.8 units and moves at a speed of 24 m/s as shown in figure below. What is the speed of the second fragments?



A. 25.6m/s
B. 21.3m/s
C. 16m/s
D. 14.8m/s

Solution: From conservation of momentum

$Pi = Pf \Rightarrow m\begin{bmatrix} u_x \\ u_y \end{bmatrix} = m_1\begin{bmatrix} v_1x \\ v_1y \end{bmatrix} + m_2\begin{bmatrix} v_2x \\ v_2y \end{bmatrix}$

$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = 0.8\begin{bmatrix} -14.4 \\ -19.2 \end{bmatrix} + 1.2\begin{bmatrix} 0.87v_2 \\ 0.5v_2 \end{bmatrix}$

$11.52 = 1.04v_{2x} \Rightarrow v_{2x} = 11.08$m/s

$15.36 = 0.6v_{2y} \Rightarrow v_{2y} = 25.6$m/s

$v_2 = \sqrt{v_{2x}^2 + v_{2y}^2}$

20. A 1 kg object explodes and breaks up into three pieces. One piece has a mass of 400g and has a velocity of $(15\vec{i}+10\vec{j})$m/s. A second piece has a mass of 250 g and has a velocity of $(25\vec{i}-10\vec{j})$m/s. What is the velocity the third piece?

A. $(-4.3i+35\vec{j})$m/s C. $(35\vec{i}-4.3\vec{j})$m/s

Telegram: https://t.me/Extremeseries

B. $(-35\vec{i}-4.3\vec{j})$m/s D. $(-35\hat{i}-4.3\hat{j})$m/s

Solution: $p_i = p_f$

$$\begin{bmatrix}0\\0\end{bmatrix}=0.4\begin{bmatrix}15\\10\end{bmatrix}+0.25\begin{bmatrix}25\\-10\end{bmatrix}+0.35\begin{bmatrix}v_x\\v_y\end{bmatrix}$$

21. Which of the following statements is **NOT** correct?
   A. Dynamic equilibrium is an equilibrium in which a body is moving at steady velocity without net force or torque acting on it.
   B. Static equilibrium is an equilibrium in which a body is at rest without net force or torque acting on it.
   C. Couples are set of forces with a resultant torque but no net force.
   D. The necessary and sufficient condition for equilibrium of a rigid body is to have the sum of all torques acting on it is zero.

Answer: D

22. A spring with a spring constant of 300N/m is stretched 20cm in 2 seconds. What is the power applied to stretch the spring?
   A. 3W B. 6W C. 30kW D. 60kW

Solution:

$$P=\frac{\omega}{t}=\frac{\frac{1}{2}kx^2}{t}=\frac{(300)(0.2)^2}{2(2)}=3W$$

Answer: A

23. A 50 kW machine is used to lift heavy objects to the top of a building under construction. If this machine is replaced with a new machine of 100kW power, then which of the following statements is **NOT** true about the new machine?
   A. The new machine can raise objects with the same mass at twice the speed of the old machine.
   B. The new machine can raise objects with twice as much mass at the same speed as the old machine.
   C. The new machine takes twice as much time to do the same work as the old machine.

   D. The new machine can do twice as much work in the same time as the old machine.

Hint: $P=\frac{W}{t}$ **Answer: C**

24. Which of the following is correct about a work done for dissipative forces?
   A. Path dependent.
   B. The same as the loss in kinetic energy.
   C. The same as the gain in potential energy.
   D. The same as the potential energy stored in spring.

Answer: A

25. A large (very large in height and cross – section) water tank open at the top has a small hole through its side near the bottom. To what height should the water be filled, to double the speed of water that flow out of the hole when it was filled to a height h?
   A. h B. 2h C. 3h D. 4h

Hint: $P_E = k_E$

$$mgh=\frac{1}{2}mv^2, v_2=2v_1$$

$$v_1=\sqrt{2gh}$$

$$v_2=\sqrt{2gh}$$

$$\frac{v_2}{v_1}=\frac{\sqrt{2gh'}}{\sqrt{2gh}}\Rightarrow\frac{v_2}{v_1}=\sqrt{\frac{h'}{h}} \text{ or } 2=\sqrt{\frac{h'}{h}}\Rightarrow h'=4h$$

Answer: A

26. Which of the following is the application of Pascal's law?
   A. Hydraulic brake in vehicles.
   B. Siphon to remove fluids.
   C. Working of submarines.
   D. Sucking of liquid into a syringe.

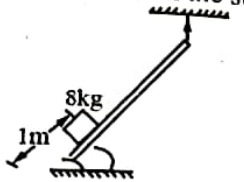
Answer: A

27. Which of the following refers to the definition of strain as a ratio?
   A. Extension to the original length.
   B. Force applied to area of the object normal to the force.
   C. Stress to Young modulus.
   D. Tensile to applied force.

Answer: A

Telegram: https://t.me/EXtremeSeries

28. A 4m long uniform board of mass 10 kg hinged on the ground at one end is hanging by a string at the other end. A block of mass 8 kg is placed at 1m from the hinge on the ground as shown in the diagram below. What is the tension in the string?



A. 10N B. 50N C. 70N D. 110N

Solution: From 2nd condition of equilibrium, taking moment at the base.

$80(1\cos 60°)v + 100(2\cos 60°) = T(4\cos 60°)$

$80(0.5) + 100(2 \times 0.5) = 2T$

$40 + 100 = 2T \Rightarrow T = 70°$

Answer: C

29. Which of the following statements is correct about the first condition of equilibrium?

A. The net force on an object must be zero.

B. The sum of all torques must be zero.

C. Clockwise moments should balance counter-clockwise moments.

D. The torque produced by a force should be perpendicular to the force.

Answer: A

30. A block of mass $m_1 = 10kg$ placed on a frictionless incline plane connected to mass $m_2 = 15kg$ using a string of negligible mass passing over a pulley of moment of inertia $I = 0.25\ kgm^2$ and radius $R = 10cm$ as shown in the diagram. What is the torque on the pulley?

A. 2.5Nm B. 4Nm C. 5Nm D. 12Nm

Solution:

For m_2, $m_2g - T = m_2a$

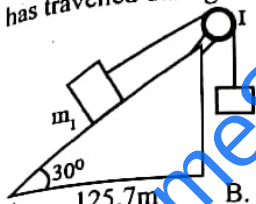
For m_1, $T - m_1g\sin\theta = m_1a$

$$\Rightarrow a = \frac{m_2g - m_1g\sin\theta}{m_1 + m_2} = \frac{150 - 50}{25} = \frac{1000}{25} = 4m/s^2$$

$$\text{And, } T = I\alpha = I\left(\frac{a}{r}\right) = (0.25)\left(\frac{4}{0.1}\right) = 10Nm$$

No answer

31. A pulley is rotating at a frequency of 20 rev/min. A motor speeds up the wheel so that 30s later the frequency of the motion is 60 rev/min. How far will a point on the pool 0.30m from its center has travelled during that period?



A. 125.7m B. 0.13m C. 37.68m D. 4.2m

Solution:

$$\theta = \left(\frac{\omega + \omega_o}{2}\right)t = \left(\frac{20+60}{2}\right)\left(\frac{1}{2}\right) = 20rev$$

$$s = r\theta = 20(2\pi)(0.3) = 20(2 \times 3.14)(0.3) = 37.68m$$

Answer: C

32. A disk of radius R with uniform distribution of mass M, rotates about an axis perpendicular to its plane at the rim with angular speed ω. The moment of inertia of the disc about the axis through the center is ½ MR^2. What is the kinetic energy of the disk?

A. $\frac{1}{4}MR^2\omega^2$ C. $\frac{1}{2}MR^2\omega^2$

B. $\frac{1}{3}MR^2\omega^2$ D. $\frac{3}{4}MR^2\omega^2$

Hint: $k_E = \frac{1}{2}I\omega^2 = \frac{1}{2}\left(\frac{1}{2}MR^2\right)\omega^2 = \frac{1}{4}MR^2\omega^2$

Answer: A

33. The activity of radioactive substance reduces from 2000Bq to 1000Bq in 10^6 seconds. What is the decay constant?

A. 6.93×10^5 decay/s C. 2.00×10^6 decay/s

B. 6.93×10^{-7} decay/s D. 5.00×10^{-7} decay/s

Answer: B

34. Which of the following phenomena clearly demonstrates the particle nature of light?

A. Refraction of light by prism.

B. The photoelectric effect

Telegram: https://t.me/EXtremeSeries

C. Diffraction of light by double slit.
D. Dispersion of white light.

35. The capacitance of a parallel plate capacitor is 10μF. If the distance between the plates is halved, keeping all other quantities constant, then what is the new capacitance of the capacitor? **Answer: B**

A. 0.2μF B. 5μF C. 10μF D. 20μF

Hint: $C = \varepsilon_r \varepsilon_o \frac{A}{d} \Rightarrow C \sim \frac{1}{d}$

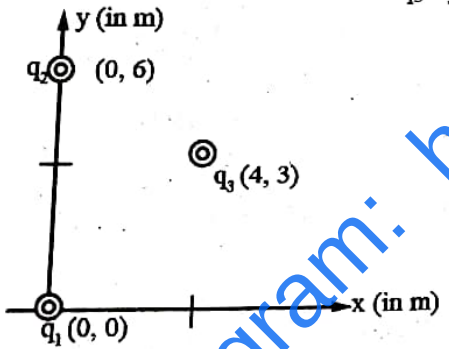
36. Two positive charges of magnitude 2.4×10^{-9}C and 1.6×10^{-9}C are separated in air by a distance of 2.0cm. How much electrostatic potential is produced by the two changes at midway between them? **Answer: D**

A. 3.6KV B. 0.72KV C. 2.16KV D. 1.44KV

Solution: $v = v_1 + v_2 = \frac{kQ_1}{r_1} + \frac{kQ_2}{r_2} = 2160 + 1440$

$v = 3600 = 3.6kv$

37. A charge of $q_1 = 10.0 \times 10^{-9}$C, $q_2 = +10.0 \times 10^{-9}$C and $q_3 = -2.5 \times 10^{-8}$C are placed on x – y plane as shown in the figure below. What is the electric force on q_3 by q_1 and q_2? **Answer: A**



A. 1.08×10^{-7}N in +ve y – direction.
B. 9.00×10^{-7}N in –ve y – direction.
C. 1.42×10^{-7}N in +ve x – direction.
D. 1.42×10^{-7}N in –ve x – direction

Solution:

$$F_{13} = \frac{kQ_1Q_3}{r^2} = \frac{(9\times10^9)(10\times10^{-9})(2.5\times10^{-8})}{(5)^2} = 9\times10^{-8}N$$

$$F_{23} = \frac{kQ_2Q_3}{r^2} = 9\times10^{-8}N$$

$F_x = F_{1x} + F_{2x}$ & $F_y = F_{1y} + F_{2y} = 0$ &

$F_y = 9 \times 10^{-8}\left(\frac{3}{5}\right) + 9 \times 10^{-8}\left(\frac{4}{5}\right)$

$F_R = F_y = 1.08 \times 10^{-7}$N in +ve y direction.

38. In a Young's experiment, there are two slits, rather than one. What do you think the reason? **Answer: A**

A. To increase the intensity.
B. One slit is for frequency, the other for wavelength.
C. To create path length difference.
D. One slit is for electric fields, and the other is for magnetic fields.

39. A light of wavelength 590nm is incident normally on a grating with spacing 1.67×10^{-6}m. How many orders of diffraction maxima can be obtained (including the zero order)? **Answer: C**

A. 1 B. 2 C. 3 D. 4

Solution: $m \le \frac{d}{\lambda}, m \le \frac{1.6\times10^{-6}}{590\times10^{-9}} = 2.8 = 2$ including the zero order, m = 3 **Answer: C**

40. Which of the following statements is correct?

A. Interference is not superposition of waves.
B. Constructive interference results in a decrease in the intensity.
C. Destructive interference results in an increase in the intensity.
D. Path difference which is integer multiple of wavelength results in constructive interference.

Answer: D

41. An electric generator has a coil of 400 turns and cross – sectional area $5m^2$ rotating at angular frequency 0.52rad/s in a uniform magnetic field 0f 0.2T. What is the induced e.m.f produced from the electric generator in each second?

A. 209.6V B. 104.8V C. 200V D. 182.4V

Solution: $\varepsilon = NAB\omega = (400)(5)(0.2)(0.524) = 209.6V$

Answer: A

Telegram: https://t.me/ExtremeSeries

42. The primary coil of a transformer is connected to a battery, a resistor, and a switch. The secondary coil is connected to an ammeter. What does the ammeter show when the switch is thrown closed?

PICTURE

A. Zero current as transformer is no working on DC.
B. A nonzero current for a short instant.
C. A steady current.
D. Emf will be induced but not current.

Answer: A

43. Which of the following statement is correct?

A. The direction of induced e.m.f. is in such as direction as to re-enforce the change produces it.
B. Self-inductance of a coil is e.m.f. divided by magnitude or rate change of current with time.
C. The induced e.m.f. in a coil is inversely proportional to its number of turns.
D. Self-inductance of a coil is inversely proportional to the cross – sectional area of the coil.

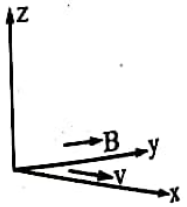
Answer: B

44. The induced electromotive force in a current carrying coil placed in a magnetic field does **NOT** depend on which of the following?

A. The speed at which the coil moves if the relative motion does not result in changing the magnetic flux.
B. The change of magnetic flux density.
C. The length of the coil that is moving in a direction perpendicular to that of the magnetic field.
D. The area of the coil outside the field.

Answer: D

45. A positivity charged particle of charge $q = 3.2 \times 10^{-19}C$ is moving parallel to x – axis with speed 1000m/s in a magnetic field of 0.5T parallel to y – axis as shown in the diagram below. What is the magnetic force on the charge?



A. 8.0×10^{-16} N in – ve z
B. 3.2×10^{-16} N in – ve x
C. 8.0×10^{-17} N in +ve x
D. 1.6×10^{-16} N in +ve z

Solution: $\vec{F} = q(\vec{v} \times \vec{B}) = 1.6 \times 10^{-16} N\hat{k}$

Answer: D

46. A charged particle moves perpendicular to a uniform magnetic field. Which statement is correct about the motion of the particle?

A. Both kinetic energy and linear momentum of the particle change.
B. The kinetic energy of the particle changes but not its linear momentum.
C. The linear momentum of the particle changes but not its kinetic energy.
D. Both the energy and momentum do not change.

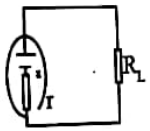
Answer: C

47. Which of the following is **NOT** correct about magnetic field?

A. Magnetic field lines are closed.
B. Magnetic monopoles exist.
C. The closer the field lines are, the stronger the magnetic field.
D. Magnetic field is produced by moving charge.

Answer: B

48. Which of the following is **NOT** correct about the circuit shown below?



A. Voltage across R_L is $\frac{R_L}{r+R_L}\varepsilon$
B. Current flowing through R_L is $\frac{\varepsilon}{r+R_L}$
C. Current flowing through r is $\frac{r}{r+R_L}\varepsilon$
D. Power drop in R_L is $\frac{R_L}{(r+R_L)^2}\varepsilon^2$

Answer: C

Hint: $I = \frac{\varepsilon}{r+R}$, both in R and r.

Telegram: https://t.me/ExtremeSeries

49. If a cell supplies a current of 0.9A when connected to a 2Ω resistor and a current of 0.3A, then what will be the internal resistance of the cell when connected to a 7Ω resistor?
A. 2Ω B. 1.2Ω C. 1Ω D. 0.5Ω

Solution:

$$I = \frac{\varepsilon}{r+R} \Rightarrow 0.9 = \frac{\varepsilon}{2+r} \text{ and, } 0.3 = \frac{\varepsilon}{7+r} \Rightarrow r = 0.5\Omega$$

Answer: D

50. A galvanometer of full – scale deflection 10mA is to be converted into a 0 – 10.0A ammeter. If its coil has a resistance of 100Ω, then what value of shunt resistance must be fitted?
A. 0.1Ω B. 10Ω C. 9.99Ω D. 0.999Ω

Solution:

$$R_S = \frac{R_C I_C}{I - I_C} = \frac{(100)(10\times10^{-3})}{10 - 10\times10^{-3}} = 0.1\Omega$$

Answer: A

1. Four vectors are listed below.

$$\vec{A} = \begin{bmatrix} 4 \\ 10 \end{bmatrix}, \vec{B} = \begin{bmatrix} 2 \\ -5 \end{bmatrix}, \vec{C} \begin{bmatrix} -6 \\ 15 \end{bmatrix}, \vec{D} = \begin{bmatrix} 15 \\ 6 \end{bmatrix}$$

Which pair of the vector is orthogonal?
A. $\vec{A}$ and $\vec{D}$ C. $\vec{C}$ and $\vec{D}$
B. $\vec{B}$ and $\vec{D}$ D. $\vec{A}$ and $\vec{B}$

Solution: Two non-zero vectors are orthogonal if their scalar product is zero.

$$\vec{A} \cdot \vec{D} = (4)(5) + (10)(-6) = 0$$

Answer: A

2. A race car is moving at a constant speed of 35 m/s. A security car was moving at a speed of 5 m/s as the race car passes by it and was accelerating at constant rate of 5 m/s^2. What was the speed of the security car when it took over the race car?
A. 35 m/s B. 60 m/s C. 5 m/s D. 65 m/s

Solution: $S_{race} = S_{security}$

$$Vt = Ut + \frac{1}{2}at^2$$

$$V = u + \frac{1}{2}at$$

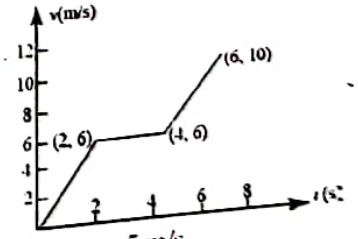
$$35 = 5 + \frac{1}{2}(5)t$$

$$t = 12 \text{ sec.}$$

At $t = 12$ sec, $V_{security} = u + at = 5\ (5)\ (112) = 65$m/s

Answer: D

3. A car start moving from rest and its motion on a straight line is shown in the figure below. What is the average velocity of the car within the first 6 seconds?



A. 5.67 m/s
B. 1.67 m/s
C. 5 m/s
D. 6 m/s

Solution: $\overline{V_{av}} = V_{av} = \frac{S_T}{t_T} = \frac{\text{Area}}{\text{time}}$

Telegram: https://t.me/ExtremeSeries

$$= \frac{\frac{1}{2}(2)(6)+(2)(6)+\frac{1}{2}(2)(4)+(2)(6)}{6}$$

$= 5.67$ m/s

Answer: A

4. Which one of the following statements is correct regarding the motion in a plane?
   A. When a body moves in a horizontal circle, its velocity is constant.
   B. When a body moves in a vertical circle, its speed is constant.
   C. The centripetal force for a body moving in either vertical or horizontal circle is toward the center.
   D. In projectile motion the horizontal component of the motion is uniformly accelerated motion.

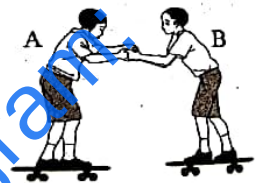
Answer: C

5. A bullet is fired at an angle of 30° above the horizontal. If air resistance is neglected, what should be the bullet's horizontal acceleration $(\vec{a}_x)$ and vertical acceleration $(\vec{a}_y)$ when it reaches its maximum height?
   A. Both $\vec{a}_x$ and $\vec{a}_y$ are zero.
   B. $\vec{a}_x$ is 10m/s$_2$ downward, and $\vec{a}_y$ is zero
   C. Both $\vec{a}_x$ and $\vec{a}_y$ are 10m/s^2, downward
   D. $\vec{a}_x$ is zero and $\vec{a}_y$ is 10m/s^2 downward.

Hint: For a projectile motion, the horizontal velocity is constant, and hence $\vec{a}_x = 0$.

Answer: D

6. Two students A and B, with masses m_A and m_B respectively, stand on a wheeled trolleys as shown in the figure below. If student A has larger mass and pushes student B, then what can be said about their accelerations?



   A. The acceleration of student B is less than that of student A.
   B. Both boys will be accelerated by the same magnitude but in opposite directions.
   C. The acceleration of student A is less than that of student B.
   D. There would be no acceleration because the forces are equal and opposite.

Explanation: Even if the force on each is the same, acceleration is inversely proportional to mass.

Answer: C

7. Which of the following statement is correct about errors in measurement?
   A. Experimental error is the same as a mistake.
   B. Errors make measured values same as accepted value.
   C. Errors occur in every scientific investigation.
   D. Errors can be avoided by using modern measuring instruments.

Hint: Errors can not be avoided in a measurement but can be minimized.

Answer: C

8. What can be concluded when the number of significant figures increases?
   A. True reading increases
   B. Precision increases
   C. Uncertainty increases
   D. Accuracy increases

Answer: B

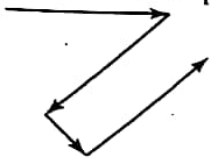
9. Two forces $\vec{F}_1 = (8\vec{i}+3\vec{j})$N and $\vec{F}_2 = (4\vec{i}+6\vec{j})$N are acting on an object. What is the magnitude and direction of the resultant force?
   A. 225 N, 143°
   B. 5 N, −37°
   C. 5 N, 143°
   D. 15 N, 37°

Solution:

$$\vec{F}_R = \vec{F}_x + \vec{F}_y$$
$$= (8+4)\mathrm{i} + (3+6)\mathrm{j}$$
$$= (12\mathrm{i} + 9\mathrm{j})\ \mathrm{N}$$
$$F_R = \sqrt{F_x^2 + F_y^2} = \sqrt{12^2 + 9^2} = 15\mathrm{N}$$
$$\sin\theta = \frac{F_y}{F} = \frac{9}{15} = 0.6$$
$$\theta = 37°$$

Answer: D

10. The diagram below shows a path taken by a flying bird.



Which one of the following vector may represent a resultant displacement?

Telegram: https://t.me/Extremeseries

A. ↗ B. → C. ↙ D. ↘

Hint: displacement is the shortest path between the start and end of a journey

Answer: D

11. A body that is released from the top of a tall building reaches the ground after six seconds. What is the ratio of the kinetic energy acquired by it at the end of the three seconds to the increment in the kinetic energy during the next three seconds?

A. 3/2 B. 1 C. 1/3 D. 2

Solution: $\frac{K_{E1}}{K_{E2}} = \frac{\frac{1}{2}mV_1^2}{\frac{1}{2}mV_2^2 - \frac{1}{2}mV_1^2}$

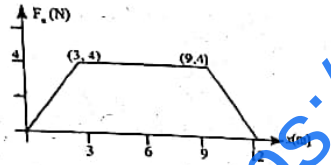
Since $V = gt$

$V_1 = 3g$ and $V_2 = 6g$

$$\frac{K_{E1}}{K_{E2}} = \frac{\frac{1}{2}m(3g)^2}{\frac{1}{2}m(6g)^2 - \frac{1}{2}m(3g)^2} = \frac{1}{3}$$

Answer: C

12. A boy pushes a block over 12 m distance within 6 seconds. The force applied by the boy varies as shown in the diagram below. What is the average power loss by the boy?



A. 4 w B. 8w C. 7w D. 6w

Solution: Power $= \frac{\text{Work}}{\text{time}} = \frac{\text{Area}}{\text{time}}$

$$= \frac{\frac{1}{2}(3)(4) + (6)(4) + \frac{1}{2}(3)(4)}{6}$$

$= 6w$

Answer: D

13. A thin rod when rotated about an axis at the end of the rod has a moment of inertia $\frac{1}{3}ML^2$. The rod has a mass of 1.5 kg and length of 1.0 m. If the rod initially at rest gains a rotational kinetic energy of 1000 J, at what speed does it rotate?

A. $10\sqrt{5}$ rad / s C. $20\sqrt{10}$ rad / s

B. 10 rad /s D. 18.8 rad/ s

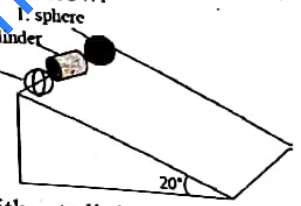
Solution: $K_{Er} = \frac{1}{2}Iw^2$, $I = \frac{1}{3}mL^2 = \frac{1}{3}(1.5)(1)^2 = 0.5$

$$1000 = \frac{1}{2}(0.5)\omega^2$$

$\omega = 20\sqrt{10}$ rad / sec

Answer: C

14. A ring, a solid cylinder and a solid sphere, all having the same mass and the same radius, are held at the top of an inclined plane as shown in the figure below.



If the objects roll without sliping and the coefficient of static friction between each object and the plane is 0.3, which one of the following shows the order in which they will get to the bottom of the inclined plane?

A. All will reach the bottom at the same time
B. Ring, solid cylinder, sphere
C. Solid cylinder, sphere, ring
D. Sphere, solid cylinder, ring

Hint: $mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$, let $I = Cmr^2$

$$gh = \frac{1}{2}V^2 + \frac{1}{2}Cmr^2\frac{V^2}{r^2}$$

$V = \sqrt{\frac{2gh}{(1+c)}}$, small $c \Rightarrow$ large V

Answer: D

15. If in a given rotational system of a body the angular momentum is increased with time, which one of the following statement is **NOT** correct?

A. The radius of rotation may increase.
B. There is no torque exerted on the system.
C. The moment of inertia may increase.
D. There is a torque exerted on the system.

Answer: B

Telegram: https://t.me/ExtremeSeries

16. Which of the following is a correct statement about the first condition of equilibrium?
A. The sum of all torques must be zero.
B. Clockwise moments should balance counter clockwise moments.
C. The net force on an object must be zero.
D. The torque produced by a force should be perpendicular to the force.

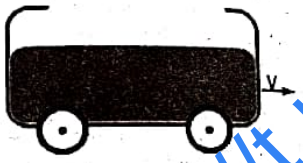
Answer: C

17. For a constant force F, if the impulse imparted to the system is large, then what can you conclude about the collision?
A. Linear momentum s conserved.
B. The contact or impact time is large.
C. Kinetic energy is conserved.
D. The charge in velocity is small.

Hint: Impulse = F (Δt)

Answer: C

18. An open trolley is rolling on a level surface without frictional loss through a vertical downpour of rain, as shown in the figure below



As the trolley rolls, an appreciable amount of rainwater accumulates in the cart. What will be the speed of the cart?
A. It increases because of conservation of mechanical energy.
B. It decreases because of conservation of mechanical energy.
C. It remains the same because the raindrops are falling perpendicular to the direction of the cart's motion.
D. It decreases because of conservation of momentum.

Explanation: momentum is conserved since there is no friction force. Hence,

$$P_i = P_f \Rightarrow m_1 u = m_2 v \Rightarrow v = \frac{m_1 u}{m_2}$$

Since $m_2 > m_1$, speed decreases

Answer: D

19. A satellite moves at a constant speed in a circular orbit about the center of the Earth at an altitude half the radius of the Earth above its surface. If g is the gravitational acceleration at the surface of the Earth and R its radius, then what is the speed of the satellite?
A. $\sqrt{2gR}$ B. $\sqrt{gR}$ C. $\sqrt{3gR/2}$ D. $\sqrt{2gR/3}$

Solution: $\frac{mv^2}{r} = \frac{GmM}{r^2}$, $r = R + \frac{R}{2} = \frac{3R}{2}$

$$V^2 = \frac{2GM}{3R}$$

And, $mg = \frac{GmM}{r^2}$

$$g = \frac{4GM}{9R^2} = \frac{2}{3R}\left(\frac{2GM}{3R}\right) = \frac{2}{3R}V^2$$

$$\Rightarrow V = \sqrt{\frac{3gR}{2}}$$

Answer: C

20. An object of mass 1 kg moving with a speed of 2 m/s was acted on by a force that produces an acceleration of 2 m/s^2. What is the change in kinetic energy of the object if the force is exerted on the object for 2 seconds in its direction of motion?
A. 1 J B. 16J C. 8J D. 4J

Solution:

$$V = u + at = 2 + 2(2) = 6$$

$$\Delta KE = \frac{1}{2}mv^2 - \frac{1}{2}mu^2 = \frac{1}{2}(1)\left[6^2 - 2^2\right] = 16J$$

Answer: B

21. In the figure below, what is the speed of water as it leaves A_2?
A. 0.9 m/s
B. 0.3 m/s
C. 3.3 m/s
D. 3.6 m/s

Solution: $A_1 V_1 = A_2 V_2$

$(4)(0.9) = 12(V_2)$

$V_2 = 0.3$ m/s

Answer: B

22. An electric immersion water heater is rated at 400 W. How long will it take to heat one kilogram of water from 10°C? (The specific heat of water is 4.2 J/g K.)
A. 45 min B. 1min C. 15 min D. 3.5 min

Solution: $\Delta Q = mc\,\Delta T$, $c = 4.2\frac{J}{gk} = 4200\frac{J}{kgk}$

$$P = \frac{\Delta Q}{\Delta t} = \frac{(1)(4200)(20)}{\Delta t}$$

Telegram: https://t.me/ExtremeSeries

$$400 = \frac{84000}{\Delta t}$$

$\Delta t = 210 \text{ sec} = 3.5 \text{ min}$ **Answer: D**

23. If the internal energy of the system decreases, then what can be concluded about the heat and the work done?
   A. Heat is removed from the system and work is done by the system.
   B. Heat is added to the system and work is done by the system.
   C. Heat is removed from the system and work is done on the system.
   D. Heat is added to the system and work is done on the system.

Hint: $\Delta u = \Delta Q + \Delta W$ **Answer: A**

24. If a thermodynamic system undergoes a process in which its internal energy increased by 400 J while doing 200 J of work on its surrounding, then what is the energy absorbed in or extracted from it in the form of heat?
   A. −600 J B. −200 J C. 200 J D. 600 J

Solution: $\Delta u = \Delta Q + \Delta W$

$400 = \Delta Q - 200$

$\Delta Q = 600 J$ **Answer: D**

25. Which of the following statement is **NOT** correct?
   A. A system can absorb heat from a hot reservoir and can convert it entirely into work without additional changes in the system and its surrounding.
   B. The spontaneous transfer of energy from a cooler body to hotter body is quite possible.
   C. No process is possible in which there is an overall decrease in the entropy of the universe.
   D. The completer conversion of energy from hot source into work is not possible.

Hint: second law of thermodynamics **Answer: A and B**

26. A heat engine operating between 100 °C and 700 °C has efficiency equal to 40% of the maximum theoretical efficiency. How much energy does this engine extract from the hot reservoir in order to do 5000 J of mechanical work?
   A. 20259.32 J C. 8104. J
   B. 81.0 J D. 20. 3J

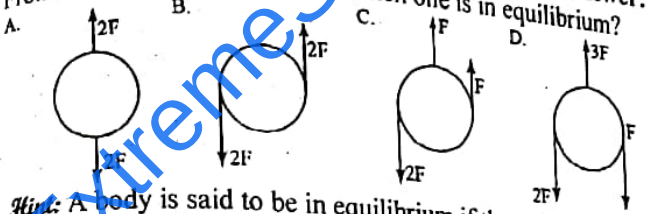
Solution: $\eta = 40\% \eta_{max}$

$$\frac{W}{Q_H} = \frac{40}{100}\left(1 - \frac{T_C}{T_H}\right), \quad T_C = 373k, \quad T_H = 973k$$

$$\frac{5000}{QN} = \frac{40}{100}\left(1 - \frac{373}{973}\right)$$

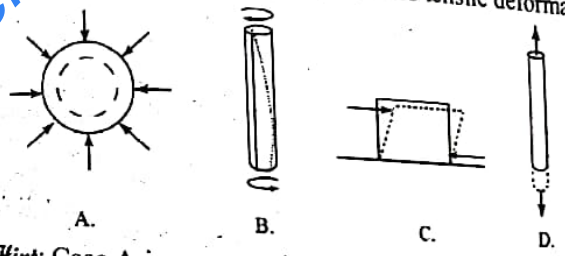
$QH = 20259.32$ J

27. From the figures given below which one is in equilibrium? **Answer: A**



Hint: A body is said to be in equilibrium if there is no net force and net torque acting on it. **Answer: A**

28. Which of the following figure indicates tensile deformation?



Hint: Case A is compression
B is Torsion
C is shear **Answer: D**

29. A beaker with water resting on a scale weighs 40 N. A block suspended on a hanging spring weighs 20 N. The spring scale reads 15 N when a block is fully submerged in the water. What is the reading of a scale on which the beaker with water rests, while the block is submerged in the water after detached from the hanging spring?
   A. 25 N B. 60 N C. 55 N D. 45 N

Hint: Since $S_{block} > S_{water}$ **Answer: B**

30. An object with a mass of 150 kg and a volume of 0.75 m^3 is floating in a liquid of density 0.8 g/cm^3. What percentage of the object's volume will be submerged below the surface of the fluid?

Telegram: https://t.me/ExtremeSeries

A. 100% B. 50% C. 75% D. 25%

Solution: Weight = bouyant force

o, object $\rho_o v_o g = \rho_f v_f g$

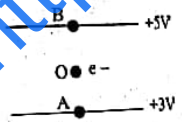
f, fluid $V_f = \frac{\rho_o v_o}{\rho_f} = \frac{150}{800} = 0.1875 m^3$

$$\frac{V_f}{V_o} \times 100\% = \left(\frac{0.1875}{0.75}\right)100\% = 25\%$$ **Answer: D**

31. Why is an interference pattern **NOT** observed between the lights produced from a pair of car headlights?
   A. The light sources are not collimated.
   B. Interference is observed only in laboratory conditions.
   C. The light sources are not coherent.
   D. The spread of light as it travels out of the source. **Answer: C**

32. Suppose that three point charges are placed along a straight line and the electrostatic force exerted on the middle charge is zero. What can we conclude about the charges?
   A. The signs of the three charges should be the same.
   B. The charges at the ends must have the same sign.
   C. The middle charge should be placed at the center between the two charges
   D. The charge at the middle and at the left end should have the same **Answer: B**

33. Two parallel plates are kept at 3 V and 5V as shown in the diagram below. If an electron is released at O, then which way will the electron move?



   A. towards B C. remains at O
   B. towards A D. parallel to the plates

Hint: Attracted by the higher potential **Answer: D**

34. A capacitor of 12μF capacitance can be fully charged after it is connected between a potential difference of 9 V. How much electrical energy is used in charging the capacitor?
   A. 1.0×10^{-3} J C. 9.72×10^{-4} J
   B. 54×10^{-4} J D. 4.86×10^{-4} J

Solution: $u = \frac{1}{2}CV^2 = \frac{1}{2}(12\times10^{-6})(9)^2$

$= 4.86 \times 10^{-4}$ J **Answer: D**

35. A copper wire of cross – sectional area 2 mm^2 carrying a current of 1.5 A. If the drift velocity of the electron is 0.00028 m/s, what is the number of charge carriers (free electrons) per unit volume in the copper wire?
   A. $1.5 \times 10^{25}/m^3$ C. $1.5 \times 10^{32}/m^3$
   B. $1.67 \times 10^{28}/m^3$ D. $1.67 \times 10^{32}/m^3$

Solution: $V_d = \frac{I}{nAq} \Rightarrow n = \frac{I}{V_d Aq}$

$$n = \frac{1.5}{(0.00028)(2\times10^{-6})(1.6\times10^{-19})}$$

$= 1.67 \times 10^{28}/m^3$ **Answer: B**

36. Which of the following statement is correct?
   A. Galvanometer can be converted to voltmeter of different scale by connecting low values resistance in parallel.
   B. Low valued internal resistance increases accuracy of voltmeter.
   C. High valued internal resistance increases the accuracy of ammeter.
   D. Galvanometer can be converted to ammeter of different scale by connecting variety of low valued shunt resistance in parallel. **Answer: D**

37. What makes the oscillatory motion a simple harmonic motion?
   A. The acceleration of the motion is directly proportional to the velocity.
   B. The acceleration of the motion is directly proportional in magnitude but opposite in direction to the displacement.
   C. The velocity of the motion is directly proportional to the displacement.
   D. The velocity of the motion is inversely proportional to the displacement. **Answer: B**

38. The two end fixed string of length 0.5 m has a mass per unit length 2 g/m. If the tension in the string is 80 N, what is the second harmonic frequency?
   A. 400 Hz B. 200 Hz C. $100\sqrt{2}$ Hz D. 600 Hz

Solution: $V = \sqrt{\frac{T}{\mu}}$, $\mu = \frac{m}{\ell} = \frac{29}{m} = \frac{2}{1000}\frac{kg}{m}$

Telegram: https://t.me/ExtremeSeries

$= \sqrt{\frac{80}{0.002}} = 200$

$f_n = \frac{nV}{2L} \Rightarrow f_2 = \frac{V}{L} = \frac{200}{0.5} = 400H_z$

Answer: A

39. Which one of the following is **NOT** the use of Doppler effect?
A. Astronomy
B. Energy transmission
C. Radar
D. Blood flow measurement

Answer: B

40. A 632.8 nm laser light produces an interference pattern on a screen of 4.0 m from a pair of slits. The slit space is equal to 0.2 mm. Calculate the fringe width.
A. 15.3 cm
B. 25.3 mm
C. 6.33 mm
D. 12.7 mm

Solution: $\frac{\lambda}{S} = \frac{W}{D} \Rightarrow W = \frac{\lambda D}{S} = \frac{(632.8\times10^{-9})(4)}{0.2\times10^{-3}}$

$= 12.7 \times 10^{-3}$m

Answer: D

41. A magnetic field of strength 5×10^{-5}T passes through an area of 20 cm^2 that is at an angle of 60° to the magnetic field. What is the magnetic flux?
A. 0 Wb
B. 5×10^{-8} Wb
C. 2.5×10^{-8} Wb
D. 8.7×10^{-8} Wb

Solution: $\emptyset = BA\cos\theta, \quad \theta = 30°$

$= (5\times10^{-5})(20\times10^{-4})(0.87)$

$= 8.7 \times 10^{-8}$ Wb

Answer: D

42. An inductor has a self inductance of 12 *mH*. The current in the circuit decreases at a rat of 2.0 *A/s*. What is the magnitude of the induced electromotive force?
A. 2.4π.V
B. 1.2π.V
C. 1.2×10^{-2} V
D. 2.4×10^{-2} V

Solution: $|\epsilon| = \left|L\frac{\Delta I}{\Delta t}\right|$

$= (12\times10^{-3})(2)$

$\varepsilon = 2.4 \times 10^{-2}$ v

Answer: D

43. Which of the following is correct about the characteristic of current and voltage in a.c. circuits?
A. If the current leads the voltage, its capacitive reactance is greater than inductive reactance.

B. If the current is in phase with the voltage, the circuit is inductive.
C. If the current leads the voltage, it is inductive circuit.
D. If the current lags the voltage, the circuit is capacitive.

Answer: A

44. An *RLC* is designed to have a resistance of 400Ω, capacitive reactance of 350 Ω and inductive reactance of 50Ω. The circuit is connected to an alternative voltage source of effective (rms) value 50 V, and frequency 50 Hz. What is the output average power?
A. 62.5 W
B. 4.0 W
C. 400 W
D. 40 W

Solution: $P_{av} = I^2_{rms}R, \quad I_{rms} = \frac{V_{rms}}{Z}$

$Z = \sqrt{R^2 + (x_L - x_c)^2} = \sqrt{400^2 + 300^2}$

$= 500\ \Omega$

$I_{rms} = \frac{50}{500} = 0.1A$

$P_{av} = (0.1)^2(400) = 4W$

Answer: B

45. Which of the following has an electron volt unit?
A. charge
B. energy
C. potential difference
D. electromagnetic intensity

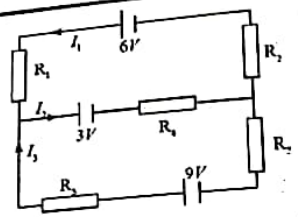
Answer: B

46. In Rutherford experiment, a beam of positively charge alpha particles was projected into a thin piece of gold foil. One of the observation of this experiment is that some charged particles deflected through a large angle (10° to 90°) are those concentrated in one place. What conclusion can be draw from this observation?
A. Most of the mass and all positive charge is in tiny central nucleus
B. Most of the atom is empty space
C. All the atom is positive charge
D. All the atom's negative charge

Answer: A

47. In the circuit shown below, each resistor has a value of 3Ω. What is the value of the current through R_4?
A. 0.75 A
B. 1.0 A
C. 0.625 A
D. 0.125 A

Telegram: https://t.me/ExtremeSeries



Solution: Applying Kirchhoff's rules

Point rule

$I_1 + I_3 = I_2$ ..... (1)

Loop – 1 (counter – clock wise traverse)

$6 - 3 = 3I_1 + 3I_2 + 3I_1$

$2I_1 + I_2 = 1$ .....(2)

Loop – 2 (clock – wise traverse)

$9 - 3 = 3I_2 + 3I_3 + 3I_3$

$2I_3 + I_2 = 2$ .....(3)

Solving the above equations, we get

$I_2 = 0.75$ A **Answer: A**

48. Suppose you obtain a piece of iron from your laboratory shelf. How can you experimentally determine, for sure, that it is magnetized?

A. If it attracts another magnet
B. If it diverges a changed electroscope
C. If it repels another magnet
D. There is no single and sure way to determine its magnetization. **Answer: C**

49. If a charged particle enters into a uniform magnetic filed perpendicular to its direction, then what will happen?

A. It moves in an elliptical path
B. The radius of its path is proportional to the magnetic field strength
C. The charged particle will not be affected by magnetic field
D. The period of the motion is inversely related to the magnetic field strength

Solution: $\frac{mV^2}{r} = qvB$

$\frac{mV}{r} = qB \quad , V = \frac{qBr}{m}$

$T = \frac{2\pi r}{V} = \frac{2\pi m}{qB}$ **Answer: D**

50. Two parallel wires each carrying 5 A current in the same direction are separated by 2.5 cm. What is the magnitude of the force per unit length acting on the wires? Is the force attractive or repulsive?

A. 2×10^{-4} N/m, attractive
B. 4×10^{-6} N/m, repulsive
C. 8×10^{-5} N/m, repulsive
D. 5×10^{-6} N/m, attractive

Solution: $F = \frac{\mu_0 I_1 I_2 \ell}{2\pi r}$

$\frac{F}{\ell} = \frac{\mu_0 I_1 I_2}{2\pi r} = \frac{4\pi \times 10^{-7}(5)(5)}{2\pi(2.5 \times 10^{-2})}$

$= 2 \times 10^{-4}$ N/m **Answer: A**

Telegram: https://t.me/Extremeseries

SECTION 4 – BIOLOGY

The questions and answers grouped in to two parts. Part one question from Grade 11 are arranged according to the units of the lesson = 42 questions part two questions from Grade 12 topics = 58 questions.

1. Which of the following steps in the scientific method comes following observation?
   A. Prediction
   B. Hypothesis
   C. Experiment
   D. Theory

Explanation:
Scientific method is a process in which scientists can approach to their work.

- This scientific method consists of several steps.
- These steps are:
  1) Asking questions about the problem.
  2) Observation/ doing the background research.
  3) Construct the hypothesis and can made a prediction.
  4) Experiment to test the hypothesis.
  5) Drawing the conclusion.
  6) Making the theory.

Answer: B Hypothesis comes after observation.

2. Which of the following field equipment is used to determine an exact location of a place?
   A. Theodolite
   B. Global positioning system
   C. Filed microscope
   D. Field pH kit

Explanation:
Biologists can grouped the basic tools (equipment) into two. These are, the laboratory equipments and field equipments.
From field equipments:

- Theodolite is used to measure the height of trees and slopes of the area.
- Field microscope is used to investigate the structure of specimen in the field.
- Field pH kit is used to measure the pH of soil or water.

Answer: B – Global positioning system (GPS) helps to makes it possible to record positions or location of a place.

3. What is a theory in biology?
   A. The outcome of the hypothesis is true.
   B. A proof that shows the hypothesis is true.

   C. A hypothesis supported by experiments and/or observations.
   D. An opinion or educated guess resulting from observations.

Explanation:
Theory in biology which can be followed those the scientific method steps.

- The most successful hypothesis become a ruling hypothesis and it may become theory.
- As a knowledge increases and techniques of investigation improve and even well-established theories.

Answer: C – A hypothesis supported by experiment and/or observation.

4. For which of the following is a theodolite used in biology education?
   A. Measuring the height of trees.
   B. Recording position where a species is found.
   C. Measuring the rate of flow of water in a cell.
   D. Measuring the pH of water or soil.

Explanation:
From biological field equipment:
1) Global positioning system (GPS) used to record the position where a species is found.
2) A flow meter equipment used to measure the rate of flow of water.
3) A pH kit is used to measure the pH of water of soil .

Answer: A – theodolite is used to measuring the height of trees or stopes of area.

5. Which of the following is a recently active area of research in biology today?
   A. Taxonic study
   B. Study about the cell theory
   C. Ecological research
   D. Stem cell research

Explanation:
Biology is the science of life and living organisms. Biologists engage in all kinds of research to try to find evidence of life on earth and other planets. To find out this evidence biology has many branches. From three branches taxonomy, cell theory and

Telegram: https://t.me/Extremeseries

ecology and their researches were earlier studies in the field of biology.

Answer: D – The stem cell research is the recently active in biology to day.

6. How many carbon atoms are there in one disaccharide molecule?

A. 6 B. 12 C. 18 D. 24

Explanation:

Biological molecules can be classified into two main groups. These are in organic and organic molecules.

- In organic molecules many contain carbon or hydrogen but not both elements.
- Organic molecules contain different combinations of carbon and hydrogen.
- Carbohydrates is one of organic molecules which is made up of carbons, hydrogen and oxygen.
- The ration of these elements are 1: 2 = 1 (CH_2O).

The different types of carbons are:

- *Monosaccharide (a single sugar molecule)* which contain from 3 – 6 carbon molecules.
- *Disaccharide molecules* are made up of two monosaccharide molecules joining together.
- *Polysaccharide molecules* are built by many hundreds of monosaccharide molecules join together.

Answer: B – Disaccharide molecules are the combination of two mono saccharide molecules.

$C_6H_{12}O_6 + C_6H_{12}O_6 \longrightarrow C_{12}H_{22}O_{11} + H_2O$

There are 12 carbon atoms in disaccharide molecule.

7. Which energy rich organic compound contains adenine in its molecule.

A. Lipids C. Carbohydrate

B. Glucose D. ATP

Explanation:

The most organic compounds are carbohydrate, lipids, proteins and nucleic acid.

- All carbohydrates contain three elements carbon, hydrogen and oxygen.

- All lipids also contain three elements carbon, hydrogen and oxygen. Their difference is the proportion of oxygen is very less in lipids.
- Proteins are made up of carbon, hydrogen oxygen nitrogen and sometimes contain sulfur.
- Nucleic acids are grouped in to DNA and RNA. Both are made up of nucleotides which have the same three compounds.

These components are:

- Phosphate group
- Pentose sugar (deoxyribose and ribose sugar)
- Nitrogenous base

ATP is energy rich organic compound made up of phosphate groups, ribose sugar and adenine from nitrogenous base.

Answer: D – ATP

8. All proteins contain carbon, hydrogen, oxygen and what other elements?

A. Chlorine B. Florine C. Nitrogen D. Sulphur

Explanation:

Proteins are the most important substances that are needed to form all living cells. Proteins are contain the elements carbon, hydrogen, oxygen like carbohydrate and lipids, but they also contain nitrogen and most contain sulphur. Plants take nitrate from the soil to make their own proteins. Animals can lat plants to make their own protein.

Answer: C – All proteins contain nitrogen rather than sulphur.

9. What makes unsaturated fatty acids different from saturated fatty acids?

A. The presence of long chain of carbon.

B. The presence of one or more double bond.

C. The presence of large number of hydrogen atoms.

D. Their occurrence as solid at room temperature.

Explanation:

- Fatty acid molecules consists of a covalently bonded hydrocarbon chain at the end of carboxyl group.
- This carboxyl group has acidic properties.
- Saturated fatty acids are the carbon – carbon bonds in the hydrocarbon chain are single bonds.

Telegram: https://t.me/Extreme Series

- Unsaturated fatty acids contain one or more carbon – carbon bonds are double bonds. **Answer: B**

10. What is the name of the sugar found in milk?
A. Glucose B. Lactose C. Maltose D. Sucrose

Explanation:

- Glucose is the most common monosaccharide which is made up during photosynthesis and used in respiration to produce energy.
- Maltose is a disaccharide formed by the combination of two glucose molecule.
- Sucrose is made of one glucose molecule and one fructose molecule – which is a disaccharide.

Glucose + fructose → sucrose (sugar cane)

Answer: B – Lactose is a disaccharide found in milk which is formed by the combinations of one glucose molecule with one galactose molecule glucose + galactose → lactose (milk sugar).

11. What can one conclude about the contents of the food staff if a mixture of Benedict solution and a food staff remain blue after heating?
A. It contains proteins C. It contain fats
B. It lacks starch D. It lacks reducing sugar

Explanation:

Different types of food staffs are tested by different types of chemical substances. Example 1: Carbohydrates are tested by Benedict (Fehling's solution) and iodine (potassium iodide).

a) *Reducing sugars – (glucose)*
The reducing sugars include all mono saccharide such as glucose, fructose and some disaccharide such as maltose. These reducing sugars are tested by Benedict solution which contain copper sulphate. Reducing sugars reduce soluble blue copper sulphate containing copper (II) ions (Cu^{2+}) to insoluble red brown copper oxide containing copper (I).

b) *Non reducing - (sucrose)*
The only common non reducing sugar is sucrose which is disaccharide. This sucrose is hydrolyzed to glucose and fructose by boiling with diluted hydrochloric acid and then tested by Benedict solution. The result is similar to reducing agents.

c) *Starch*

Starch is a polymer of a glucose. It is a major fuel of plants and not found in animals. Starch is tested by using of iodine or potassium iodide and observed a blue black colouration.

Example – 2 – lipids: Lipids include oils (olive oil) and fats. To test lipids, we can use emulsion test and grease spot test. Emulsion test is adding of oil or fat and ethanol in test tube and then add cold water, the result is a cloudy white suspension. Grease spot test is rub a drop of the sample (oil) into a place of paper. The result is a permanent transparent spot on the paper.

Example – 3 – proteins – A soluble protein such as egg albumin is tested by:

1) *Million's test*
Add protein solution (egg abbamin) in to a test tube and add million's regent and boil. The result is a white suspensioin (precipitate) forms which coagulates on heating and turns red or salmon pink color.

2) *Biuret test*
Adding protein solution (egg albamin) to a test tube add equal volume of potassium hydroxide solution and mix. Add two drops of copper sulphate and mix without heating. The result is purple color develops slowly. Therefore the answer is – D. It lacks reducing sugar. Because the Benedict's solution which contain copper (II) an not reduced into copper (I).

12. In which of its contents RNA differs from DNA?
A. Deoxyribose and guanine
B. Ribose and uracil
C. Ribose and thymine
D. Phosphate and adenine

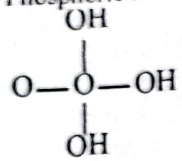
Explanation:

DNA and RNA are the nucleic acids of organic compounds. These nucleic acids are made up of nucleotides. A nucleotide has three components:

1) Pentose (five carbon) sugar
In DNA deoxyl – ribose sugar reducing of one oxygen on the second carbon chain.
In RNA – ribose sugar contains oxygen on the second carbon.

Telegram: https://t.me/ExtremeSeries

2) Phosphoric acid (phosphate)



3) Nitrogenous bases:
Each nucleic acid (RNA & DNA) contains four different bases. Two of them derived from purine and two of them derived from pyrimidine's. Purines have two rings and pyrimidine have one ring in their structure. The two purine are Adenine (A) and Guanine (G). The two pyrimidine are Thymine (T) and cytosine (c). In RNA instead of thymine it replaced by uracil. Therefore the answer is B – RNA is differs from DNA by its Ribose sugar and uracil its nitrogenous bases.

13. Which of the following groups of substances are all in organic?
A. Water, sugar, calcium, carbonate
B. Sugar, fatty acid, amino acid
C. Water, calcium carbonate, carbon dioxide
D. Carbon dioxide, amino acid, fatty acid

Explanation:
Biological molecules can be classified in to two main types. These are inorganic molecules and organic molecules. In organic molecules are contain carbon or hydrogen. Example: H_2O no carbon (water), CO_2 (no hydrogen), $CaCO_3$. Organic molecules contain both carbon and hydrogen elements which join together. Example – Sugar ($C_{12}H_{22}O_{11}$) fatty acid ($C_{17}H_{35}COOH$), amino acid ($NH_2 - CH_2 - COOH$). Then choice letter A contains sugar which is organic. Choice letter B, all are organic molecules. Choice letter D amino acid and fatty acids are organic molecules. **Therefore, the answer is C** all are inorganic molecules.

14. In which of the following compounds are both members of the pair are polymers of carbohydrates that naturally occur in plants?
A. Starch and chitin
B. Starch and cellulose
C. Glycogen and cellulose
D. Glycogen and chitin

Explanation:
Polymers are a complex organic molecules which are formed from monomers (simple molecules). The polymers of carbohydrates are:

- Starch is a polymer of alpha (α) glucose and stored in plants.
- Cellulose is a polymer of Beta (β) glucose all plant cell walls are made up of cellulose.
- Glycogen is the animal equipment to starch which is made up of alpha (α) glucose.
- Chitin – is closely related to cellulose in structure and function. It is occur in some fungi in some animals particularly in Arthropods. Then choice letter A chitin is not found in plants. Choice letter C glycogen is not found in plants. Choice letter D both glycogen and chitin are not occur in plant cell. Therefore, the answer is letter B, both starch and cellulose are naturally occur in plants.

15. Which of the following is generally expected to give better health benefit when present in human diet?
A. Animal fat
B. Saturated fatty acid
C. Mono unsaturated fatty acid
D. Poly unsaturated fatty acid

Explanation:

- Approximately 95 percent of the lipids in human diet are triglycerides (three fatty acids attached to a glycerol molecule).
- The remaining 5 percent are mainly cholesterol and the phospholipid.
- *Fatty acids* are divided into salurated (single bonds) and unsaturated (double bonds).
- Saturated fatty acids are contain too much of hydrogen than unsaturated fatty acid.
- *Unsaturated fatty acids* have only one double bond is mono saturated and more than one double bond are poly unsaturated fatty acids. Saturated fatty acids are solid at room temperature mostly found in animal body. Unsaturated fatty acids are liquids at room temperature and found in plant cells. They are very important used to

Telegram: https://t.me/Extremeseries

present cholesterol being laid down in the lining of arteries and helps to prevent heart disease.

- *Animal fat and saturated fatty acids* are contain to much of hydrogen and can not give better health benefits because they are solids. Mono unsaturated fatty acid is found in plans and do not reduce hydrogen. **Therefore the answer is – D** – *poly unsaturated fatty acid* has to give better health benefit, because of having less amount of hydrogen which forms oil in plants.

16. Which of the following classes of enzymes digests carbohydrates?
A. Amylase
B. Lipades
C. Protease
D. Nucleases

Explanation:
- Lipases are enzymes that acts on lipids.
- Protease are enzymes that acts on proteins.
- Nucleases may catalyze nucleic acids.

Therefore, the answer must be A amylase because Amylase involve in the digestion of carbohydrates.

17. One of the following molecules is building units of an enzyme molecule.
A. Amino acids
B. Glucose
C. Nucleic tides
D. Fatty acids

Explanation:
- All enzymes are proteins.
- Glucose in monomer of carbohydrate.
- Nucleotide is nucleic acid and fatty acids are lipids do not contain enzymes.

Therefore, **the answer A**. Because amino acid is a monomers of protein.

18. Which of the following pairs of molecules are known to have catalytic activity?
A. Lipids and proteins
B. Carbohydrates and proteins
C. Proteins and RNAs
D. Proteins and DNAs

Explanation:
- Catalyst is a substance that speeds up a chemical reactions and remains unchanged at the end of the reaction.

- An enzyme is a globular protein with a uniquely shaped active site and acts as biological catalyst for a specific reaction. Lipids carbohydrates and DNA molecules have no catalytic activity in biological reactions. **Therefore answer is C** proteins and RNAs are biological catalysts in biological reactions.

19. To which class of enzymes do the digestive enzymes belong?
A. Esterase's
B. Transferases
C. Hydrolases
D. Isomerizes

Explanation:
Classification of also based on their catalytic reactions. For example:
1) Esterase's enzyme acts on the hydrolysis of lipids to break ester bond.
2) Transfereses – catalyze the transfer of a chemical group from one compound to another compound.
3) Isomerases – catalyze the transformation of one isomer in to another. For instance the conversion of glucose 1, 6 – diphosphate into fructose 1, 6 – diphosphate.
4) Hydorlases – These catalyze hydrolysis (splitting of molecules by use of water) reactions. Most of digestive enzymes are hydrolases.

Therefore, the answer is C hydrolases.

20. Which of the following types of enzyme inhibition can be removed when the end product of the metabolic pathway is depleted?
A. Allosteric inhibition
B. Non reversible inhibition
C. Completive inhibition
D. Reversible competitive inhibition

Explanation:
- Inhibitors are substances which slow down or stop enzyme action that binds to enzymes and prevent them from forming enzyme – substrate complexes. The two main types of inhibitors are:
  - Irreversible inhibitors and
  - Reversible inhibitors

Telegram: https://t.me/ExtremeSeries

- Irreversible (nonreversible) inhibitors are bend strongly and the bond that helds them breaks easily and releasing the inhibitors.

1. Competitive inhibitors is a molecule that inhibits enzyme actively by competing with the substrate for the active site.
2. Noncompetitive inhibitor is a molecule that alters the conformation of the active site by binding with the allosteric site of the enzyme.
   - Allosteric inhibitors are designed to change the shape allo means different and steric shape.
   - Allosteric inhibitors are regulated by compounds which act as non – competitive inhibitors.

   These noncompetitive inhibitors produced in a cell as a result of metabolic bath ways in cells. Therefore, the answer is A allosteric inhibitors, because it regulates by noncompetitive inhibitors.

21. The main component of the plant cell wall is:
A. Starch B. Cellulose C. Protein D. Chitin

Explanation:

- Starch is the stored polymers in the internal part of plant cells.
- Proteins are important to form the structure of plasma membrane, immune system, the structure of chromosomes, enzymes etc.
- Chitin has similar function to plant cellulose which is found in fungi.
- Many cellulose molecules lies side by side by hydrogen bond to each other.
- Cellulose has fibrous in nature that give the cell walls their strength.

Therefore the answer is B – Because plant cell wall is made up of cellulose.

22. Which one of the following is the main constituent of biological membranes?
A. Phospholipids C. Glycolipids
B. Glycoproteins D. Cholesterols

Explanation:

Biological membrane or cell membranes separate the content of cells from their external environment. It is used to controlling exchange of materials is and out from the cell.

Cell membranes are contain:

1. Phospholipids bilayer consists of phosphate glycerol and two fatty acid molecules.
   - The phosphate groups are hydrophilic polar heads.
   - The glycerol and fatty acid molecules are hydrophobic non polar tail sides of bilayer.
2. Protein molecules are consists of integral proteins and glycoproteins (the combination of glycogen and protein) molecules.
3. Glycolipids are molecules which contain lipids combined with carbohydrates.
4. Cholesterol is closely related to lipids and is slightly polar at one end.

Therefore, the answer is A, because for all biological membrane the main constituents are phospholipid bilayers.

23. Which unit is the best to use for measuring the smallest cells and organelles?
A. Micro meter C. millimeter
B. Milliliter D. Nanometer

Explanation:

The measuring units commonly used to describe microscope objects are the micrometer (mm) and the nanometer (nm).

- Starting with similar unit, the millimeter (mm).
- One thousands (1/100) of one millimeter is one micrometer (μm).
- One thousand (1/100) micrometer or one millions (1/1000,000) millimeter is one nanometer. The micrometer is used to describe cells and organelles in the microscope.

Therefore the answer is – A – micrometer.

24. If a suspension of a mixture of cellular organelles is spun in a centrifuge which organelle settles to the bottom first:
A. Mitochondria C. Chloroplasts
B. Nuclei D. Ribosomes

Telegram: https://t.me/ExtremeSeries